

Project Data Page:





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DESIGN CALCULATION

In Accordance with ASME Section VIII Division 1

ASME Code Version : 2025

Analysis Performed by : SL User

Job File : ...\\401-V-32004 - Copy - 7.pvdb

Date of Analysis : Jul 7, 2026 2:00pm

PV Elite 28 U1, March 2026

Vessel Design Summary:

ASME Code, Section VIII Division 1, 2025

Diameter Spec : 2200.000 x 2244.000 mm. OD
 Vessel Design Length, Tangent to Tangent 6701.60 mm.
 Distance of Bottom Tangent above Grade 6300.00 mm.
 Distance of Base above Grade 300.00 mm.
 Specified Datum Line Distance 6000.80 mm.
 Internal Design Temperature 80 °C
 Internal Design Pressure 23.000 bars
 External Design Temperature 80 °C
 External Design Pressure 1.013 bars
 Maximum Allowable Working Pressure 24 bars
 External Max. Allowable Working Pressure 4 bars
 Shop Test Pressure 31.048 bars
 Required Minimum Design Metal Temperature -38.0 °C
 Warmest Computed Minimum Design Metal Temperature -39.0 °C
 Wind Design Code IS 875 (Part 3):2015 Amd. 1&2 2020
 Earthquake Design Code Response Spectrum

Materials of Construction:

Component Type	Material	Class	Thickness	UNS #	Material Treatment Options
Shell	SA-516 70	...	...	K02700	+IT
Head	SA-516 70	...	...	K02700	+IT
Skirt	SA-285 C	...	...	K02801	
Skirt	SA-516 70	...	...	K02700	
Nozzle	SA-333 6	...	...	K03006	+IT
Nozzle	SA-350 LF2	1	...	K03011	+IT
Nozzle	SA-516 70	...	...	K02700	+IT
Re-Pad	SA-516 70	...	...	K02700	+IT
Nozzle Flg	SA-350 LF2	1	...	K03011	+IT
Basing	SA-285 C	...	...	K02801	
Base Bolting	SA-36	...	...	K02600	

+FG - Fine Grained, +R - rolled, +T - Tempered, +N - Normalized
 +Q - quenched, +CBV - Cb & V added, +IT - Impact tested

Element Pressures and MAWP (bars & mm.):

Element Description or Type	Design Pressure + Stat. head	Ext. Press.	Element M.A.W.P	Total Corrosion Allowance	Str. Flg. Gov.	In Creep Range
Skirt 3	...	...	...	1.4000	N/A	No
Skirt 2	...	...	...	1.4500	N/A	No
Skirt 1	...	...	...	1.2500	N/A	No
bottom D end	23.271	1.01	24.000	1.5000	No	No
Shell 1	23.231	1.01	25.000	1.5000	N/A	No
Shell 2	23.112	1.01	25.000	1.5000	N/A	No
Shell 3	23.000	1.01	25.000	1.5000	N/A	No
Top D end	23.000	1.01	24.000	1.5000	No	No

Liquid Level: 4100.80 mm. Dens.: 0.001 kg./cm³ Sp. Gr.: 0.673

Element Types and Properties:

Element	"To" Elev	Element	Nominal	Finished	Reqd Thk	Reqd Thk	Long	Circ
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Type	mm.	Length mm.	Thickness mm.	Thickness mm.	Internal mm.	External mm.	Eff	Eff
Skirt	-3500.8	2500.0	12.0	12.0	...	...	0.70	0.70
Skirt	-1000.8	2500.0	12.0	12.0	...	...	0.70	0.70
Skirt	-0.8	1000.0	12.0	12.0	...	...	0.70	0.70
Ellipse	50.0	50.8	25.0	21.0	20.1	7.2	1.00	1.00
Cylinder	1850.0	1800.0	22.0	22.0	20.2	13.0	1.00	1.00
Cylinder	4050.0	2200.0	22.0	22.0	20.1	13.0	1.00	1.00
Cylinder	6650.0	2600.0	22.0	22.0	20.1	13.0	1.00	1.00
Ellipse	6700.8	50.8	25.0	21.0	19.9	7.2	1.00	1.00

Local Stress Analysis Results:

Description	Analysis Type	Max Stress Ratio	High Stress Location	Pass Fail
N6_3"	WRC-107/537	0.857	n/a	Passed
N4_8"	WRC-297	0.906	Shell	Passed
N7_3"	WRC-297	0.906	Shell	Passed
N5_3"	WRC-297	0.906	Shell	Passed
N3_1"	WRC-297	0.906	Shell	Passed
N1_16"	WRC-297	0.899	Shell	Passed
N2_16"	WRC-107/537	0.847	n/a	Passed

Wind/Earthquake Shear, Bending:

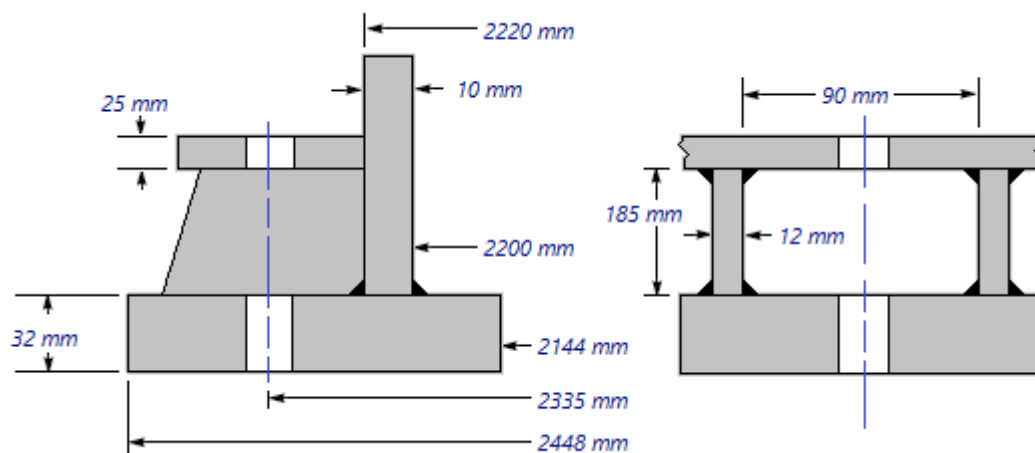
From	To	Distance to Support mm.	Cumulative Wind Shear Kgf	Earthquake Shear Kgf	Wind Bending Kg-m.	Earthquake Bending Kg-m.
10	20	1250	11138	7408.37	95699.4	79916.3
20	30	3750	9441.49	7366.95	69974.5	61456.6
30	40	5500	8379.67	7298.52	47697.6	43134.3
40	50	6025.4	7954.94	7180.59	39530.2	35892.4
50	60	6950.8	7931.83	6673	39126.6	35530.5
60	70	8950.8	6496.83	5589.91	26140.6	23569.7
70	80	11350.8	5495.2	4305.73	12949.1	11283.4
80	90	12676.2	3638.22	1738.14	1075.43	88.2994

Abs Max of the all of the Stress Ratio's : 0.4639

Basing Data : Continuous Top Ring W/Gussets

Thickness of Basing	32.0000	mm.
Inside Diameter of Basing	2144.0000	mm.
Outside Diameter of Basing	2448.0000	mm.
Nominal Diameter of Bolts	30.0000	mm.
Diameter of Bolt Circle	2335.0000	mm.
Number of Bolts	20	
Thickness of Gusset Plates	12.0000	mm.
Average Width of Gusset Plates	102.0000	mm.
Height of Gussets	185.0000	mm.
Distance between Gussets	90.0000	mm.
Thickness of Top Plate or Ring	25.0000	mm.
Radial Width of the Top Plate	120.0000	mm.
Number of Gussets	40	

Basing Sketch for Cylindrical Skirts:



Basing Cross Section View

Loads for Foundation/Support Design:

Total Wind Shear on Support	11138. Kg
Total Earthquake Shear on Support	7408. Kg
Wind Moment on Support	101428. Kg-m.
Earthquake Moment on Support	85645. Kg-m.
Maximum Initial Bolt Load	6653. Kg

Wind and Earthquake moments include the effects of user defined forces and moments if any exist in the job and were specified to act (compute loads and stresses) during these cases. Also included are moment effects due to eccentric weights if any are present in the input.

Weights:

Fabricated - Bare w/o removable internals	19725.7 kg.
Shop Test - Fabricated + water (full)	49536.2 kg.
Shipping - Fab. + rem. intls.+ shipping app.	39134.3 kg.
Erected - Fab. + rem. intls.+ insul. (etc)	40694.3 kg.
Empty - Fab. + intls. + details + wghts.	40694.3 kg.
Operating - Empty + operating liquid (no ca)	51986.5 kg.

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Nozzle Calculation Summary (bars & mm.):

Description	MAWP	Ext	MAPNC	UG-45	[tr]	Weld Path	Areas or Stresses
N6_3"	24	OK	...	OK	6.30	OK	Passed
N4_8"	24	OK	...	OK	8.66	OK	Passed
N7_3"	25	OK	...	OK	6.30	OK	Passed
N5_3"	25	OK	...	OK	6.30	OK	Passed
K1B_3"	25	OK	...	OK	7.23	OK	Passed
K4B_3"	25	OK	...	OK	7.23	OK	Passed
K3F_3"	25	OK	...	OK	7.23	OK	Passed
K2B_1.5"	25	...	...	OK	6.02	OK	No Calc[*]
K3B_3"	25	OK	...	OK	7.23	OK	Passed
K3D_3"	25	OK	...	OK	7.23	OK	Passed
N8_2"	25	...	...	OK	6.30	OK	No Calc[*]
K2A_1.5"	25	...	...	OK	6.02	OK	No Calc[*]
K3A_3"	25	OK	...	OK	7.23	OK	Passed
K3C_3"	25	OK	...	OK	7.23	OK	Passed
K1A_3"	25	OK	...	OK	7.23	OK	Passed
K4A_3"	25	OK	...	OK	7.23	OK	Passed
K3E_3"	25	OK	...	OK	7.23	OK	Passed
MH1_24"	24	OK	...	OK	9.83	OK	Passed
N3_1"	25	...	...	OK	4.92	OK	No Calc[*]
N1_16"	25	OK	...	OK	9.83	OK	Passed
N2_16"	24	OK	...	OK	9.83	OK	Passed

Nozzle MAWP Summary:

Minimum MAWP Nozzles : 24 Nozzle : MH124"
 Minimum MAWP Shells/Flanges : 24 Element : bottom D end
 Minimum MAPnc Shells/Flanges : 26 Element : Top D end

Computed Vessel M.A.W.P. : 24 bars

[*] - This was a small opening and the areas were not computed or the MAWP of this connection could not be computed because the longitudinal bending stress was greater than the hoop stress.

Note: MAWPs (Internal Case) shown above are at the High Point.

Multiple output lines for the same nozzle indicates required Code calculations in both the longitudinal and circumferential planes of reinforcement where applicable.

Check the spatial relationship between the nozzles:

From Node	Nozzle Description	Y Coordinate mm.	Layout Angle deg	Dia. Limit mm.
40	N6_3"	0.000	0.000	157.130
50	N4_8"	400.800	0.000	399.700
50	N7_3"	500.800	180.000	157.130
50	N5_3"	500.800	90.000	157.130
50	K1B_3"	150.800	90.000	158.400
50	K4B_3"	150.800	180.000	158.400
50	K3F_3"	150.800	270.000	158.400
50	K2B_1.5"	150.800	120.000	111.104
50	K3B_3"	150.800	145.000	158.400
50	K3D_3"	150.800	215.000	158.400
50	N8_2"	250.800	250.000	125.328
60	K2A_1.5"	3650.800	0.000	111.104
60	K3A_3"	3650.800	90.000	158.400
60	K3C_3"	3650.800	180.000	158.400
60	K1A_3"	3650.800	120.000	158.400
60	K4A_3"	3650.800	160.000	158.400
60	K3E_3"	3650.800	270.000	158.400
70	MH1_24"	5800.800	0.000	1225.200
70	N3_1"	5150.801	90.000	94.848
70	N1_16"	4750.801	180.000	778.800



80	N2_16"	0.000	0.000	778.800
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The nozzle spacing is computed by the following:

$= \sqrt{ll^2 + lc^2}$ where

ll - Arc length along the inside vessel surface in the long. direction.

lc - Arc length along the inside vessel surface in the circ. direction

If any interferences/violations are found, they will be noted below.

No interference violations have been detected!

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Nozzle Schedule:

Description	Nominal or Actual Size	Schd or FVC Type	Flg Type	Nozzle O/Dia in	Wall Thk mm	Reinforcing Pad Diameter	Pad Thk mm	Cut Length mm	Flg Class
N3_1"	1.000 in	Actual	LWN	2.120	14.224	...	...	222.35	300
K2B_1.5"	1.500 in	Actual	LWN	2.760	16.002	...	...	230.52	300
K2A_1.5"	1.500 in	Actual	LWN	2.760	16.002	...	...	230.52	300
N8_2"	2.000 in	Actual	LWN	3.320	16.764	...	...	229.18	300
N6_3"	3.000 in	80	WNF	3.500	7.620	140.00	22.00	239.77	300
N7_3"	3.000 in	80	WNF	3.500	7.620	140.00	22.00	241.17	300
N5_3"	3.000 in	80	WNF	3.500	7.620	140.00	22.00	241.17	300
K1B_3"	3.000 in	Actual	LWN	4.620	20.574	...	...	223.59	300
K4B_3"	3.000 in	Actual	LWN	4.620	20.574	...	...	223.59	300
K3F_3"	3.000 in	Actual	LWN	4.620	20.574	...	...	223.59	300
K3B_3"	3.000 in	Actual	LWN	4.620	20.574	...	...	223.59	300
K3D_3"	3.000 in	Actual	LWN	4.620	20.574	...	...	223.59	300
K3A_3"	3.000 in	Actual	LWN	4.620	20.574	...	...	223.59	300
K3C_3"	3.000 in	Actual	LWN	4.620	20.574	...	...	223.59	300
K1A_3"	3.000 in	Actual	LWN	4.620	20.574	...	...	223.59	300
K4A_3"	3.000 in	Actual	LWN	4.620	20.574	...	...	223.59	300
K3E_3"	3.000 in	Actual	LWN	4.620	20.574	...	...	223.59	300
N4_8"	8.000 in	80	WNF	8.625	12.700	350.00	22.00	245.74	300
N1_16"	16.000 in	Actual	WNF	16.000	10.000	812.80	22.00	124.41	300
N2_16"	16.000 in	Actual	WNF	16.000	10.000	700.00	22.00	415.04	300
MH1_24"	24.000 in	Actual	WNF	24.787	10.000	1100.00	22.00	229.38	300

General Notes for the above table:

The cut length is the outside projection + inside projection + drop + in-plane shell thickness. This value does not include weld gaps, nor does it account for shrinkage.

In the case of oblique nozzles, the outside diameter must be increased. The re-pad width around the nozzle is calculated as follows:

Width of Pad = (Pad Outside Dia. (per above) - Nozzle Outside Dia.)/2

For hub nozzles, the thickness and diameter shown are those of the smaller and thinner section.

Nozzle Material and Weld Fillet Leg Size Details (mm.):

Description	Material	Shl Grve Weld	Noz Shl/Pad Weld	Pad OD Weld	Pad Grve Weld	Inside Weld
N3_1"	SA-350 LF2	22.000	10.000	...	...	...
K2B_1.5"	SA-350 LF2	22.000	10.000	...	...	...
K2A_1.5"	SA-350 LF2	22.000	10.000	...	...	...
N8_2"	SA-350 LF2	22.000	10.000	...	...	...
N6_3"	SA-333 6	21.000	10.000	18.000	22.000	...
N7_3"	SA-333 6	22.000	10.000	18.000	22.000	...
N5_3"	SA-333 6	22.000	10.000	18.000	22.000	...
K1B_3"	SA-350 LF2	22.000	10.000	...	...	...
K4B_3"	SA-350 LF2	22.000	10.000	...	...	...
K3F_3"	SA-350 LF2	22.000	10.000	...	...	...
K3B_3"	SA-350 LF2	22.000	10.000	...	...	...
K3D_3"	SA-350 LF2	22.000	10.000	...	...	...
K3A_3"	SA-350 LF2	22.000	10.000	...	...	...
K3C_3"	SA-350 LF2	22.000	10.000	...	...	...
K1A_3"	SA-350 LF2	22.000	10.000	...	...	...
K4A_3"	SA-350 LF2	22.000	10.000	...	...	...
K3E_3"	SA-350 LF2	22.000	10.000	...	...	...
N4_8"	SA-333 6	22.000	10.000	18.000	22.000	...
N1_16"	SA-516 70	22.000	10.000	18.000	22.000	...
N2_16"	SA-516 70	21.000	10.000	18.000	22.000	...
MH1_24"	SA-516 70	22.000	10.000	18.000	22.000	...

Note: The Outside projections below do not include the flange thickness.

Nozzle Miscellaneous Data:



Description	Elev/Distance From Datum mm.	Layout Angle deg	Proj Outside mm.	Proj Inside mm.	Installed in Component
N3_1"	5150.000	90.0	200.02	0.00	Shell 3
K2B_1.5"	150.000	120.0	207.96	0.00	Shell 1
K2A_1.5"	3650.000	0.0	207.96	0.00	Shell 2
N8_2"	249.999	250.0	206.38	0.00	Shell 1
N6_3"	...	0.0	218.27	0.00	bottom D end
N7_3"	499.999	180.0	218.27	0.00	Shell 1
N5_3"	499.999	90.0	218.27	0.00	Shell 1
K1B_3"	150.000	90.0	200.02	0.00	Shell 1
K4B_3"	150.000	180.0	200.02	0.00	Shell 1
K3F_3"	150.000	270.0	200.02	0.00	Shell 1
K3B_3"	150.000	145.0	200.02	0.00	Shell 1
K3D_3"	150.000	215.0	200.02	0.00	Shell 1
K3A_3"	3650.000	90.0	200.02	0.00	Shell 2
K3C_3"	3650.000	180.0	200.02	0.00	Shell 2
K1A_3"	3650.000	120.0	200.02	0.00	Shell 2
K4A_3"	3650.000	160.0	200.02	0.00	Shell 2
K3E_3"	3650.000	270.0	200.02	0.00	Shell 2
N4_8"	399.999	0.0	218.27	0.00	Shell 1
N1_16"	4750.000	180.0	83.47	0.00	Shell 3
N2_16"	...	0.0	383.67	0.00	Top D end
MH1_24"	5800.000	0.0	161.38	0.00	Shell 3



Bill of Materials:

QTY	DESCRIPTION	MATERIAL
2	ELLIPTICAL HEAD: 2.0 X 1, 25.0mm. THK X 2200.0mm. ID X 50.8mm.	SA-516 70
3	PLATFORM: 1200mm X 1500mm WIDE	
2	INSULATION: 600mm X 70mm THK	CC
1	INSULATION: 1800mm X 70mm THK	CC
1	INSULATION: 2200mm X 70mm THK	CC
1	INSULATION: 2600mm X 70mm THK	CC
1	PLATFORM: 1200mm X 8245mm WIDE	
1	LADDER: 13500.0mm	...
1	CLASS 300 GR 1.1, 24.0" BLIND FLANGE(S)	SA-350 LF2
1	BASERING: 2448.0mm. OD X 2144.0 ID X 32.0mm.	SA-285 C
20	BASERING BOLTS: 30.0mm. DIA	SA-36
20	BASERING BOLT NUTS: 30.0mm. DIA	...
1	BASERING TOP RING: 2464.0mm. OD X 2224.0 ID X 25.0mm.	...
40	BASERING GUSSET PLATES: 102.0mm. X 102.0mm. X 185.0mm. X 12.0mm.	...
4	CYLINDRICAL SKIRT SEGMENT 12.0mm. THK X 3052.5mm. X 2500.0mm.	SA-285 C
2	CYLINDRICAL SKIRT SEGMENT 12.0mm. THK X 806.3mm. X 2500.0mm.	SA-285 C
2	CYLINDRICAL SKIRT SEGMENT 12.0mm. THK X 3052.5mm. X 1000.0mm.	SA-516 70
1	CYLINDRICAL SKIRT SEGMENT 12.0mm. THK X 806.3mm. X 1000.0mm.	SA-516 70
2	CYLINDRICAL SEGMENT 22.0mm. THK X 3455.7mm. X 1800.0mm.	SA-516 70
2	CYLINDRICAL SEGMENT 22.0mm. THK X 3455.7mm. X 2200.0mm.	SA-516 70
2	CYLINDRICAL SEGMENT 22.0mm. THK X 3455.7mm. X 2600.0mm.	SA-516 70
1	NAMEPLATE	...



Minimum Design Metal Temperature Results Summary (°C):

Description	Curve	Basic MDMT	Reduced MDMT	UG-20(f) MDMT	Thickness ratio	Gov Thk mm.	E*	PWHT reqd
Notes								
bottom D end [10]	!	-38	-39		0.989	21.000	1.00	No
bottom D end [7]	!	-38	-47		0.829	25.000	1.00	No
Shell 1 [8]	!	-38	-41		0.949	22.000	1.00	No
Shell 2 [8]	!	-38	-41		0.944	22.000	1.00	No
Shell 3 [8]	!	-38	-41		0.941	22.000	1.00	No
Top D end [10]	!	-38	-39		0.978	21.000	1.00	No
Top D end [7]	!	-38	-48		0.820	25.000	1.00	No
N63"	[1]	-38	-46		0.858	21.000	1.00	No
Nozzle Flg [5]	!	-46	-84					
N48"	[1]	-38	-43		0.913	22.000	1.00	No
Nozzle Flg [5]	!	-46	-85					
N73"	[1]	-38	-43		0.913	22.000	1.00	No
Nozzle Flg [5]	!	-46	-85					
N53"	[1]	-38	-43		0.913	22.000	1.00	No
Nozzle Flg [5]	!	-46	-85					
K1B3"	[1]	-46	-104		0.035	20.574	1.00	No
Nozzle Flg [5]	!	-46	-46					
K4B3"	[1]	-46	-104		0.035	20.574	1.00	No
Nozzle Flg [5]	!	-46	-46					
K3F3"	[1]	-46	-104		0.035	20.574	1.00	No
Nozzle Flg [5]	!	-46	-46					
K2B1.5"	[1]	-46	-104		0.024	16.002	1.00	No
Nozzle Flg [5]	!	-46	-46					
K3B3"	[1]	-46	-104		0.035	20.574	1.00	No
Nozzle Flg [5]	!	-46	-46					
K3D3"	[1]	-46	-104		0.035	20.574	1.00	No
Nozzle Flg [5]	!	-46	-46					
N82"	[1]	-46	-104		0.030	16.764	1.00	No
Nozzle Flg [5]	!	-46	-46					
K2A1.5"	[1]	-46	-104		0.024	16.002	1.00	No
Nozzle Flg [5]	!	-46	-46					
K3A3"	[1]	-46	-104		0.035	20.574	1.00	No
Nozzle Flg [5]	!	-46	-46					
K3C3"	[1]	-46	-104		0.035	20.574	1.00	No
Nozzle Flg [5]	!	-46	-46					
K1A3"	[1]	-46	-104		0.035	20.574	1.00	No
Nozzle Flg [5]	!	-46	-46					
K4A3"	[1]	-46	-104		0.035	20.574	1.00	No
Nozzle Flg [5]	!	-46	-46					
K3E3"	[1]	-46	-104		0.035	20.574	1.00	No
Nozzle Flg [5]	!	-46	-46					
MH124"	[1]	-38	-43		0.905	22.000	1.00	No
Nozzle Flg [5]	!	-46	-85					
N31"	[1]	-46	-104		0.019	14.224	1.00	No
Nozzle Flg [5]	!	-46	-46					
N116"	[1]	-38	-43		0.905	22.000	1.00	No
Nozzle Flg [5]	!	-46	-85					
N216"	[1]	-38	-46		0.848	21.000	1.00	No
Nozzle Flg [5]	!	-46	-85					
Nozl. Bolting[22]		-48			1.000	64.000	1.00	No

Warmest MDMT: -38 -39

Required Minimum Design Metal Temperature -38.0 °C

Warmest Computed Minimum Design Metal Temperature -39.0 °C

Notes:

- [!] - This was an impact tested material.
- [1] - Governing Nozzle Weld.
- [4] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-c).
- [5] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-b).
- [6] - MDMT Calculations at the Shell/Head Joint.
- [7] - MDMT Calculations for the Straight Flange.



- [8] - Cylinder/Cone/Flange Junction MDMT.
- [9] - Calculations in the Spherical Portion of the Head.
- [10] - Calculations in the Knuckle Portion of the Head.
- [11] - Calculated (Body Flange) Flange MDMT.
- [12] - Calculated Flat Head MDMT per UCS-66.3
- [13] - Tubesheet MDMT, shell side, if applicable
- [14] - Tubesheet MDMT, tube side, if applicable
- [15] - Nozzle Material
- [16] - Shell or Head Material
- [17] - Impact Testing required
- [18] - Impact Testing not required, see UCS-66(b)(3)
- [20] - Cylinder/Cone Junction MDMT based on Longitudinal Stress considerations
- [21] - Body Flange Bolting Material
- [22] - Nozzle Flange Bolting Material
- [23] - Stiffening Ring to Shell Weld
- [24] - Saddle to Shell Weld

UG-84(b)(2) was not considered.

UCS-66(g) was not considered.

UCS-66(i) was not considered.

Notes:

Impact test temps were not entered in and not considered in the analysis.

UCS-66(i) applies to impact tested materials not by specification and

UCS-66(g) applies to materials impact tested per UG-84.1 General Note (c).

The Basic MDMT includes the (30F) PWHT credit if applicable.

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Class From To : Basic Element Checks.

Note 30 50 There is a high jump in the Joint Eff.

Class From To: Check of Additional Element Data

There were no geometry errors or warnings.

PV Elite performs all calculations internally in Imperial Units to remain compliant with the ASME Code and any built in assumptions in the ASME Code formulas. The finalized results are reflected to show the set of selected units for this analysis.

Information Regarding Nozzle Loads

Please note that nozzle loadings, if included, are assumed to be local in nature and will not contribute to or create a net section bending moment. Therefore, the addition of nozzle loads will not affect the support load calculation. If you wish to create a load on the support(s) from nozzle loads, you can enable that feature in tools->configuration->Job Specific Settings.



One or more nozzles in this model was specified using the actual thickness basis and a standard flange is also specified. In many cases a matching nominal diameter cannot be determined and the weight of the flange and optional blind flange cannot be determined. When actual thickness basis is specified, you must calculate the weight of the assembly (by hand) and enter it into the 'Overriding Weight' field in the nozzle dialog when there is a standard flange specified. However, for FVC nozzles that have their nominal diameter specified in the FVC dialog, entering the overriding weight for that nozzle is not required.



The level of precision is set to a value of less than 3. At this level you may not obtain proper resolution on some results. This value can be changed in Tools->Set Configuration Parameters->Default Values->Level of Precision.



The ASME Code Section VIII Division 1 does not explicitly require stresses on elements to be computed under test conditions. Please see interpretation VIII-1-10-12 for more information. PV Elite also computes weight due to test fluid and uses the weight in various calculations such as saddle and leg loads. PV Elite also computes static pressure due to operating and test liquids. The static pressure is used in conjunction with the design and test pressures to determine a total pressure in the determination of required thickness, MAWP etc. This ensures compliance with UG-22 requirements.

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Units used in this Analysis (barsUniCode):

Name	System Unit	Constant	User Unit
Length	Feet	304.8000	mm.
Force	Pounds	0.4536	Kgf
Mass	Pounds	0.4536	kg.
Area	sq. inches	6.4516	cm2
Moment	ft. lbs.	0.1383	Kg-m.
Stress	lbs./sq.in.	0.0069	N./mm2
Temperature	Degrees F	0.5556	°C
Pressure	psig	0.0689	bars
Elast. Modulus	lbs./sq.in.	6.8946	KPa.
Pipe Density	lbs./cu.in.	0.0277	kg./cm3
Ins. Density	lbs./cu.ft.	0.1600E-04	kg./cm3
Fluid Density	lbs./cu.ft.	0.1600E-04	kg./cm3
Wind Speed	miles/hr	1.6093	Km/hr
Tray Weight	lbs./sq.ft.	0.0005	Kgs/cm2
Inertia	in.4	41.6231	cm4
G Load	G's	1.0000	g's
Wind Load	lbs./sq.ft.	4.8825	Kgs/m2
Elevation	Feet	304.8000	mm.
Volume	in.3	16.3900	cm3
Diameter	inches	25.4000	mm.
Thickness	inches	25.4000	mm.

PV Elite Vessel Analysis Program: Input Data

Design Internal Pressure (for Hydrotest) 23 bars
 Design Internal Temperature 80.0 °C
 Projection of Nozzle from Vessel Top 0 mm.
 Projection of Nozzle from Vessel Bottom 0 mm.
 Minimum Design Metal Temperature -38.0 °C
 Type of Construction Welded
 Special Service None
 Degree of Radiography RT-1
 Use Higher Longitudinal Stresses (Flag) Y
 Select t for Internal Pressure (Flag) N
 Select t for External Pressure (Flag) N
 Select t for Axial Stress (Flag) N
 Select Location for Stiff. Rings (Flag) N
 Consider Vortex Shedding N

Shop Pressure Test:

Type of Pressure Test UG-99(b)
 Pressure Test Position Horizontal
 Test Performed in Corroded Condition No

Load Case 1 NP+EW+1.1WI+FW+BW
 Load Case 2 NP+EW+EE+FS+BS
 Load Case 3 NP+OW+1.1WI+FW+BW
 Load Case 4 NP+OW+EQ+FS+BS
 Load Case 5 NP+HW+HI
 Load Case 6 NP+HW+HE
 Load Case 7 IP+OW+1.1WI+FW+BW
 Load Case 8 IP+OW+EQ+FS+BS
 Load Case 9 EP+OW+1.1WI+FW+BW
 Load Case 10 EP+OW+EQ+FS+BS
 Load Case 11 HP+HW+HI
 Load Case 12 HP+HW+HE
 Load Case 13 IP+WE+EW
 Load Case 14 IP+WF+CW
 Load Case 15 IP+VO+OW
 Load Case 16 IP+VE+EW
 Load Case 17 NP+VO+OW
 Load Case 18 FS+BS+IP+OW
 Load Case 19 FS+BS+EP+OW
 Load Case 20 BL+IP+OW



Wind Design Code	IS 875 (PART 3):2015-2020	
Basic Wind Speed for IS-875	180	Km/hr
Wind Zone Number	5	
Base Elevation	300	mm.
Percent Wind for Hydrotest	33.0	
Risk Factor	[k1]	1.0
Terrain Category		2
Equipment Class		Class A
Topography Factor	[k3]	1.0
Importance Factor	[k4]	1.15
Damping Factor (Beta) for Wind (Ope)		0.0100
Damping Factor (Beta) for Wind (Empty)		0.0000
Damping Factor (Beta) for Wind (Filled)		0.0000
Seismic Design Code		Spectrum
Range Type		Period
Ordinate Type		Acceleration
Include Missing Mass Components		Yes
Shock Scale Factor X Direction		1.0000
Shock Scale Factor Y Direction		0.6700
Combination Method		Group
Spectrum Name		User Defined

User Defined Spectrum Data Points:

Acceleration	Period
0.0700	0.0100
0.1170	0.0500
0.1750	0.1000
0.1750	0.1300
0.1750	0.2000
0.1750	0.2700
0.1750	0.3000
0.1750	0.4000
0.1400	0.5000
0.1270	0.5500
0.1170	0.6000
0.1050	0.6700
0.0870	0.8000
0.0700	1.0000
0.0580	1.2000
0.0500	1.4000
0.0440	1.6000
0.0390	1.8000
0.0350	2.0000
0.0290	2.4000

Design Pressure + Static Head	Y
Consider MAP New and Cold in Noz. Design	N
Consider External Loads for Nozzle Des.	Y
Use ASME VIII-1 Appendix 1-9	N
Perform Blast Load Analysis	No
Material Database Year	Current w/Addenda or Code Year

Configuration Directives:

g0 is taken as min(g0, attached shell thk)	Yes
Do not use Nozzle MDMT Interpretation VIII-1 01-37	No
Use Table G instead of exact equation for "A"	Yes
Shell Head Joints are Tapered	Yes
Compute "K" in corroded condition	Yes
Use Code Case 2286	No
Use the MAWP to compute the MDMT	Yes
For thickness ratios <= 0.35, MDMT will be -155F (-104C)	Yes
For PWHT & P1 Materials the MDMT can be < -55F (-48C)	No
Using Metric Material Databases, ASME II D	Yes



Calculate B31.3 type stress for Nozzles with Loads	Yes
Reduce the MDMT due to lower membrane stress	Yes
Consider Longitudinal Stress in MDMT Calculations	Yes

Complete Listing of Vessel Elements and Details:

Element#	1/8
Element From Node	10
Element To Node	20
Element Type	Skirt Support
Description	Skirt 3
Distance "FROM" to "TO"	2500 mm.
Skirt Inside Diameter	2200 mm.
Diameter of Skirt at Base	2200 mm.
Skirt Thickness	12 mm.
Internal Corrosion Allowance	0.7 mm.
Nominal Thickness	12 mm.
External Corrosion Allowance	0.7 mm.
Design Temperature Internal Pressure	80 °C
Design Temperature External Pressure	80 °C
Effective Diameter Multiplier	1.2
Material Name	SA-285 C
Allowable Stress, Ambient	108 N./mm2
Allowable Stress, Operating	108 N./mm2
Allowable Stress, Hydrotest	140.4 N./mm2
Material Density	0.00775 kg./cm3
P Number Thickness	32.004 mm.
Yield Stress, Operating	192.43 N./mm2
UCS-66 Chart Curve Designation	A
External Pressure Chart Name	CS-2
UNS Number	K02801
Product Form	Plate
Efficiency, Longitudinal Seam	0.7
Efficiency, Head-to-Skirt or Circ. Seam	0.7
Weld is pre-Heated	No

Element From Node	10
Detail Type	Platform
Detail ID	PLAT:[1 OF 1]
Dist. from "FROM" Node / Offset dist	2500 mm.
Platform Start Angle (degrees)	0.0
Platform End Angle (degrees)	90.0
Platform Wind Area	18000 cm2
Platform Weight	838.39 Kg
Platform Railing Weight	0 Kg/mm.
Platform Grating Weight	0.017 Kgs/cm2
Platform Width	1500 mm.
Platform Height	1200 mm.
Platform Clearance or End Offset	230 mm.
Platform Force Coefficient	1.0
Ladder Layout Angle	0.0
Ladder Start Elevation	0 mm.
Ladder End Elevation	0 mm.
Unit Weight of Ladder	0 Kg/mm.
Platform Length (top head platform)	0 mm.

Element From Node	10
Detail Type	Weight
Detail ID	Skirt opening
Dist. from "FROM" Node / Offset dist	1500 mm.
Miscellaneous Weight	200 Kg
Offset from Element Centerline	0 mm.

Element#	2/8
Element From Node	20
Element To Node	30
Element Type	Skirt Support
Description	Skirt 2



Distance "FROM" to "TO"	2500	mm.
Skirt Inside Diameter	2200	mm.
Diameter of Skirt at Base	2200	mm.
Skirt Thickness	12	mm.
Internal Corrosion Allowance	0.75	mm.
Nominal Thickness	12	mm.
External Corrosion Allowance	0.7	mm.
Design Temperature Internal Pressure	80	°C
Design Temperature External Pressure	80	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-285 C	
Efficiency, Longitudinal Seam	0.7	
Efficiency, Head-to-Skirt or Circ. Seam	0.7	
Weld is pre-Heated	No	

Element#	3/8	
Element From Node	30	
Element To Node	40	
Element Type	Skirt Support	
Description	Skirt 1	
Distance "FROM" to "TO"	1000	mm.
Skirt Inside Diameter	2200	mm.
Diameter of Skirt at Base	2200	mm.
Skirt Thickness	12	mm.
Internal Corrosion Allowance	0.55	mm.
Nominal Thickness	12	mm.
External Corrosion Allowance	0.7	mm.
Design Temperature Internal Pressure	80	°C
Design Temperature External Pressure	80	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-516 70	
Allowable Stress, Ambient	138	N./mm ²
Allowable Stress, Operating	138	N./mm ²
Allowable Stress, Hydrotest	179.4	N./mm ²
Material Density	0.00775	kg./cm ³
P Number Thickness	32.004	mm.
Yield Stress, Operating	243	N./mm ²
UCS-66 Chart Curve Designation	B	
External Pressure Chart Name	CS-2	
UNS Number	K02700	
Product Form	Plate	
Efficiency, Longitudinal Seam	0.7	
Efficiency, Head-to-Skirt or Circ. Seam	0.7	
Weld is pre-Heated	No	

Element#	4/8	
Element From Node	40	
Element To Node	50	
Element Type	Elliptical	
Description	bottom D end	
Distance "FROM" to "TO"	50.8	mm.
Inside Diameter	2200	mm.
Element Thickness	21	mm.
Internal Corrosion Allowance	1.5	mm.
Nominal Thickness	25	mm.
External Corrosion Allowance	0	mm.
Design Internal Pressure	23	bars
Design Temperature Internal Pressure	80	°C
Design External Pressure	1.0132	bars
Design Temperature External Pressure	80	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-516 70 +IT	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Elliptical Head Factor	2.0	
Weld is pre-Heated	No	



Element From Node 40
 Detail Type Liquid
 Detail ID Liquid: 40
 Dist. from "FROM" Node / Offset dist -550 mm.
 Height/Length of Liquid 600.8 mm.
 Liquid Density 0.0006722 kg./cm3

Element From Node 40
 Detail Type Insulation
 Detail ID Ins: 40
 Dist. from "FROM" Node / Offset dist -550 mm.
 Height/Length of Insulation 600.8 mm.
 Thickness of Insulation 70 mm.
 Density 0.40000E-04 kg./cm3

Element From Node 40
 Detail Type Nozzle
 Detail ID N63"
 Dist. from "FROM" Node / Offset dist 0 mm.
 Nozzle Diameter 3 in.
 Nozzle Schedule 80
 Nozzle Class 300
 Layout Angle 0.0
 Blind Flange (Y/N) N
 Weight of Nozzle (Used if > 0) 33.097 Kg
 Grade of Attached Flange GR 1.1
 Nozzle Matl SA-333 6 +IT

Element From Node 40
 Detail Type Weight
 Detail ID Vortex Breaker
 Dist. from "FROM" Node / Offset dist -200 mm.
 Miscellaneous Weight 5 Kg
 Offset from Element Centerline 0 mm.

Element# 5/8
 Element From Node 50
 Element To Node 60
 Element Type Cylinder
 Description Shell 1
 Distance "FROM" to "TO" 1800 mm.
 Inside Diameter 2200 mm.
 Element Thickness 22 mm.
 Internal Corrosion Allowance 1.5 mm.
 Nominal Thickness 22 mm.
 External Corrosion Allowance 0 mm.
 Design Internal Pressure 23 bars
 Design Temperature Internal Pressure 80 °C
 Design External Pressure 1.0132 bars
 Design Temperature External Pressure 80 °C
 Effective Diameter Multiplier 1.2
 Material Name SA-516 70 +IT
 Efficiency, Longitudinal Seam 1.0
 Efficiency, Circumferential Seam 1.0
 Weld is pre-Heated No

Element From Node 50
 Detail Type Platform
 Detail ID PLAT:[1 OF 1]
 Dist. from "FROM" Node / Offset dist 0.48768E-04 mm.
 Platform Start Angle (degrees) 0.0
 Platform End Angle (degrees) 360.0
 Platform Wind Area 36000 cm2
 Platform Weight 3369.6 Kg
 Platform Railing Weight 0 Kg/mm.
 Platform Grating Weight 0.017 Kgs/cm2
 Platform Width 1500 mm.



Platform Height	1200	mm.
Platform Clearance or End Offset	230	mm.
Platform Force Coefficient	1.0	
Ladder Layout Angle	0.0	
Ladder Start Elevation	0	mm.
Ladder End Elevation	0	mm.
Unit Weight of Ladder	0	Kgf/mm.
Platform Length (top head platform)	0	mm.

Element From Node	50	
Detail Type	Liquid	
Detail ID	Liquid: 50	
Dist. from "FROM" Node / Offset dist	0	mm.
Height/Length of Liquid	1800	mm.
Liquid Density	0.0006722	kg./cm3

Element From Node	50	
Detail Type	Insulation	
Detail ID	Ins: 50	
Dist. from "FROM" Node / Offset dist	0	mm.
Height/Length of Insulation	1800	mm.
Thickness of Insulation	70	mm.
Density	0.40000E-04	kg./cm3

Element From Node	50	
Detail Type	Nozzle	
Detail ID	N48"	
Dist. from "FROM" Node / Offset dist	350	mm.
Nozzle Diameter	8	in.
Nozzle Schedule	80	
Nozzle Class	300	
Layout Angle	0.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	0	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-333 6 +IT	

Element From Node	50	
Detail Type	Nozzle	
Detail ID	N73"	
Dist. from "FROM" Node / Offset dist	450	mm.
Nozzle Diameter	3	in.
Nozzle Schedule	80	
Nozzle Class	300	
Layout Angle	180.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	12.91	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-333 6 +IT	

Element From Node	50	
Detail Type	Nozzle	
Detail ID	N53"	
Dist. from "FROM" Node / Offset dist	450	mm.
Nozzle Diameter	3	in.
Nozzle Schedule	80	
Nozzle Class	300	
Layout Angle	90.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	12.91	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-333 6 +IT	

Element From Node	50	
Detail Type	Nozzle	
Detail ID	K1B3"	
Dist. from "FROM" Node / Offset dist	100	mm.
Nozzle Diameter	3	in.
Nozzle Schedule	None	
Nozzle Class	300	



Layout Angle	90.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	18.919	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-350 LF2	+IT
Element From Node	50	
Detail Type	Nozzle	
Detail ID	K4B3"	
Dist. from "FROM" Node / Offset dist	100	mm.
Nozzle Diameter	3	in.
Nozzle Schedule	None	
Nozzle Class	300	
Layout Angle	180.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	18.919	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-350 LF2	+IT
Element From Node	50	
Detail Type	Nozzle	
Detail ID	K3F3"	
Dist. from "FROM" Node / Offset dist	100	mm.
Nozzle Diameter	3	in.
Nozzle Schedule	None	
Nozzle Class	300	
Layout Angle	270.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	18.919	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-350 LF2	+IT
Element From Node	50	
Detail Type	Nozzle	
Detail ID	K2B1.5"	
Dist. from "FROM" Node / Offset dist	100	mm.
Nozzle Diameter	1.5	in.
Nozzle Schedule	None	
Nozzle Class	300	
Layout Angle	120.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	9.4306	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-350 LF2	+IT
Element From Node	50	
Detail Type	Nozzle	
Detail ID	K3B3"	
Dist. from "FROM" Node / Offset dist	100	mm.
Nozzle Diameter	3	in.
Nozzle Schedule	None	
Nozzle Class	300	
Layout Angle	145.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	18.919	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-350 LF2	+IT
Element From Node	50	
Detail Type	Nozzle	
Detail ID	K3D3"	
Dist. from "FROM" Node / Offset dist	100	mm.
Nozzle Diameter	3	in.
Nozzle Schedule	None	
Nozzle Class	300	
Layout Angle	215.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	18.919	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-350 LF2	+IT



Element From Node	50
Detail Type	Nozzle
Detail ID	N82"
Dist. from "FROM" Node / Offset dist	200 mm.
Nozzle Diameter	2 in.
Nozzle Schedule	None
Nozzle Class	300
Layout Angle	250.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	10.577 Kgf
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-350 LF2 +IT

Element From Node	50
Detail Type	Weight
Detail ID	elbow Pipe
Dist. from "FROM" Node / Offset dist	400 mm.
Miscellaneous Weight	120 Kgf
Offset from Element Centerline	0 mm.

Element From Node	50
Detail Type	Weight
Detail ID	N40VHPP
Dist. from "FROM" Node / Offset dist	350 mm.
Miscellaneous Weight	275 Kgf
Offset from Element Centerline	2364 mm.

Element#	6/8
Element From Node	60
Element To Node	70
Element Type	Cylinder
Description	Shell 2
Distance "FROM" to "TO"	2200 mm.
Inside Diameter	2200 mm.
Element Thickness	22 mm.
Internal Corrosion Allowance	1.5 mm.
Nominal Thickness	22 mm.
External Corrosion Allowance	0 mm.
Design Internal Pressure	23 bars
Design Temperature Internal Pressure	80 °C
Design External Pressure	1.0132 bars
Design Temperature External Pressure	80 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70 +IT
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No

Element From Node	60
Detail Type	Liquid
Detail ID	Liquid: 60
Dist. from "FROM" Node / Offset dist	0 mm.
Height/Length of Liquid	1700 mm.
Liquid Density	0.0006722 kg./cm3

Element From Node	60
Detail Type	Insulation
Detail ID	Ins: 60
Dist. from "FROM" Node / Offset dist	0 mm.
Height/Length of Insulation	2200 mm.
Thickness of Insulation	70 mm.
Density	0.40000E-04 kg./cm3

Element From Node	60
Detail Type	Nozzle
Detail ID	K2A1.5"
Dist. from "FROM" Node / Offset dist	1800 mm.



Nozzle Diameter 1.5 in.
 Nozzle Schedule None
 Nozzle Class 300
 Layout Angle 0.0
 Blind Flange (Y/N) N
 Weight of Nozzle (Used if > 0) 9.0093 Kg
 Grade of Attached Flange GR 1.1
 Nozzle Matl SA-350 LF2 +IT

Element From Node 60
 Detail Type Nozzle
 Detail ID K3A3"
 Dist. from "FROM" Node / Offset dist 1800 mm.
 Nozzle Diameter 3 in.
 Nozzle Schedule None
 Nozzle Class 300
 Layout Angle 90.0
 Blind Flange (Y/N) N
 Weight of Nozzle (Used if > 0) 18.919 Kg
 Grade of Attached Flange GR 1.1
 Nozzle Matl SA-350 LF2 +IT

Element From Node 60
 Detail Type Nozzle
 Detail ID K3C3"
 Dist. from "FROM" Node / Offset dist 1800 mm.
 Nozzle Diameter 3 in.
 Nozzle Schedule None
 Nozzle Class 300
 Layout Angle 180.0
 Blind Flange (Y/N) N
 Weight of Nozzle (Used if > 0) 18.919 Kg
 Grade of Attached Flange GR 1.1
 Nozzle Matl SA-350 LF2 +IT

Element From Node 60
 Detail Type Nozzle
 Detail ID K1A3"
 Dist. from "FROM" Node / Offset dist 1800 mm.
 Nozzle Diameter 3 in.
 Nozzle Schedule None
 Nozzle Class 300
 Layout Angle 120.0
 Blind Flange (Y/N) N
 Weight of Nozzle (Used if > 0) 18.919 Kg
 Grade of Attached Flange GR 1.1
 Nozzle Matl SA-350 LF2 +IT

Element From Node 60
 Detail Type Nozzle
 Detail ID K4A3"
 Dist. from "FROM" Node / Offset dist 1800 mm.
 Nozzle Diameter 3 in.
 Nozzle Schedule None
 Nozzle Class 300
 Layout Angle 160.0
 Blind Flange (Y/N) N
 Weight of Nozzle (Used if > 0) 18.919 Kg
 Grade of Attached Flange GR 1.1
 Nozzle Matl SA-350 LF2 +IT

Element From Node 60
 Detail Type Nozzle
 Detail ID K3E3"
 Dist. from "FROM" Node / Offset dist 1800 mm.
 Nozzle Diameter 3 in.
 Nozzle Schedule None
 Nozzle Class 300
 Layout Angle 270.0
 Blind Flange (Y/N) N



Weight of Nozzle (Used if > 0) 18.919 Kgf
 Grade of Attached Flange GR 1.1
 Nozzle Matl SA-350 LF2 +IT

Element From Node 60
 Detail Type Weight
 Detail ID Lifting Wt.0
 Dist. from "FROM" Node / Offset dist 1500 mm.
 Miscellaneous Weight 450 Kgf
 Offset from Element Centerline 0 mm.

Element From Node 60
 Detail Type Weight
 Detail ID Clip Wt
 Dist. from "FROM" Node / Offset dist 1300 mm.
 Miscellaneous Weight 1150 Kgf
 Offset from Element Centerline 0 mm.

Element# 7/8
 Element From Node 70
 Element To Node 80
 Element Type Cylinder
 Description Shell 3
 Distance "FROM" to "TO" 2600 mm.
 Element Outside Diameter 2244 mm.
 Element Thickness 22 mm.
 Internal Corrosion Allowance 1.5 mm.
 Nominal Thickness 22 mm.
 External Corrosion Allowance 0 mm.
 Design Internal Pressure 23 bars
 Design Temperature Internal Pressure 80 °C
 Design External Pressure 1.0132 bars
 Design Temperature External Pressure 80 °C
 Effective Diameter Multiplier 1.2
 Material Name SA-516 70 +IT
 Efficiency, Longitudinal Seam 1.0
 Efficiency, Circumferential Seam 1.0
 Weld is pre-Heated No

Element From Node 70
 Detail Type Platform
 Detail ID PLAT:[1 OF 1]
 Dist. from "FROM" Node / Offset dist 0.0002896 mm.
 Platform Start Angle (degrees) 0.0
 Platform End Angle (degrees) 180.0
 Platform Wind Area 36000 cm2
 Platform Weight 1702.4 Kgf
 Platform Railing Weight 0 Kgf/mm.
 Platform Grating Weight 0.017 Kgs/cm2
 Platform Width 1500 mm.
 Platform Height 1200 mm.
 Platform Clearance or End Offset 230 mm.
 Platform Force Coefficient 1.0
 Ladder Layout Angle 0.0
 Ladder Start Elevation 0 mm.
 Ladder End Elevation 0 mm.
 Unit Weight of Ladder 0 Kgf/mm.
 Platform Length (top head platform) 0 mm.

Element From Node 70
 Detail Type Insulation
 Detail ID Ins: 70
 Dist. from "FROM" Node / Offset dist 0 mm.
 Height/Length of Insulation 2600 mm.
 Thickness of Insulation 70 mm.
 Density 0.40000E-04 kg./cm3

Element From Node 70



Detail Type	Nozzle
Detail ID	MH124"
Dist. from "FROM" Node / Offset dist	1750 mm.
Nozzle Diameter	24 in.
Nozzle Schedule	None
Nozzle Class	300
Layout Angle	0.0
Blind Flange (Y/N)	Y
Weight of Nozzle (Used if > 0)	379.79 Kg
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-516 70 +IT

Element From Node	70
Detail Type	Nozzle
Detail ID	N31"
Dist. from "FROM" Node / Offset dist	1100 mm.
Nozzle Diameter	1 in.
Nozzle Schedule	None
Nozzle Class	300
Layout Angle	90.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	7.4114 Kg
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-350 LF2 +IT

Element From Node	70
Detail Type	Nozzle
Detail ID	N116"
Dist. from "FROM" Node / Offset dist	700 mm.
Nozzle Diameter	16 in.
Nozzle Schedule	None
Nozzle Class	300
Layout Angle	180.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	193.7 Kg
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-516 70 +IT

Element From Node	70
Detail Type	Weight
Detail ID	Demister
Dist. from "FROM" Node / Offset dist	2200 mm.
Miscellaneous Weight	150 Kg
Offset from Element Centerline	0 mm.

Element From Node	70
Detail Type	Weight
Detail ID	Distributor
Dist. from "FROM" Node / Offset dist	750 mm.
Miscellaneous Weight	100 Kg
Offset from Element Centerline	0 mm.

Element From Node	70
Detail Type	Weight
Detail ID	N10VHPP
Dist. from "FROM" Node / Offset dist	700 mm.
Miscellaneous Weight	772 Kg
Offset from Element Centerline	2399 mm.

Element From Node	70
Detail Type	Weight
Detail ID	Manway Davit
Dist. from "FROM" Node / Offset dist	1800 mm.
Miscellaneous Weight	360 Kg
Offset from Element Centerline	1700 mm.

Element From Node	70
Detail Type	Weight
Detail ID	Pipe Davit
Dist. from "FROM" Node / Offset dist	2400 mm.



Miscellaneous Weight 560 Kgf
 Offset from Element Centerline 1700 mm.

Element# 8/8
 Element From Node 80
 Element To Node 90
 Element Type Elliptical
 Description Top D end
 Distance "FROM" to "TO" 50.8 mm.
 Inside Diameter 2200 mm.
 Element Thickness 21 mm.
 Internal Corrosion Allowance 1.5 mm.
 Nominal Thickness 25 mm.
 External Corrosion Allowance 0 mm.
 Design Internal Pressure 23 bars
 Design Temperature Internal Pressure 80 °C
 Design External Pressure 1.0132 bars
 Design Temperature External Pressure 80 °C
 Effective Diameter Multiplier 1.2
 Material Name SA-516 70 +IT
 Efficiency, Longitudinal Seam 1.0
 Efficiency, Circumferential Seam 1.0
 Elliptical Head Factor 2.0
 Weld is pre-Heated No

Element From Node 80
 Detail Type Platform
 Detail ID PLAT:[1 OF 1]
 Dist. from "FROM" Node / Offset dist 900 mm.
 Platform Start Angle (degrees) 0.0
 Platform End Angle (degrees) 0.0
 Platform Wind Area 153804 cm2
 Platform Weight 12062 Kgf
 Platform Railing Weight 0 Kgf/mm.
 Platform Grating Weight 0.017 Kgs/cm2
 Platform Width 8245 mm.
 Platform Height 1200 mm.
 Platform Clearance or End Offset 4122.5 mm.
 Platform Force Coefficient 1.0
 Ladder Layout Angle 45.0
 Ladder Start Elevation 0 mm.
 Ladder End Elevation 13500 mm.
 Unit Weight of Ladder 0.037 Kgf/mm.
 Platform Length (top head platform) 8245 mm.

Element From Node 80
 Detail Type Insulation
 Detail ID Ins: 80
 Dist. from "FROM" Node / Offset dist 0 mm.
 Height/Length of Insulation 600.8 mm.
 Thickness of Insulation 70 mm.
 Density 0.40000E-04 kg./cm3

Element From Node 80
 Detail Type Nozzle
 Detail ID N216"
 Dist. from "FROM" Node / Offset dist 0 mm.
 Nozzle Diameter 16 in.
 Nozzle Schedule None
 Nozzle Class 300
 Layout Angle 0.0
 Blind Flange (Y/N) N
 Weight of Nozzle (Used if > 0) 199.72 Kgf
 Grade of Attached Flange GR 1.1
 Nozzle Matl SA-516 70 +IT

Element From Node 80
 Detail Type Weight



Detail ID	N2QVHPP	
Dist. from "FROM" Node / Offset dist	0	mm.
Miscellaneous Weight	513	Kgf
Offset from Element Centerline	2424	mm.
Element From Node	80	
Detail Type	Weight	
Detail ID	PSV	
Dist. from "FROM" Node / Offset dist	600	mm.
Miscellaneous Weight	1000	Kgf
Offset from Element Centerline	4122.5	mm.



XY Coordinate Calculations:

From	To	X (Horiz.) mm.	Y (Vert.) mm.	DX (Horiz.) mm.	DY (Vert.) mm.
Skirt 3	...	...	2500	...	2500
Skirt 2	...	...	5000	...	2500
Skirt 1	...	...	6000	...	1000
bottom D end	...	...	6050.8	...	50.8
Shell 1	...	...	7850.8	...	1800
Shell 2	...	...	10050.8	...	2200
Shell 3	...	...	12650.8	...	2600
Top D end	...	...	12701.6	...	50.8

The calculated dimensions are based on the given element lengths. Due to variability in manufacturing (weld gaps etc.), the Tangent to Tangent length may not be exact. The length includes gasket thicknesses and raised face heights.

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Internal Pressure Results Summary:

Element Thickness, Pressure, Diameter and Allowable Stress :

From	To	Int. Press + Liq. Hd bars	Nominal Thickness mm.	Total Corr Allowance mm.	Element Diameter mm.	Allowable Stress(SE) N./mm ²	Element#
Skirt 3		...	12	1.4	2200	...	1
Skirt 2		...	12	1.45	2200	...	2
Skirt 1		...	12	1.25	2200	...	3
bottom D end		23.271	25	1.5	2200	138	4
Shell 1		23.231	22	1.5	2200	138	5
Shell 2		23.112	22	1.5	2200	138	6
Shell 3		23	22	1.5	2244	138	7
Top D end		23	25	1.5	2200	138	8

Element Required Thickness and MAWP :

From	To	Design Pressure bars	M.A.W.P. Corroded bars	M.A.W.P. New & Cold bars	Minimum Thickness mm.	Required Thickness mm.	Element#
Skirt 3		...	No Calc	No Calc	12	No Calc	1
Skirt 2		...	No Calc	No Calc	12	No Calc	2
Skirt 1		...	No Calc	No Calc	12	No Calc	3
bottom D end		23	24.1587	26.2932	21	20.0735	4
Shell 1		23	25.1665	27.2705	22	20.2334	5
Shell 2		23	25.2853	27.2705	22	20.1366	6
Shell 3		23	25.3975	27.2705	22	20.0776	7
Top D end		23	24.4294	26.2932	21	19.8571	8

Minimum

24

26

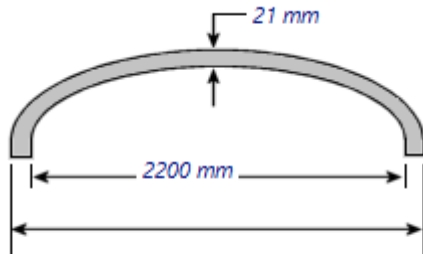
MAWP: 24 bars, limited by: Nozzle Reinforcement.

Elements Suitable for Design Internal Pressure.

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Internal Pressure Calculation Results:

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Internal Pressure Results for: bottom D end

Elliptical Head Design Information:							
Design Pressure	P	23.271 bars		Int. Design Temperature		80 °C	
Material		SA-516 70		Ext. Design Temperature		80 °C	
Length		50.800 mm.		External Pressure Chart		CS-2	
				UNS Number		K02700	
Int. Corr. All.	c	1.5000 mm.		Allowable Stress (ope)	S	138.0 N./mm^2	
Ext. Corr. All.	ce	0.0000 mm.		Allowable Stress (amb)	Sa	138.0 N./mm^2	
Inside Diameter	Di	2200.000 mm.		Actual Stress		131.5 N./mm^2	
Spec. Min. Thk t _e , or s _a		21.0000 mm.		Spec. Nominal Thk.		25.000 mm.	
Surface Area		58467.6 cm^2		SG of Contents		0.673	
Element Volume		1587211.8 cm3		Weight of Contents		1068.0 Kgf	
Empty Weight		1216.9 Kgf		Operating Weight		2284.9 Kgf	
				Comp. All. Ope. Stress		126.7 N./mm^2	
				Aspect Ratio		2.00	

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 27.500 mm.
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 13.750 mm.

Material treatment options: +IT

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot D \cdot K_{cor}) / (2 \cdot S \cdot E - 0.2 \cdot P) \text{ Appendix 1-4(c)}$$

$$= (23.271 \cdot 2203.0 \cdot 0.998) / (2 \cdot 138.0 \cdot 1.0 - 0.2 \cdot 23.271)$$

$$= 18.5735 + 1.5000 = 20.0735 \text{ mm.}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.271 bars

$$= (2 \cdot S \cdot E \cdot t) / (K_{cor} \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138.0 \cdot 1.0 \cdot 19.5) / (0.998 \cdot 2203.0 + 0.2 \cdot 19.5)$$

$$= 24 - 0 = 24 \text{ bars}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (2 \cdot S \cdot E \cdot t) / (K \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138.0 \cdot 1.0 \cdot 21.0) / (1.0 \cdot 2200.0 + 0.2 \cdot 21.0)$$

$$= 26 \text{ bars}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (K_{cor} \cdot D + 0.2 \cdot t)) / (2 \cdot E \cdot t)$$

$$= (23.271 \cdot (0.998 \cdot 2203.0 + 0.2 \cdot 19.5)) / (2 \cdot 1.0 \cdot 19.5)$$



= 131.451 N./mm²

Straight Flange Required Thickness:

$$\begin{aligned} &= (P \cdot R) / (S \cdot E - 0.6 \cdot P) + c \quad \text{per UG-27 (c)(1)} \\ &= (23.271 \cdot 1101.5) / (138.0 \cdot 1.0 - 0.6 \cdot 23.271) + 1.5 \\ &= 20.266 \text{ mm.} \end{aligned}$$

Straight Flange Maximum Allowable Working Pressure:

Less Operating Hydrostatic Head Pressure of 0.234 bars

$$\begin{aligned} &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \quad \text{per UG-27 (c)(1)} \\ &= (138.0 \cdot 1.0 \cdot 23.5) / (1101.5 + 0.6 \cdot 23.5) \\ &= 29 - 0 = 29 \text{ bars} \end{aligned}$$

Factor K, corroded condition [Kcor]:

$$\begin{aligned} &= (2 + (\text{Inside Diameter} / (2 \cdot \text{Inside Head Depth})))^2 / 6 \\ &= (2 + (2203.0 / (2 \cdot 551.5)))^2 / 6 \\ &= 0.998188 \end{aligned}$$

Percent Elong. per UCS-79, VIII-1-01-57 $(75 \cdot t_{\text{nom}} / R_f) \cdot (1 - R_f / R_o)$ 4.851 %

MDMT Calculations in the Knuckle Portion:

Note: This Element/Detail was specified as being Impact Tested.

MDMT Calculations in the Head Straight Flange:

Note: This Element/Detail was specified as being Impact Tested.

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 27.500 mm.
Head inner surface maximum deviation inside the specified shape, 0.625% of D: 13.750 mm.

Material treatment options: +IT

EXTERNAL PRESSURE RESULTS for: bottom D end

Elliptical Head From 40 to 50 Ext. Chart: CS-2 at 80 °C

Elastic Modulus from Chart: CS-2 at 80 °C: 199943392 KPa.

Results for Maximum Allowable Ext. Pressure				MAEP
Tca	Outer Dia	Do/t	Factor A	Factor B
19.500	2242.00	114.97	0.0012080	90.90

$$\text{MAEP} = B / (K_0 \cdot \text{Do}/t) = 90.8975 / (0.9 \cdot 114.9744) = 8.7838 \text{ bars}$$

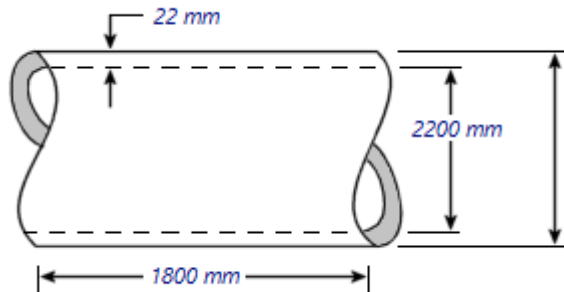
Results for Required Thickness				Tca
Tca	Outer Dia	Do/t	Factor A	Factor B
5.746	2242.00	390.20	0.0003559	35.59

$$\text{MAEP} = B / (K_0 \cdot \text{Do}/t) = 35.5863 / (0.9 \cdot 390.1999) = 1.0133 \text{ bars}$$

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Internal Pressure Calculation Results:

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Internal Pressure Results for: Shell 1

Cylindrical Shell Design Information:					
Design Pressure	P	23.231 bars	Int. Design Temperature		80 °C
Material		SA-516 70	Ext. Design Temperature		80 °C
Length		1800.000 mm.	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	1.5000 mm.	Allowable Stress (ope)	S	138.0 N./mm^2
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	138.0 N./mm^2
Inside Diameter	Di	2200.000 mm.	Actual Stress		126.2 N./mm^2
Spec. Min. Thk t _e , or sa		22.0000 mm.	Spec. Nominal Thk.		22.000 mm.
Surface Area		126895.2 cm^2	SG of Contents		0.673
Element Volume		6843614.5 cm3	Weight of Contents		4605.0 Kg
Empty Weight		2142.5 Kg	Operating Weight		6747.5 Kg
			Comp. All. Ope. Stress		128.0 N./mm^2

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Material treatment options: +IT

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (23.231 \cdot 1101.5) / (138.0 \cdot 1.0 - 0.6 \cdot 23.231)$$

$$= 18.7334 + 1.5000 = 20.2334 \text{ mm.}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.231 bars

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 20.5) / (1101.5 + 0.6 \cdot 20.5)$$

$$= 25 - 0 = 25 \text{ bars}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 22.0) / (1100.0 + 0.6 \cdot 22.0)$$

$$= 27 \text{ bars}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (23.231 \cdot (1101.5 + 0.6 \cdot 20.5)) / (1.0 \cdot 20.5)$$

$$= 126.225 \text{ N./mm}^2$$

% Elongation per Table UG-79-1 (50 * t_{nom} / R_f * (1 - R_f / R_o)) 0.990 %

Minimum Design Metal Temperature Results:



Note: This Element/Detail was specified as being Impact Tested.

Material treatment options: +IT

EXTERNAL PRESSURE RESULTS for: Shell 1

Cylindrical Shell From 50 to 60 Ext. Chart: CS-2 at 80 °C

Elastic Modulus from Chart: CS-2 at 80 °C: 199943392 KPa.

Results for Maximum Allowable Ext. Pressure					MAEP	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
20.500	2244.00	7068.27	109.46	3.1499	0.0003485	34.84

$$MAEP = (4*B)/(3*(Do/t)) = (4*34.8399)/(3*109.4634) = 4.2435 \text{ bars}$$

Results for Required Thickness					Tca	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
11.524	2244.00	7068.27	194.72	3.1499	0.0001480	14.80

$$MAEP = (4*B)/(3*(Do/t)) = (4*14.798)/(3*194.7208) = 1.0132 \text{ bars}$$

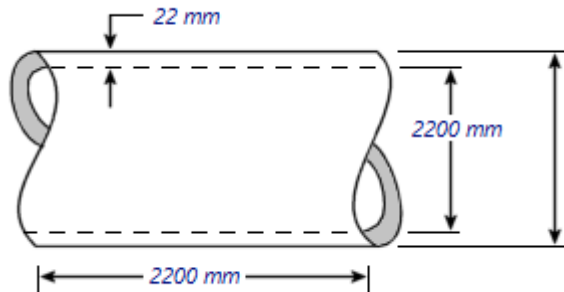
Results for Maximum Stiffened Length					Slen	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
20.500	2244.00	148885.61	109.46	50.0000	0.0000934	9.34

$$MAEP = (4*B)/(3*(Do/t)) = (4*9.3385)/(3*109.4634) = 1.1374 \text{ bars}$$

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Internal Pressure Calculation Results:

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Internal Pressure Results for: Shell 2

Cylindrical Shell Design Information:					
Design Pressure	P	23.112 bars	Int. Design Temperature	80 °C	
Material		SA-516 70	Ext. Design Temperature	80 °C	
Length		2200.000 mm.	External Pressure Chart	CS-2	
			UNS Number	K02700	
Int. Corr. All.	c	1.5000 mm.	Allowable Stress (ope)	S	138.0 N./mm^2
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	138.0 N./mm^2
Inside Diameter	Di	2200.000 mm.	Actual Stress	125.6 N./mm^2	
Spec. Min. Thk t _e , or s _a		22.0000 mm.	Spec. Nominal Thk.	22.000 mm.	
Surface Area		155094.1 cm^2	SG of Contents	0.673	
Element Volume		8364418.0 cm3	Weight of Contents	4349.2 Kg	
Empty Weight		2618.6 Kg	Operating Weight	6967.8 Kg	
			Comp. All. Ope. Stress	128.0 N./mm^2	

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Material treatment options: +IT

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (23.112 \cdot 1101.5) / (138.0 \cdot 1.0 - 0.6 \cdot 23.112)$$

$$= 18.6367 + 1.5000 = 20.1366 \text{ mm.}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.112 bars

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 20.5) / (1101.5 + 0.6 \cdot 20.5)$$

$$= 25 - 0 = 25 \text{ bars}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 22.0) / (1100.0 + 0.6 \cdot 22.0)$$

$$= 27 \text{ bars}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (23.112 \cdot (1101.5 + 0.6 \cdot 20.5)) / (1.0 \cdot 20.5)$$

$$= 125.580 \text{ N./mm}^2$$

% Elongation per Table UG-79-1 (50 * t_{nom} / R_f * (1 - R_f / R_o)) 0.990 %

Minimum Design Metal Temperature Results:



Note: This Element/Detail was specified as being Impact Tested.

Material treatment options: +IT

EXTERNAL PRESSURE RESULTS for: Shell 2

Cylindrical Shell From 60 to 70 Ext. Chart: CS-2 at 80 °C

Elastic Modulus from Chart: CS-2 at 80 °C: 199943392 KPa.

Results for Maximum Allowable Ext. Pressure					MAEP	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
20.500	2244.00	7068.27	109.46	3.1499	0.0003485	34.84

$$MAEP = (4*B)/(3*(Do/t)) = (4*34.8399)/(3*109.4634) = 4.2435 \text{ bars}$$

Results for Required Thickness					Tca	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
11.524	2244.00	7068.27	194.72	3.1499	0.0001480	14.80

$$MAEP = (4*B)/(3*(Do/t)) = (4*14.798)/(3*194.7208) = 1.0132 \text{ bars}$$

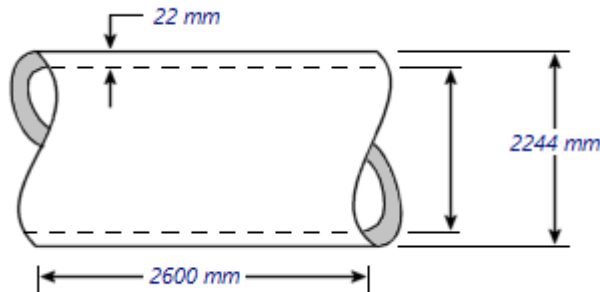
Results for Maximum Stiffened Length					Slen	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
20.500	2244.00	148885.61	109.46	50.0000	0.0000934	9.34

$$MAEP = (4*B)/(3*(Do/t)) = (4*9.3385)/(3*109.4634) = 1.1374 \text{ bars}$$

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Internal Pressure Calculation Results:

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Internal Pressure Results for: Shell 3

Cylindrical Shell Design Information:					
Design Pressure	P	23.000 bars	Int. Design Temperature		80 °C
Material		SA-516 70	Ext. Design Temperature		80 °C
Length		2600.000 mm.	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	1.5000 mm.	Allowable Stress (ope)	S	138.0 N./mm^2
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	138.0 N./mm^2
Outside Diameter	Do	2244.000 mm.	Actual Stress		125.0 N./mm^2
Spec. Min. Thk t _e , or sa		22.0000 mm.	Spec. Nominal Thk.		22.000 mm.
Surface Area		183293.1 cm^2	SG of Contents		0.000
Element Volume		9885221.0 cm3	Weight of Contents		0.0 Kgf
Empty Weight		3094.7 Kgf	Operating Weight		3094.7 Kgf
			Comp. All. Ope. Stress		128.0 N./mm^2

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Material treatment options: +IT

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R_o) / (S \cdot E + 0.4 \cdot P) \text{ per Appendix 1-1 (a)(1)}$$

$$= (23.0 \cdot 1122.0) / (138.0 \cdot 1.0 + 0.4 \cdot 23.0)$$

$$= 18.5776 + 1.5000 = 20.0776 \text{ mm.}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

$$= (S \cdot E \cdot t) / (R_o - 0.4 \cdot t) \text{ per Appendix 1-1 (a)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 20.5) / (1122.0 - 0.4 \cdot 20.5)$$

$$= 25 \text{ bars}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R_o - 0.4 \cdot t) \text{ per Appendix 1-1 (a)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 22.0) / (1122.0 - 0.4 \cdot 22.0)$$

$$= 27 \text{ bars}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R_o - 0.4 \cdot t)) / (E \cdot t)$$

$$= (23.0 \cdot (1122.0 - 0.4 \cdot 20.5)) / (1.0 \cdot 20.5)$$

$$= 124.970 \text{ N./mm}^2$$

% Elongation per Table UG-79-1 (50 * t_{nom} / R_f * (1 - R_f / R_o)) 0.990 %

Minimum Design Metal Temperature Results:



Note: This Element/Detail was specified as being Impact Tested.

Material treatment options: +IT

EXTERNAL PRESSURE RESULTS for: Shell 3

Cylindrical Shell From 70 to 80 Ext. Chart: CS-2 at 80 °C

Elastic Modulus from Chart: CS-2 at 80 °C: 199943392 KPa.

Results for Maximum Allowable Ext. Pressure					MAEP	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
20.500	2244.00	7068.27	109.46	3.1499	0.0003485	34.84

$$MAEP = (4*B)/(3*(Do/t)) = (4*34.8399)/(3*109.4634) = 4.2435 \text{ bars}$$

Results for Required Thickness					Tca	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
11.524	2244.00	7068.27	194.72	3.1499	0.0001480	14.80

$$MAEP = (4*B)/(3*(Do/t)) = (4*14.798)/(3*194.7208) = 1.0132 \text{ bars}$$

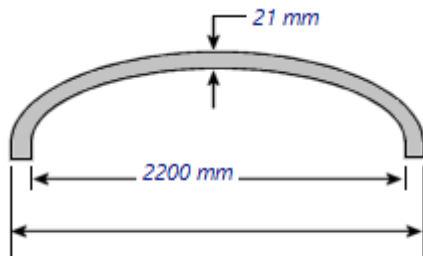
Results for Maximum Stiffened Length					Slen	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
20.500	2244.00	148885.61	109.46	50.0000	0.0000934	9.34

$$MAEP = (4*B)/(3*(Do/t)) = (4*9.3385)/(3*109.4634) = 1.1374 \text{ bars}$$

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Internal Pressure Calculation Results:

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Internal Pressure Results for: Top D end

Elliptical Head Design Information:					
Design Pressure	P	23.000 bars	Int. Design Temperature		80 °C
Material		SA-516 70	Ext. Design Temperature		80 °C
Length		50.800 mm.	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	1.5000 mm.	Allowable Stress (ope)	S	138.0 N./mm^2
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	138.0 N./mm^2
Inside Diameter	Di	2200.000 mm.	Actual Stress		129.9 N./mm^2
Spec. Min. Thk t,e, or sa		21.0000 mm.	Spec. Nominal Thk.		25.000 mm.
Surface Area		58467.6 cm^2	SG of Contents		0.000
Element Volume		1587211.8 cm3	Weight of Contents		0.0 Kg
Empty Weight		1216.9 Kg	Operating Weight		1216.9 Kg
			Comp. All. Ope. Stress		126.7 N./mm^2
			Aspect Ratio		2.00

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 27.500 mm.
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 13.750 mm.

Material treatment options: +IT

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot D \cdot K_{cor}) / (2 \cdot S \cdot E - 0.2 \cdot P) \text{ Appendix 1-4(c)}$$

$$= (23.0 \cdot 2203.0 \cdot 0.998) / (2 \cdot 138.0 \cdot 1.0 - 0.2 \cdot 23.0)$$

$$= 18.3571 + 1.5000 = 19.8571 \text{ mm.}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

$$= (2 \cdot S \cdot E \cdot t) / (K_{cor} \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138.0 \cdot 1.0 \cdot 19.5) / (0.998 \cdot 2203.0 + 0.2 \cdot 19.5)$$

$$= 24 \text{ bars}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (2 \cdot S \cdot E \cdot t) / (K \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138.0 \cdot 1.0 \cdot 21.0) / (1.0 \cdot 2200.0 + 0.2 \cdot 21.0)$$

$$= 26 \text{ bars}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (K_{cor} \cdot D + 0.2 \cdot t)) / (2 \cdot E \cdot t)$$

$$= (23.0 \cdot (0.998 \cdot 2203.0 + 0.2 \cdot 19.5)) / (2 \cdot 1.0 \cdot 19.5)$$

$$= 129.923 \text{ N./mm}^2$$

Straight Flange Required Thickness:
 $= (P \cdot R) / (S \cdot E - 0.6 \cdot P) + c$ per UG-27 (c)(1)
 $= (23.0 \cdot 1101.5) / (138.0 \cdot 1.0 - 0.6 \cdot 23.0) + 1.5$
 $= 20.045 \text{ mm.}$

Straight Flange Maximum Allowable Working Pressure:
 $= (S \cdot E \cdot t) / (R + 0.6 \cdot t)$ per UG-27 (c)(1)
 $= (138.0 \cdot 1.0 \cdot 23.5) / (1101.5 + 0.6 \cdot 23.5)$
 $= 29 \text{ bars}$

Factor K, corroded condition [Kcor]:
 $= (2 + (\text{Inside Diameter} / (2 \cdot \text{Inside Head Depth}))^2) / 6$
 $= (2 + (2203.0 / (2 \cdot 551.5))^2) / 6$
 $= 0.998188$

Percent Elong. per UCS-79, VIII-1-01-57 $(75 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 4.851 %

MDMT Calculations in the Knuckle Portion:

Note: This Element/Detail was specified as being Impact Tested.

MDMT Calculations in the Head Straight Flange:

Note: This Element/Detail was specified as being Impact Tested.

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 27.500 mm.
Head inner surface maximum deviation inside the specified shape, 0.625% of D: 13.750 mm.

Material treatment options: +IT

EXTERNAL PRESSURE RESULTS for: Top D end

Elliptical Head From 80 to 90 Ext. Chart: CS-2 at 80 °C

Elastic Modulus from Chart: CS-2 at 80 °C: 199943392 KPa.

Results for Maximum Allowable Ext. Pressure				MAEP
Tca	Outer Dia	Do/t	Factor A	Factor B
19.500	2242.00	114.97	0.0012080	90.90

$$MAEP = B / (K_0 \cdot Do / t) = 90.8975 / (0.9 \cdot 114.9744) = 8.7838 \text{ bars}$$

Results for Required Thickness				Tca
Tca	Outer Dia	Do/t	Factor A	Factor B
5.746	2242.00	390.20	0.0003559	35.59

$$MAEP = B / (K_0 \cdot Do / t) = 35.5863 / (0.9 \cdot 390.1999) = 1.0133 \text{ bars}$$

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Hydrostatic Test Pressure Results:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	31.048 bars
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	29.900 bars
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	33.965 bars
UG100	= 1.10 * M.A.W.P. * Sa/S	26.271 bars
PED	= max(1.43*DP, 1.25*DP*ratio)	32.890 bars
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	32.890 bars
App 27-4	= M.A.W.P.	23.883 bars

UG-99(b), Test Pressure Calculation:
 = Test Factor • MAWP • Stress Ratio
 = 1.3 • 23.883 • 1.0
 = 31.048 bars

Horizontal Test performed per: UG-99b

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (N./mm² & bars):

From To	Stress	Allowable	Ratio	Pressure
bottom D end	164.1	179.4	0.915	31.26
Shell 1	158.2	179.4	0.882	31.26
Shell 2	158.2	179.4	0.882	31.26
Shell 3	158.2	179.4	0.882	31.27
Top D end	164.1	179.4	0.915	31.26

Stress ratios for Nozzle and Pad Materials (N./mm²):

Description	Pad/Nozzle	Ambient	Operating	Ratio
N6_3"	Nozzle	118.00	118.00	1.000
N6_3"	Pad	138.00	138.00	1.000
N4_8"	Nozzle	118.00	118.00	1.000
N4_8"	Pad	138.00	138.00	1.000
N7_3"	Nozzle	118.00	118.00	1.000
N7_3"	Pad	138.00	138.00	1.000
N5_3"	Nozzle	118.00	118.00	1.000
N5_3"	Pad	138.00	138.00	1.000
K1B_3"	Nozzle	138.00	138.00	1.000
K4B_3"	Nozzle	138.00	138.00	1.000
K3F_3"	Nozzle	138.00	138.00	1.000
K2B_1.5"	Nozzle	138.00	138.00	1.000
K3B_3"	Nozzle	138.00	138.00	1.000
K3D_3"	Nozzle	138.00	138.00	1.000
N8_2"	Nozzle	138.00	138.00	1.000
K2A_1.5"	Nozzle	138.00	138.00	1.000
K3A_3"	Nozzle	138.00	138.00	1.000
K3C_3"	Nozzle	138.00	138.00	1.000
K1A_3"	Nozzle	138.00	138.00	1.000
K4A_3"	Nozzle	138.00	138.00	1.000
K3E_3"	Nozzle	138.00	138.00	1.000
MH1_24"	Nozzle	138.00	138.00	1.000
MH1_24"	Pad	138.00	138.00	1.000
N3_1"	Nozzle	138.00	138.00	1.000
N1_16"	Nozzle	138.00	138.00	1.000
N1_16"	Pad	138.00	138.00	1.000
N2_16"	Nozzle	138.00	138.00	1.000
N2_16"	Pad	138.00	138.00	1.000

Minimum

1.000

Stress ratios for Pressurized Vessel Elements (N./mm²):



Description	Ambient	Operating	Ratio
bottom D end	138.00	138.00	1.000
Shell 1	138.00	138.00	1.000
Shell 2	138.00	138.00	1.000
Shell 3	138.00	138.00	1.000
Top D end	138.00	138.00	1.000

Minimum

1.000

Hoop Stress in Nozzle Wall during Pressure Test (N./mm2):

Description	Ambient	Operating	Ratio
N6_3"	19.59	153.40	0.128
N4_8"	29.57	153.40	0.193
N7_3"	19.59	153.40	0.128
N5_3"	19.59	153.40	0.128
K1B_3"	7.67	179.40	0.043
K4B_3"	7.67	179.40	0.043
K3F_3"	7.67	179.40	0.043
K2B_1.5"	5.60	179.40	0.031
K3B_3"	7.67	179.40	0.043
K3D_3"	7.67	179.40	0.043
N8_2"	6.61	179.40	0.037
K2A_1.5"	5.60	179.40	0.031
K3A_3"	7.67	179.40	0.043
K3C_3"	7.67	179.40	0.043
K1A_3"	7.67	179.40	0.043
K4A_3"	7.67	179.40	0.043
K3E_3"	7.67	179.40	0.043
MH1_24"	97.19	179.40	0.542
N3_1"	4.67	179.40	0.026
N1_16"	62.29	179.40	0.347
N2_16"	62.28	179.40	0.347

Elements Suitable for Test Pressure.

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External Pressure Calculation Results Summary:

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External Pressure Calculations:

From	To	Section Length mm.	Outside Diameter mm.	Corroded Thickness mm.	Factor A	Factor B N./mm ²
10	20	6000	...	...	No Calc	No Calc
20	30	6000	...	...	No Calc	No Calc
30	40	6000	...	...	No Calc	No Calc
40	50	No Calc	2242	19.5	0.001208	90.8975
50	60	7068.27	2244	20.5	0.00034848	34.8399
60	70	7068.27	2244	20.5	0.00034848	34.8399
70	80	7068.27	2244	20.5	0.00034848	34.8399
80	90	No Calc	2242	19.5	0.001208	90.8975

External Pressure Calculations:

From	To	External Actual T. mm.	External Required T. mm.	External Design Pressure bars	External M.A.W.P. bars
10	20	...	No Calc	...	No Calc
20	30	...	No Calc	...	No Calc
30	40	...	No Calc	...	No Calc
40	50	21	7.24577	1.0132	8.78382
50	60	22	13.0242	1.0132	4.24347
60	70	22	13.0242	1.0132	4.24347
70	80	22	13.0242	1.0132	4.24347
80	90	21	7.24577	1.0132	8.78382

Minimum

4

External Pressure Calculations:

From	To	Actual Length Bet. Stiffeners mm.	Allowable Length Bet. Stiffeners mm.	Ring Inertia Required cm ⁴	Ring Inertia Available cm ⁴
10	20	6000	No Calc	No Calc	No Calc
20	30	6000	No Calc	No Calc	No Calc
30	40	6000	No Calc	No Calc	No Calc
40	50	No Calc	No Calc	No Calc	No Calc
50	60	7068.27	148886	No Calc	No Calc
60	70	7068.27	148886	No Calc	No Calc
70	80	7068.27	148886	No Calc	No Calc
80	90	No Calc	No Calc	No Calc	No Calc

Elements Suitable for External Pressure.

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Element and Detail Weights:

From	To	Element Metal Wgt. kg.	Element ID Volume cm3	Corroded Metal Wgt. kg.	Corroded ID Volume cm3	Extra due Misc % kg.
10	20	2129.15	...	1940.65	...	149.04
20	30	1615.8	...	1420.6	...	113.106
30	40	646.322	...	578.96	...	45.2425
40	50	1216.88	1587211	1143.87	1593449	85.1816
50	60	2142.49	6843614	1997.76	6862292	149.975
60	70	2618.6	8364418	2441.71	8387245	183.302
70	80	3094.71	9885221	2885.65	9912199	216.63
80	90	1216.88	1587211	1143.87	1593449	85.1816
Total		14680	28267676.00	13553	28348636.00	1027

Weight of Details:

From	Type	Weight of Detail kg.	X Offset, Dtl. Cent. mm.	Y Offset, Dtl. Cent. mm.	Z Offset, Dtl. Cent. mm.	Description
10	Plat	897.075	1381.56	2500	-1381.56	PLAT:[1 OF 1]
10	Wght	200	...	1500	...	Skirt opening
40	Liqd	1068.02	...	-275	...	Liquid: 40
40	Insl	21.8175	...	-249.6	...	Ins: 40
40	Nozl	35.4139	...	-699.431	...	N6_3"
40	Wght	5	...	-200	...	Vortex Breaker
50	Plat	3605.45	0.19216E-04	0.48768E-04	...	PLAT:[1 OF 1]
50	Liqd	4605	...	900	...	Liquid: 50
50	Insl	36.6819	...	900	...	Ins: 50
50	Nozl	60.4452	1209.13	350	...	N4_8"
50	Nozl	13.814	-1209.13	450	...	N7_3"
50	Nozl	13.814	...	450	-1209.13	N5_3"
50	Nozl	20.2435	...	100	-1200.01	K1B_3"
50	Nozl	20.2435	-1200.01	100	...	K4B_3"
50	Nozl	20.2435	...	100	1200.01	K3F_3"
50	Nozl	10.0907	-601.987	100	-1042.67	K2B_1.5"
50	Nozl	20.2435	-982.993	100	-688.299	K3B_3"
50	Nozl	20.2435	-982.993	100	688.299	K3D_3"
50	Nozl	11.3175	-411.514	200	1130.63	N8_2"
50	Wght	120	...	400	...	elbow Pipe
50	Wght	275	2364	350	...	N4_OVH_PP
60	Liqd	4349.16	...	850	...	Liquid: 60
60	Insl	44.8334	...	1100	...	Ins: 60
60	Nozl	9.63995	1203.97	1800	...	K2A_1.5"
60	Nozl	20.2435	...	1800	-1200.01	K3A_3"
60	Nozl	20.2435	-1200.01	1800	...	K3C_3"
60	Nozl	20.2435	-600.006	1800	-1039.24	K1A_3"
60	Nozl	20.2435	-1127.64	1800	-410.428	K4A_3"
60	Nozl	20.2435	...	1800	1200.01	K3E_3"
60	Wght	450	...	1500	...	Lifting Wt.
60	Wght	1150	...	1300	...	Clip Wt
70	Plat	1821.59	...	0.00028956	-1381.56	PLAT:[1 OF 1]
70	Insl	52.985	...	1300	...	Ins: 70
70	Nozl	406.373	1180.68	1750	...	MH1_24"
70	Nozl	7.93021	...	1100	-1200.01	N3_1"
70	Nozl	207.257	-1141.73	700	...	N1_16"
70	Wght	150	...	2200	...	Demister
70	Wght	100	...	750	...	Distributor
70	Wght	772	-2399	700	...	N1_OVH_PP
70	Wght	360	1700	1800	...	Manway Davit
70	Wght	560	1700	2400	...	Pipe Davit
80	Plat	12906.3	...	900	...	PLAT:[1 OF 1]
80	Insl	21.8175	...	300.4	...	Ins: 80
80	Nozl	213.705	...	782.133	...	N2_16"
80	Wght	513	2424	...	...	N2_OVH_PP
80	Wght	1000	4122.5	600	...	PSV



Total Weight of Each Detail Type:

Platforms	19230.5
Liquid	10022.2
Insulation	178.1
Nozzles	1192.2
Weights	5655.0
Sum of the Detail Weights	36278.0 kg.

Weight Summation Results: (kg.)

	Fabricated	Shop Test	Shipping	Erected	Empty	Operating
Main Elements	15708.5	15708.5	15708.5	15708.5	15708.5	15708.5
Nozzles	1192.2	1192.2	1192.2	1192.2	1192.2	1192.2
Wld Weights	2825.0	2825.0	2825.0	2825.0	2825.0	2825.0
Platforms	...	...	19230.5	19230.5	19230.5	19230.5
Insulation	...	...	178.1	178.1	178.1	178.1
Empty Weights	...	...	...	1560.0	1560.0	...
Test Weights	...	1560.0	...	...	...	...
Ope Weights	...	...	...	...	...	2830.0
Ope. Liquid	...	...	...	...	...	10022.2
Test Liquid	...	28250.4	...	...	...	...
Totals	19725.7	49536.2	39134.3	40694.3	40694.3	51986.5

Field Installation Options:

Miscellaneous Weight Percent: 7.0 %

Note that the above value for the miscellaneous weight percent has been applied to the shells/heads/flange/tubesheets/tubes etc. in the weight calculations for metallic components.

Weight Summary:

Fabricated Wt.	- Bare weight without removable internals	19725.7 kg.
Shop Test Wt.	- Fabricated weight + water (full)	49536.2 kg.
Shipping Wt.	- Fab. weight + removable intls.+ shipping app.	39134.3 kg.
Erected Wt.	- Fab. wt + or - loose items (trays,platforms etc.)	40694.3 kg.
Ope. Wt. no Liq	- Fab. weight + internals. + details + weights	40694.3 kg.
Operating Wt.	- Empty weight + operating liq. uncorroded	51986.5 kg.

Weights used in Vortex Shedding Calculations:

Mass of the Upper 1/3 of the Vertical Vessel (operating)	29798.5 kg.
Mass of the Upper 1/3 of the Vertical Vessel (filled w/water)	41796.0 kg.
Mass of the Upper 1/3 of the Vertical Vessel (empty)	26315.7 kg.

Surface Areas of Elements:

From	To	Outside Surface Area cm^2	Inside Surface Area cm^2
10	20	174673	172788
20	30	174673	172788
30	40	69869	69115
40	50	58467.6	55975.9
50	60	126895	124407
60	70	155094	152053
70	80	183293	179699
80	90	58467.6	55975.9

Total	1001431.8	982801.2 cm2
Total	1077.9	1057.9 ft^2



Element and Detail Weights:

From	To	Total Ele. Empty Wgt. kg.	Total. Ele. Oper. Wgt. kg.	Total. Ele. Hydro. Wgt. kg.	Total Dtl. Offset Mom. Kg-m.	Oper. Wgt. No Liquid kg.
10	20	3375.26	3375.26	3375.26	1239.38	3375.26
20	30	1728.91	1728.91	1728.91	...	1728.91
30	40	691.564	691.564	691.564	...	691.564
40	50	1364.29	2432.31	2950.54	...	1364.29
50	60	6420.3	11145.3	13259.7	631.673	6540.3
60	70	4557.6	8906.76	12916.9	47.6609	4557.6
70	80	7599.47	7749.47	17478.7	44.8627	7749.47
80	90	14956.9	15956.9	16543.2	5366.12	15956.9

Cumulative Vessel Weight

From	To	Cumulative Ope Wgt. No Liquid kg.	Cumulative Oper. Wgt. kg.	Cumulative Hydro. Wgt. kg.
10	20	41964.3	51986.5	68944.8
20	30	38589.1	48611.2	65569.5
30	40	36860.2	46882.3	63840.6
40	50	36168.6	46190.8	63149
50	60	34804.3	43758.5	60198.5
60	70	28264	32613.2	46938.7
70	80	23706.4	23706.4	34021.8
80	90	15956.9	15956.9	16543.2

The cumulative operating weights no liquid in the column above are the cumulative operating weights minus the operating liquid weight minus any weights absent in the empty condition.

Cumulative Vessel Moment

From	To	Cumulative Empty Mom. Kg-m.	Cumulative Oper. Mom. Kg-m.	Cumulative Hydro. Mom. Kg-m.
10	20	7329.7	7329.7	7329.7
20	30	6090.31	6090.31	6090.31
30	40	6090.31	6090.31	6090.31
40	50	6090.31	6090.31	6090.31
50	60	6090.31	6090.31	6090.31
60	70	5458.64	5458.64	5458.64
70	80	5410.98	5410.98	5410.98
80	90	5366.12	5366.12	5366.12

Nozzle Flange MAWP Results: (bars & °C)

Nozzle Description	Flange Ope.	Rating Ambient	Des. Temp	Class	Grade/Group	Equiv. Press	- - - UG-44	Max Pressure 50%	DNV
N6_3"	47.94	51.00	80	300	GR 1.1	43.965	27.946	47.941	27.946
N4_8"	47.94	51.00	80	300	GR 1.1	27.731	44.180	47.941	44.180
N7_3"	47.94	51.00	80	300	GR 1.1	...	...	...	...
N5_3"	47.94	51.00	80	300	GR 1.1	...	...	...	...
K1B_3"	47.94	51.00	80	300	GR 1.1	...	...	...	...
K4B_3"	47.94	51.00	80	300	GR 1.1	...	...	...	...
K3F_3"	47.94	51.00	80	300	GR 1.1	...	...	...	...
K2B_1.5"	47.94	51.00	80	300	GR 1.1	...	...	...	...
K3B_3"	47.94	51.00	80	300	GR 1.1	...	...	...	...
K3D_3"	47.94	51.00	80	300	GR 1.1	...	...	...	...
N8_2"	47.94	51.00	80	300	GR 1.1	...	...	...	...
K2A_1.5"	47.94	51.00	80	300	GR 1.1	...	...	...	...
K3A_3"	47.94	51.00	80	300	GR 1.1	...	...	...	...
K3C_3"	47.94	51.00	80	300	GR 1.1	...	...	...	...
K1A_3"	47.94	51.00	80	300	GR 1.1	...	...	...	...
K4A_3"	47.94	51.00	80	300	GR 1.1	...	...	...	...
K3E_3"	47.94	51.00	80	300	GR 1.1	...	...	...	...
MH1_24"	47.94	51.00	80	300	GR 1.1	...	...	...	...
N3_1"	47.94	51.00	80	300	GR 1.1	...	...	...	...
N1_16"	47.94	51.00	80	300	GR 1.1	19.479	47.941	47.941	47.941
N2_16"	47.94	51.00	80	300	GR 1.1	19.479	47.941	47.941	47.941
Min Rating	27.946	51.000, [for Core Elements]					...	...	...

Selected Method for Derating ASME B16.5 Flange MAWP: ASME UG-44(b)

The UG-44(b) Method is based on the paper PVP 2013-97814. PV Elite uses the maximum loads from each load category to determine ME and FE. In many cases, the computed maximum allowable pressure will be greater than the flange rating. In these cases, the minimum of the rating from the table and the UG-44(b) method will be used. SA-193 B8 Cl. 2 bolts or ones with higher allowable stresses at the specified bolt size shall be used.

The axial force FE must put the nozzle in tension (-P for 107/297; +Fr for PD 5500 AnnexG) to be considered.

Nozzle Description	Gasket Dimension G mm.	Fm Value	Resolved Moment ME Kg-m.	Axial Force FE Kgf	MAWP PD bars	Maximum Pressure Pmax bars	Pass Fail
N6_3"	111.125	0.50	115	205	23.883	27.946	Pass
N4_8"	248.539	0.50	818	555	23.883	44.180	Pass
N1_16"	442.976	0.50	3268	1105	23.883	47.941	Pass
N2_16"	442.976	0.50	3268	1105	23.883	47.941	Pass

Evaluating Flange on Nozzle [N63"]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 \cdot [(PR - PD) + FM \cdot PR]$$

$$= 16(115) + 4 \cdot \text{abs}(205)111.125 \leq \pi \cdot 111.125^3 \cdot [(47.941 - 23.883) + .500 \cdot 47.941]$$

$$= 0.19329E+04 \leq 0.21115E+04 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle N63" [Pmax]:

$$= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$$

$$= \min(47.941, 0.500 \cdot 47.941 + 47.941 - 1932.9 / (\pi \cdot 111.125^3))$$

$$= \min(47.941, 27.946)$$

$$= 27.946 \text{ bars}$$

Evaluating Flange on Nozzle [N48"]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 \cdot [(PR - PD) + FM \cdot PR]$$

$$= 16(818) + 4 \cdot \text{abs}(555)248.539 \leq \pi \cdot 248.539^3 \cdot [(47.941 - 23.883) + .500 \cdot 47.941]$$

$$= 0.13640E+05 \leq 0.23623E+05 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle N48" [Pmax]:



$$= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$$
$$= \min(47.941, 0.500 \cdot 47.941 + 47.941 - 13639.8 / (\pi \cdot 248.539^3))$$
$$= \min(47.941, 44.180)$$
$$= 44.180 \text{ bars}$$

Evaluating Flange on Nozzle [N116"]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 [(PR - PD) + FM \cdot PR]$$
$$= 16(3268) + 4 \cdot \text{abs}(1105) 442.976 \leq \pi \cdot 442.976^3 [(47.941 - 23.883) + .500 \cdot 47.941]$$
$$= 0.54246E+05 \leq 0.13375E+06 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle N116" [Pmax]:

$$= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$$
$$= \min(47.941, 0.500 \cdot 47.941 + 47.941 - 54245.9 / (\pi \cdot 442.976^3))$$
$$= \min(47.941, 52.432)$$
$$= 47.941 \text{ bars}$$

Evaluating Flange on Nozzle [N216"]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 [(PR - PD) + FM \cdot PR]$$
$$= 16(3268) + 4 \cdot \text{abs}(1105) 442.976 \leq \pi \cdot 442.976^3 [(47.941 - 23.883) + .500 \cdot 47.941]$$
$$= 0.54246E+05 \leq 0.13375E+06 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle N216" [Pmax]:

$$= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$$
$$= \min(47.941, 0.500 \cdot 47.941 + 47.941 - 54245.9 / (\pi \cdot 442.976^3))$$
$$= \min(47.941, 52.432)$$
$$= 47.941 \text{ bars}$$

Flange Pressure Ratings are per ASME B16.5 2025 Metric Edition

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The Natural Frequencies for the vessel have been computed iteratively by solving a system of matrices. These matrices describe the mass and the stiffness of the vessel. This is the generalized eigenvalue/eigenvector problem and is referenced in some mathematical texts.

The Natural Frequency for the Vessel (Empty.) is 3.71066 Hz.

The Natural Frequency for the Vessel (Ope...) is 3.52735 Hz.

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Wind Load Calculation Results:

Wind Load Calculations per India Std. IS 875 (Part 3):2015, Amd. 1&2 (2020):

Input Values:

Wind Design Code	IS 875 (PART 3):2015-2020	
Basic Wind Speed for IS-875	180	Km/hr
Wind Zone Number	5	
Base Elevation	300	mm.
Percent Wind for Hydrotest	33.0	
Risk Factor	[k1]	1.0
Terrain Category		2
Equipment Class		Class A
Topography Factor	[k3]	1.0
Importance Factor	[k4]	1.15
Damping Factor (Beta) for Wind (Ope)		0.0100
Damping Factor (Beta) for Wind (Empty)		0.0000
Damping Factor (Beta) for Wind (Filled)		0.0000
Use Gust Response Factor (Dynamic Analysis)		No

Calculated Values:

Vessel Height to Diameter ratio: 5.780

Checking the requirement for dynamic effect of wind (9.1):

Vessel Operating Natural Frequency: 3.527 Hz

Note: Vessel Natural frequency is ≥ 1 Hz.

Note: Vessel height to diameter ratio (5.780) is > 5 .

Note:

As per Section 9.1, Dynamic effects of wind should be considered. The Cross-Wind dynamic effects can be considered using the Vortex Shedding which can be invoked from Design Constraints tab. While the Along-Wind effects can be considered using Gust Response Factor method, which can be invoked from the Wind Load tab.

Design Wind Speed [Vz]:

$$= \text{Basic Wind Speed} \cdot k_1 \cdot k_{2,1} \cdot k_3 \cdot k_4 \quad (\text{for the 1st element})$$

$$= 50.0 \cdot 1.0 \cdot 1.0 \cdot 1.0 \cdot 1.15$$

$$= 57.496 \text{ meters/s}$$

Design Wind Pressure per 4.5.2 [pd]:

$$= \max(0.7 \cdot p_z, K_d \cdot K_a \cdot K_c \cdot 0.6 \cdot V_z^2)$$

$$= \max(0.7 \cdot 1983.503, 1.0 \cdot 0.88 \cdot 1.0 \cdot 0.6 \cdot 57.5^2)$$

$$= 1739.5 \text{ N/m}^2$$

Element Wind Load (operating condition):

$$= \text{Wind Area} \cdot C_f \cdot p_d$$

$$= 6.7 \cdot 0.816 \cdot 1739.475$$

$$= 9465.7 \text{ N}$$

From	Height mm.	k1	k2	k3	Vz m/sec	Cf
10	1550.00	1.0000	1.0000	1.0000	57.50	0.8156
20	4050.00	1.0000	1.0000	1.0000	57.50	0.8156
30	5800.00	1.0000	1.0000	1.0000	57.50	0.8156
40	6325.40	1.0000	1.0000	1.0000	57.50	0.8156
50	7250.80	1.0000	1.0000	1.0000	57.50	0.8156
60	9250.80	1.0000	1.0000	1.0000	57.50	0.8156
70	11650.80	1.0000	1.0165	1.0000	58.45	0.8156
80	13246.38	1.0000	1.0325	1.0000	59.36	0.8156

Wind Vibration Calculations



This evaluation is based on work by Kanti Mahajan and Ed Zorilla

Nomenclature

Cf - Correction factor for natural frequency
 D - Average outer insulated diameter of vessel, mm.
 Df - Damping Factor < 0.75 Unstable, > 0.95 Stable
 Dr - Average outer insulated of top half of vessel, mm.
 f - Natural frequency of vibration (Hertz)
 f1 - Natural frequency of bare vessel based on a unit value of $(D/L^2)(10^4)$
 L - Total height of structure mm.
 Lc - Total length of conical section(s) of vessel, mm.
 tb - Uncorroded plate thickness at bottom of vessel, mm.
 V30 - Design Wind Speed provided by user, Km/hr
 Vc - Critical wind velocity, Km/hr
 Vw - Maximum wind speed at top of structure, Km/hr
 W - Total corroded weight of structure, Kgf
 Ws - Cor. vessel weight excl. weight of parts which do not effect stiff. Kgf
 Z - Maximum amplitude of vibration at top of vessel, mm.
 D1 - Logarithmic decrement (taken as 0.03 for Welded Structures)
 Vp - Vib. Chance, $\leq 0.320E-06$ (High); $0.320E-06 < 0.400E-06$ (Probable)
 P30 - wind pressure 30 feet above the base

Check other Conditions and Basic Assumptions:

:: Total Cone Length / Total Length < 0.5

:: $0.0/12701.6 = 0.0$

:: $[D / L^2] \cdot 10^4 < 8$ (English Units, ft.)

:: $[7.57/41.67^2] \cdot 10^4 = 43.612$ [Geometry Violation]

Compute the vibration possibility. If $V_p > 0.400E-06$ no chance. [Vp]:

= $W / (L \cdot Dr^2)$

= $50780 / (12701.6 \cdot 2337.317^2)$

= $0.73181E-06$

Since V_p is $> 0.400E-06$ no further vibration analysis is required !

Platform Load Calculations

ID	Wind Area cm2	Elevation mm.	Pressure Kgs/m2	Force Kgf	Cf
PLAT:[1 OF 1]	18000.00	2800.00	177.45	319.42	1.00
PLAT:[1 OF 1]	36000.00	6350.80	177.45	638.83	1.00
PLAT:[1 OF 1]	36000.00	10350.80	180.16	648.58	1.00
PLAT:[1 OF 1]	153804.00	13850.80	190.47	2929.48	1.00

Wind Loads on Masses/Equipment/Piping

ID	Wind Area cm2	Elevation mm.	Pressure Kgs/m2	Force Kgf
Skirt opening	0.00	1800.00	177.45	0.00
Vortex Breake	0.00	6100.00	177.45	0.00
elbow Pipe	0.00	6750.80	177.45	0.00
N40VHPP	0.00	6700.80	177.45	0.00
Lifting Wt.	0.00	9650.80	178.44	0.00
Clip Wt	0.00	9450.80	177.94	0.00
Demister	0.00	12550.80	186.63	0.00
Distributor	0.00	11100.80	182.01	0.00
N10VHPP	0.00	11050.80	181.88	0.00
Manway Davit	0.00	12150.80	185.18	0.00
Pipe Davit	0.00	12750.80	187.36	0.00
N20VHPP	0.00	12950.80	188.09	0.00
PSV	0.00	13550.80	190.28	0.00

The Natural Frequency for the Vessel (Ope...) is 3.52735 Hz.

Wind Load Calculation:

From	To	Wind Height mm.	Wind Diameter mm.	Wind Area cm^2	Wind Pressure Kgs/m^2	Element Wind Load Kgf
10	20	1550	2668.8	66720	177.452	1696.53
20	30	4050	2668.8	66720	177.452	1061.82
30	40	5800	2668.8	26688	177.452	424.729
40	50	6325.4	2858.4	1452.07	177.452	23.1091
50	60	7250.8	2860.8	51494.4	177.452	1435
60	70	9250.8	2860.8	62937.6	177.452	1001.63
70	80	11650.8	2860.8	74380.8	183.36	1856.98
80	90	13246.4	2858.4	15842.4	189.161	3638.22

The Wind Loads calculated and printed in the Wind Load calculation report have been factored by the input scalar/load reduction factor of: 1.100.

Be sure the wind speed is in accordance with the specified wind design code.

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Earthquake Load Calculation Results:

Input Values:

Seismic Design Code	Spectrum
Range Type	Period
Ordinate Type	Acceleration
Include Missing Mass Components	Yes
Shock Scale Factor X Direction	1.0000
Shock Scale Factor Y Direction	0.6700
Combination Method	Group
Spectrum Name	User Defined

User Defined Spectrum Data Points:

Acceleration	Period
0.0700	0.0100
0.1170	0.0500
0.1750	0.1000
0.1750	0.1300
0.1750	0.2000
0.1750	0.2700
0.1750	0.3000
0.1750	0.4000
0.1400	0.5000
0.1270	0.5500
0.1170	0.6000
0.1050	0.6700
0.0870	0.8000
0.0700	1.0000
0.0580	1.2000
0.0500	1.4000
0.0440	1.6000
0.0390	1.8000
0.0350	2.0000
0.0290	2.4000

Computed Natural Frequencies (OPE): 5

Mode	Freq (Hz)	Freq (Rad/Sec)	Period (Sec)
1	3.6340	22.8331	0.2752
2	25.2528	158.6677	0.0396
3	33.4504	210.1753	0.0299
4	81.1500	509.8804	0.0123
5	127.1724	799.0479	0.0079

Mass Participation Factors: 5

Mode	X	Y
1	2.01116165	0.00000000
2	0.00993469	-0.00000000
3	0.00000000	0.00908664
4	0.00018455	0.00000000
5	-0.00000000	-0.00006519

Computed EigenVectors, Mass Normalized: 5

Node	Dx	Dy	Rz
Mode # 1	Freq (HZ) = 3.6340		
10	0.000000	0.000000	-0.000000
20	0.005791	0.000000	-0.000090
30	0.019292	0.000000	-0.000157
40	0.026321	0.000000	-0.000177
50	0.026688	0.000000	-0.000177
60	0.040211	-0.000000	-0.000192
70	0.057758	-0.000000	-0.000202
80	0.079127	0.000000	-0.000206
90	0.079543	0.000000	-0.000206



Mode # 2	Freq (HZ) = 25.2528		
10	0.000000	-0.000000	-0.000000
20	0.039025	-0.000000	-0.000261
30	0.080312	-0.000000	-0.000088
40	0.087791	-0.000000	0.000091
50	0.087787	-0.000000	0.000097
60	0.074600	0.000000	0.000316
70	0.030788	0.000000	0.000543
80	-0.041918	-0.000000	0.000648
90	-0.043298	-0.000000	0.000648

Mode # 3	Freq (HZ) = 33.4504		
10	0.000000	0.000001	-0.000000
20	0.000000	0.020510	0.000000
30	-0.000000	0.040776	-0.000000
40	-0.000000	0.048604	-0.000000
50	-0.000000	0.048830	-0.000000
60	0.000000	0.055527	0.000000
70	-0.000000	0.061816	0.000000
80	0.000000	0.067178	-0.000000
90	0.000000	0.067222	-0.000000

Mode # 4	Freq (HZ) = 81.1500		
10	0.000000	0.000000	-0.000001
20	0.085666	0.000000	-0.000222
30	0.101483	0.000000	0.000423
40	0.076488	0.000000	0.000757
50	0.074613	0.000000	0.000766
60	-0.036483	-0.000000	0.000693
70	-0.093610	0.000000	-0.000130
80	0.024323	0.000000	-0.000771
90	0.026396	0.000000	-0.000771

Mode # 5	Freq (HZ) = 127.1724		
10	-0.000000	-0.000001	0.000000
20	-0.000000	-0.041009	0.000000
30	-0.000000	-0.072555	0.000000
40	0.000000	-0.081420	0.000000
50	-0.000000	-0.081531	0.000000
60	0.000000	-0.062994	0.000000
70	-0.000000	-0.009264	0.000000
80	0.000000	0.058724	0.000000
90	-0.000000	0.059289	0.000000

Mass Percentages:			
Included(X)	Included(Y)	Added(X)	Added(Y)
95.78	95.60	4.22	4.40

NOTE: In the following reports, "Max Contrib" displays the contribution of the Mode/Load combination having the maximum impact on the total (and names that Mode/Load combination).

Restraint Loads:			
Node	Fx(Kgf)	Fy(Kgf)	Mz(Kg-m.)
10	7409.7	3025.3	79916.3
Max	7374.9	3023.1	79894.0
Contrib	1 (X)	3 (Y)	1 (X)

Element Forces and Moments:			
Node	Fx(Kgf)	Fy(Kgf)	Mz(Kg-m.)
10	7408.4	3023.7	79916.3
Max	7374.9	3023.1	79894.0
Contrib	1 (X)	3 (Y)	1 (X)
20	7408.4	3023.7	61456.6



Max	7374.9	3023.1	61456.4
Contrib	1 (X)	3 (Y)	1 (X)
20	7366.9	2974.4	61456.6
Max	7338.8	2974.2	61456.4
Contrib	1 (X)	3 (Y)	1 (X)
30	7366.9	2974.4	43134.3
Max	7338.8	2974.2	43109.1
Contrib	1 (X)	3 (Y)	1 (X)
30	7298.5	2923.1	43134.3
Max	7275.4	2922.9	43109.1
Contrib	1 (X)	3 (Y)	1 (X)
40	7298.5	2923.1	35892.4
Max	7275.4	2922.9	35833.6
Contrib	1 (X)	3 (Y)	1 (X)
40	7180.6	2844.1	35892.4
Max	7163.7	2844.0	35833.6
Contrib	1 (X)	3 (Y)	1 (X)
50	7180.6	2844.1	35530.5
Max	7163.7	2844.0	35469.7
Contrib	1 (X)	3 (Y)	1 (X)
50	6673.0	2500.0	35530.5
Max	6671.7	2499.3	35469.7
Contrib	1 (X)	3 (Y)	1 (X)
60	6673.0	2500.0	23569.7
Max	6671.7	2499.3	23460.5
Contrib	1 (X)	3 (Y)	1 (X)
60	5589.9	1924.2	23569.7
Max	5576.8	1920.6	23460.5
Contrib	1 (X)	3 (Y)	1 (X)
70	5589.9	1924.2	11283.4
Max	5576.8	1920.6	11191.2
Contrib	1 (X)	3 (Y)	1 (X)
70	4305.7	1391.2	11283.4
Max	4270.5	1385.4	11191.2
Contrib	1 (X)	3 (Y)	1 (X)
80	4305.7	1391.2	88.3
Max	4270.5	1385.4	87.6
Contrib	1 (X)	3 (Y)	1 (X)
80	1738.1	559.9	88.3
Max	1723.5	557.6	87.6
Contrib	1 (X)	3 (Y)	1 (X)
90	1738.1	559.9	0.0
Max	1723.5	557.6	0.0
Contrib	1 (X)	3 (Y)	

Displacements:

Node	Dx(mm.)	Dy(mm.)	Rz(Deg)
10	0.0000	0.0000	0.0000
Max	0.0000	0.0000	0.0000
Contrib	1 (X)	3 (Y)	1 (X)
20	0.2960	0.0047	0.0104
Max	0.2958	0.0047	0.0104
Contrib	1 (X)	3 (Y)	1 (X)
30	0.9857	0.0094	0.0181
Max	0.9855	0.0094	0.0181
Contrib	1 (X)	3 (Y)	1 (X)
40	1.3448	0.0112	0.0203
Max	1.3446	0.0112	0.0203
Contrib	1 (X)	3 (Y)	1 (X)



50	1.3635	0.0113	0.0204
Max	1.3633	0.0113	0.0204
Contrib	1 (X)	3 (Y)	1 (X)
60	2.0542	0.0128	0.0221
Max	2.0541	0.0128	0.0221
Contrib	1 (X)	3 (Y)	1 (X)
70	2.9505	0.0143	0.0233
Max	2.9505	0.0143	0.0233
Contrib	1 (X)	3 (Y)	1 (X)
80	4.0421	0.0155	0.0238
Max	4.0421	0.0155	0.0238
Contrib	1 (X)	3 (Y)	1 (X)
90	4.0634	0.0155	0.0238
Max	4.0634	0.0155	0.0238
Contrib	1 (X)	3 (Y)	1 (X)



The following table is for the Operating Case.

Wind/Earthquake Shear, Bending:

From	To	Distance to Support mm.	Cumulative Wind Shear Kgf	Earthquake Shear Kgf	Wind Bending Kg-m.	Earthquake Bending Kg-m.
10	20	1250	11138	7408.37	95699.4	79916.3
20	30	3750	9441.49	7366.95	69974.5	61456.6
30	40	5500	8379.67	7298.52	47697.6	43134.3
40	50	6025.4	7954.94	7180.59	39530.2	35892.4
50	60	6950.8	7931.83	6673	39126.6	35530.5
60	70	8950.8	6496.83	5589.91	26140.6	23569.7
70	80	11350.8	5495.2	4305.73	12949.1	11283.4
80	90	12676.2	3638.22	1738.14	1075.43	88.2994

Note:

The Wind Shears/Moments and the Earthquake Shears/Moments calculated and printed in the Wind/Earthquake Shear and Bending report have been factored by the input Scalar/Load reductions factors of;
 Wind: 1.100; Earthquake: 1.000.

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Wind Deflection Calculations:

The following table is for Wind Deflection; Operating Case.

Wind Load Deflection:

From	To	Cumulative Shear Kg	Centroid Deflection mm.	Elem. End Deflection mm.	Elem. Ang. Rotation
10	20	11138	0.072938	0.27775	0.00021122
20	30	9441.49	0.59477	1.00573	0.00036164
30	40	8379.67	1.19244	1.39024	0.00040599
40	50	7954.94	1.40055	1.41089	0.00040716
50	60	7931.83	1.78561	2.17491	0.00043959
60	70	6496.83	2.66647	3.17102	0.00046327
70	80	5495.2	3.77828	4.39165	0.00047282
80	90	3638.22	4.40366	4.41567	0.00047282

Critical Wind Velocity for Tower Vibration:

From	To	1st Crit. Wind Speed Km/hr	2nd Crit. Wind Speed Km/hr
10	20	168.992	1056.2
20	30	168.992	1056.2
30	40	168.992	1056.2
40	50	180.998	1131.24
50	60	181.15	1132.18
60	70	181.15	1132.18
70	80	181.15	1132.18
80	90	180.998	1131.24

Allowable deflection at the Tower Top (Ope)(6.000"/100ft. Criteria)
 Allowable deflection : 63.508 Actual deflection : 4.416 mm.

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Longitudinal Stress Constants:

From	To	Metal Area New cm ²	Metal Area Corroded cm ²	Section Modulus New mm. ³	Section Modulus Corroded mm. ³
10	20	833.907	736.626	45867572	40542040
20	30	833.907	733.167	45867572	40353480
30	40	833.907	746.998	45867572	41107356
40	50	1465.27	1361.52	80604176	74997544
50	60	1535.74	1431.99	84482104	78880464
60	70	1535.74	1431.99	84482104	78880464
70	80	1535.74	1431.99	84482104	78880464
80	90	1465.27	1361.52	80604176	74997544

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Longitudinal Allowable Stresses:

From	To	Tensile N./mm ²	Hydrotest Tensile N./mm ²	Compressive N./mm ²	Hydrotest Compressive N./mm ²
10	20	90.7184	117.934	-108.628	-112.764
20	30	90.7184	117.934	-108.466	-112.764
30	40	115.918	150.693	-109.11	-112.764
40	50	165.596	215.276	-126.716	-128.59
50	60	165.596	215.276	-127.967	-129.705
60	70	165.596	215.276	-127.967	-129.705
70	80	165.596	215.276	-127.967	-129.705
80	90	165.596	215.276	-126.716	-128.59

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Longitudinal Stress Report



Longitudinal Operating and Empty Stresses are computed in the corroded condition. Stresses due to loads in the hydrostatic test cases have been computed in the new and cold condition.

Longitudinal Pressure Stresses due to:

From	To	Longitudinal Stress Internal Pressure N./mm ²	Longitudinal Stress External Pressure N./mm ²	Longitudinal Stress Hydrotest Pressure N./mm ²
10	20	...	...	...
20	30	...	...	...
30	40	...	...	...
40	50	64.504	-2.93803	80.6989
50	60	61.335	-2.79843	77.0025
60	70	61.335	-2.79843	77.0025
70	80	61.335	-2.79843	77.0025
80	90	64.504	-2.93803	80.6989

Longitudinal Stresses due to Weight Loads for these Conditions:

From	To	Wght. Str. Empty N./mm ²	Wght. Str. Operating N./mm ²	Wght. Str. Hydrotest N./mm ²	Wght. Str. Emp. Mom. N./mm ²	Wght. Str. Opr. Mom. N./mm ²
10	20	-5.58678	-6.92105	...	1.77297	1.77297
20	30	-5.16167	-6.50223	...	1.48006	1.48006
30	40	-4.83912	-6.15487	...	1.45292	1.45292
40	50	-2.60516	-2.60516	...	0.79637	0.79637
50	60	-2.38353	-2.38353	...	0.75716	0.75716
60	70	-1.93563	-1.93563	...	0.67863	0.67863
70	80	-1.6235	-1.6235	...	0.67271	0.67271
80	90	-1.14935	-1.14935	...	0.70167	0.70167

Longitudinal Stresses due to Weight Loads and Bending for these Conditions:

From	To	Wght. Str. Hyd. Mom. N./mm ²	Bend. Str. Oper. Wind N./mm ²	Bend. Str. Oper. Equ. N./mm ²	Bend. Str. Hyd. Wind N./mm ²	Bend. Str. Hyd. Equ. N./mm ²
10	20	...	23.1485	19.3308	...	...
20	30	...	17.0051	14.9351	...	...
30	40	...	11.3788	10.2902	...	...
40	50	...	5.16894	4.69327	...	...
50	60	...	4.86433	4.41725	...	...
60	70	...	3.24987	2.93025	...	...
70	80	...	1.60987	1.40279	...	...
80	90	...	0.14062	0.011546	...	...

Longitudinal Stresses due to these Conditions:

From	To	Vortex Shedding Operating Case N./mm ²	Vortex Shedding Empty Case N./mm ²	Vortex Shedding Test Case N./mm ²	Earthquake Empty Case N./mm ²
10	20	...	...	...	15.6209
20	30	...	...	...	12.2064
30	40	...	...	...	8.58487
40	50	...	...	...	3.96964
50	60	...	...	...	3.73914
60	70	...	...	...	2.55384
70	80	...	...	...	1.25634



80 90 ... 0.010249

Longitudinal Stresses due to Applied Axial Forces:

From	To	Longitudinal Stress Y Forces Wind N./mm ²	Longitudinal Stress Y Forces Seismic N./mm ²
10	20	...	-0.40255
20	30	...	-0.39786
30	40	...	-0.38376
40	50	...	-0.20486
50	60	...	-0.17121
60	70	...	-0.13177
70	80	...	-0.095275
80	90	...	-0.040332

Longitudinal Stresses due to User Forces and Moments:

From	To	Wind For/Mom Corroded N./mm ²	Earthquake For/Mom Corroded N./mm ²	Wind For/Mom No Corrosion N./mm ²	Earthquake For/Mom No Corrosion N./mm ²
10	20	...	...	...	...
20	30	...	...	...	...
30	40	...	...	...	...
40	50	...	...	...	...
50	60	...	...	...	...
60	70	...	...	...	...
70	80	...	...	...	...
80	90	...	...	...	...

Bending Stresses due to Blast Load:

From	To	Bending Stress N./mm ²
10	20	...
20	30	...
30	40	...
40	50	...
50	60	...
60	70	...
70	80	...
80	90	...



Stress Combination Load Cases for Vertical Vessels:

Load Case Definition Key

IP = Longitudinal Stress due to Internal Pressure
 EP = Longitudinal Stress due to External Pressure
 HP = Longitudinal Stress due to Hydrotest Pressure
 NP = No Pressure
 EW = Longitudinal Stress due to Weight (No Liquid)
 OW = Longitudinal Stress due to Weight (Operating)
 HW = Longitudinal Stress due to Weight (Hydrotest)
 WI = Bending Stress due to Wind Moment (Operating)
 EQ = Bending Stress due to Earthquake Moment (Operating)
 EE = Bending Stress due to Earthquake Moment (Empty)
 HI = Bending Stress due to Wind Moment (Hydrotest)
 HE = Bending Stress due to Earthquake Moment (Hydrotest)
 WE = Bending Stress due to Wind Moment (Empty) (no CA)
 WF = Bending Stress due to Wind Moment (Filled) (no CA)
 CW = Longitudinal Stress due to Weight (Empty) (no CA)
 VO = Bending Stress due to Vortex Shedding Loads (Ope)
 VE = Bending Stress due to Vortex Shedding Loads (Emp)
 VF = Bending Stress due to Vortex Shedding Loads (Test No CA.)
 FW = Axial Stress due to Vertical Forces for the Wind Case
 FS = Axial Stress due to Vertical Forces for the Seismic Case
 BW = Bending Stress due to Lat. Forces for the Wind Case, Corroded
 BS = Bending Stress due to Lat. Forces for the Seismic Case, Corroded
 BN = Bending Stress due to Lat. Forces for the Wind Case, UnCorroded
 BU = Bending Stress due to Lat. Forces for the Seismic Case, UnCorroded
 BL = Bending Stress due to Blast Loads, Corroded

General Notes:

Case types HI and HE are in the Un-Corroded condition.

Case types WE, WF, and CW are in the Un-Corroded condition.

A blank stress and stress ratio indicates that the corresponding stress comprising those components that did not contribute to that type of stress.

An asterisk (*) in the final column denotes overstress.

Analysis of Load Case 1 : NP+EW+WI+FW+BW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	19.33	90.72	-30.51	108.63	0.2131	0.2809
20	13.32	90.72	-23.65	108.47	0.1469	0.2180
30	7.99	115.92	-17.67	109.11	0.0690	0.1620
40	3.36	165.60	-8.57	126.72	0.0203	0.0676
50	3.24	165.60	-8.01	127.97	0.0196	0.0626
60	1.99	165.60	-5.86	127.97	0.0120	0.0458
70	0.66	165.60	-3.91	127.97	0.0040	0.0305
80		165.60	-1.99	126.72		0.0157

Analysis of Load Case 2 : NP+EW+EE+FS+BS

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	11.40	90.72	-23.38	108.63	0.1257	0.2153
20	8.13	90.72	-19.25	108.47	0.0896	0.1774
30	4.81	115.92	-15.26	109.11	0.0415	0.1399
40	1.96	165.60	-7.58	126.72	0.0118	0.0598
50	1.94	165.60	-7.05	127.97	0.0117	0.0551
60	1.17	165.60	-5.30	127.97	0.0070	0.0414
70	0.21	165.60	-3.65	127.97	0.0013	0.0285
80		165.60	-1.90	126.72		0.0150

Analysis of Load Case 3 : NP+OW+WI+FW+BW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
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Node	Stress	Stress	Stress	Stress	Ratio	Ratio
10	18.00	90.72	-31.84	108.63	0.1984	0.2931
20	11.98	90.72	-24.99	108.47	0.1321	0.2304
30	6.68	115.92	-18.99	109.11	0.0576	0.1740
40	3.36	165.60	-8.57	126.72	0.0203	0.0676
50	3.24	165.60	-8.01	127.97	0.0196	0.0626
60	1.99	165.60	-5.86	127.97	0.0120	0.0458
70	0.66	165.60	-3.91	127.97	0.0040	0.0305
80		165.60	-1.99	126.72		0.0157

Analysis of Load Case 4 : NP+OW+EQ+FS+BS

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	13.78	90.72	-28.43	108.63	0.1519	0.2617
20	9.52	90.72	-23.32	108.47	0.1049	0.2150
30	5.20	115.92	-18.28	109.11	0.0449	0.1676
40	2.68	165.60	-8.30	126.72	0.0162	0.0655
50	2.62	165.60	-7.73	127.97	0.0158	0.0604
60	1.54	165.60	-5.68	127.97	0.0093	0.0444
70	0.36	165.60	-3.79	127.97	0.0022	0.0297
80		165.60	-1.90	126.72		0.0150

Analysis of Load Case 5 : NP+HW+HI

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	0.00	117.93	0.00	112.76	0.0000	-0.0000
20	0.00	117.93	0.00	112.76	0.0000	-0.0000
30	0.00	150.69	0.00	112.76	0.0000	-0.0000
40	0.00	215.28	0.00	128.59	0.0000	-0.0000
50	0.00	215.28	0.00	129.70	0.0000	-0.0000
60	0.00	215.28	0.00	129.70	0.0000	-0.0000
70	0.00	215.28	0.00	129.70	0.0000	-0.0000
80	0.00	215.28	0.00	128.59	0.0000	-0.0000

Analysis of Load Case 6 : NP+HW+HE

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	0.00	117.93	0.00	112.76	0.0000	-0.0000
20	0.00	117.93	0.00	112.76	0.0000	-0.0000
30	0.00	150.69	0.00	112.76	0.0000	-0.0000
40	0.00	215.28	0.00	128.59	0.0000	-0.0000
50	0.00	215.28	0.00	129.70	0.0000	-0.0000
60	0.00	215.28	0.00	129.70	0.0000	-0.0000
70	0.00	215.28	0.00	129.70	0.0000	-0.0000
80	0.00	215.28	0.00	128.59	0.0000	-0.0000

Analysis of Load Case 7 : IP+OW+WI+FW+BW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	18.00	90.72	-31.84	108.63	0.1984	0.2931
20	11.98	90.72	-24.99	108.47	0.1321	0.2304
30	6.68	115.92	-18.99	109.11	0.0576	0.1740
40	67.86	165.60		126.72	0.4098	
50	64.57	165.60		127.97	0.3899	
60	63.33	165.60		127.97	0.3824	
70	61.99	165.60		127.97	0.3744	
80	64.20	165.60		126.72	0.3877	

Analysis of Load Case 8 : IP+OW+EQ+FS+BS

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	13.78	90.72	-28.43	108.63	0.1519	0.2617
20	9.52	90.72	-23.32	108.47	0.1049	0.2150
30	5.20	115.92	-18.28	109.11	0.0449	0.1676



40	67.18	165.60	126.72	0.4057
50	63.95	165.60	127.97	0.3862
60	62.88	165.60	127.97	0.3797
70	61.69	165.60	127.97	0.3725
80	64.03	165.60	126.72	0.3866

Analysis of Load Case 9 : EP+OW+WI+FW+BW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	18.00	90.72	-31.84	108.63	0.1984	0.2931
20	11.98	90.72	-24.99	108.47	0.1321	0.2304
30	6.68	115.92	-18.99	109.11	0.0576	0.1740
40	0.42	165.60	-11.51	126.72	0.0025	0.0908
50	0.44	165.60	-10.80	127.97	0.0027	0.0844
60		165.60	-8.66	127.97		0.0677
70		165.60	-6.70	127.97		0.0524
80		165.60	-4.93	126.72		0.0389

Analysis of Load Case 10 : EP+OW+EQ+FS+BS

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	13.78	90.72	-28.43	108.63	0.1519	0.2617
20	9.52	90.72	-23.32	108.47	0.1049	0.2150
30	5.20	115.92	-18.28	109.11	0.0449	0.1676
40		165.60	-11.24	126.72		0.0887
50		165.60	-10.53	127.97		0.0823
60		165.60	-8.47	127.97		0.0662
70		165.60	-6.59	127.97		0.0515
80		165.60	-4.84	126.72		0.0382

Analysis of Load Case 11 : HP+HW+HI

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	0.00	117.93	0.00	112.76	0.0000	-0.0000
20	0.00	117.93	0.00	112.76	0.0000	-0.0000
30	0.00	150.69	0.00	112.76	0.0000	-0.0000
40	80.70	215.28		128.59	0.3749	
50	77.00	215.28		129.70	0.3577	
60	77.00	215.28		129.70	0.3577	
70	77.00	215.28		129.70	0.3577	
80	80.70	215.28		128.59	0.3749	

Analysis of Load Case 12 : HP+HW+HE

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	0.00	117.93	0.00	112.76	0.0000	-0.0000
20	0.00	117.93	0.00	112.76	0.0000	-0.0000
30	0.00	150.69	0.00	112.76	0.0000	-0.0000
40	80.70	215.28		128.59	0.3749	
50	77.00	215.28		129.70	0.3577	
60	77.00	215.28		129.70	0.3577	
70	77.00	215.28		129.70	0.3577	
80	80.70	215.28		128.59	0.3749	

Analysis of Load Case 13 : IP+WE+EW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		90.72	-7.36	108.63		0.0678
20		90.72	-6.64	108.47		0.0612
30		115.92	-6.29	109.11		0.0577
40	62.70	165.60		126.72	0.3786	
50	59.71	165.60		127.97	0.3606	
60	60.08	165.60		127.97	0.3628	
70	60.38	165.60		127.97	0.3646	
80	64.06	165.60		126.72	0.3868	



Analysis of Load Case 14 : IP+WF+CW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		90.72	-6.11	108.63		0.0563
20		90.72	-5.72	108.47		0.0527
30		115.92	-5.51	109.11		0.0505
40	62.08	165.60		126.72	0.3749	
50	59.11	165.60		127.97	0.3570	
60	59.53	165.60		127.97	0.3595	
70	59.82	165.60		127.97	0.3612	
80	63.44	165.60		126.72	0.3831	

Analysis of Load Case 15 : IP+VO+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		90.72	-8.69	108.63		0.0800
20		90.72	-7.98	108.47		0.0736
30		115.92	-7.61	109.11		0.0697
40	62.70	165.60		126.72	0.3786	
50	59.71	165.60		127.97	0.3606	
60	60.08	165.60		127.97	0.3628	
70	60.38	165.60		127.97	0.3646	
80	64.06	165.60		126.72	0.3868	

Analysis of Load Case 16 : IP+VE+EW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		90.72	-7.36	108.63		0.0678
20		90.72	-6.64	108.47		0.0612
30		115.92	-6.29	109.11		0.0577
40	62.70	165.60		126.72	0.3786	
50	59.71	165.60		127.97	0.3606	
60	60.08	165.60		127.97	0.3628	
70	60.38	165.60		127.97	0.3646	
80	64.06	165.60		126.72	0.3868	

Analysis of Load Case 17 : NP+VO+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		90.72	-8.69	108.63		0.0800
20		90.72	-7.98	108.47		0.0736
30		115.92	-7.61	109.11		0.0697
40		165.60	-3.40	126.72		0.0268
50		165.60	-3.14	127.97		0.0245
60		165.60	-2.61	127.97		0.0204
70		165.60	-2.30	127.97		0.0179
80		165.60	-1.85	126.72		0.0146

Analysis of Load Case 18 : FS+BS+IP+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		75.60	-9.10	90.52		0.1005
20		75.60	-8.38	90.39		0.0927
30		96.60	-7.99	90.93		0.0879
40	62.49	138.00		105.60	0.4528	
50	59.54	138.00		106.64	0.4314	
60	59.95	138.00		106.64	0.4344	
70	60.29	138.00		106.64	0.4369	
80	64.02	138.00		105.60	0.4639	

Analysis of Load Case 19 : FS+BS+EP+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
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10	75.60	-9.10	90.52	0.1005
20	75.60	-8.38	90.39	0.0927
30	96.60	-7.99	90.93	0.0879
40	138.00	-6.54	105.60	0.0620
50	138.00	-6.11	106.64	0.0573
60	138.00	-5.54	106.64	0.0520
70	138.00	-5.19	106.64	0.0487
80	138.00	-4.83	105.60	0.0457

Analysis of Load Case 20 : BL+IP+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		90.72	-8.69	108.63		0.0800
20		90.72	-7.98	108.47		0.0736
30		115.92	-7.61	109.11		0.0697
40	62.70	165.60		126.72	0.3786	
50	59.71	165.60		127.97	0.3606	
60	60.08	165.60		127.97	0.3628	
70	60.38	165.60		127.97	0.3646	
80	64.06	165.60		126.72	0.3868	

Absolute Maximum of the all of the Stress Ratio's 0.4639

Governing Element: Top D end

Governing Load Case 18 : FS+BS+IP+OW

Element Description	Long Pressure Stress	Weight + Axial Load Stress	Eccentric Moment Stress	Wind or Earthquake Stress	Combined Stress	Which Side
Skirt 3	...	-7.3	1.8	...	...	+ Tens
Skirt 3	...	-7.3	-1.8	...	-9.1	- Comp
Skirt 2	...	-6.9	1.5	...	...	+ Tens
Skirt 2	...	-6.9	-1.5	...	-8.4	- Comp
Skirt 1	...	-6.5	1.5	...	...	+ Tens
Skirt 1	...	-6.5	-1.5	...	-8.0	- Comp
bottom D end	64.5	-2.8	0.8	...	62.5	+ Tens
bottom D end	64.5	-2.8	-0.8	...	...	- Comp
Shell 1	61.3	-2.6	0.8	...	59.5	+ Tens
Shell 1	61.3	-2.6	-0.8	...	...	- Comp
Shell 2	61.3	-2.1	0.7	...	59.9	+ Tens
Shell 2	61.3	-2.1	-0.7	...	...	- Comp
Shell 3	61.3	-1.7	0.7	...	60.3	+ Tens
Shell 3	61.3	-1.7	-0.7	...	...	- Comp
Top D end	64.5	-1.2	0.7	...	64.0	+ Tens
Top D end	64.5	-1.2	-0.7	...	...	- Comp

Skirt Support Analysis Per EN 13445-3 Section 16.12:

The analysis of this skirt has been conducted referencing the EN 13445-3 design code. It is important to note that the selected design code does not explicitly cover the analysis of this component. Therefore, the results provided herein should be regarded as informational only.

Item: Skirt 1

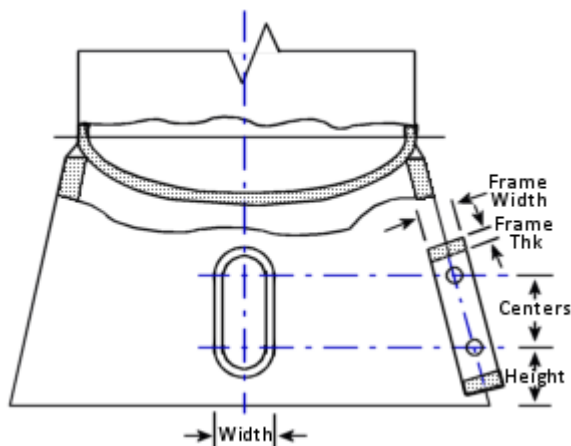
Total vessel (empty) weight	Fg	37573.051 Kgf
Head Weight below section 2-2	DeltaFg	1364.293 Kgf
Weight of vessel contents (liquid)	Ff	10022.178 Kgf
Vessel Gross Weight	Fg + Ff	47595.230 Kgf
Skirt Mean Diameter	Dz	2211.850 mm.
Skirt Analysis Thickness (Corroded)	ez	10.750 mm.
Skirt Height	H	1000.000 mm.
Additional Axial Force at section 1-1	F1	0.000 Kgf
Moment at section 1-1	M1	39530.172 Kg-m.
Applied Moment - Bottom of Skirt	Mb	47697.637 Kg-m.
Weight at Bottom of Skirt	W	46882.332 Kgf

Structural Type B Skirt per EN 13445 Figure 16.12-2:

Type of Head:	Elliptical
Head Mean Diameter at Skirt Top	Db 2222.500 mm.
Head Corroded Thickness	eb 19.500 mm.

Per EN 13445 Section 7.5.4 [R, r]:

$$\begin{aligned}
 h_i &= 551.5 \text{ mm.} \\
 K &= D_i / (2 * h_i) = 2203.0 / (2 * 551.5) = 1.9973 \\
 R &= D_i (0.44 * K + 0.02) = 2203.0 * (0.44 * 1.997 + 0.02) = 1980.0636 \text{ mm.} \\
 r &= D_i ((0.5 / K) - 0.08) = 2203.0 * ((0.5 / 1.997) - 0.08) = 375.26 \text{ mm.}
 \end{aligned}$$



There are no Access Openings in the Skirt - Plain Skirt Considered

Moment of Inertia	Ixx =	4568094.000 cm ⁴
Moment of Inertia	Iyy =	4568094.000 cm ⁴
Section Modulus	W =	41305676.000 mm. ³
Section Metal Area	A =	746.989 cm ²

Analysis of the Section at weakest section Modulus [Z min]:

Radius of Gyration of Skirt at 4-4 [rg]:

$$\begin{aligned}
 &= \sqrt{(\min(I_{xx}, I_{yy}) / \text{Area})} \\
 &= \sqrt{(\min(4568094, 4568094) / 746.989)} \\
 &= 782.007 \text{ mm.}
 \end{aligned}$$

Skirt Slenderness Ratio [rg/L]:

$$\begin{aligned} &= r_g / L \\ &= 782.007/1000.0 \\ &= 0.782 \end{aligned}$$

Correct the Moment [M4]:

top of skirt is the critical end
Controlling Moment [M4] = 39530.168 Kg-m.

Axial Stress at Residual (Weakest) Surface Area [Sigma]:

Downwind Side:

$$\begin{aligned} &= -M / Z - W / A \\ &= -39530.168 / 16522E+09 - 46882.332 / 746.98877 \\ &= -8.501 \text{ N./mm}^2 \end{aligned}$$

Upwind Side:

$$\begin{aligned} &= +(M) / Z - W / A \\ &= +39530.168 / 16522E+09 - 46882.332 / 746.98877 \\ &= -3.809 \text{ N./mm}^2 \end{aligned}$$

16.12.3.4 Checking at connection areas:

Force at the joint, 16.12-1 [Fzp]:

$$\begin{aligned} &= F_1 - F_g - F_f + 4 \cdot M_1 / D_z \\ &= 0.0 - 37573.051 - 10022.178 + 4 \cdot 39530.168 / 2211.85 \\ &= 23891.3 \text{ Kg} \end{aligned}$$

Force at the joint, 16.12-2 [Fzq]:

$$\begin{aligned} &= F_1 - F_g - F_f - 4 \cdot M_1 / D_z \\ &= 0.0 - 37573.051 - 10022.178 - 4 \cdot 39530.168 / 2211.85 \\ &= -119081.8 \text{ Kg} \end{aligned}$$

16.12.3.4.1 Membrane stresses:

Membrane stress at 1-1, 16.12-3 & 4:

Stress [σ_{1pm}]:

$$\begin{aligned} &= (F_{zp} + \Delta F_g) / (\pi \cdot D_b \cdot e_b) + P \cdot D_b / (4 \cdot e_b) \\ &= (23891.328 + 1364.293 + 10022.178) / (\pi \cdot 2222.5 \cdot 19.5) + 23.0 \cdot 2222.5 / (4 \cdot 19.5) \\ &= 68.08 \text{ N./mm}^2 \end{aligned}$$

Stress [σ_{1qm}]:

$$\begin{aligned} &= (F_{zq} + \Delta F_g + F_f) / (\pi \cdot D_b \cdot e_b) + P \cdot D_b / (4 \cdot e_b) \\ &= (-0.1E+06 + 1364.293 + 10022.178) / (\pi \cdot 2222.5 \cdot 19.5) + 23.0 \cdot 2222.5 / (4 \cdot 19.5) \\ &= 57.78 \text{ N./mm}^2 \end{aligned}$$

Membrane stress at 2-2, 16.12-9:

Stress [σ_{2pm} & σ_{2qm}]:

$$\begin{aligned} &= (F_f + \Delta F_g) / (\pi \cdot D_b \cdot e_b) + P \cdot D_b / (4 \cdot e_b) \\ &= (10022.178 + 1364.293) / (\pi \cdot 2222.5 \cdot 19.5) + 23.0 \cdot 2222.5 / (4 \cdot 19.5) \\ &= 66.36 \text{ N./mm}^2 \end{aligned}$$

Membrane stress at 3-3, 16.12 12 & 13:

Stress [σ_{3pm}]:

$$\begin{aligned} &= F_{zp} / (\pi \cdot D_z \cdot e_z) \\ &= 23891.328 / (\pi \cdot 2211.85 \cdot 10.75) \\ &= 3.14 \text{ N./mm}^2 \end{aligned}$$

Stress [σ_{3qm}]:

$$\begin{aligned} &= F_{zq} / (\pi \cdot D_z \cdot e_z) \\ &= -119081.789 / (\pi \cdot 2211.85 \cdot 10.75) \\ &= -15.63 \text{ N./mm}^2 \end{aligned}$$

16.12.3.4.2 Bending stresses:

Cosine of angle, 16.12-25 [$\cos(\gamma)$]:

$$\begin{aligned} &= 1 - 0.5(D_b + e_b - D_z + e_z) / (r + e_b) \\ &= 1 - 0.5(2222.5 + 19.5 - 2211.85 + 10.75) / (375.26 + 19.5) \\ &= 0.9482 \end{aligned}$$



$$\gamma = \arccos(\cos(\gamma)) = \arccos(0.948) = 18.523 \text{ deg}$$

Dimension [a]:

$$= 0.5 \cdot \sqrt{(eb^2 + ez^2 + 2 \cdot eb \cdot ez \cdot \cos(\gamma))}$$

$$= 0.5 \cdot \sqrt{(19.5^2 + 10.75^2 + 2 \cdot 19.5 \cdot 10.75 \cdot 0.948)}$$

$$= 14.944 \text{ mm.}$$

Bending moment at verification points p and q [Mp & Mq]:

$$M_p = a \cdot F_{zp} = 14.94 \cdot 23891.3 = 357.05 \text{ Kg-m.}$$

$$M_q = a \cdot F_{zq} = 14.94 \cdot -119081.8 = -1779.64 \text{ Kg-m.}$$

Correction factor [C]:

$$= 0.63 - 0.057 \left(\frac{eb}{ez} \right)^2$$

$$= 0.63 - 0.057 \left(\frac{19.5}{10.75} \right)^2$$

$$= 0.442$$

Bending stresses in sections 1-1 to 3-3 at the outer surface (o), 16.12-26 to 29:

Stress [$\sigma_{1pb(o)}$ & $\sigma_{2pb(o)}$]:

$$= C \cdot 6 \cdot M_p / (\pi \cdot D_b \cdot eb^2)$$

$$= 0.442 \cdot 6 \cdot 357.049 / (\pi \cdot 2222.5 \cdot 19.5^2)$$

$$= 3.50 \text{ N./mm}^2$$

Stress [$\sigma_{1qb(o)}$ & $\sigma_{2qb(o)}$]:

$$= C \cdot 6 \cdot M_q / (\pi \cdot D_b \cdot eb^2)$$

$$= 0.442 \cdot 6 \cdot -1779.645 / (\pi \cdot 2222.5 \cdot 19.5^2)$$

$$= -17.45 \text{ N./mm}^2$$

Stress [$\sigma_{3pb(o)}$]:

$$= C \cdot 6 \cdot M_p / (\pi \cdot D_z \cdot ez^2)$$

$$= 0.442 \cdot 6 \cdot 357.049 / (\pi \cdot 2211.85 \cdot 10.75^2)$$

$$= 11.58 \text{ N./mm}^2$$

Stress [$\sigma_{3qb(o)}$]:

$$= C \cdot 6 \cdot M_q / (\pi \cdot D_z \cdot ez^2)$$

$$= 0.442 \cdot 6 \cdot -1779.645 / (\pi \cdot 2211.85 \cdot 10.75^2)$$

$$= -57.70 \text{ N./mm}^2$$

Geometry variables, 16.12-32 to 36:

$$eb/D_b = 0.768/87.5 = 0.0088$$

$$y = 125 \cdot eb/D_b = 125 \cdot 0.0088 = 1.0967$$

As this an elliptical head, it will be treated as a Korbbogen type head

Geometric value [α]:

$$= 4.2 - 0.2 \cdot y = 4.2 - 0.2 \cdot 1.097 = 3.9807$$

Bending stress in sections 1-1 and 2-2 Equation 16.12-31 [$\sigma_{1b(p)}$ & $\sigma_{2b(p)}$]:

$$= (P + Ph) \cdot D_b \cdot ((y / y_A) \cdot \alpha - 1) / (4 \cdot eb)$$

$$= (23.0 + 0.0) \cdot 2222.5 \cdot ((18.52/40.0) \cdot 3.981 - 1) / (4 \cdot 19.5)$$

$$= 55.27 \text{ N./mm}^2$$

16.12.3.4.3 Total stresses and strength conditions:

Structural shape B and C:

Stress [σ_{1pitot}]:

$$= \sigma_{1pm} - \sigma_{1pb(o)} + \sigma_{1b(p)}$$

$$= 68.08 - 3.501 + 55.271$$

$$= 119.85 \text{ N./mm}^2$$

Stress [σ_{1potot}]:

$$= \sigma_{1pm} + \sigma_{1pb(o)} - \sigma_{1b(p)}$$

$$= 68.08 + 3.501 - 55.271$$

$$= 16.31 \text{ N./mm}^2$$

Stress [σ_{1qitot}]:

$$= \sigma_{1qm} - \sigma_{1qb(o)} + \sigma_{1b(p)}$$

$$= 57.782 - -17.45 + 55.271$$



$$= 130.50 \text{ N./mm}^2$$

Stress [σ_{1qotot}]:
 $= \sigma_{1qm} + \sigma_{1qb(o)} - \sigma_{1b(p)}$
 $= 57.782 + -17.45 - 55.271$
 $= -14.94 \text{ N./mm}^2$

Stress [σ_{2pitot}]:
 $= \sigma_{2pm} + \sigma_{2pb(o)} + \sigma_{2b(p)}$
 $= 66.359 + 3.501 + 55.271$
 $= 125.13 \text{ N./mm}^2$

Stress [σ_{2potot}]:
 $= \sigma_{2pm} - \sigma_{2pb(o)} - \sigma_{2b(p)}$
 $= 66.359 - 3.501 - 55.271$
 $= 7.59 \text{ N./mm}^2$

Stress [σ_{2qitot}]:
 $= \sigma_{2qm} + \sigma_{2qb(o)} + \sigma_{2b(p)}$
 $= 66.359 + -17.45 + 55.271$
 $= 104.18 \text{ N./mm}^2$

Stress [σ_{2qotot}]:
 $= \sigma_{2pm} - \sigma_{2pb(o)} - \sigma_{2b(p)}$
 $= 66.359 - -17.45 - 55.271$
 $= 28.54 \text{ N./mm}^2$

Stress [σ_{3pitot}]:
 $= \sigma_{3pm} - \sigma_{3pb}$
 $= 3.137 - 11.575$
 $= -8.44 \text{ N./mm}^2$

Stress [σ_{3potot}]:
 $= \sigma_{3pm} + \sigma_{3pb}$
 $= 3.137 + 11.575$
 $= 14.71 \text{ N./mm}^2$

Stress [σ_{3qitot}]:
 $= \sigma_{3qm} - \sigma_{3qb}$
 $= -15.634 - -57.696$
 $= 42.06 \text{ N./mm}^2$

Stress [σ_{3qotot}]:
 $= \sigma_{3qm} + \sigma_{3qb}$
 $= -15.634 + -57.696$
 $= -73.33 \text{ N./mm}^2$

Allowable stresses:

For σ_{1pitot} and σ_{1potot} , the stress must not exceed:
 $= fs(3 - (1 / 1.5)) \cdot (\sigma_{1pm} / f)^2$
 $= 137.997(3 - (1/1.5)) \cdot (68.08/137.997)^2$
 $= 391.60 \text{ N./mm}^2$

For σ_{1qitot} and σ_{1qotot} , the stress must not exceed:
 $= fs(3 - (1 / 1.5)) \cdot (\sigma_{1qm} / f)^2$
 $= 137.997(3 - (1/1.5)) \cdot (57.782/137.997)^2$
 $= 397.86 \text{ N./mm}^2$

For σ_{2pitot} and σ_{2potot} , the stress must not exceed:
 $= fs(3 - (1 / 1.5)) \cdot (\sigma_{2pm} / f)^2$
 $= 137.997(3 - (1/1.5)) \cdot (66.359/137.997)^2$
 $= 392.72 \text{ N./mm}^2$

For σ_{2qitot} and σ_{2qotot} , the stress must not exceed:
 $= fs(3 - (1 / 1.5)) \cdot (\sigma_{2qm} / f)^2$
 $= 137.997(3 - (1/1.5)) \cdot (66.359/137.997)^2$
 $= 392.72 \text{ N./mm}^2$

For σ_{3pitot} & σ_{3potot} , the stress must not exceed:



$$= fs(3 - (1 / 1.5) \cdot (\sigma_{3pm} / f)^2)$$

$$= 137.997(3 - (1/1.5) \cdot (3.137/137.997)^2)$$

$$= 413.94 \text{ N./mm}^2$$

For σ_{3qitot} & σ_{3qotot} , the stress must not exceed:

$$= fs(3 - (1 / 1.5) \cdot (\sigma_{3qm} / f)^2)$$

$$= 137.997(3 - (1/1.5) \cdot (-15.634/137.997)^2)$$

$$= 412.81 \text{ N./mm}^2$$

Please note: None of the following stresses in the table below are in the region where there is an access opening.

Summary of Stresses: (N./mm2)

Stress	Actual	Allowed	Result	Equation
Stot1pi	119.85	391.60	Pass	46
Stot1po	16.31	391.60	Pass	47
Stot1qi	130.50	397.86	Pass	48
Stot1qo	-14.94	397.86	Pass	49
Stot2pi	125.13	392.72	Pass	50
Stot2po	7.59	392.72	Pass	51
Stot2qi	104.18	392.72	Pass	52
Stot2qo	28.54	392.72	Pass	53
Stot3pi	-8.44	413.94	Pass	54
Stot3po	14.71	413.94	Pass	55
Stot3qi	42.06	412.81	Pass	56
Stot3qo	-73.33	412.81	Pass	57

The stresses are satisfactory in the region without openings.

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Shop/Field Installation Options :

Platform(s) installed in the Shop.
Insulation is installed in the Shop.

Note : The CG is computed from the first Element From Node

Center of Gravity of the Platforms	11297.610 mm.
Center of Gravity of the Liquid	7579.591 mm.
Center of Gravity of the Insulation	9350.801 mm.
Center of Gravity of the Nozzles	10536.739 mm.
Center of Gravity of the Added Weights (Operating)	10573.177 mm.
Center of Gravity of the Added Weights (Empty)	10017.971 mm.
Center of Gravity of Bare Shell New and Cold	7389.980 mm.
Center of Gravity of Bare Shell Corroded	7439.295 mm.
Vessel CG in the Operating Condition	9356.569 mm.
Vessel CG in the Fabricated (Shop/Empty) Condition	9620.517 mm.
Vessel CG in the Test Condition	9510.001 mm.

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Skirt Data :

Skirt Outside Diameter at Base	SOD, D6	2224.0000	mm.
Skirt Thickness	STHK	12.0000	mm.
Skirt Internal Corrosion Allowance	SCA	0.7000	mm.
Skirt External Corrosion Allowance		0.7000	mm.
Skirt Material		SA-285 C	

Basing Input: Type of Geometry: Continuous Top Ring W/Gussets

Thickness of Basing	TBA, ea4	32.0000	mm.
Design Temperature of the Basing		38.00	°C
Basing Material		SA-285 C	
Basing Ope. All. Stress	BASOPE, f4	108.00	N./mm2
Basing Yield Stress		207.00	N./mm2
Inside Diameter of Basing	DI, D4	2144.0000	mm.
Outside Diameter of Basing	DOU, D5	2448.0000	mm.
Nominal Diameter of Bolts	BND, d(b0)	30.0000	mm.
Bolt Corrosion Allowance	BCA	1.5000	mm.
Root Area of a Single Bolt	Area	5.4000	cm2
Bolt Material		SA-36	
Bolt Operating Allowable Stress	SA, fB	137.30	N./mm2
Number of Bolts	RN, nb	20	
Diameter of Bolt Circle	DC, DBC	2335.0000	mm.
Bolt Allowable Shear Stress		107.876	N./mm2
Ultimate Comp. Strength of Concrete	FPC	35.0	N./mm2
Allowable Comp. Strength of Concrete	FC	8.8	N./mm2
Modular ratio Steel/Concrete		9.833	
Thickness of Gusset Plates	TGA, e7	12.0000	mm.
Width of Gussets at Top Plate	TWDT	102.0000	mm.
Width of Gussets at Base Plate	BWDT	102.0000	mm.
Gusset Plate Elastic Modulus	E	201734000.0	KPa.
Gusset Plate Yield Stress	SY	207.0	N./mm2
Height of Chair	h1	242.0000	mm.
Distance between Gussets	RG, b6	90.0000	mm.
Dist. from Bolt Center to Gusset (Rg/2)	CG	45.0000	mm.
Number of Gussets per bolt	NG	2	
Thickness of Top Plate or Ring	TTA, e5	25.0000	mm.
Radial Width of the Top Plate	TOPWTH, b5	120.0000	mm.
Anchor Bolt Hole Dia. in Top Plate	dia,H	57.0000	mm.
External Corrosion Allowance	CA	1.5000	mm.
Dead Weight of Vessel	DW	40694.3	Kgf
Operating Weight of Vessel	ROW	51986.5	Kgf
Earthquake Moment on Basing	EQMOM	85644.5	Kg-m.
Wind Moment on Basing	WIMOM	101427.6	Kg-m.
Percent Bolt Preload	ppl	100.0	
Use AISC A5.2 Increase in Fc and Bolt Stress		No	
Use Allowable Weld Stress per AISC J2.5		No	
Factor for Increase of Allowables	Fact	1.0000	

Results for Brownell and Young Basing Analysis : Analyze Option

Note: This analysis is based on Neutral Axis shift method for Steel on Concrete (or a material with significantly different Young's modulus).

Calculation of Bolt Loads for Multiple Load Conditions:

Bolt Load, Earthquake Operating Case:

$$= ((4 \cdot M/DC) - W) / nb \quad \text{per Jawad \& Farr, Eq. 12.3}$$

$$= ((4 \cdot 85645/2335.0) - 51987) / 20$$

$$= 4736.2 \text{ Kg}$$



Bolt Load, Windload Empty Case:

$$= ((4 \cdot M/DC) - W) / nb \quad \text{per Jawad \& Farr, Eq. 12.3}$$

$$= ((4 \cdot 101428/2335.0) - 40694) / 20$$

$$= 6652.7 \text{ Kgf}$$

Bolt Load, Windload Vortex Shedding Case:

$$= ((4 \cdot M/DC) - W) / nb \quad \text{per Jawad \& Farr, Eq. 12.3}$$

$$= ((4 \cdot 0/2335.0) - 51987) / 20$$

$$= -2599.3 \text{ Kgf}$$

Governing Bolt Load Condition, Wind + Dead Weight Condition:

Area Available in 1 Bolt (corroded)	Abss :	5.4000	cm ²
Area Available in all the Bolts	Abss * RN :	108.0000	cm ²

Trial#	k	knew	Cc	Ct	z	j	Ft	Fc	Fs
2	0.173	0.259	1.129	2.721	0.465	0.773	31713.9	72408.2	4374.5
4	0.302	0.281	1.516	2.437	0.437	0.781	32828.4	73522.7	5056.1
6	0.270	0.265	1.427	2.507	0.444	0.780	32527.9	73222.2	4869.1
8	0.262	0.263	1.404	2.525	0.446	0.779	32454.9	73149.3	4824.2
10	0.263	0.262	1.406	2.524	0.446	0.779	32461.0	73155.3	4827.9
11	0.262	0.262	1.405	2.524	0.446	0.779	32458.0	73152.3	4826.0

The Actual Stress in a Single Bolt [Sbolt]:

$$= 2 \cdot Ft / (T1 \cdot Dc \cdot Ct)$$

$$= 2 \cdot 32458.0 / (1.472 \cdot 2335.0 \cdot 2.524)$$

$$= 73.359 \text{ N./mm}^2, \text{ Should be } \leq 137.3 \text{ N./mm}^2$$

Thickness of the Band of Bolting Steel [T1]

$$= RN \cdot \text{Bolt Area} / (\pi \cdot Dc)$$

$$= 20 \cdot 5.4 / (\pi \cdot 2335.0)$$

$$= 1.472 \text{ mm.}$$

Check the Bearing Stress in the Concrete [fc(max)]

$$= fc[(2kd + t3) / (2kd)]$$

$$= 2.651[(2 \cdot 0.2622 \cdot 2335.0 + 152.0) / (2 \cdot 0.2622 \cdot 2335.0)]$$

$$= 2.98 \text{ N./mm}^2, \text{ Should be } \leq 8.8 \text{ N./mm}^2$$

Values for table 10.3, l = 112.0, b = 183.39, l/b = 0.610718

Maximum Moment per unit width [Mmax]:

$$= \max(Mx, My) = \max(479.387, 983.036) = 983.036 \text{ Kgf}$$

Reqd Thickness of Basing, Brownell & Young Method [T]:

$$= \sqrt{(6 \cdot Mmax / fallow)} + Ca$$

$$= \sqrt{(6 \cdot 983.036 / 136.6)} + 1.5$$

$$= 22.076 \text{ mm.}$$

Nomenclature:

a = (Dc - Ds) / 2	Skirt Distance to Bolt Circle
P = Sa * Abss	Maximum Load on one Bolt
l	Radial Top Plate Width
lavg	Average Gusset Plate Width
g1 = Gamma 1	Constant Term f(b/l)
g2 = Gamma 2	Constant Term f(b/l)
g = Flat distance / 2	Nut 1/2 Dimension (from Tema)
Fb	Allowable Bending Stress

Values for table 10.6, l = 120.0, b = 90.0, b/l = 0.75

As b/l (0.75) is less than 1, inverting b/l = 1.333.

Moment Term, based on geometry [Mo]:

$$= P / (4 \cdot \pi) [1.3(\ln((2l \sin(\pi \cdot a/l)) / (\pi \cdot g))) + 1] - [(0.7 - g2)P / (4\pi)]$$

$$= 7560.24 / (4 \cdot \pi) [1.3(\ln((2 \cdot 120.0 \cdot \sin(\pi \cdot 55.5/120.0)) / (\pi \cdot 25.4))) + 1] - [(0.7 - 0.091) \cdot 7560.24 / (4 \cdot \pi)]$$

$$= 1090.8739 \text{ Kgf}$$

Required Thickness of Continuous Top Ring [Tc]:

$$= \sqrt{(6 \cdot \text{Abs}(Mo) / Fb)} + Ca$$

$$= \sqrt{(6 \cdot \text{Abs}(1090.87) / 138.0)} + 1.5$$

= 23.0671 mm.

Required Gusset Plate Thickness [tg]:
 $= P / (\text{Stress Term} \cdot l_{avg}) + C_a$
 $= 7560.24 / (124.11 \cdot 102.0) + 1.5$
 $= 11.025$ (not less than $9.525 + 1.5$) mm.

Bolt spacing [m]:
 $= \pi \cdot \text{Bolt Circle Diameter} / \text{number of Bolts}$
 $= \pi \cdot 2335.0 / 20 = 366.781$ mm.

Req. Skirt Thk. to withstand Local Bending, (Brownell and Young) [t]:
 $= 1.76(P \cdot a / (m \cdot (h + t_{ba}) \cdot 1.5 \cdot S_{ktope}))^{2/3} \cdot r^{1/3} + C_a$
 $= 1.76(7560 \cdot 55.5 / (366.78 \cdot 217.0 \cdot 162))^{2/3} \cdot 1112.01^{1/3} + C_a$
 $= 8.515 + 1.4 = 9.915$ mm.

Shear Stress in a Single Bolt [tb]:
 $= \text{Shear Force} / (\text{Bolt Area} \cdot \text{Number of Bolts})$
 $= 11138.0 / (5.4 \cdot 20)$
 $= 10.1$ N./mm². Must be less than 107.9 N./mm².

Summary of Basing Thickness Calculations

Required Basing Thickness (tension)	22.0765	mm.
Actual Basing Thickness as entered by user	32.0000	mm.
Required Thickness of Chair Cap	23.0671	mm.
Actual Top Ring Thickness as entered by user	25.0000	mm.
Required Gusset thickness, + CA	11.0250	mm.
Actual Gusset Thickness as entered by user	12.0000	mm.
Required Thickness of Skirt for Local Stress	9.9154	mm.
Given Thickness of Skirt	12.0000	mm.
Required Gusset Height to meet local stress	126.4823	mm.

Weld Size Calculations per Steel Plate Engineering Data - Vol. 2

Compute the Weld load at the Skirt/Base Junction [W]:
 $= \text{SkirtStress} \cdot (\text{SkirtThickness} - C_A)$
 $= 31.843 \cdot (12.0 - 1.4)$
 $= 34.42$ Kg/mm.

Results for Computed Minimum Basing Weld Size [BWeld]:
 $= W / [(0.4 \cdot \text{Yield}) \cdot 2 \cdot 0.707]$
 $= 34 / [(0.4 \cdot 192) \cdot 2 \cdot 0.707]$
 $= 3.101$ mm.

Results for Computed Minimum Gusset and Top Plate to Skirt Weld Size

Vertical Plate Load [Wv]:
 $= \text{Bolt Load} / (C_{mwth} + 2 \cdot (H_g + T_{ta}))$
 $= 7560.2 / (182.0 + 2 \cdot (185.0 + 25.0))$
 $= 12.559$ Kg/mm.

Horizontal Plate Load [Wh]:
 $= \text{Bolt Load} \cdot e / (C_{mwth} \cdot (H_g + T_{ta}) + 0.6667 \cdot (H_g + T_{ta})^2)$
 $= 7560.2 \cdot 55.5 / (182.0 \cdot (210.0) + 0.6667 \cdot (210.0)^2)$
 $= 6.205$ Kg/mm.

Resultant Weld Load [Wr]:
 $= \sqrt{ W_v^2 + W_h^2 }$
 $= \sqrt{ 12.56^2 + 6.21^2 }$
 $= 14.008$ Kg/mm.

Results for Computed Min Gusset and Top Plate to Skirt Weld Size [GsWeld]:
 $= W_r / [(0.4 \cdot \text{Yield}) \cdot 2 \cdot 0.707]$
 $= 14.01 / [(0.4 \cdot 192) \cdot 2 \cdot 0.707]$
 $= 1.262$ mm.



Results for Computed Minimum Gusset to Top Plate Weld Size

Weld Load [Wv]:
= Bolt Load / (2 • TopWth)
= 7560.2 / (2 • 102.0)
= 37.06 Kgf/mm.

Weld Load [Wh]:
= Bolt Load • e / (2 • Hgt • TopWth)
= 7560.2 • 55.5 / (2 • 210.0 • 102.0)
= 9.794 Kgf/mm.

Resultant Weld Load [Wr]:
= $\sqrt{Wv^2 + Wh^2}$
= $\sqrt{37.06^2 + 9.79^2}$
= 38.332 Kgf/mm.

Results for Computed Min Gusset to Top Plate Weld Size [GtpWeld]:
= Wr / [(0.4 • Yield) • 2 • 0.707]
= 38.33 / [(0.4 • 192) • 2 • 0.707]
= 3.454 mm.

Note: The calculated weld sizes need not exceed the component thickness framing into the weld. At the same time, the weld must meet a minimum size specification which is 3/16 in. (4.76 mm) or 1/4 in. (6.35 mm), depending on the component thickness.

Summary of Required Weld Sizes:

Required Basing to Skirt Double Fillet Weld Size	4.7625	mm.
Required Gusset to Skirt Double Fillet Weld Size	4.7625	mm.
Required Top Plate to Skirt Weld Size	6.3500	mm.
Required Gusset to Top Plate Double Fillet Weld Size	3.4540	mm.

Review notes about nozzle loadings and supports in the Warnings report.

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Input, Nozzle Desc: N63"

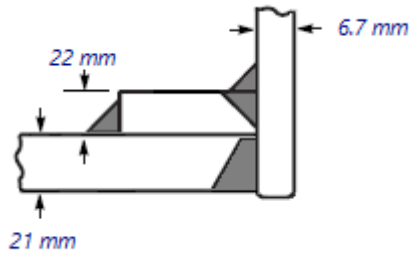
From: 40

Pressure for Reinforcement Calculations	P	23.271	bars
Temperature for Internal Pressure	Temp	80	°C
Design External Pressure	Pext	1.01	bars
Temperature for External Pressure	Tempex	80	°C
Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	138.00	N./mm2
Parent Allowable Stress At Ambient	Sva	138.00	N./mm2
Inside Diameter of Elliptical Head	D	2200.00	mm.
Aspect Ratio of Elliptical Head	Ar	2.00	
Head Finished (Minimum) Thickness	t	21.0000	mm.
Head Internal Corrosion Allowance	cas	1.5000	mm.
Head External Corrosion Allowance	co	0.0000	mm.
Distance from Head Centerline	L1	0.0000	mm.
User Entered Minimum Design Metal Temperature		-38.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-333 6	+IT
Material UNS Number		K03006	
Material Specification/Type	Smls. & wld. pipe		
Allowable Stress at Temperature	Sn	118.00	N./mm2
Allowable Stress At Ambient	Sna	118.00	N./mm2
Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		3.0000	in.
Size and Thickness Basis		Minimum	
Nominal Thickness		80	
Flange Material		SA-350 LF2	+IT
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	1.5000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	218.2720	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	21.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Pad Material		SA-516 70	+IT
Pad Allowable Stress at Temperature	Sp	138.00	N./mm2
Pad Allowable Stress At Ambient	Spa	138.00	N./mm2
Diameter of Pad along vessel surface	Dp	140.0000	mm.
Thickness of Pad	te	22.0000	mm.
Weld leg size between Pad and Shell	Wp	18.0000	mm.
Groove weld depth between Pad and Nozzle	Wgpn	22.0000	mm.
Reinforcing Pad Width		25.5500	mm.
Flange Class		300	
Flange Grade		GR 1.1	
Piping Exit Angle [North Clockwise reference]		90.0	deg
Bend Radius Multiplier		1.50	
Horizontal Run Length		1350.0	mm.
Centerline Distance to Tangent		950.0	mm.

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: N63"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	2.975 in.
Actual Thickness Used in Calculation	0.263 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]
 $= P \cdot D \cdot K1 / (2 \cdot Sv \cdot E - 0.2 \cdot P)$ per Appendix 1-4(c)
 $= 23.27 \cdot 2203.0 \cdot 0.8988 / (2 \cdot 138.0 \cdot 1.0 - 0.2 \cdot 23.27)$
 $= 16.7247 \text{ mm.}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (Sn \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 23.27 \cdot 39.28 / (118 \cdot 1.0 - 0.6 \cdot 23.27)$
 $= 0.7840 \text{ mm.}$

Required Nozzle thickness under External Pressure per UG-28 : 0.4107 mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	157.1300 mm.
Parallel to Vessel Wall, opening length	d	78.5650 mm.
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		34.9188 mm.

Weld Strength Reduction Factor [fr1]:
 $= \min(1, Sn/Sv)$
 $= \min(1, 118.0/138.0)$
 $= 0.855$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, Sn/Sv)$
 $= \min(1, 118.0/138.0)$
 $= 0.855$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, Sp/Sv)$
 $= \min(1, 138.0/138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(0.855, 1.0)$
 $= 0.855$

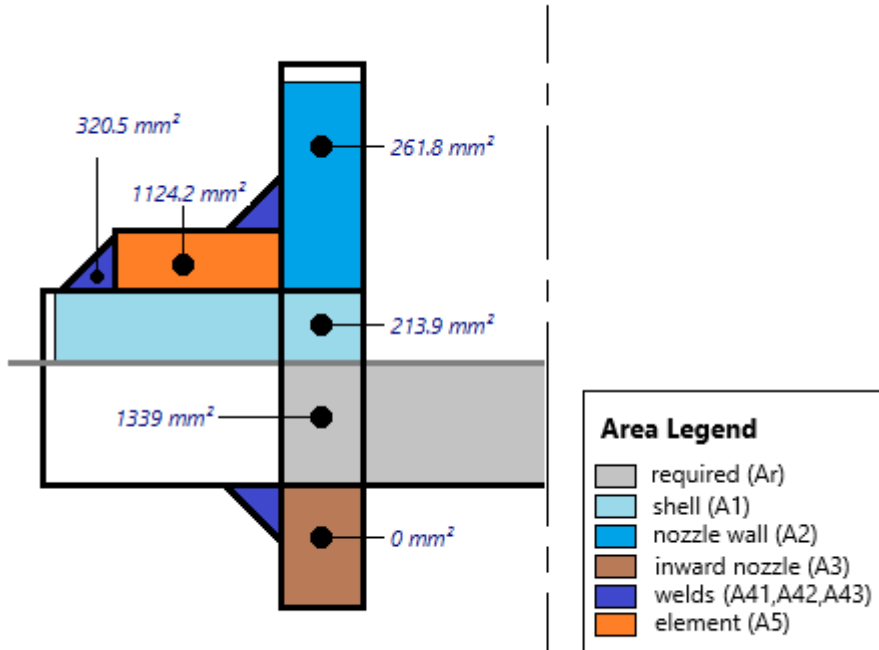


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm2)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	13.390	2.300	NA
Area in Shell	A1	2.139	10.600	NA
Area in Nozzle Wall	A2	2.618	2.841	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	3.205	3.205	NA
Area in Element	A5	11.242	11.242	NA
TOTAL AREA AVAILABLE	Atot	19.203	27.887	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Insufficient.
 The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS: Diameter Thickness
 Based on given Pad Thickness: 113.5770 22.0000 mm.
 Based on given Pad Diameter: 140.0000 10.6242 mm.
 Based on the Estimated Diameter Limit: 155.5425 8.1464 mm.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$
 $= (78.565 \cdot 16.7247 \cdot 1.0 + 2 \cdot 5.1675 \cdot 16.7247 \cdot 1.0 \cdot (1 - 0.86))$
 $= 13.390 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 78.565(1.0 \cdot 19.5 - 1.0 \cdot 16.725) - 2 \cdot 5.168$
 $(1.0 \cdot 19.5 - 1.0 \cdot 16.7247) \cdot (1 - 0.855)$
 $= 2.139 \text{ cm}^2$

Area Available in Nozzle Wall Projecting Outward [A2]:
 $= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$
 $= (2 \cdot 34.92) \cdot (5.17 - 0.78) \cdot 0.8551$



= 2.618 cm²

Area Available in Welds [A41 + A42 + A43]:
= (Wo² - Ar Lost)•Fr3+((Wi-can/0.707)² - Ar Lost)•fr2 + Trapfr4
= (1.) • 0.86 + (0.0) • 0.86 + 234.9806² • 1.0
= 3.205 cm²

Area Available in Element [A5]:
= (min(Dp,DL)-(Nozzle OD))•(min(tp,Tlwp,te)) • fr4
= (140.0 - 88.9) • 22.0 • 1.0
= 11.242 cm²

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures ta = 2.2840 mm.
Wall Thickness per UG16(b), tr16b = 3.0000 mm.
Wall Thickness, shell/head, internal pressure trb1 = 20.0735 mm.
Wall Thickness tb1 = max(trb1, tr16b) = 20.0735 mm.
Wall Thickness, shell/head, external pressure trb2 = 2.3074 mm.
Wall Thickness tb2 = max(trb2, tr16b) = 3.0000 mm.
Wall Thickness per table UG-45 tb3 = 6.3000 mm.

Determine Nozzle Thickness candidate [tb]:
= min[tb3, max(tb1,tb2)]
= min[6.3, max(20.0735, 3.0)]
= 6.3000 mm.

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
= max(ta, tb)
= max(2.284, 6.3)
= 6.3000 mm.

Available Nozzle Neck Thickness = 6.6675 mm. -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	:	48.8,	Allowable	:	118.0 N./mm ²	Passed
Expansion	:	0.0,	Allowable	:	246.2 N./mm ²	Passed
Occasional	:	8.3,	Allowable	:	156.9 N./mm ²	Passed
Shear	:	29.3,	Allowable	:	82.6 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld (Impact tested) :

Note:

This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification	-46 °C
Calculated Minimum Design Metal Temperature	-104 °C

Nozzle Neck to Pad Weld for the Nozzle (Impact tested) :

Note:

This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification	-46 °C
Calculated Minimum Design Metal Temperature	-104 °C

Nozzle Neck to Pad Weld for Reinf. pad (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Shell to Pad Weld at Pad OD for pad (Impact tested) :



Note: This Element/Detail was specified as being Impact Tested.

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note:

This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification	-46 °C
Calculated Minimum Design Metal Temperature	-104 °C
Gov. MDMT of the Nozzle	: -104 °C
Gov. MDMT of the Reinforcement Pad	: -46 °C
Gov. MDMT of the nozzle to shell joint welded assembly	: -46 °C

Flange MDMT including Temperature reduction per UCS-66.1:

MDMT of ASME B16.5/47 flange per Matl. Specification	-46 °C
Flange MDMT with Temp reduction per UCS-66(i)(2)	-84 °C

Where the Stress Reduction Ratio per UCS-66(i)(2) is :

$$\text{Design Pressure/Ambient Rating} = 23.27/51.00 = 0.456$$

Weld Size Calculations, Description: N63"

Intermediate Calc. for nozzle/shell Welds	Tmin	5.1675 mm.
Intermediate Calc. for pad/shell Welds	TminPad	19.0000 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$3.6173 = 0.7 * t_{min}$	$7.0700 = 0.7 * W_o$ mm.
Pad Weld	$9.5000 = 0.5 * T_{minPad}$	$12.7260 = 0.7 * W_p$ mm.

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$\begin{aligned} &= \max(0, (A-A1+2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - tr)) \cdot S_v) \\ &= \max(0, (13.3903 - 2.1388 + 2 \cdot 5.1675 \cdot 0.8551 \cdot (1.0 \cdot 19.5 - 16.7247)) \cdot 138) \\ &= 16177.65 \text{ Kgf} \end{aligned}$$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$\begin{aligned} &= (A2+A5+A4-(W_i-Can/.707)^2 \cdot f_{r2}) \cdot S_v \\ &= (2.6177 + 11.242 + 3.2049 - 0.0 \cdot 0.86) \cdot 138 \\ &= 24012.39 \text{ Kgf} \end{aligned}$$

Weld Load [W2]:

$$\begin{aligned} &= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v \\ &= (2.6177 + 0.0 + 0.8551 + (1.7233)) \cdot 138 \\ &= 7311.54 \text{ Kgf} \end{aligned}$$

Weld Load [W3]:

$$\begin{aligned} &= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S \\ &= (2.6177 + 0.0 + 3.2049 + 11.242 + (1.7233)) \cdot 138 \\ &= 26437.27 \text{ Kgf} \end{aligned}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$\begin{aligned} &= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw} \\ &= (3.1416/2.0) \cdot 88.9 \cdot 10.0 \cdot 0.49 \cdot 118 \\ &= 8233. \text{ Kgf} \end{aligned}$$

Shear, Pad Element Weld [Spew]:

$$\begin{aligned} &= (\pi/2) \cdot D_P \cdot W_P \cdot 0.49 \cdot S_{EW} \\ &= (3.1416/2.0) \cdot 140.0 \cdot 18.0 \cdot 0.49 \cdot 138 \end{aligned}$$

= 27293. Kgf

Shear, Nozzle Wall [Snw]:

$$\begin{aligned} &= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn \\ &= (3.1416 \cdot 41.8662) \cdot (6.6675 - 1.5) \cdot 0.7 \cdot 118 \\ &= 5725. Kgf \end{aligned}$$

Tension, Pad Groove Weld [Tpgw]:

$$\begin{aligned} &= (\pi/2) \cdot Dlo \cdot Wgpn \cdot 0.74 \cdot Seg \\ &= (3.1416/2) \cdot 88.9 \cdot 22.0 \cdot 0.74 \cdot 138 \\ &= 31990. Kgf \end{aligned}$$

Tension, Shell Groove Weld [Tngw]:

$$\begin{aligned} &= (\pi/2) \cdot Dlo \cdot (T - cas) \cdot 0.74 \cdot Sng \\ &= (3.1416/2.0) \cdot 88.9 \cdot (21.0 - 1.5) \cdot 0.74 \cdot 138 \\ &= 28355. Kgf \end{aligned}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SPEW + SNW) = (27293 + 5725) = 33018 \text{ Kgf}$$

Failure Path [Path 2-2]:

$$\begin{aligned} &= (Sonw + Tpgw + Tngw + Sinw) \\ &= (8233 + 31990 + 28355 + 0) = 68578 \text{ Kgf} \end{aligned}$$

Failure Path [Path 3-3]:

$$\begin{aligned} &= (Spew + Tngw + Sinw) \\ &= (27293 + 28355 + 0) = 55648 \text{ Kgf} \end{aligned}$$

Summary of Failure Path Calculations:

Path 1-1 = 33017 Kgf, must exceed W = 16177 Kgf	or W1 = 24012 Kgf
Path 2-2 = 68578 Kgf, must exceed W = 16177 Kgf	or W2 = 7311 Kgf
Path 3-3 = 55648 Kgf, must exceed W = 16177 Kgf	or W3 = 26437 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 24 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 1 bars

The Drop for this Nozzle is : 0.4947 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 239.7668 mm.

Input Echo, WRC107/537 Item 1, Description: N63" :

Diameter Basis for Vessel	Vbasis	ID
Cylindrical or Spherical Vessel	Cylsph	Spherical
Internal Corrosion Allowance	Cas	1.5000 mm.
Vessel Diameter	Dv	3960.000 mm.
Vessel Thickness	Tv	21.000 mm.
Design Temperature	T1	80.0 C
Vessel Material		SA-516 70
Vessel UNS Number		K02700
Vessel Cold S.I. Allowable	Smc	138.00 N./mm2
Vessel Hot S.I. Allowable	Smh	138.00 N./mm2

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Attachment Type	Type	Round
WRC107 Attachment Classification	Holsol	Hollow

Diameter Basis for Nozzle	Nbasis	ID
Corrosion Allowance for Nozzle	Can	1.5000 mm.
Nozzle Diameter	Dn	75.565 mm.



Nozzle Thickness	Tn	6.668	mm.
Nozzle Material		SA-333 6	
Nozzle UNS Number		K03006	
Nozzle Cold S.I. Allowable	SNmc	118.00	N./mm2
Nozzle Hot S.I. Allowable	SNmh	118.00	N./mm2
Thickness of Reinforcing Pad	Tpad	22.000	mm.
Diameter of Reinforcing Pad	Dpad	140.000	mm.
Design Internal Pressure	Dp	23.271	bars
Include Pressure Thrust		No	

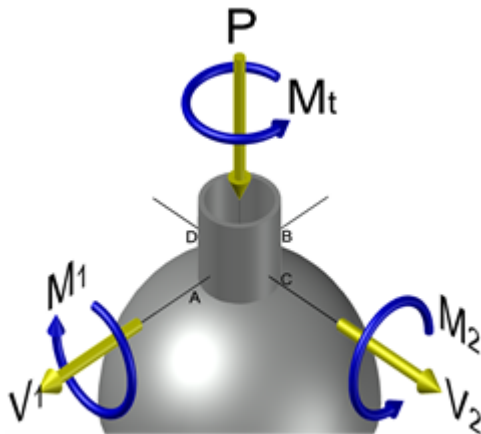
External Forces and Moments in WRC 107/537 Convention:

Radial Load	(SUS)	P	-205.0	Kgf
Longitudinal Shear	(SUS)	(V1) V1	310.0	Kgf
Circumferential Shear	(SUS)	(Vc) V2	310.0	Kgf
Circumferential Moment	(SUS)	(Mc) M1	95.0	Kg-m.
Longitudinal Moment	(SUS)	(M1) M2	65.0	Kg-m.
Torsional Moment	(SUS)	Mt	115.0	Kg-m.

Use Interactive Control No
 WRC107 Version Version March 1979

Include Pressure Stress Indices per Div. 2 No
 Compute Pressure Stress per WRC-368 No
 Local Loads applied at end of Nozzle/Attachment No

WRC Bulletin 537 provides equations for the dimensionless curves found in bulletin 107. As noted in the foreword to bulletin 537, "537 is equivalent to WRC 107". Where 107 is printed in the results below, "537" can be interchanged with "107".



WRC 107/537 has dimensional limitations that can be found in the bulletin's Foreword and which can also be inferred by review of the non-dimensional curve limits.

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))}$$

$$= 88.9 + 2 \cdot 1.65 \cdot \sqrt{(1991.25 (21.0 - 1.5))}$$

$$= 739.171 \text{ mm.}$$

WRC 107 Stress Calculation for SUStained loads:

Radial Load		P	-205.0	Kgf
Circumferential Shear	(VC)	V2	310.0	Kgf
Longitudinal Shear	(VL)	V1	310.0	Kgf
Circumferential Moment	(MC)	M1	95.0	Kg-m.
Longitudinal Moment	(ML)	M2	65.0	Kg-m.
Torsional Moment		MT	115.0	Kg-m.

Dimensionless Param: U = 0.15 TAU = 8.10 RHO = 4.00 (8.03)

Dimensionless Loads for Spherical Shells at Attachment Junction:



Curves read for 1979	Figure	Value	Location
N(x) * T / P	SP 4	0.09028	(A,B,C,D)
M(x) / P	SP 4	0.03871	(A,B,C,D)
N(x) * T * SQRT(Rm * T) / MC	SM 4	0.34861	(A,B,C,D)
M(x) * SQRT(Rm * T) / MC	SM 4	0.16811	(A,B,C,D)
N(x) * T * SQRT(Rm * T) / ML	SM 4	0.34861	(A,B,C,D)
M(x) * SQRT(Rm * T) / ML	SM 4	0.16811	(A,B,C,D)
N(y) * T / P	SP 4	0.34693	(A,B,C,D)
M(y) / P	SP 4	0.24010	(A,B,C,D)
N(y) * T * SQRT(Rm * T) / MC	SM 4	0.10222	(A,B,C,D)
M(y) * SQRT(Rm * T) / MC	SM 4	1.16355	(A,B,C,D)
N(y) * T * SQRT(Rm * T) / ML	SM 4	0.10222	(A,B,C,D)
M(y) * SQRT(Rm * T) / ML	SM 4	1.16355	(A,B,C,D)

Stress Concentration Factors: Kn = 1.00, Kb = 1.00

Stresses in the Vessel at the Attachment Junction (N./mm2)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad. Memb. P		0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1
Rad. Bend. P		0.3	-0.3	0.3	-0.3	0.3	-0.3	0.3	-0.3
Rad. Memb. MC		0.0	0.0	0.0	0.0	-0.7	-0.7	0.7	0.7
Rad. Bend. MC		0.0	0.0	0.0	0.0	-1.9	1.9	1.9	-1.9
Rad. Memb. ML		-0.4	-0.4	0.4	0.4	0.0	0.0	0.0	0.0
Rad. Bend. ML		-1.3	1.3	1.3	-1.3	0.0	0.0	0.0	0.0
Tot. Rad. Str.		-1.4	0.7	2.1	-1.0	-2.2	1.1	2.9	-1.4
Tang. Memb. P		0.4	0.4	0.4	0.4	0.4	0.4	0.4	0.4
Tang. Bend. P		1.7	-1.7	1.7	-1.7	1.7	-1.7	1.7	-1.7
Tang. Memb. MC		0.0	0.0	0.0	0.0	-0.2	-0.2	0.2	0.2
Tang. Bend. MC		0.0	0.0	0.0	0.0	-13.1	13.1	13.1	-13.1
Tang. Memb. ML		-0.1	-0.1	0.1	0.1	0.0	0.0	0.0	0.0
Tang. Bend. ML		-9.0	9.0	9.0	-9.0	0.0	0.0	0.0	0.0
Tot. Tang. Str.		-7.0	7.6	11.2	-10.1	-11.2	11.6	15.4	-14.2
Shear VC		0.5	0.5	-0.5	-0.5	0.0	0.0	0.0	0.0
Shear VL		0.0	0.0	0.0	0.0	-0.5	-0.5	0.5	0.5
Shear MT		2.2	2.2	2.2	2.2	2.2	2.2	2.2	2.2
Tot. Shear		2.7	2.7	1.7	1.7	1.7	1.7	2.7	2.7
Str. Int.		8.1	8.8	11.5	10.4	11.5	11.9	15.9	14.7

Unitless Prm: U = 0.36 TAU = 0.00 (13.05) RHO = 0.00 (3.77)

Dimensionless Loads for Spherical Shells at Pad edge:

Curves read for 1979	Figure	Value	Location
N(x) * T / P	SR 2	0.17299	(A,B,C,D)
M(x) / P	SR 2	0.11959	(A,B,C,D)
N(x) * T * SQRT(Rm * T) / MC	SR 3	0.16067	(A,B,C,D)
M(x) * SQRT(Rm * T) / MC	SR 3	0.38978	(A,B,C,D)
N(x) * T * SQRT(Rm * T) / ML	SR 3	0.16067	(A,B,C,D)
M(x) * SQRT(Rm * T) / ML	SR 3	0.38978	(A,B,C,D)
N(y) * T / P	SR 2	0.05187	(A,B,C,D)
M(y) / P	SR 2	0.03591	(A,B,C,D)
N(y) * T * SQRT(Rm * T) / MC	SR 3	0.04936	(A,B,C,D)
M(y) * SQRT(Rm * T) / MC	SR 3	0.11640	(A,B,C,D)
N(y) * T * SQRT(Rm * T) / ML	SR 3	0.04936	(A,B,C,D)
M(y) * SQRT(Rm * T) / ML	SR 3	0.11640	(A,B,C,D)



Stress Concentration Factors: $K_n = 1.00$, $K_b = 1.00$

Stresses in the Vessel at the Edge of Reinforcing Pad (N./mm²)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad. Memb. P		0.9	0.9	0.9	0.9	0.9	0.9	0.9	0.9
Rad. Bend. P		3.8	-3.8	3.8	-3.8	3.8	-3.8	3.8	-3.8
Rad. Memb. MC		0.0	0.0	0.0	0.0	-2.0	-2.0	2.0	2.0
Rad. Bend. MC		0.0	0.0	0.0	0.0	-29.1	29.1	29.1	-29.1
Rad. Memb. ML		-1.4	-1.4	1.4	1.4	0.0	0.0	0.0	0.0
Rad. Bend. ML		-19.9	19.9	19.9	-19.9	0.0	0.0	0.0	0.0
Tot. Rad. Str.		-16.6	15.6	26.0	-21.4	-26.4	24.2	35.8	-30.0
Tang. Memb. P		0.3	0.3	0.3	0.3	0.3	0.3	0.3	0.3
Tang. Bend. P		1.1	-1.1	1.1	-1.1	1.1	-1.1	1.1	-1.1
Tang. Memb. MC		0.0	0.0	0.0	0.0	-0.6	-0.6	0.6	0.6
Tang. Bend. MC		0.0	0.0	0.0	0.0	-8.7	8.7	8.7	-8.7
Tang. Memb. ML		-0.4	-0.4	0.4	0.4	0.0	0.0	0.0	0.0
Tang. Bend. ML		-5.9	5.9	5.9	-5.9	0.0	0.0	0.0	0.0
Tot. Tang. Str.		-4.9	4.7	7.8	-6.4	-7.9	7.2	10.7	-8.9
Shear VC		0.7	0.7	-0.7	-0.7	0.0	0.0	0.0	0.0
Shear VL		0.0	0.0	0.0	0.0	-0.7	-0.7	0.7	0.7
Shear MT		1.9	1.9	1.9	1.9	1.9	1.9	1.9	1.9
Tot. Shear		2.6	2.6	1.2	1.2	1.2	1.2	2.6	2.6
Str. Int.		17.1	16.2	26.0	21.5	26.4	24.3	36.0	30.3

WRC 107/537 Stress Summations:

Vessel Stress Summation at Attachment Junction (N./mm²)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad. Pm (SUS)		55.6	55.6	55.6	55.6	55.6	55.6	55.6	55.6
Rad. Pl (SUS)		-0.3	-0.3	0.6	0.6	-0.5	-0.5	0.8	0.8
Rad. Q (SUS)		-1.0	1.0	1.6	-1.6	-1.6	1.6	2.2	-2.2
Long. Pm (SUS)		55.6	55.6	55.6	55.6	55.6	55.6	55.6	55.6
Long. Pl (SUS)		0.3	0.3	0.5	0.5	0.2	0.2	0.6	0.6
Long. Q (SUS)		-7.3	7.3	10.6	-10.6	-11.4	11.4	14.8	-14.8
Shear Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear Pl (SUS)		0.5	0.5	-0.5	-0.5	-0.5	-0.5	0.5	0.5
Shear Q (SUS)		2.2	2.2	2.2	2.2	2.2	2.2	2.2	2.2
Pm (SUS)		55.6	55.6	55.6	55.6	55.6	55.6	55.6	55.6
Pm+Pl (SUS)		56.1	56.1	56.6	56.6	56.1	56.1	56.8	56.8
Pm+Pl+Q (Total)		55.3	64.1	67.1	54.9	53.7	67.5	71.5	54.7

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	55.57	138.00	Passed
Pm+Pl (SUS)	56.78	207.00	Passed
Pm+Pl+Q (TOTAL)	71.52	413.99	Passed



Because only sustained loads were specified, the $Pm+Pl+Q$ allowable was $3 * Smh$.

WRC 107/537 Stress Summations:

Vessel Stress Summation at Reinforcing Pad Edge (N./mm²)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad.	Pm (SUS)	118.2	118.2	118.2	118.2	118.2	118.2	118.2	118.2
Rad.	Pl (SUS)	-0.5	-0.5	2.3	2.3	-1.1	-1.1	2.9	2.9
Rad.	Q (SUS)	-16.1	16.1	23.7	-23.7	-25.3	25.3	32.9	-32.9
Long.	Pm (SUS)	118.2	118.2	118.2	118.2	118.2	118.2	118.2	118.2
Long.	Pl (SUS)	-0.1	-0.1	0.7	0.7	-0.3	-0.3	0.9	0.9
Long.	Q (SUS)	-4.8	4.8	7.1	-7.1	-7.5	7.5	9.8	-9.8
Shear	Pm (SUS)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear	Pl (SUS)	0.7	0.7	-0.7	-0.7	-0.7	-0.7	0.7	0.7
Shear	Q (SUS)	1.9	1.9	1.9	1.9	1.9	1.9	1.9	1.9
Pm (SUS)		118.2	118.2	118.2	118.2	118.2	118.2	118.2	118.2
Pm+Pl (SUS)		118.7	118.7	120.8	120.8	118.3	118.3	121.4	121.4
Pm+Pl+Q (Total)		113.8	134.5	144.3	112.0	110.4	142.5	154.3	109.6

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	118.25	138.00	Passed
Pm+Pl (SUS)	121.38	207.00	Passed
Pm+Pl+Q (TOTAL)	154.29	413.99	Passed

Because only sustained loads were specified, the $Pm+Pl+Q$ allowable was $3 * Smh$.

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Input, Nozzle Desc: N48"

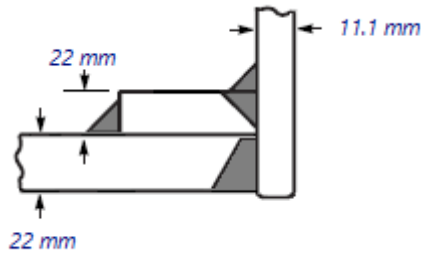
From: 50

Pressure for Reinforcement Calculations	P	23.208	bars
Temperature for Internal Pressure	Temp	80	°C
Design External Pressure	Pext	1.01	bars
Temperature for External Pressure	Tempex	80	°C
Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	138.00	N./mm2
Parent Allowable Stress At Ambient	Sva	138.00	N./mm2
Inside Diameter of Cylindrical Shell	D	2200.00	mm.
Design Length of Section	L	7068.2671	mm.
Shell Finished (Minimum) Thickness	t	22.0000	mm.
Shell Internal Corrosion Allowance	cas	1.5000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		400.80	mm.
User Entered Minimum Design Metal Temperature		-38.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-333 6	+IT
Material UNS Number		K03006	
Material Specification/Type	Smls. & wld. pipe		
Allowable Stress at Temperature	Sn	118.00	N./mm2
Allowable Stress At Ambient	Sna	118.00	N./mm2
Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		8.0000	in.
Size and Thickness Basis		Minimum	
Nominal Thickness		80	
Flange Material		SA-350 LF2	+IT
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	1.5000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	218.2720	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	22.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Pad Material		SA-516 70	+IT
Pad Allowable Stress at Temperature	Sp	138.00	N./mm2
Pad Allowable Stress At Ambient	Spa	138.00	N./mm2
Diameter of Pad along vessel surface	Dp	350.0000	mm.
Thickness of Pad	te	22.0000	mm.
Weld leg size between Pad and Shell	Wp	18.0000	mm.
Groove weld depth between Pad and Nozzle	Wgpn	22.0000	mm.
Reinforcing Pad Width		65.4625	mm.
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: N48"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	7.750 in.
Actual Thickness Used in Calculation	0.438 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 23.21 \cdot 1101.5 / (138 \cdot 1.0 - 0.6 \cdot 23.21)$
 $= 18.7146 \text{ mm.}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 23.21 \cdot 99.92 / (118 \cdot 1.0 - 0.6 \cdot 23.21)$
 $= 1.9889 \text{ mm.}$

Required Nozzle thickness under External Pressure per UG-28 : 0.6957 mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	399.7000 mm.
Parallel to Vessel Wall, opening length	d	199.8500 mm.
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		46.0312 mm.

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 118.0 / 138.0)$
 $= 0.855$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 118.0 / 138.0)$
 $= 0.855$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, S_p / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(0.855, 1.0)$
 $= 0.855$

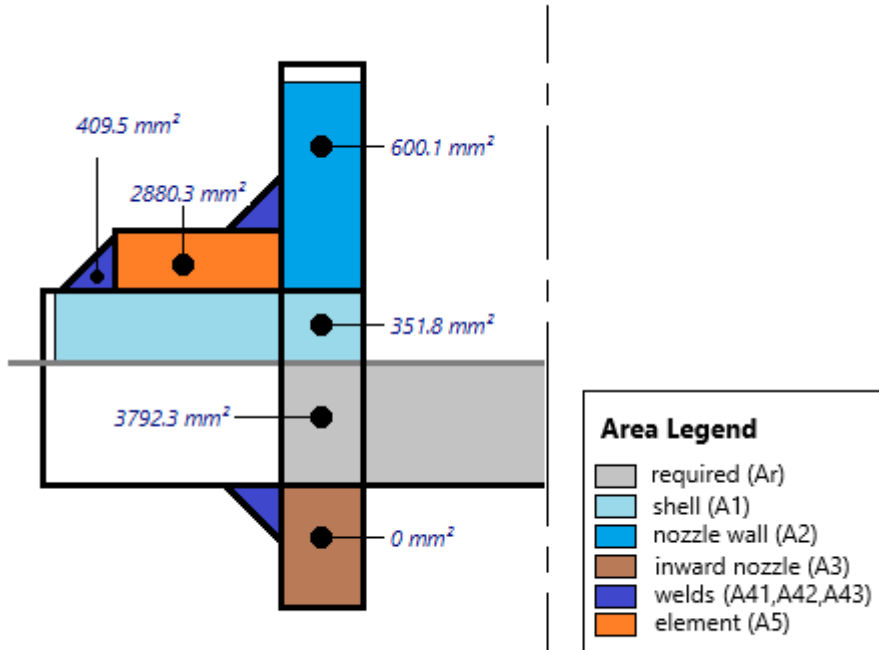


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm2)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	37.923	11.676	NA
Area in Shell	A1	3.518	17.688	NA
Area in Nozzle Wall	A2	6.001	7.019	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	4.095	4.095	NA
Area in Element	A5	28.804	28.804	NA
TOTAL AREA AVAILABLE	Atot	42.418	57.606	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Insufficient.
 The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS: Diameter Thickness
 Based on given Pad Thickness: 329.5650 22.0000 mm.
 Based on given Pad Diameter: 350.0000 18.5662 mm.
 Based on the Estimated Diameter Limit: 398.1125 13.5769 mm.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1))$ UG-37(c)
 $= (199.85 \cdot 18.7146 \cdot 1.0 + 2 \cdot 9.6125 \cdot 18.7146 \cdot 1.0 \cdot (1 - 0.86))$
 $= 37.923 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 199.85(1.0 \cdot 20.5 - 1.0 \cdot 18.715) - 2 \cdot 9.613$
 $(1.0 \cdot 20.5 - 1.0 \cdot 18.7146) \cdot (1 - 0.855)$
 $= 3.518 \text{ cm}^2$

Area Available in Nozzle Wall Projecting Outward [A2]:
 $= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$
 $= (2 \cdot 46.03) \cdot (9.61 - 1.99) \cdot 0.8551$



= 6.001 cm²

Area Available in Welds [A41 + A42 + A43]:
= $W_o^2 \cdot fr_3 + (W_i - can/0.707)^2 \cdot fr_2 + W_p^2 \cdot fr_4$
= $10.0^2 \cdot 0.86 + (0.0)^2 \cdot 0.86 + 18.0^2 \cdot 1.0$
= 4.095 cm²

Area Available in Element [A5]:
= $(\min(D_p, DL) - (\text{Nozzle OD})) \cdot (\min(t_p, Tlwp, te)) \cdot fr_4$
= $(350.0 - 219.075) \cdot 22.0 \cdot 1.0$
= 28.804 cm²

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 3.4889 mm.
Wall Thickness per UG16(b),	tr16b = 3.0000 mm.
Wall Thickness, shell/head, internal pressure	trb1 = 20.2146 mm.
Wall Thickness	tb1 = max(trb1, tr16b) = 20.2146 mm.
Wall Thickness, shell/head, external pressure	trb2 = 2.3091 mm.
Wall Thickness	tb2 = max(trb2, tr16b) = 3.0000 mm.
Wall Thickness per table UG-45	tb3 = 8.6600 mm.

Determine Nozzle Thickness candidate [tb]:
= min[tb3, max(tb1, tb2)]
= min[8.66, max(20.2146, 3.0)]
= 8.6600 mm.

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
= max(ta, tb)
= max(3.4889, 8.66)
= 8.6600 mm.

Available Nozzle Neck Thickness = 11.1125 mm. -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 35.9	Allowable	: 118.0 N./mm ²	Passed
Expansion	: 0.0	Allowable	: 259.1 N./mm ²	Passed
Occasional	: 11.5	Allowable	: 156.9 N./mm ²	Passed
Shear	: 17.1	Allowable	: 82.6 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld (Impact tested) :

Note:

This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification	-46 °C
Calculated Minimum Design Metal Temperature	-104 °C

Nozzle Neck to Pad Weld for the Nozzle (Impact tested) :

Note:

This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification	-46 °C
Calculated Minimum Design Metal Temperature	-104 °C

Nozzle Neck to Pad Weld for Reinf. pad (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Shell to Pad Weld at Pad OD for pad (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note:

This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification	-46 °C
Calculated Minimum Design Metal Temperature	-104 °C
Gov. MDMT of the Nozzle	: -104 °C
Gov. MDMT of the Reinforcement Pad	: -43 °C
Gov. MDMT of the nozzle to shell joint welded assembly	: -43 °C

Flange MDMT including Temperature reduction per UCS-66.1:

MDMT of ASME B16.5/47 flange per Matl. Specification	-46 °C
Flange MDMT with Temp reduction per UCS-66(i)(2)	-85 °C

Where the Stress Reduction Ratio per UCS-66(i)(2) is :

$$\text{Design Pressure/Ambient Rating} = 23.21/51.00 = 0.455$$

Weld Size Calculations, Description: N48"

Intermediate Calc. for nozzle/shell Welds	Tmin	9.6125 mm.
Intermediate Calc. for pad/shell Welds	TminPad	19.0000 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm.
Pad Weld	9.5000 = 0.5*TminPad	12.7260 = 0.7 * Wp mm.

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$= \max(0, (A-A1+2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - t_r)) \cdot S_v) \\ = \max(0, (37.9226 - 3.5184 + 2 \cdot 9.6125 \cdot 0.8551 \cdot (1.0 \cdot 20.5 - 18.7146)) \cdot 138) \\ = 48824.91 \text{ Kg}$$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$= (A2+A5+A4-(W_i \cdot \text{Can}/.707)^2 \cdot f_{r2}) \cdot S_v \\ = (6.0013 + 28.8035 + 4.0951 - 0.0 \cdot 0.86) \cdot 138 \\ = 54738.05 \text{ Kg}$$

Weld Load [W2]:

$$= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v \\ = (6.0013 + 0.0 + 0.8551 + (3.37)) \cdot 138 \\ = 14390.03 \text{ Kg}$$

Weld Load [W3]:

$$= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S \\ = (6.0013 + 0.0 + 4.0951 + 28.8035 + (3.37)) \cdot 138 \\ = 59480.09 \text{ Kg}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw} \\ = (3.1416/2.0) \cdot 219.075 \cdot 10.0 \cdot 0.49 \cdot 118 \\ = 20289. \text{ Kg}$$

Shear, Pad Element Weld [Spew]:

$$= (\pi/2) \cdot D_P \cdot W_P \cdot 0.49 \cdot S_{EW} \\ = (3.1416/2.0) \cdot 350.0 \cdot 18.0 \cdot 0.49 \cdot 138$$

= 68233. Kgf

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (D_{lr} + D_{lo}) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot S_n$$

$$= (3.1416 \cdot 104.7313) \cdot (11.1125 - 1.5) \cdot 0.7 \cdot 118$$

$$= 26638. Kgf$$

Tension, Pad Groove Weld [Tpgw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_{gpn} \cdot 0.74 \cdot S_{eg}$$

$$= (3.1416/2) \cdot 219.075 \cdot 22.0 \cdot 0.74 \cdot 138$$

$$= 78833. Kgf$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot D_{lo} \cdot (T - cas) \cdot 0.74 \cdot S_{ng}$$

$$= (3.1416/2.0) \cdot 219.075 \cdot (22.0 - 1.5) \cdot 0.74 \cdot 138$$

$$= 73458. Kgf$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SPEW + SNW) = (68233 + 26638) = 94872 Kgf$$

Failure Path [Path 2-2]:

$$= (Sonw + Tpgw + Tngw + Sinw)$$

$$= (20289 + 78833 + 73458 + 0) = 172580 Kgf$$

Failure Path [Path 3-3]:

$$= (Spew + Tngw + Sinw)$$

$$= (68233 + 73458 + 0) = 141692 Kgf$$

Summary of Failure Path Calculations:

Path 1-1 = 94871 Kgf, must exceed W = 48824 Kgf	or W1 = 54738 Kgf
Path 2-2 = 172579 Kgf, must exceed W = 48824 Kgf	or W2 = 14390 Kgf
Path 3-3 = 141691 Kgf, must exceed W = 48824 Kgf	or W3 = 59480 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 24 bars

Nozzle is O.K. for the External Pressure: 1 bars

The Drop for this Nozzle is : 5.4674 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 245.7394 mm.

Input Echo, WRC297 Item 1, Description: N48" :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	1.5000	mm.
Shell External Corrosion Allowance	0.0000	mm.
Shell Diameter	2200.000	mm.
Shell Thickness	22.0000	mm.
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc	138.00 N./mm ²
Vessel Hot S.I. Allowable	Smh	138.00 N./mm ²

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	1.5000	mm.
Nozzle Diameter	196.850	mm.
Nozzle Thickness	11.1125	mm.
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-333 6	
Nozzle Cold S.I. Allowable	SNmc	118.00 N./mm ²
Nozzle Hot S.I. Allowable	SNmh	118.00 N./mm ²

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Diameter of Reinforcing Pad 350.000 mm.
 Thickness of Reinforcing Pad 22.0000 mm.

Note:

*External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUStained loads.*

Design Internal Pressure	Dp	23.21	bars
Radial Load	P	-555.00	Kgf
Circumferential Shear	Vc	820.00	Kgf
Longitudinal Shear	Vl	820.00	Kgf
Circumferential Moment	Mc	490.00	Kg-m.
Longitudinal Moment	Ml	655.00	Kg-m.
Torsional Moment	Mt	820.00	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))}$
 $= 219.075 + 2 \cdot 1.65 \cdot \sqrt{(1122.75(22.0 - 1.5))}$
 $= 719.723 \text{ mm.}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	2245.500 mm.
Nozzle Outside Diameter	(d) =	219.075 mm.
Vessel Thickness used	(T) =	42.500 mm.
Nozzle Thickness used	(t) =	9.613 mm.
T / t	=	4.421
d / t	=	22.791

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 0.709$$

Nr/P	=	0.091
Mr/P	=	0.079
M0/P	=	0.137
N0/P	=	0.298
MrD/Mc	=	0.123
NrDL/Mc	=	0.121
M0d/Mc	=	0.267
N0DL/Mc	=	0.125
MrD/Ml	=	0.116
NrDL/Ml	=	0.114
M0D/Ml	=	0.237
N0DL/Ml	=	0.292

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	0	0	0	0
Outplane Bending (P)	2	-2	2	-2
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (Ml)	-4	-4	4	4
Outplane Bending (Ml)	-23	23	23	-23
Normal Pressure Stress	59	61	59	61
Outplane Stress Summary	34	78	90	41

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
--------------------	----	----	----	----



(Stresses parallel to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Inplane Membrane (P)	0	0	0	0
Inplane Bending (P)	1	-1	1	-1
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-1	-1	1	1
Inplane Bending (ML)	-11	11	11	-11
Inplane Pressure Stress	29	29	29	29
Inplane Stress Summary	18	37	44	18

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	2	2	2	2
Shear Stress Summary	3	3	1	1

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	35	78	90	41

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	0	0	0	0
Outplane Bending (P)	2	-2	2	-2
Outplane Membrane (Mc)	-1	-1	1	1
Outplane Bending (Mc)	-19	19	19	-19
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	29	29	29	29
Outplane Stress Summary	11	45	53	9

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Inplane Membrane (P)	0	0	0	0
Inplane Bending (P)	1	-1	1	-1
Inplane Membrane (Mc)	-1	-1	1	1
Inplane Bending (Mc)	-8	8	8	-8
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	59	61	59	61
Inplane Stress Summary	50	67	71	52

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Torsional Shear (Mt)	2	2	2	2



Shear Stress Summary 1 1 3 3

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	50	67	71	52

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au	Al	Bu	Bl
	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside
Hoop Membrane (P)	0	0	0	0
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-4	-4	4	4
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	27	30	27	30

Hoop Stress Summary 23 26 33 36

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au	Al	Bu	Bl
	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	11	-11	11	-11
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-20	-20	20	20
Axial Bending (ML)	-111	111	111	-111
Axial Pressure Stress	13	13	13	13

Axial Stress Summary -105 94 158 -88

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress)	Au	Al	Bu	Bl
	Outside	Inside	Outside	Inside
Shear due to (Vc)	2	2	-2	-2
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	11	11	11	11

Shear Stress Summary 13 13 8 8

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities	Au	Al	Bu	Bl
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	132	97	158	125

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Hoop Membrane (P)	0	0	0	0
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-1	-1	1	1
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	27	30	27	30

Hoop Stress Summary 27 29 30 32

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	11	-11	11	-11
Axial Membrane (Mc)	-15	-15	15	15
Axial Bending (Mc)	-89	89	89	-89
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	13	13	13	13
Axial Stress Summary	-78	77	130	-71

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-2	-2	2	2
Shear due to Torsion	11	11	11	11
Shear Stress Summary	8	8	13	13

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	106	78	132	107

Stress Computations at the Edge of the Reinforcing Pad:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - ca)}$$

$$= 219.075 + 2 \cdot 1.65 \cdot \sqrt{1111.75(22.0 - 1.5)}$$

$$= 717.264 \text{ mm.}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter (D) = 2223.500 mm.
 Nozzle Outside Diameter (d) = 350.000 mm.
 Vessel Thickness used (T) = 20.500 mm.
 Nozzle Thickness used (t) = 75.075 mm.
 $T / t = 0.273$
 $d / t = 10.000$

$$\lambda = [(d/D) \cdot \sqrt{D/T}] = 1.639$$

Note: Re-pad thickness is added to nozzle thickness to simulate the Re-pad.

Nr/P = 0.079
 Mr/P = 0.166
 M0/P = 0.048
 N0/P = 0.094
 MrD/Mc = 0.296
 NrDL/Mc = 0.088
 M0d/Mc = 0.091
 N0DL/Mc = 0.068
 MrD/Ml = 0.132
 NrDL/Ml = 0.089
 M0D/Ml = 0.041
 N0DL/Ml = 0.114

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
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Outplane Membrane (P)	1	1	1	1
Outplane Bending (P)	3	-3	3	-3
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-4	-4	4	4
Outplane Bending (ML)	-10	10	10	-10
Normal Pressure Stress	123	125	123	125

Outplane Stress Summary	112	129	144	117
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Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses parallel to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Inplane Membrane (P)	1	1	1	1
Inplane Bending (P)	12	-12	12	-12
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-3	-3	3	3
Inplane Bending (ML)	-34	34	34	-34
Inplane Pressure Stress	61	61	61	61

Inplane Stress Summary	37	80	114	19
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Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress normal to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	2	2	2	2

Shear Stress Summary	2	2	1	1
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Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Two * Max Shear Stress	112	129	144	117

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside
Outplane Membrane (P)	1	1	1	1
Outplane Bending (P)	3	-3	3	-3
Outplane Membrane (Mc)	-2	-2	2	2
Outplane Bending (Mc)	-17	17	17	-17
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	61	61	61	61

Outplane Stress Summary	46	74	86	43
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses parallel to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside
Inplane Membrane (P)	1	1	1	1
Inplane Bending (P)	12	-12	12	-12
Inplane Membrane (Mc)	-2	-2	2	2
Inplane Bending (Mc)	-57	57	57	-57
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	123	125	123	125

Inplane Stress Summary 76 169 198 58

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)		Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Shear (Vc)		0	0	0	0
Outplane Shear (Vl)		0	0	0	0
Torsional Shear (Mt)		2	2	2	2

Shear Stress Summary 1 1 2 2

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)		Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
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Two * Max Shear Stress 76 169 198 59

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	59	61	59	61	59	61	59	61
Circ. Pl (SUS)	-4	-4	4	4	-1	-1	1	1
Circ. Q (SUS)	-21	21	25	-25	-7	7	9	-9
Long. Pm (SUS)	29	29	29	29	29	29	29	29
Long. Pl (SUS)	-1	-1	1	1	-1	-1	1	1
Long. Q (SUS)	-10	10	12	-12	-17	17	21	-21
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	2	2	2	2	2	2	2	2
Pm (SUS)	59.0	61.0	59.0	61.0	59.0	61.0	59.0	61.0
Pm+Pl (SUS)	55.0	57.0	63.0	65.0	58.0	60.0	60.0	62.0
Pm+Pl+Q (Total)	34.2	78.1	88.1	40.2	51.1	67.2	69.2	53.1

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	61.00	138.00	Passed
Pm+Pl (SUS)	65.00	207.00	Passed
Pm+Pl+Q (TOTAL)	88.09	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Reinforcing Pad Edge (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	123	125	123	125	123	125	123	125
Circ. Pl (SUS)	-3	-3	5	5	-1	-1	3	3
Circ. Q (SUS)	-7	7	13	-13	-45	45	69	-69

Long. Pm (SUS)	61	61	61	61	61	61	61	61
Long. Pl (SUS)	-2	-2	4	4	-1	-1	3	3
Long. Q (SUS)	-22	22	46	-46	-14	14	20	-20
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	2	2	2	2	2	2	2	2
Pm (SUS)	123.0	125.0	123.0	125.0	123.0	125.0	123.0	125.0
Pm+Pl (SUS)	120.0	122.0	128.0	130.0	122.0	124.0	126.0	128.0
Pm+Pl+Q (Total)	113.1	129.1	141.1	117.0	77.1	169.0	195.0	59.3

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	125.00	138.00	Passed
Pm+Pl (SUS)	130.00	207.00	Passed
Pm+Pl+Q (TOTAL)	195.04	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	27	30	27	30	27	30	27	30
Circ. Pl (SUS)	-4	-4	4	4	-1	-1	1	1
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	13	13	13	13	13	13	13	13
Long. Pl (SUS)	-20	-20	20	20	-15	-15	15	15
Long. Q (SUS)	-100	100	122	-122	-78	78	100	-100
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	2	2	-2	-2	-2	-2	2	2
Shear Q (SUS)	11	11	11	11	11	11	11	11
Pm (SUS)	27.0	30.0	27.0	30.0	27.0	30.0	27.0	30.0
Pm+Pl (SUS)	30.3	33.2	34.2	35.6	28.3	31.3	30.0	32.0
Pm+Pl+Q (Total)	132.6	95.4	155.6	124.3	107.5	77.7	129.7	106.2

Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	30.00	118.00	Passed
Pm+Pl (SUS)	35.56	177.00	Passed
Pm+Pl+Q (TOTAL)	155.65	353.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.



Input, Nozzle Desc: N73"

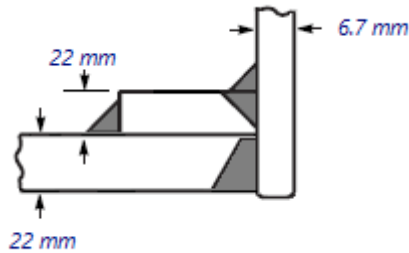
From: 50

Pressure for Reinforcement Calculations	P	23.201	bars
Temperature for Internal Pressure	Temp	80	°C
Design External Pressure	Pext	1.01	bars
Temperature for External Pressure	Tempex	80	°C
Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	138.00	N./mm2
Parent Allowable Stress At Ambient	Sva	138.00	N./mm2
Inside Diameter of Cylindrical Shell	D	2200.00	mm.
Design Length of Section	L	7068.2671	mm.
Shell Finished (Minimum) Thickness	t	22.0000	mm.
Shell Internal Corrosion Allowance	cas	1.5000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		500.80	mm.
User Entered Minimum Design Metal Temperature		-38.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-333 6	+IT
Material UNS Number		K03006	
Material Specification/Type	Smls. & wld. pipe		
Allowable Stress at Temperature	Sn	118.00	N./mm2
Allowable Stress At Ambient	Sna	118.00	N./mm2
Diameter Basis (for tr calc only)		Inside	
Layout Angle		180.00	deg
Diameter		3.0000	in.
Size and Thickness Basis		Minimum	
Nominal Thickness		80	
Flange Material		SA-350 LF2	+IT
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	1.5000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	218.2720	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	22.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Pad Material		SA-516 70	+IT
Pad Allowable Stress at Temperature	Sp	138.00	N./mm2
Pad Allowable Stress At Ambient	Spa	138.00	N./mm2
Diameter of Pad along vessel surface	Dp	140.0000	mm.
Thickness of Pad	te	22.0000	mm.
Weld leg size between Pad and Shell	Wp	18.0000	mm.
Groove weld depth between Pad and Nozzle	Wgpn	22.0000	mm.
Reinforcing Pad Width		25.5500	mm.
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: N73"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	2.975 in.
Actual Thickness Used in Calculation	0.263 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 23.2 \cdot 1101.5 / (138 \cdot 1.0 - 0.6 \cdot 23.2)$
 $= 18.7092 \text{ mm.}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 23.2 \cdot 39.28 / (118 \cdot 1.0 - 0.6 \cdot 23.2)$
 $= 0.7817 \text{ mm.}$

Required Nozzle thickness under External Pressure per UG-28 : 0.4107 mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	157.1300 mm.
Parallel to Vessel Wall, opening length	d	78.5650 mm.
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		34.9188 mm.

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 118.0 / 138.0)$
 $= 0.855$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 118.0 / 138.0)$
 $= 0.855$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, S_p / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(0.855, 1.0)$
 $= 0.855$

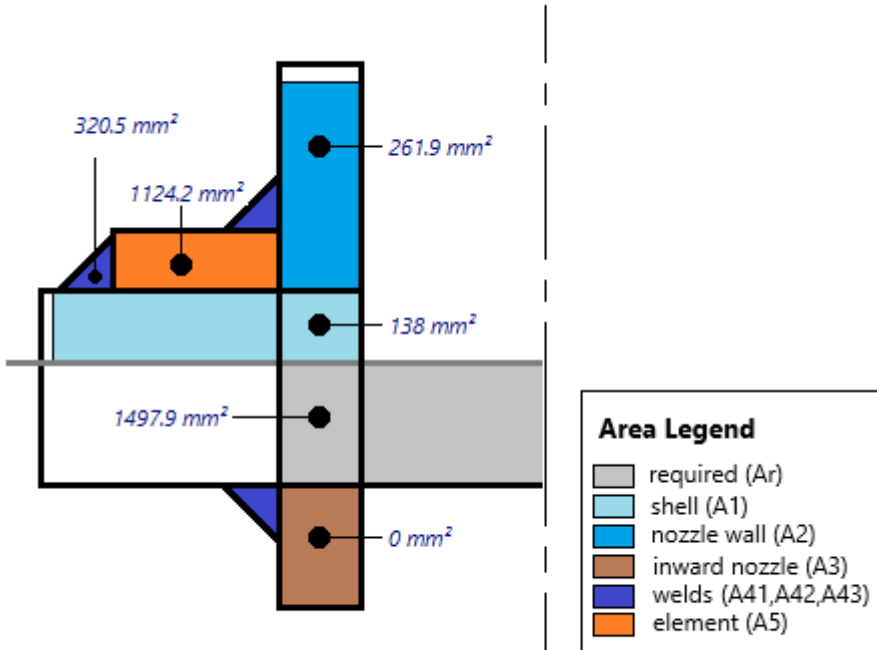


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm2)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	14.979	4.613	NA
Area in Shell	A1	1.380	6.917	NA
Area in Nozzle Wall	A2	2.619	2.841	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	3.205	3.205	NA
Area in Element	A5	11.242	11.242	NA
TOTAL AREA AVAILABLE	Atot	18.446	24.205	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Insufficient.
 The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS: Diameter Thickness
 Based on given Pad Thickness: 124.2413 22.0000 mm.
 Based on given Pad Diameter: 140.0000 15.2154 mm.
 Based on the Estimated Diameter Limit: 155.5425 11.6669 mm.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$
 $= (78.565 \cdot 18.7092 \cdot 1.0 + 2 \cdot 5.1675 \cdot 18.7092 \cdot 1.0 \cdot (1 - 0.86))$
 $= 14.979 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 78.565(1.0 \cdot 20.5 - 1.0 \cdot 18.709) - 2 \cdot 5.168$
 $(1.0 \cdot 20.5 - 1.0 \cdot 18.7092) \cdot (1 - 0.855)$
 $= 1.380 \text{ cm}^2$

Area Available in Nozzle Wall Projecting Outward [A2]:
 $= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$
 $= (2 \cdot 34.92) \cdot (5.17 - 0.78) \cdot 0.8551$



= 2.619 cm²

Area Available in Welds [A41 + A42 + A43]:
= (Wo² - Ar Lost)•Fr3+((Wi-can/0.707)² - Ar Lost)•fr2 + Trapfr4
= (1.) • 0.86 + (0.0) • 0.86 + 234.9806² • 1.0
= 3.205 cm²

Area Available in Element [A5]:
= (min(Dp,DL)-(Nozzle OD))•(min(tp,Tlwp,te)) • fr4
= (140.0 - 88.9) • 22.0 • 1.0
= 11.242 cm²

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 2.2817 mm.
Wall Thickness per UG16(b),	tr16b = 3.0000 mm.
Wall Thickness, shell/head, internal pressure	trb1 = 20.2092 mm.
Wall Thickness	tb1 = max(trb1, tr16b) = 20.2092 mm.
Wall Thickness, shell/head, external pressure	trb2 = 2.3091 mm.
Wall Thickness	tb2 = max(trb2, tr16b) = 3.0000 mm.
Wall Thickness per table UG-45	tb3 = 6.3000 mm.

Determine Nozzle Thickness candidate [tb]:
= min[tb3, max(tb1,tb2)]
= min[6.3, max(20.2092, 3.0)]
= 6.3000 mm.

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
= max(ta, tb)
= max(2.2817, 6.3)
= 6.3000 mm.

Available Nozzle Neck Thickness = 6.6675 mm. -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 48.8,	Allowable	: 118.0 N./mm ²	Passed
Expansion	: 0.0,	Allowable	: 246.2 N./mm ²	Passed
Occasional	: 8.3,	Allowable	: 156.9 N./mm ²	Passed
Shear	: 29.3,	Allowable	: 82.6 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld (Impact tested) :

Note:

This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification	-46 °C
Calculated Minimum Design Metal Temperature	-104 °C

Nozzle Neck to Pad Weld for the Nozzle (Impact tested) :

Note:

This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification	-46 °C
Calculated Minimum Design Metal Temperature	-104 °C

Nozzle Neck to Pad Weld for Reinf. pad (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Shell to Pad Weld at Pad OD for pad (Impact tested) :



Note: This Element/Detail was specified as being Impact Tested.

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note:

This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification	-46 °C
Calculated Minimum Design Metal Temperature	-104 °C
Gov. MDMT of the Nozzle	: -104 °C
Gov. MDMT of the Reinforcement Pad	: -43 °C
Gov. MDMT of the nozzle to shell joint welded assembly	: -43 °C

Flange MDMT including Temperature reduction per UCS-66.1:

MDMT of ASME B16.5/47 flange per Matl. Specification	-46 °C
Flange MDMT with Temp reduction per UCS-66(i)(2)	-85 °C

Where the Stress Reduction Ratio per UCS-66(i)(2) is :

$$\text{Design Pressure/Ambient Rating} = 23.20/51.00 = 0.455$$

Weld Size Calculations, Description: N73"

Intermediate Calc. for nozzle/shell Welds	Tmin	5.1675 mm.
Intermediate Calc. for pad/shell Welds	TminPad	19.0000 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$3.6173 = 0.7 \cdot t_{min}$	$7.0700 = 0.7 \cdot W_o$ mm.
Pad Weld	$9.5000 = 0.5 \cdot T_{minPad}$	$12.7260 = 0.7 \cdot W_p$ mm.

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$= \max(0, (A-A1+2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t_{tr})) \cdot S_v)$$

$$= \max(0, (14.9791 - 1.3801 + 2 \cdot 5.1675 \cdot 0.8551 \cdot (1.0 \cdot 20.5 - 18.7092)) \cdot 138)$$

$$= 19358.59 \text{ Kg}$$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$= (A2+A5+A4-(W_i \cdot \text{Can}/.707)^2 \cdot f_{r2}) \cdot S_v$$

$$= (2.6191 + 11.242 + 3.2049 - 0.0 \cdot 0.86) \cdot 138$$

$$= 24014.38 \text{ Kg}$$

Weld Load [W2]:

$$= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v$$

$$= (2.6191 + 0.0 + 0.8551 + (1.8116)) \cdot 138$$

$$= 7437.88 \text{ Kg}$$

Weld Load [W3]:

$$= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S$$

$$= (2.6191 + 0.0 + 3.2049 + 11.242 + (1.8116)) \cdot 138$$

$$= 26563.61 \text{ Kg}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$$

$$= (3.1416/2.0) \cdot 88.9 \cdot 10.0 \cdot 0.49 \cdot 118$$

$$= 8233. \text{ Kg}$$

Shear, Pad Element Weld [Spew]:

$$= (\pi/2) \cdot D_P \cdot W_P \cdot 0.49 \cdot S_{EW}$$

$$= (3.1416/2.0) \cdot 140.0 \cdot 18.0 \cdot 0.49 \cdot 138$$

= 27293. Kgf

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$$

$$= (3.1416 \cdot 41.8662) \cdot (6.6675 - 1.5) \cdot 0.7 \cdot 118$$

$$= 5725. Kgf$$

Tension, Pad Groove Weld [Tpgw]:

$$= (\pi/2) \cdot Dlo \cdot Wgpn \cdot 0.74 \cdot Seg$$

$$= (3.1416/2) \cdot 88.9 \cdot 22.0 \cdot 0.74 \cdot 138$$

$$= 31990. Kgf$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot Dlo \cdot (T - cas) \cdot 0.74 \cdot Sng$$

$$= (3.1416/2.0) \cdot 88.9 \cdot (22.0 - 1.5) \cdot 0.74 \cdot 138$$

$$= 29809. Kgf$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SPEW + SNW) = (27293 + 5725) = 33018 Kgf$$

Failure Path [Path 2-2]:

$$= (Sonw + Tpgw + Tngw + Sinw)$$

$$= (8233 + 31990 + 29809 + 0) = 70032 Kgf$$

Failure Path [Path 3-3]:

$$= (Spew + Tngw + Sinw)$$

$$= (27293 + 29809 + 0) = 57102 Kgf$$

Summary of Failure Path Calculations:

Path 1-1 = 33017 Kgf, must exceed W = 19358 Kgf	or W1 = 24014 Kgf
Path 2-2 = 70032 Kgf, must exceed W = 19358 Kgf	or W2 = 7437 Kgf
Path 3-3 = 57102 Kgf, must exceed W = 19358 Kgf	or W3 = 26563 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 25 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 1 bars

The Drop for this Nozzle is : 0.8985 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 241.1705 mm.

Input Echo, WRC297 Item 1, Description: N73" :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	1.5000	mm.
Shell External Corrosion Allowance	0.0000	mm.
Shell Diameter	2200.000	mm.
Shell Thickness	22.0000	mm.
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc	138.00 N./mm ²
Vessel Hot S.I. Allowable	Smh	138.00 N./mm ²

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	1.5000	mm.
Nozzle Diameter	75.565	mm.
Nozzle Thickness	6.6675	mm.
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-333 6	
Nozzle Cold S.I. Allowable	SNmc	118.00 N./mm ²
Nozzle Hot S.I. Allowable	SNmh	118.00 N./mm ²



See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Diameter of Reinforcing Pad 140.000 mm.
 Thickness of Reinforcing Pad 22.0000 mm.

Note:

*External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUSTained loads.*

Design Internal Pressure	Dp	23.20	bars
Radial Load	P	-205.00	Kgf
Circumferential Shear	Vc	310.00	Kgf
Longitudinal Shear	Vl	310.00	Kgf
Circumferential Moment	Mc	65.00	Kg-m.
Longitudinal Moment	Ml	95.00	Kg-m.
Torsional Moment	Mt	115.00	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

*Warning - The ratio, Dn/Tv, (4.041) is < 5
 check the Limitations of WRC bulletin 297.*

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))}$$

$$= 88.9 + 2 \cdot 1.65 \cdot \sqrt{(1122.75(22.0 - 1.5))}$$

$$= 589.548 \text{ mm.}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	2245.500 mm.
Nozzle Outside Diameter	(d) =	88.900 mm.
Vessel Thickness used	(T) =	42.500 mm.
Nozzle Thickness used	(t) =	5.168 mm.
T / t	=	8.224
d / t	=	17.204

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 0.288$$

Nr/P	=	0.127
Mr/P	=	0.079
M0/P	=	0.304
N0/P	=	0.349
MrD/Mc	=	0.093
NrDL/Mc	=	0.108
M0d/Mc	=	0.319
N0DL/Mc	=	-0.142
MrD/Ml	=	0.094
NrDL/Ml	=	0.109
M0D/Ml	=	0.312
N0DL/Ml	=	-0.095

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	0	0	0	0
Outplane Bending (P)	2	-2	2	-2
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (Ml)	0	0	0	0
Outplane Bending (Ml)	-10	10	10	-10
Normal Pressure Stress	58	61	58	61



Outplane Stress Summary 51 71 71 48

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE
 (Stresses parallel to longitudinal plane)

	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
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Inplane Membrane (P)	0	0	0	0
Inplane Bending (P)	0	0	0	0
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	-3	3	3	-3
Inplane Pressure Stress	29	29	29	29

Inplane Stress Summary 26 31 34 26

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE
 (Shear stress normal to longitudinal plane)

	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
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Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	2	2	2	2

Shear Stress Summary 2 2 1 1

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE
 (Stress Intensities)

	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
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Two * Max Shear Stress 51 71 71 48

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE
 (Stresses Normal to circumferential plane)

	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
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Outplane Membrane (P)	0	0	0	0
Outplane Bending (P)	2	-2	2	-2
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	-7	7	7	-7
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	29	29	29	29

Outplane Stress Summary 24 36 38 19

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE
 (Stresses parallel to circumferential plane)

	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
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Inplane Membrane (P)	0	0	0	0
Inplane Bending (P)	0	0	0	0
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	-2	2	2	-2
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	58	61	58	61

Inplane Stress Summary 57 62 62 59

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE
 (Shear stress normal to circumferential plane)

	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
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Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Torsional Shear (Mt)	2	2	2	2
Shear Stress Summary	1	1	2	2

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Two * Max Shear Stress	57	62	62	59

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Hoop Membrane (P)	0	0	0	0
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	19	22	19	22
Hoop Stress Summary	20	23	19	22

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Axial Membrane (P)	1	1	1	1
Axial Bending (P)	6	-6	6	-6
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-34	-34	34	34
Axial Bending (ML)	-91	91	91	-91
Axial Pressure Stress	9	9	9	9
Axial Stress Summary	-108	61	144	-52

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)	4	4	-4	-4
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	17	17	17	17
Shear Stress Summary	21	21	13	13

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	136	71	145	79

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	0	0	0	0
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0



Hoop Pressure Stress	19	22	19	22
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Hoop Stress Summary	20	23	19	22
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Axial Membrane (P)	1	1	1	1
Axial Bending (P)	6	-6	6	-6
Axial Membrane (Mc)	-23	-23	23	23
Axial Bending (Mc)	-63	63	63	-63
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	9	9	9	9
Axial Stress Summary	-68	44	104	-34

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-4	-4	4	4
Shear due to Torsion	17	17	17	17
Shear Stress Summary	13	13	21	21

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	93	50	110	71

Stress Computations at the Edge of the Reinforcing Pad:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))}$
 $= 88.9 + 2 \cdot 1.65 \cdot \sqrt{(1111.75(22.0 - 1.5))}$
 $= 587.089 \text{ mm.}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	2223.500 mm.
Nozzle Outside Diameter	(d) =	140.000 mm.
Vessel Thickness used	(T) =	20.500 mm.
Nozzle Thickness used	(t) =	30.718 mm.
T / t	=	0.667
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 0.656$$

Note: Re-pad thickness is added to nozzle thickness to simulate the Re-pad.

Nr/P	=	0.136
Mr/P	=	0.271
M0/P	=	0.084
N0/P	=	0.155
MrD/Mc	=	0.304
NrDL/Mc	=	0.104
M0d/Mc	=	0.104
N0DL/Mc	=	-0.042
MrD/Ml	=	0.244
NrDL/Ml	=	0.103
M0D/Ml	=	0.084
N0DL/Ml	=	0.071

Vessel Stresses (N./mm²):



LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	0	0	0	0
Outplane Bending (P)	2	-2	2	-2
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-1	-1	1	1
Outplane Bending (ML)	-7	7	7	-7
Normal Pressure Stress	123	125	123	125
Outplane Stress Summary	117	131	135	117

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	0	0	0	0
Inplane Bending (P)	7	-7	7	-7
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-1	-1	1	1
Inplane Bending (ML)	-23	23	23	-23
Inplane Pressure Stress	61	61	61	61
Inplane Stress Summary	45	76	95	33

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	1	1	1	1
Shear Stress Summary	2	2	1	1

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	117	131	135	117

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	0	0	0	0
Outplane Bending (P)	2	-2	2	-2
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	-6	6	6	-6
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	61	61	61	61
Outplane Stress Summary	58	67	71	52

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Inplane Membrane (P)	0	0	0	0
Inplane Bending (P)	7	-7	7	-7
Inplane Membrane (Mc)	-1	-1	1	1

Inplane Bending (Mc)	-19	19	19	-19
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	123	125	123	125

Inplane Stress Summary	111	137	152	100
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE				
(Shear stress normal to circumferential plane)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Torsional Shear (Mt)	1	1	1	1
Shear Stress Summary	1	1	2	2

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE				
(Stress Intensities	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	111	137	152	100

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	58	61	58	61	58	61	58	61
Circ. Pl (SUS)	0	0	0	0	0	0	0	0
Circ. Q (SUS)	-8	8	12	-12	-2	2	2	-2
Long. Pm (SUS)	29	29	29	29	29	29	29	29
Long. Pl (SUS)	0	0	0	0	0	0	0	0
Long. Q (SUS)	-3	3	3	-3	-5	5	9	-9
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	2	2	2	2	2	2	2	2
Pm (SUS)	58.0	61.0	58.0	61.0	58.0	61.0	58.0	61.0
Pm+Pl (SUS)	58.0	61.0	58.0	61.0	58.0	61.0	58.0	61.0
Pm+Pl+Q (Total)	50.2	69.1	70.1	49.2	56.1	63.1	60.2	59.1

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	61.00	138.00	Passed
Pm+Pl (SUS)	61.00	207.00	Passed
Pm+Pl+Q (TOTAL)	70.10	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Reinforcing Pad Edge (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl

Circ. Pm (SUS)	123	125	123	125	123	125	123	125
Circ. Pl (SUS)	-1	-1	1	1	-1	-1	1	1
Circ. Q (SUS)	-5	5	9	-9	-12	12	26	-26
Long. Pm (SUS)	61	61	61	61	61	61	61	61
Long. Pl (SUS)	-1	-1	1	1	0	0	0	0
Long. Q (SUS)	-16	16	30	-30	-4	4	8	-8
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	1	1	1	1	1	1	1	1
Pm (SUS)	123.0	125.0	123.0	125.0	123.0	125.0	123.0	125.0
Pm+Pl (SUS)	122.0	124.0	124.0	126.0	122.0	124.0	124.0	126.0
Pm+Pl+Q (Total)	117.0	129.0	133.0	117.0	110.0	136.0	150.0	100.0

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	125.00	138.00	Passed
Pm+Pl (SUS)	126.00	207.00	Passed
Pm+Pl+Q (TOTAL)	150.01	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	19	22	19	22	19	22	19	22
Circ. Pl (SUS)	0	0	0	0	0	0	0	0
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	9	9	9	9	9	9	9	9
Long. Pl (SUS)	-33	-33	35	35	-22	-22	24	24
Long. Q (SUS)	-85	85	97	-97	-57	57	69	-69
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	4	4	-4	-4	-4	-4	4	4
Shear Q (SUS)	17	17	17	17	17	17	17	17
Pm (SUS)	19.0	22.0	19.0	22.0	19.0	22.0	19.0	22.0
Pm+Pl (SUS)	43.7	46.7	44.6	44.7	33.0	35.9	34.1	34.3
Pm+Pl+Q (Total)	134.7	70.2	142.4	79.4	92.7	50.0	107.0	71.6

Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	22.00	118.00	Passed
Pm+Pl (SUS)	46.69	177.00	Passed
Pm+Pl+Q (TOTAL)	142.37	353.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.



Input, Nozzle Desc: N53"

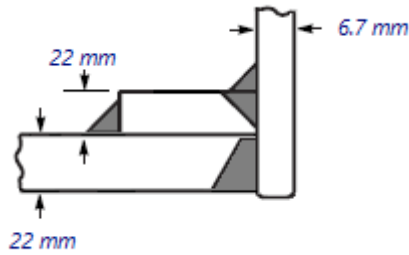
From: 50

Pressure for Reinforcement Calculations	P	23.201	bars
Temperature for Internal Pressure	Temp	80	°C
Design External Pressure	Pext	1.01	bars
Temperature for External Pressure	Tempex	80	°C
Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	138.00	N./mm2
Parent Allowable Stress At Ambient	Sva	138.00	N./mm2
Inside Diameter of Cylindrical Shell	D	2200.00	mm.
Design Length of Section	L	7068.2671	mm.
Shell Finished (Minimum) Thickness	t	22.0000	mm.
Shell Internal Corrosion Allowance	cas	1.5000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		500.80	mm.
User Entered Minimum Design Metal Temperature		-38.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-333 6	+IT
Material UNS Number		K03006	
Material Specification/Type	Smls. & wld. pipe		
Allowable Stress at Temperature	Sn	118.00	N./mm2
Allowable Stress At Ambient	Sna	118.00	N./mm2
Diameter Basis (for tr calc only)		Inside	
Layout Angle		90.00	deg
Diameter		3.0000	in.
Size and Thickness Basis		Minimum	
Nominal Thickness		80	
Flange Material		SA-350 LF2	+IT
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	1.5000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	218.2720	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	22.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Pad Material		SA-516 70	+IT
Pad Allowable Stress at Temperature	Sp	138.00	N./mm2
Pad Allowable Stress At Ambient	Spa	138.00	N./mm2
Diameter of Pad along vessel surface	Dp	140.0000	mm.
Thickness of Pad	te	22.0000	mm.
Weld leg size between Pad and Shell	Wp	18.0000	mm.
Groove weld depth between Pad and Nozzle	Wgpn	22.0000	mm.
Reinforcing Pad Width		25.5500	mm.
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: N53"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	2.975 in.
Actual Thickness Used in Calculation	0.263 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 23.2 \cdot 1101.5 / (138 \cdot 1.0 - 0.6 \cdot 23.2)$
 $= 18.7092 \text{ mm.}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 23.2 \cdot 39.28 / (118 \cdot 1.0 - 0.6 \cdot 23.2)$
 $= 0.7817 \text{ mm.}$

Required Nozzle thickness under External Pressure per UG-28 : 0.4107 mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	157.1300 mm.
Parallel to Vessel Wall, opening length	d	78.5650 mm.
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		34.9188 mm.

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 118.0 / 138.0)$
 $= 0.855$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 118.0 / 138.0)$
 $= 0.855$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, S_p / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(0.855, 1.0)$
 $= 0.855$

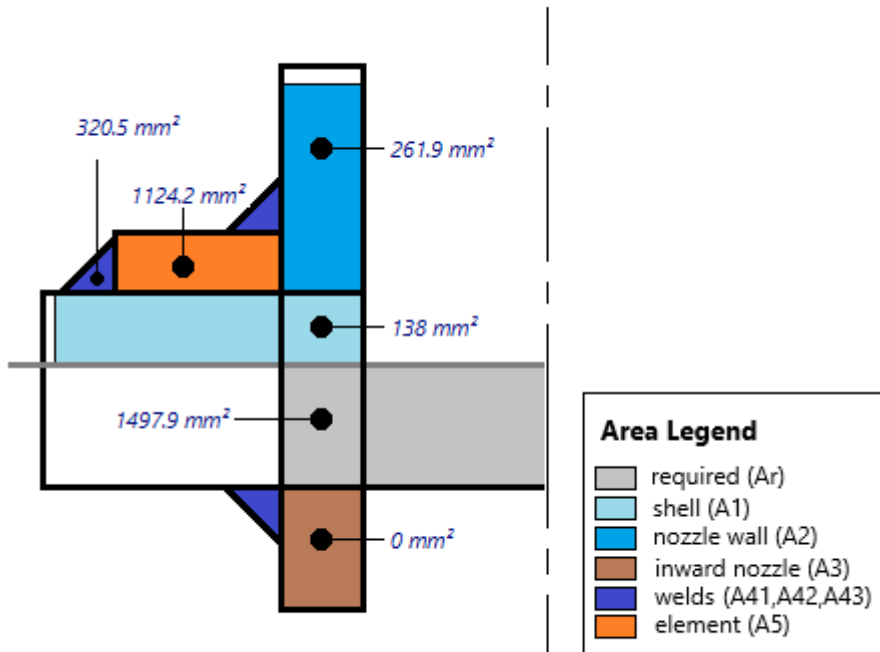


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm2)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	14.979	4.613	NA
Area in Shell	A1	1.380	6.917	NA
Area in Nozzle Wall	A2	2.619	2.841	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	3.205	3.205	NA
Area in Element	A5	11.242	11.242	NA
TOTAL AREA AVAILABLE	Atot	18.446	24.205	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Insufficient.
 The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS: Diameter Thickness
 Based on given Pad Thickness: 124.2413 22.0000 mm.
 Based on given Pad Diameter: 140.0000 15.2154 mm.
 Based on the Estimated Diameter Limit: 155.5425 11.6669 mm.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$
 $= (78.565 \cdot 18.7092 \cdot 1.0 + 2 \cdot 5.1675 \cdot 18.7092 \cdot 1.0 \cdot (1 - 0.86))$
 $= 14.979 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 78.565(1.0 \cdot 20.5 - 1.0 \cdot 18.709) - 2 \cdot 5.168$
 $(1.0 \cdot 20.5 - 1.0 \cdot 18.7092) \cdot (1 - 0.855)$
 $= 1.380 \text{ cm}^2$

Area Available in Nozzle Wall Projecting Outward [A2]:
 $= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$
 $= (2 \cdot 34.92) \cdot (5.17 - 0.78) \cdot 0.8551$



= 2.619 cm²

Area Available in Welds [A41 + A42 + A43]:
= (Wo² - Ar Lost)•Fr3+((Wi-can/0.707)² - Ar Lost)•fr2 + Trapfr4
= (1.) • 0.86 + (0.0) • 0.86 + 234.9806² • 1.0
= 3.205 cm²

Area Available in Element [A5]:
= (min(Dp,DL)-(Nozzle OD))•(min(tp,Tlwp,te)) • fr4
= (140.0 - 88.9) • 22.0 • 1.0
= 11.242 cm²

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 2.2817 mm.
Wall Thickness per UG16(b),	tr16b = 3.0000 mm.
Wall Thickness, shell/head, internal pressure	trb1 = 20.2092 mm.
Wall Thickness	tb1 = max(trb1, tr16b) = 20.2092 mm.
Wall Thickness, shell/head, external pressure	trb2 = 2.3091 mm.
Wall Thickness	tb2 = max(trb2, tr16b) = 3.0000 mm.
Wall Thickness per table UG-45	tb3 = 6.3000 mm.

Determine Nozzle Thickness candidate [tb]:
= min[tb3, max(tb1,tb2)]
= min[6.3, max(20.2092, 3.0)]
= 6.3000 mm.

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
= max(ta, tb)
= max(2.2817, 6.3)
= 6.3000 mm.

Available Nozzle Neck Thickness = 6.6675 mm. -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 48.8,	Allowable	: 118.0 N./mm ²	Passed
Expansion	: 0.0,	Allowable	: 246.2 N./mm ²	Passed
Occasional	: 8.3,	Allowable	: 156.9 N./mm ²	Passed
Shear	: 29.3,	Allowable	: 82.6 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld (Impact tested) :

Note:

This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification	-46 °C
Calculated Minimum Design Metal Temperature	-104 °C

Nozzle Neck to Pad Weld for the Nozzle (Impact tested) :

Note:

This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification	-46 °C
Calculated Minimum Design Metal Temperature	-104 °C

Nozzle Neck to Pad Weld for Reinf. pad (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Shell to Pad Weld at Pad OD for pad (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note:

This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification	-46 °C
Calculated Minimum Design Metal Temperature	-104 °C
Gov. MDMT of the Nozzle	: -104 °C
Gov. MDMT of the Reinforcement Pad	: -43 °C
Gov. MDMT of the nozzle to shell joint welded assembly	: -43 °C

Flange MDMT including Temperature reduction per UCS-66.1:

MDMT of ASME B16.5/47 flange per Matl. Specification	-46 °C
Flange MDMT with Temp reduction per UCS-66(i)(2)	-85 °C

Where the Stress Reduction Ratio per UCS-66(i)(2) is :

$$\text{Design Pressure/Ambient Rating} = 23.20/51.00 = 0.455$$

Weld Size Calculations, Description: N53"

Intermediate Calc. for nozzle/shell Welds	Tmin	5.1675 mm.
Intermediate Calc. for pad/shell Welds	TminPad	19.0000 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$3.6173 = 0.7 * t_{min}$	$7.0700 = 0.7 * W_o$ mm.
Pad Weld	$9.5000 = 0.5 * T_{minPad}$	$12.7260 = 0.7 * W_p$ mm.

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$= \max(0, (A-A1+2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t_{tr})) \cdot S_v)$$

$$= \max(0, (14.9791 - 1.3801 + 2 \cdot 5.1675 \cdot 0.8551 \cdot (1.0 \cdot 20.5 - 18.7092)) \cdot 138)$$

$$= 19358.59 \text{ Kg}$$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$= (A2+A5+A4-(W_i \cdot C_{an}/707)^2 \cdot f_{r2}) \cdot S_v$$

$$= (2.6191 + 11.242 + 3.2049 - 0.0 \cdot 0.86) \cdot 138$$

$$= 24014.38 \text{ Kg}$$

Weld Load [W2]:

$$= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v$$

$$= (2.6191 + 0.0 + 0.8551 + (1.8116)) \cdot 138$$

$$= 7437.88 \text{ Kg}$$

Weld Load [W3]:

$$= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S$$

$$= (2.6191 + 0.0 + 3.2049 + 11.242 + (1.8116)) \cdot 138$$

$$= 26563.61 \text{ Kg}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$$

$$= (3.1416/2.0) \cdot 88.9 \cdot 10.0 \cdot 0.49 \cdot 118$$

$$= 8233. \text{ Kg}$$

Shear, Pad Element Weld [Spew]:

$$= (\pi/2) \cdot D_P \cdot W_P \cdot 0.49 \cdot S_{EW}$$

$$= (3.1416/2.0) \cdot 140.0 \cdot 18.0 \cdot 0.49 \cdot 138$$

= 27293. Kgf

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (D_{lr} + D_{lo}) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot S_n$$

$$= (3.1416 \cdot 41.8662) \cdot (6.6675 - 1.5) \cdot 0.7 \cdot 118$$

$$= 5725. Kgf$$

Tension, Pad Groove Weld [Tpgw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_{gpn} \cdot 0.74 \cdot S_{eg}$$

$$= (3.1416/2) \cdot 88.9 \cdot 22.0 \cdot 0.74 \cdot 138$$

$$= 31990. Kgf$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot D_{lo} \cdot (T - cas) \cdot 0.74 \cdot S_{ng}$$

$$= (3.1416/2.0) \cdot 88.9 \cdot (22.0 - 1.5) \cdot 0.74 \cdot 138$$

$$= 29809. Kgf$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SPEW + SNW) = (27293 + 5725) = 33018 Kgf$$

Failure Path [Path 2-2]:

$$= (Sonw + Tpgw + Tngw + Sinw)$$

$$= (8233 + 31990 + 29809 + 0) = 70032 Kgf$$

Failure Path [Path 3-3]:

$$= (Spew + Tngw + Sinw)$$

$$= (27293 + 29809 + 0) = 57102 Kgf$$

Summary of Failure Path Calculations:

Path 1-1 = 33017 Kgf, must exceed W = 19358 Kgf	or W1 = 24014 Kgf
Path 2-2 = 70032 Kgf, must exceed W = 19358 Kgf	or W2 = 7437 Kgf
Path 3-3 = 57102 Kgf, must exceed W = 19358 Kgf	or W3 = 26563 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 25 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 1 bars

The Drop for this Nozzle is : 0.8985 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 241.1705 mm.

Input Echo, WRC297 Item 1, Description: N53" :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	1.5000	mm.
Shell External Corrosion Allowance	0.0000	mm.
Shell Diameter	2200.000	mm.
Shell Thickness	22.0000	mm.
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc	138.00 N./mm ²
Vessel Hot S.I. Allowable	Smh	138.00 N./mm ²

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	1.5000	mm.
Nozzle Diameter	75.565	mm.
Nozzle Thickness	6.6675	mm.
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-333 6	
Nozzle Cold S.I. Allowable	SNmc	118.00 N./mm ²
Nozzle Hot S.I. Allowable	SNmh	118.00 N./mm ²



See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Diameter of Reinforcing Pad 140.000 mm.
 Thickness of Reinforcing Pad 22.0000 mm.

Note:

*External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUSTAINED loads.*

Design Internal Pressure	Dp	23.20	bars
Radial Load	P	-205.00	Kgf
Circumferential Shear	Vc	310.00	Kgf
Longitudinal Shear	Vl	310.00	Kgf
Circumferential Moment	Mc	65.00	Kg-m.
Longitudinal Moment	Ml	95.00	Kg-m.
Torsional Moment	Mt	115.00	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

*Warning - The ratio, Dn/Tv, (4.041) is < 5
 check the Limitations of WRC bulletin 297.*

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))}$$

$$= 88.9 + 2 \cdot 1.65 \cdot \sqrt{(1122.75(22.0 - 1.5))}$$

$$= 589.548 \text{ mm.}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	2245.500 mm.
Nozzle Outside Diameter	(d) =	88.900 mm.
Vessel Thickness used	(T) =	42.500 mm.
Nozzle Thickness used	(t) =	5.168 mm.
T / t	=	8.224
d / t	=	17.204

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 0.288$$

Nr/P	=	0.127
Mr/P	=	0.079
M0/P	=	0.304
N0/P	=	0.349
MrD/Mc	=	0.093
NrDL/Mc	=	0.108
M0d/Mc	=	0.319
N0DL/Mc	=	-0.142
MrD/Ml	=	0.094
NrDL/Ml	=	0.109
M0D/Ml	=	0.312
N0DL/Ml	=	-0.095

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	0	0	0	0
Outplane Bending (P)	2	-2	2	-2
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (Ml)	0	0	0	0
Outplane Bending (Ml)	-10	10	10	-10
Normal Pressure Stress	58	61	58	61



Outplane Stress Summary 51 71 71 48

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE
 (Stresses parallel to longitudinal plane)

	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
--	----------------------	---------------------	-------------------------	------------------------

Inplane Membrane (P)	0	0	0	0
Inplane Bending (P)	0	0	0	0
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	-3	3	3	-3
Inplane Pressure Stress	29	29	29	29

Inplane Stress Summary 26 31 34 26

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE
 (Shear stress normal to longitudinal plane)

	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
--	----------------------	---------------------	-------------------------	------------------------

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	2	2	2	2

Shear Stress Summary 2 2 1 1

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE
 (Stress Intensities)

	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
--	----------------------	---------------------	-------------------------	------------------------

Two * Max Shear Stress 51 71 71 48

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE
 (Stresses Normal to circumferential plane)

	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
--	-----------------------	----------------------	------------------------	-----------------------

Outplane Membrane (P)	0	0	0	0
Outplane Bending (P)	2	-2	2	-2
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	-7	7	7	-7
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	29	29	29	29

Outplane Stress Summary 24 36 38 19

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE
 (Stresses parallel to circumferential plane)

	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
--	-----------------------	----------------------	------------------------	-----------------------

Inplane Membrane (P)	0	0	0	0
Inplane Bending (P)	0	0	0	0
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	-2	2	2	-2
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	58	61	58	61

Inplane Stress Summary 57 62 62 59

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE
 (Shear stress normal to circumferential plane)

	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
--	-----------------------	----------------------	------------------------	-----------------------



Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Torsional Shear (Mt)	2	2	2	2
Shear Stress Summary	1	1	2	2

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Two * Max Shear Stress	57	62	62	59

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Hoop Membrane (P)	0	0	0	0
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	19	22	19	22
Hoop Stress Summary	20	23	19	22

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Axial Membrane (P)	1	1	1	1
Axial Bending (P)	6	-6	6	-6
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-34	-34	34	34
Axial Bending (ML)	-91	91	91	-91
Axial Pressure Stress	9	9	9	9
Axial Stress Summary	-108	61	144	-52

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)	4	4	-4	-4
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	17	17	17	17
Shear Stress Summary	21	21	13	13

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	136	71	145	79

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	0	0	0	0
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0



Hoop Pressure Stress	19	22	19	22
----------------------	----	----	----	----

Hoop Stress Summary	20	23	19	22
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
---	-----------------------	----------------------	------------------------	-----------------------

Axial Membrane (P)	1	1	1	1
Axial Bending (P)	6	-6	6	-6
Axial Membrane (Mc)	-23	-23	23	23
Axial Bending (Mc)	-63	63	63	-63
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	9	9	9	9

Axial Stress Summary	-68	44	104	-34
----------------------	-----	----	-----	-----

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
---	---------------	--------------	---------------	--------------

Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-4	-4	4	4
Shear due to Torsion	17	17	17	17

Shear Stress Summary	13	13	21	21
----------------------	----	----	----	----

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities	Cu Outside	Cl Inside	Du Outside	Dl Inside
--	---------------	--------------	---------------	--------------

Two * Max Shear Stress	93	50	110	71
------------------------	----	----	-----	----

Stress Computations at the Edge of the Reinforcing Pad:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))}$
 $= 88.9 + 2 \cdot 1.65 \cdot \sqrt{(1111.75(22.0 - 1.5))}$
 $= 587.089 \text{ mm.}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	2223.500 mm.
Nozzle Outside Diameter	(d) =	140.000 mm.
Vessel Thickness used	(T) =	20.500 mm.
Nozzle Thickness used	(t) =	30.718 mm.
T / t	=	0.667
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 0.656$$

Note: Re-pad thickness is added to nozzle thickness to simulate the Re-pad.

Nr/P	=	0.136
Mr/P	=	0.271
M0/P	=	0.084
N0/P	=	0.155
MrD/Mc	=	0.304
NrDL/Mc	=	0.104
M0d/Mc	=	0.104
N0DL/Mc	=	-0.042
MrD/Ml	=	0.244
NrDL/Ml	=	0.103
M0D/Ml	=	0.084
N0DL/Ml	=	0.071

Vessel Stresses (N./mm²):



LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	0	0	0	0
Outplane Bending (P)	2	-2	2	-2
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-1	-1	1	1
Outplane Bending (ML)	-7	7	7	-7
Normal Pressure Stress	123	125	123	125
Outplane Stress Summary	117	131	135	117

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	0	0	0	0
Inplane Bending (P)	7	-7	7	-7
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-1	-1	1	1
Inplane Bending (ML)	-23	23	23	-23
Inplane Pressure Stress	61	61	61	61
Inplane Stress Summary	45	76	95	33

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	1	1	1	1
Shear Stress Summary	2	2	1	1

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	117	131	135	117

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	0	0	0	0
Outplane Bending (P)	2	-2	2	-2
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	-6	6	6	-6
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	61	61	61	61
Outplane Stress Summary	58	67	71	52

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Inplane Membrane (P)	0	0	0	0
Inplane Bending (P)	7	-7	7	-7
Inplane Membrane (Mc)	-1	-1	1	1



Inplane Bending (Mc)	-19	19	19	-19
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	123	125	123	125

Inplane Stress Summary	111	137	152	100
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE				
(Shear stress normal to circumferential plane)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Torsional Shear (Mt)	1	1	1	1
Shear Stress Summary	1	1	2	2

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE				
(Stress Intensities	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	111	137	152	100

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	58	61	58	61	58	61	58	61
Circ. Pl (SUS)	0	0	0	0	0	0	0	0
Circ. Q (SUS)	-8	8	12	-12	-2	2	2	-2
Long. Pm (SUS)	29	29	29	29	29	29	29	29
Long. Pl (SUS)	0	0	0	0	0	0	0	0
Long. Q (SUS)	-3	3	3	-3	-5	5	9	-9
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	2	2	2	2	2	2	2	2
Pm (SUS)	58.0	61.0	58.0	61.0	58.0	61.0	58.0	61.0
Pm+Pl (SUS)	58.0	61.0	58.0	61.0	58.0	61.0	58.0	61.0
Pm+Pl+Q (Total)	50.2	69.1	70.1	49.2	56.1	63.1	60.2	59.1

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	61.00	138.00	Passed
Pm+Pl (SUS)	61.00	207.00	Passed
Pm+Pl+Q (TOTAL)	70.10	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Reinforcing Pad Edge (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl

Circ. Pm (SUS)	123	125	123	125	123	125	123	125
Circ. Pl (SUS)	-1	-1	1	1	-1	-1	1	1
Circ. Q (SUS)	-5	5	9	-9	-12	12	26	-26
Long. Pm (SUS)	61	61	61	61	61	61	61	61
Long. Pl (SUS)	-1	-1	1	1	0	0	0	0
Long. Q (SUS)	-16	16	30	-30	-4	4	8	-8
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	1	1	1	1	1	1	1	1
Pm (SUS)	123.0	125.0	123.0	125.0	123.0	125.0	123.0	125.0
Pm+Pl (SUS)	122.0	124.0	124.0	126.0	122.0	124.0	124.0	126.0
Pm+Pl+Q (Total)	117.0	129.0	133.0	117.0	110.0	136.0	150.0	100.0

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	125.00	138.00	Passed
Pm+Pl (SUS)	126.00	207.00	Passed
Pm+Pl+Q (TOTAL)	150.01	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	19	22	19	22	19	22	19	22
Circ. Pl (SUS)	0	0	0	0	0	0	0	0
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	9	9	9	9	9	9	9	9
Long. Pl (SUS)	-33	-33	35	35	-22	-22	24	24
Long. Q (SUS)	-85	85	97	-97	-57	57	69	-69
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	4	4	-4	-4	-4	-4	4	4
Shear Q (SUS)	17	17	17	17	17	17	17	17
Pm (SUS)	19.0	22.0	19.0	22.0	19.0	22.0	19.0	22.0
Pm+Pl (SUS)	43.7	46.7	44.6	44.7	33.0	35.9	34.1	34.3
Pm+Pl+Q (Total)	134.7	70.2	142.4	79.4	92.7	50.0	107.0	71.6

Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	22.00	118.00	Passed
Pm+Pl (SUS)	46.69	177.00	Passed
Pm+Pl+Q (TOTAL)	142.37	353.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.



Input, Nozzle Desc: K1B3"

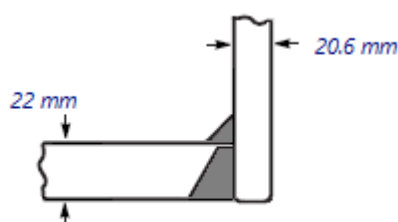
From: 50

Pressure for Reinforcement Calculations	P	23.224	bars
Temperature for Internal Pressure	Temp	80	°C
Design External Pressure	Pext	1.01	bars
Temperature for External Pressure	Tempex	80	°C
Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	138.00	N./mm2
Parent Allowable Stress At Ambient	Sva	138.00	N./mm2
Inside Diameter of Cylindrical Shell	D	2200.00	mm.
Design Length of Section	L	7068.2671	mm.
Shell Finished (Minimum) Thickness	t	22.0000	mm.
Shell Internal Corrosion Allowance	cas	1.5000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		150.80	mm.
User Entered Minimum Design Metal Temperature		-38.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-350 LF2	+IT
Material UNS Number		K03011	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm2
Allowable Stress At Ambient	Sna	138.00	N./mm2
Diameter Basis (for tr calc only)		Inside	
Layout Angle		90.00	deg
Diameter		3.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	20.5740	mm.
Flange Material		SA-350 LF2	+IT
Flange Type		Long Weld Neck	
Corrosion Allowance	can	1.5000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	200.0250	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	22.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: K1B3"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45



Actual Inside Diameter Used in Calculation 3.000 in.
 Actual Thickness Used in Calculation 0.810 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 23.22 \cdot 1101.5 / (138 \cdot 1.0 - 0.6 \cdot 23.22)$
 $= 18.7280$ mm.

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 23.22 \cdot 39.6 / (138 \cdot 1.0 - 0.6 \cdot 23.22)$
 $= 0.6733$ mm.

Required Nozzle thickness under External Pressure per UG-28 : 0.4669 mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	158.4000	mm.
Parallel to Vessel Wall, opening length	d	79.2000	mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	47.6850	mm.

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1.0, 1.0)$
 $= 1.000$

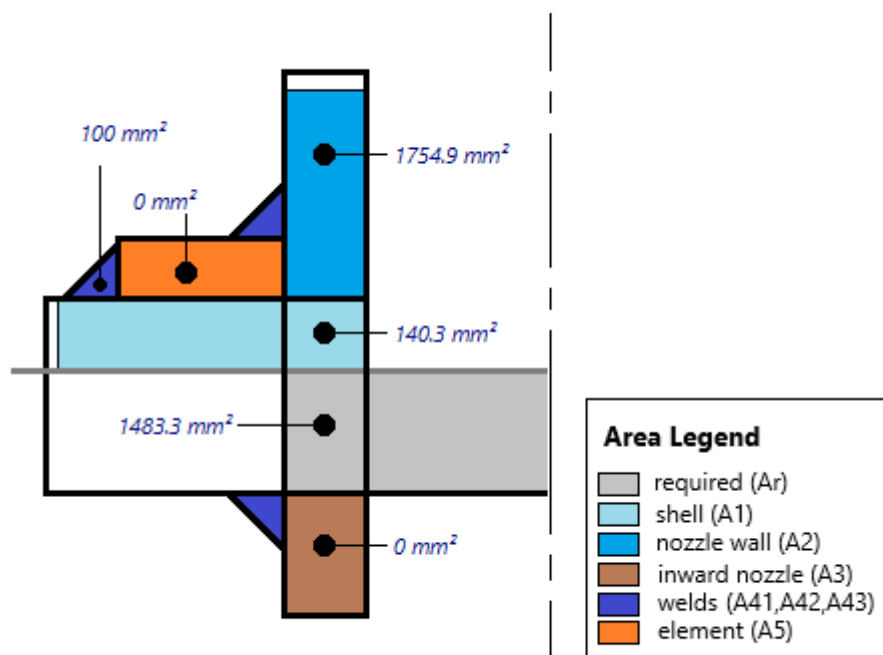


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm2)

AREA AVAILABLE, A1 to A5	Design	External	Mapnc
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Area Required	Ar	14.833	4.564	NA
Area in Shell	A1	1.403	7.109	NA
Area in Nozzle Wall	A2	17.549	17.746	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	1.000	1.000	NA
Area in Element	A5	0.000	0.000	NA
TOTAL AREA AVAILABLE	Atot	19.952	25.854	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$
 $= (79.2 \cdot 18.728 \cdot 1.0 + 2 \cdot 19.074 \cdot 18.728 \cdot 1.0 \cdot (1 - 1.0))$
 $= 14.833 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 79.2(1.0 \cdot 20.5 - 1.0 \cdot 18.728) - 2 \cdot 19.074$
 $(1.0 \cdot 20.5 - 1.0 \cdot 18.728) \cdot (1 - 1.0)$
 $= 1.403 \text{ cm}^2$

Area Available in Nozzle Projecting Outward [A2]:
 $= (2 \cdot tlnp)(tn - trn)fr2$
 $= (2 \cdot 47.68)(19.07 - 0.67)1.0$
 $= 17.549 \text{ cm}^2$

Area Available in Inward Weld + Outward Weld [A41 + A43]:
 $= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$
 $= 10.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0$
 $= 1.000 \text{ cm}^2$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures $ta = 2.1733 \text{ mm.}$
 Wall Thickness per UG16(b), $tr16b = 3.0000 \text{ mm.}$
 Wall Thickness, shell/head, internal pressure $trb1 = 20.2280 \text{ mm.}$
 Wall Thickness $tb1 = \max(trb1, tr16b) = 20.2280 \text{ mm.}$
 Wall Thickness, shell/head, external pressure $trb2 = 2.3091 \text{ mm.}$
 Wall Thickness $tb2 = \max(trb2, tr16b) = 3.0000 \text{ mm.}$
 Wall Thickness per table UG-45 $tb3 = 7.2300 \text{ mm.}$

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[7.23, \max(20.228, 3.0)]$
 $= 7.2300 \text{ mm.}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(2.1733, 7.23)$
 $= 7.2300 \text{ mm.}$

Available Nozzle Neck Thickness = 20.5740 mm. -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note:
This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification -46 °C

Calculated Minimum Design Metal Temperature -104 °C



Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Weld Size Calculations, Description: K1B3"

Intermediate Calc. for nozzle/shell Welds Tmin 19.0000 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm.

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
= max(0, (A-A1+2•tn•fr1•(E1•t-tr))•Sv)
= max(0, (14.8326 - 1.4034 + 2 • 19.074 • 1.0 •
(1.0 • 20.5 - 18.728))138)
= 19848.14 Kgf

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
= (A2+A5+A4-(Wi-Can/.707)²•fr2)•Sv
= (17.5488 + 0.0 + 1. - 0.0 • 1.0) • 138
= 26100.91 Kgf

Weld Load [W2]:
= (A2 + A3 + A4 + (2 • tn • t • fr1)) • Sv
= (17.5488 + 0.0 + 1. + (7.8203)) • 138
= 37105.32 Kgf

Weld Load [W3]:
= (A2+A3+A4+A5+(2•tn•t•fr1))•S
= (17.5488 + 0.0 + 1. + 0.0 + (7.8203)) • 138
= 37105.32 Kgf

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
= (π/2) • Dlo • Wo • 0.49 • Snw
= (3.1416/2.0) • 117.348 • 10.0 • 0.49 • 138
= 12710. Kgf

Shear, Nozzle Wall [Snw]:
= (π • (Dlr + Dlo)/4) • (Thk - Can) • 0.7 • Sn
= (3.1416 • 49.137) • (20.574 - 1.5) • 0.7 • 138
= 29003. Kgf

Tension, Shell Groove Weld [Tngw]:
= (π/2) • Dlo • (T - cas) • 0.74 • Sng
= (3.1416/2.0) • 117.348 • (22.0 - 1.5) • 0.74 • 138
= 39348. Kgf

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
= (SONW + SNW) = (12710 + 29003) = 41712 Kgf

Failure Path [Path 2-2]:
= (Sonw + Tpgw + Tngw + Sinw)
= (12710 + 0 + 39348 + 0) = 52058 Kgf

Failure Path [Path 3-3]:
= (Sonw + Tngw + Sinw)
= (12710 + 39348 + 0) = 52058 Kgf

Summary of Failure Path Calculations:

Path 1-1 = 41712 Kgf, must exceed W = 19848 Kgf or W1 = 26100 Kgf



Path 2-2 = 52057 Kgf, must exceed W = 19848 Kgf or W2 = 37105 Kgf
Path 3-3 = 52057 Kgf, must exceed W = 19848 Kgf or W3 = 37105 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 25 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 1 bars

The Drop for this Nozzle is : 1.5659 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 223.5909 mm.

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Input, Nozzle Desc: K4B3"

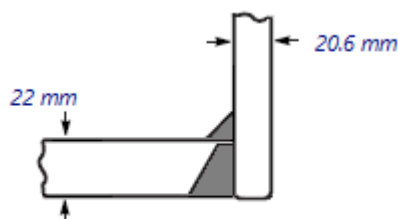
From: 50

Pressure for Reinforcement Calculations	P	23.224	bars
Temperature for Internal Pressure	Temp	80	°C
Design External Pressure	Pext	1.01	bars
Temperature for External Pressure	Tempex	80	°C
Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	138.00	N./mm2
Parent Allowable Stress At Ambient	Sva	138.00	N./mm2
Inside Diameter of Cylindrical Shell	D	2200.00	mm.
Design Length of Section	L	7068.2671	mm.
Shell Finished (Minimum) Thickness	t	22.0000	mm.
Shell Internal Corrosion Allowance	cas	1.5000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		150.80	mm.
User Entered Minimum Design Metal Temperature		-38.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-350 LF2	+IT
Material UNS Number		K03011	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm2
Allowable Stress At Ambient	Sna	138.00	N./mm2
Diameter Basis (for tr calc only)		Inside	
Layout Angle		180.00	deg
Diameter		3.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	20.5740	mm.
Flange Material		SA-350 LF2	+IT
Flange Type		Long Weld Neck	
Corrosion Allowance	can	1.5000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	200.0250	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel Wgnv		22.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: K4B3"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45



Actual Inside Diameter Used in Calculation 3.000 in.
 Actual Thickness Used in Calculation 0.810 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 23.22 \cdot 1101.5 / (138 \cdot 1.0 - 0.6 \cdot 23.22)$
 $= 18.7280$ mm.

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 23.22 \cdot 39.6 / (138 \cdot 1.0 - 0.6 \cdot 23.22)$
 $= 0.6733$ mm.

Required Nozzle thickness under External Pressure per UG-28 : 0.4669 mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	158.4000	mm.
Parallel to Vessel Wall, opening length	d	79.2000	mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	47.6850	mm.

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1.0, 1.0)$
 $= 1.000$

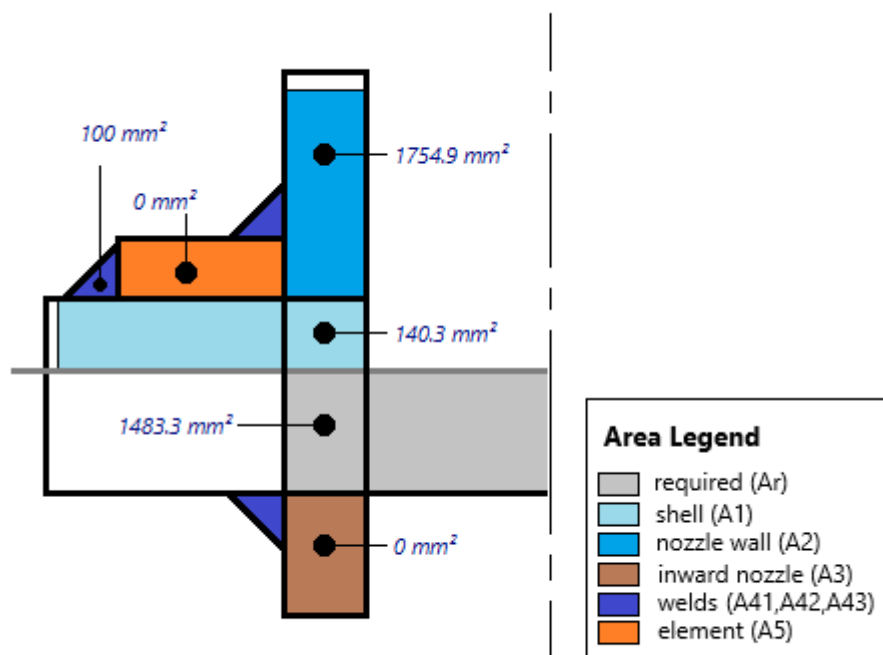


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm2)

AREA AVAILABLE, A1 to A5	Design	External	Mapnc
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Area Required	Ar	14.833	4.564	NA
Area in Shell	A1	1.403	7.109	NA
Area in Nozzle Wall	A2	17.549	17.746	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	1.000	1.000	NA
Area in Element	A5	0.000	0.000	NA
TOTAL AREA AVAILABLE	Atot	19.952	25.854	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$
 $= (79.2 \cdot 18.728 \cdot 1.0 + 2 \cdot 19.074 \cdot 18.728 \cdot 1.0 \cdot (1 - 1.0))$
 $= 14.833 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 79.2(1.0 \cdot 20.5 - 1.0 \cdot 18.728) - 2 \cdot 19.074$
 $(1.0 \cdot 20.5 - 1.0 \cdot 18.728) \cdot (1 - 1.0)$
 $= 1.403 \text{ cm}^2$

Area Available in Nozzle Projecting Outward [A2]:
 $= (2 \cdot tlnp)(tn - trn)fr2$
 $= (2 \cdot 47.68)(19.07 - 0.67)1.0$
 $= 17.549 \text{ cm}^2$

Area Available in Inward Weld + Outward Weld [A41 + A43]:
 $= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$
 $= 10.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0$
 $= 1.000 \text{ cm}^2$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures $ta = 2.1733 \text{ mm.}$
 Wall Thickness per UG16(b), $tr16b = 3.0000 \text{ mm.}$
 Wall Thickness, shell/head, internal pressure $trb1 = 20.2280 \text{ mm.}$
 Wall Thickness $tb1 = \max(trb1, tr16b) = 20.2280 \text{ mm.}$
 Wall Thickness, shell/head, external pressure $trb2 = 2.3091 \text{ mm.}$
 Wall Thickness $tb2 = \max(trb2, tr16b) = 3.0000 \text{ mm.}$
 Wall Thickness per table UG-45 $tb3 = 7.2300 \text{ mm.}$

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[7.23, \max(20.228, 3.0)]$
 $= 7.2300 \text{ mm.}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(2.1733, 7.23)$
 $= 7.2300 \text{ mm.}$

Available Nozzle Neck Thickness = 20.5740 mm. -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note:
This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification -46 °C

Calculated Minimum Design Metal Temperature -104 °C



Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Weld Size Calculations, Description: K4B3"

Intermediate Calc. for nozzle/shell Welds Tmin 19.0000 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm.

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
= max(0, (A-A1+2•tn•fr1•(E1•t-tr))•Sv)
= max(0, (14.8326 - 1.4034 + 2 • 19.074 • 1.0 •
(1.0 • 20.5 - 18.728))138)
= 19848.14 Kgf

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
= (A2+A5+A4-(Wi-Can/.707)²•fr2)•Sv
= (17.5488 + 0.0 + 1. - 0.0 • 1.0) • 138
= 26100.91 Kgf

Weld Load [W2]:
= (A2 + A3 + A4 + (2 • tn • t • fr1)) • Sv
= (17.5488 + 0.0 + 1. + (7.8203)) • 138
= 37105.32 Kgf

Weld Load [W3]:
= (A2+A3+A4+A5+(2•tn•t•fr1))•S
= (17.5488 + 0.0 + 1. + 0.0 + (7.8203)) • 138
= 37105.32 Kgf

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
= (π/2) • Dlo • Wo • 0.49 • Snw
= (3.1416/2.0) • 117.348 • 10.0 • 0.49 • 138
= 12710. Kgf

Shear, Nozzle Wall [Snw]:
= (π • (Dlr + Dlo)/4) • (Thk - Can) • 0.7 • Sn
= (3.1416 • 49.137) • (20.574 - 1.5) • 0.7 • 138
= 29003. Kgf

Tension, Shell Groove Weld [Tngw]:
= (π/2) • Dlo • (T - cas) • 0.74 • Sng
= (3.1416/2.0) • 117.348 • (22.0 - 1.5) • 0.74 • 138
= 39348. Kgf

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
= (SONW + SNW) = (12710 + 29003) = 41712 Kgf

Failure Path [Path 2-2]:
= (Sonw + Tpgw + Tngw + Sinw)
= (12710 + 0 + 39348 + 0) = 52058 Kgf

Failure Path [Path 3-3]:
= (Sonw + Tngw + Sinw)
= (12710 + 39348 + 0) = 52058 Kgf

Summary of Failure Path Calculations:

Path 1-1 = 41712 Kgf, must exceed W = 19848 Kgf or W1 = 26100 Kgf



Path 2-2 = 52057 Kgf, must exceed W = 19848 Kgf or W2 = 37105 Kgf
Path 3-3 = 52057 Kgf, must exceed W = 19848 Kgf or W3 = 37105 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 25 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 1 bars

The Drop for this Nozzle is : 1.5659 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 223.5909 mm.

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Input, Nozzle Desc: K3F3"

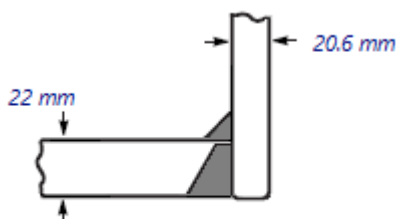
From: 50

Pressure for Reinforcement Calculations	P	23.224	bars
Temperature for Internal Pressure	Temp	80	°C
Design External Pressure	Pext	1.01	bars
Temperature for External Pressure	Tempex	80	°C
Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	138.00	N./mm2
Parent Allowable Stress At Ambient	Sva	138.00	N./mm2
Inside Diameter of Cylindrical Shell	D	2200.00	mm.
Design Length of Section	L	7068.2671	mm.
Shell Finished (Minimum) Thickness	t	22.0000	mm.
Shell Internal Corrosion Allowance	cas	1.5000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		150.80	mm.
User Entered Minimum Design Metal Temperature		-38.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-350 LF2	+IT
Material UNS Number		K03011	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm2
Allowable Stress At Ambient	Sna	138.00	N./mm2
Diameter Basis (for tr calc only)		Inside	
Layout Angle		270.00	deg
Diameter		3.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	20.5740	mm.
Flange Material		SA-350 LF2	+IT
Flange Type		Long Weld Neck	
Corrosion Allowance	can	1.5000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	200.0250	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel Wgnv		22.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: K3F3"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45



Actual Inside Diameter Used in Calculation 3.000 in.
 Actual Thickness Used in Calculation 0.810 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 23.22 \cdot 1101.5 / (138 \cdot 1.0 - 0.6 \cdot 23.22)$
 $= 18.7280$ mm.

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 23.22 \cdot 39.6 / (138 \cdot 1.0 - 0.6 \cdot 23.22)$
 $= 0.6733$ mm.

Required Nozzle thickness under External Pressure per UG-28 : 0.4669 mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	158.4000	mm.
Parallel to Vessel Wall, opening length	d	79.2000	mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	47.6850	mm.

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1.0, 1.0)$
 $= 1.000$

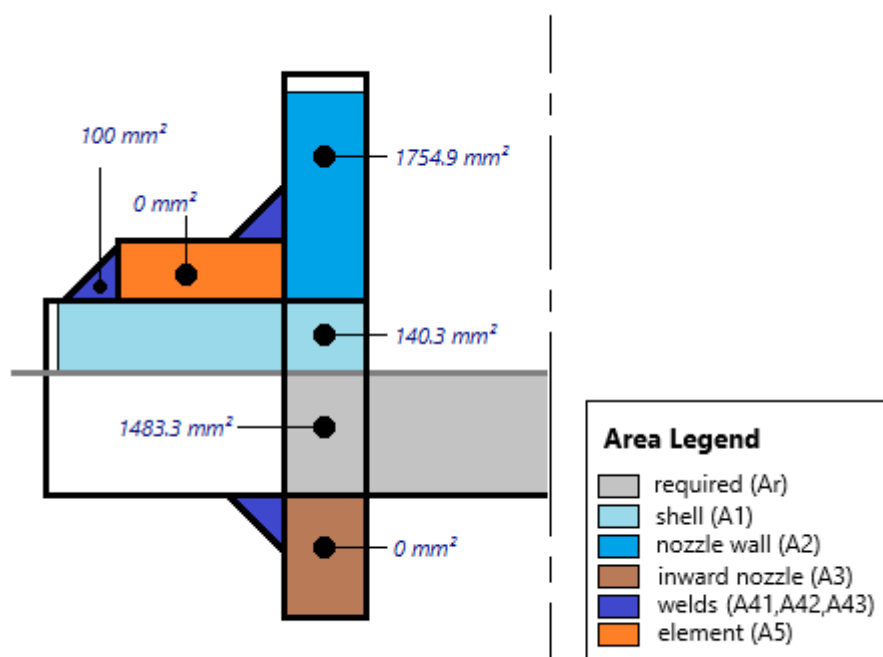


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm2)

AREA AVAILABLE, A1 to A5	Design	External	Mapnc
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Area Required	Ar	14.833	4.564	NA
Area in Shell	A1	1.403	7.109	NA
Area in Nozzle Wall	A2	17.549	17.746	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	1.000	1.000	NA
Area in Element	A5	0.000	0.000	NA
TOTAL AREA AVAILABLE	Atot	19.952	25.854	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$
 $= (79.2 \cdot 18.728 \cdot 1.0 + 2 \cdot 19.074 \cdot 18.728 \cdot 1.0 \cdot (1 - 1.0))$
 $= 14.833 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 79.2(1.0 \cdot 20.5 - 1.0 \cdot 18.728) - 2 \cdot 19.074$
 $(1.0 \cdot 20.5 - 1.0 \cdot 18.728) \cdot (1 - 1.0)$
 $= 1.403 \text{ cm}^2$

Area Available in Nozzle Projecting Outward [A2]:
 $= (2 \cdot tlnp)(tn - trn)fr2$
 $= (2 \cdot 47.68)(19.07 - 0.67)1.0$
 $= 17.549 \text{ cm}^2$

Area Available in Inward Weld + Outward Weld [A41 + A43]:
 $= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$
 $= 10.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0$
 $= 1.000 \text{ cm}^2$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures $ta = 2.1733 \text{ mm.}$
 Wall Thickness per UG16(b), $tr16b = 3.0000 \text{ mm.}$
 Wall Thickness, shell/head, internal pressure $trb1 = 20.2280 \text{ mm.}$
 Wall Thickness $tb1 = \max(trb1, tr16b) = 20.2280 \text{ mm.}$
 Wall Thickness, shell/head, external pressure $trb2 = 2.3091 \text{ mm.}$
 Wall Thickness $tb2 = \max(trb2, tr16b) = 3.0000 \text{ mm.}$
 Wall Thickness per table UG-45 $tb3 = 7.2300 \text{ mm.}$

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[7.23, \max(20.228, 3.0)]$
 $= 7.2300 \text{ mm.}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(2.1733, 7.23)$
 $= 7.2300 \text{ mm.}$

Available Nozzle Neck Thickness = 20.5740 mm. -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note:
This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification -46 °C

Calculated Minimum Design Metal Temperature -104 °C



Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Weld Size Calculations, Description: K3F3"

Intermediate Calc. for nozzle/shell Welds Tmin 19.0000 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm.

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
= max(0, (A-A1+2•tn•fr1•(E1•t-tr))•Sv)
= max(0, (14.8326 - 1.4034 + 2 • 19.074 • 1.0 •
(1.0 • 20.5 - 18.728))138)
= 19848.14 Kgf

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
= (A2+A5+A4-(Wi-Can/.707)²•fr2)•Sv
= (17.5488 + 0.0 + 1. - 0.0 • 1.0) • 138
= 26100.91 Kgf

Weld Load [W2]:
= (A2 + A3 + A4 + (2 • tn • t • fr1)) • Sv
= (17.5488 + 0.0 + 1. + (7.8203)) • 138
= 37105.32 Kgf

Weld Load [W3]:
= (A2+A3+A4+A5+(2•tn•t•fr1))•S
= (17.5488 + 0.0 + 1. + 0.0 + (7.8203)) • 138
= 37105.32 Kgf

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
= (π/2) • Dlo • Wo • 0.49 • Snw
= (3.1416/2.0) • 117.348 • 10.0 • 0.49 • 138
= 12710. Kgf

Shear, Nozzle Wall [Snw]:
= (π • (Dlr + Dlo)/4) • (Thk - Can) • 0.7 • Sn
= (3.1416 • 49.137) • (20.574 - 1.5) • 0.7 • 138
= 29003. Kgf

Tension, Shell Groove Weld [Tngw]:
= (π/2) • Dlo • (T - cas) • 0.74 • Sng
= (3.1416/2.0) • 117.348 • (22.0 - 1.5) • 0.74 • 138
= 39348. Kgf

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
= (SONW + SNW) = (12710 + 29003) = 41712 Kgf

Failure Path [Path 2-2]:
= (Sonw + Tpgw + Tngw + Sinw)
= (12710 + 0 + 39348 + 0) = 52058 Kgf

Failure Path [Path 3-3]:
= (Sonw + Tngw + Sinw)
= (12710 + 39348 + 0) = 52058 Kgf

Summary of Failure Path Calculations:

Path 1-1 = 41712 Kgf, must exceed W = 19848 Kgf or W1 = 26100 Kgf



Path 2-2 = 52057 Kgf, must exceed W = 19848 Kgf or W2 = 37105 Kgf
Path 3-3 = 52057 Kgf, must exceed W = 19848 Kgf or W3 = 37105 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 25 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 1 bars

The Drop for this Nozzle is : 1.5659 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 223.5909 mm.

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Input, Nozzle Desc: K2B1.5"

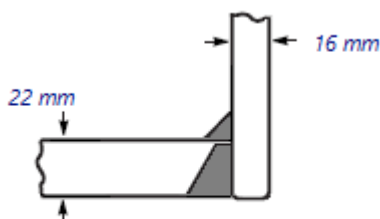
From: 50

Pressure for Reinforcement Calculations	P	23.224	bars
Temperature for Internal Pressure	Temp	80	°C
Design External Pressure	Pext	1.01	bars
Temperature for External Pressure	Tempex	80	°C
Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	138.00	N./mm2
Parent Allowable Stress At Ambient	Sva	138.00	N./mm2
Inside Diameter of Cylindrical Shell	D	2200.00	mm.
Design Length of Section	L	7068.2671	mm.
Shell Finished (Minimum) Thickness	t	22.0000	mm.
Shell Internal Corrosion Allowance	cas	1.5000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		150.80	mm.
User Entered Minimum Design Metal Temperature		-38.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-350 LF2	+IT
Material UNS Number		K03011	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm2
Allowable Stress At Ambient	Sna	138.00	N./mm2
Diameter Basis (for tr calc only)		Inside	
Layout Angle		120.00	deg
Diameter		1.5000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.0020	mm.
Flange Material		SA-350 LF2	+IT
Flange Type		Long Weld Neck	
Corrosion Allowance	can	1.5000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	207.9620	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel Wgnv		22.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: K2B1.5"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45



Actual Inside Diameter Used in Calculation 1.500 in.
Actual Thickness Used in Calculation 0.630 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
= $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $23.22 \cdot 1101.5 / (138 \cdot 1.0 - 0.6 \cdot 23.22)$
= 18.7280 mm.

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
= $R \cdot \exp([P / (S_n \cdot E)] - 1)$ per Appendix 1-2 (a)(1)
= $20.55 \cdot \exp([23.22 / (138.0 \cdot 1.0)] - 1)$
= 0.3488 mm.

Required Nozzle thickness under External Pressure per UG-28 : 0.3519 mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	111.1040 mm.
Parallel to Vessel Wall	Rn+tn+t	55.5520 mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	36.2550 mm.

*Taking a UG-36(c)(3)(a) exemption for nozzle: K2B1.5".
This calculation is valid for nozzles that meet all the requirements of
paragraph UG-36. Please check the Code carefully, especially for nozzles
that are not isolated or do not meet Code spacing requirements. To force
the computation of areas for small nozzles go to Tools->Configuration
and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 1.8519 mm.
Wall Thickness per UG16(b),	tr16b	= 3.0000 mm.
Wall Thickness, shell/head, internal pressure	trb1	= 20.2280 mm.
Wall Thickness	tb1 = max(trb1, tr16b)	= 20.2280 mm.
Wall Thickness, shell/head, external pressure	trb2	= 2.3091 mm.
Wall Thickness	tb2 = max(trb2, tr16b)	= 3.0000 mm.
Wall Thickness per table UG-45	tb3	= 6.0200 mm.

Determine Nozzle Thickness candidate [tb]:
= min[tb3, max(tb1, tb2)]
= min[6.02, max(20.228, 3.0)]
= 6.0200 mm.

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
= max(ta, tb)
= max(1.8519, 6.02)
= 6.0200 mm.

Available Nozzle Neck Thickness = 16.0020 mm. -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

*Note:
This Material was specified as being an Impact Tested (Low Temperature) Material.*

Impact Test Temperature provided per Specification	-46 °C
Calculated Minimum Design Metal Temperature	-104 °C
Gov. MDMT of the nozzle to shell joint welded assembly :	-104 °C

Weld Size Calculations, Description: K2B1.5"

Intermediate Calc. for nozzle/shell Welds	Tmin	14.5020 mm.
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Results Per UW-16.1:

Required Thickness	Actual Thickness
--------------------	------------------

PV Elite 28 U1 Licensee: SL User
FileName : 401-V-32004 - Copy - 7
Nozzle Calcs.: K2B_1.5"

Noz1: 50 2:00pm Jul 7, 2026

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HEXAGON

Nozzle Weld $6.0000 = \text{Min per Code } 7.0700 = 0.7 * W_o \text{ mm.}$

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 0.5586 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 230.5206 mm.

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Input, Nozzle Desc: K3B3"

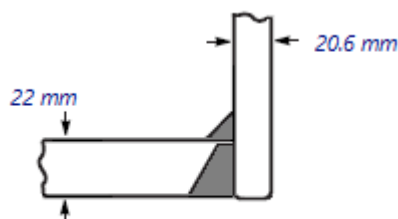
From: 50

Pressure for Reinforcement Calculations	P	23.224	bars
Temperature for Internal Pressure	Temp	80	°C
Design External Pressure	Pext	1.01	bars
Temperature for External Pressure	Tempex	80	°C
Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	138.00	N./mm2
Parent Allowable Stress At Ambient	Sva	138.00	N./mm2
Inside Diameter of Cylindrical Shell	D	2200.00	mm.
Design Length of Section	L	7068.2671	mm.
Shell Finished (Minimum) Thickness	t	22.0000	mm.
Shell Internal Corrosion Allowance	cas	1.5000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		150.80	mm.
User Entered Minimum Design Metal Temperature		-38.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-350 LF2	+IT
Material UNS Number		K03011	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm2
Allowable Stress At Ambient	Sna	138.00	N./mm2
Diameter Basis (for tr calc only)		Inside	
Layout Angle		145.00	deg
Diameter		3.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	20.5740	mm.
Flange Material		SA-350 LF2	+IT
Flange Type		Long Weld Neck	
Corrosion Allowance	can	1.5000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	200.0250	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel Wgnv		22.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: K3B3"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45



Actual Inside Diameter Used in Calculation 3.000 in.
 Actual Thickness Used in Calculation 0.810 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 23.22 \cdot 1101.5 / (138 \cdot 1.0 - 0.6 \cdot 23.22)$
 $= 18.7280$ mm.

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 23.22 \cdot 39.6 / (138 \cdot 1.0 - 0.6 \cdot 23.22)$
 $= 0.6733$ mm.

Required Nozzle thickness under External Pressure per UG-28 : 0.4669 mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	158.4000	mm.
Parallel to Vessel Wall, opening length	d	79.2000	mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	47.6850	mm.

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1.0, 1.0)$
 $= 1.000$

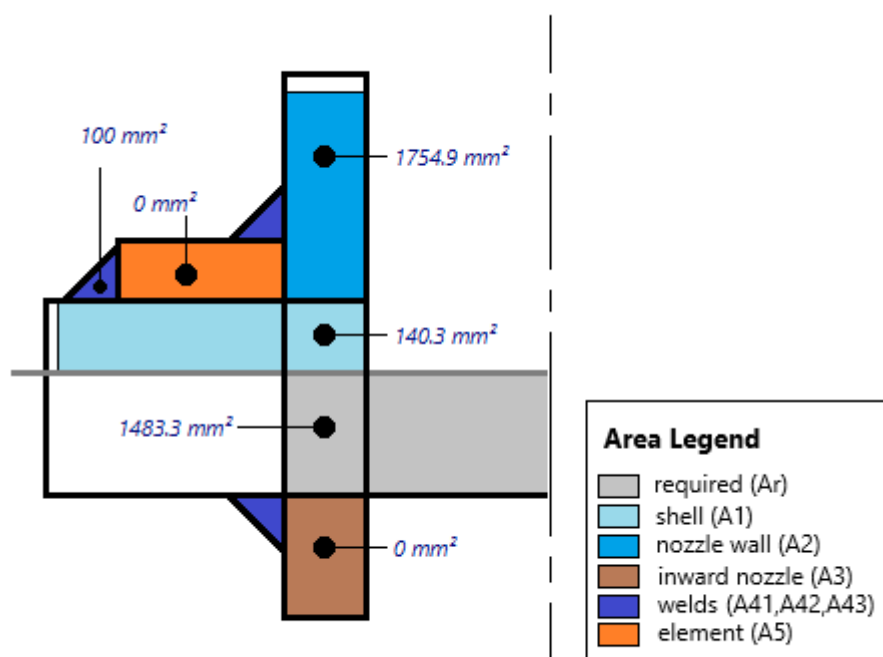


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm2)

AREA AVAILABLE, A1 to A5	Design	External	Mapnc
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Area Required	Ar	14.833	4.564	NA
Area in Shell	A1	1.403	7.109	NA
Area in Nozzle Wall	A2	17.549	17.746	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	1.000	1.000	NA
Area in Element	A5	0.000	0.000	NA
TOTAL AREA AVAILABLE	Atot	19.952	25.854	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$
 $= (79.2 \cdot 18.728 \cdot 1.0 + 2 \cdot 19.074 \cdot 18.728 \cdot 1.0 \cdot (1 - 1.0))$
 $= 14.833 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 79.2(1.0 \cdot 20.5 - 1.0 \cdot 18.728) - 2 \cdot 19.074$
 $(1.0 \cdot 20.5 - 1.0 \cdot 18.728) \cdot (1 - 1.0)$
 $= 1.403 \text{ cm}^2$

Area Available in Nozzle Projecting Outward [A2]:
 $= (2 \cdot tlnp)(tn - trn)fr2$
 $= (2 \cdot 47.68)(19.07 - 0.67)1.0$
 $= 17.549 \text{ cm}^2$

Area Available in Inward Weld + Outward Weld [A41 + A43]:
 $= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$
 $= 10.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0$
 $= 1.000 \text{ cm}^2$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures $ta = 2.1733 \text{ mm.}$
 Wall Thickness per UG16(b), $tr16b = 3.0000 \text{ mm.}$
 Wall Thickness, shell/head, internal pressure $trb1 = 20.2280 \text{ mm.}$
 Wall Thickness $tb1 = \max(trb1, tr16b) = 20.2280 \text{ mm.}$
 Wall Thickness, shell/head, external pressure $trb2 = 2.3091 \text{ mm.}$
 Wall Thickness $tb2 = \max(trb2, tr16b) = 3.0000 \text{ mm.}$
 Wall Thickness per table UG-45 $tb3 = 7.2300 \text{ mm.}$

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[7.23, \max(20.228, 3.0)]$
 $= 7.2300 \text{ mm.}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(2.1733, 7.23)$
 $= 7.2300 \text{ mm.}$

Available Nozzle Neck Thickness = 20.5740 mm. -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note:
This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification -46 °C

Calculated Minimum Design Metal Temperature -104 °C



Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Weld Size Calculations, Description: K3B3"

Intermediate Calc. for nozzle/shell Welds Tmin 19.0000 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm.

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot tn \cdot fr1 \cdot (E1 \cdot t-tr)) \cdot Sv)$
 $= \max(0, (14.8326 - 1.4034 + 2 \cdot 19.074 \cdot 1.0 \cdot (1.0 \cdot 20.5 - 18.728)) \cdot 138)$
 $= 19848.14 \text{ Kgf}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A5+A4-(Wi-Can/.707)^2 \cdot fr2) \cdot Sv$
 $= (17.5488 + 0.0 + 1. - 0.0 \cdot 1.0) \cdot 138$
 $= 26100.91 \text{ Kgf}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot tn \cdot t \cdot fr1)) \cdot Sv$
 $= (17.5488 + 0.0 + 1. + (7.8203)) \cdot 138$
 $= 37105.32 \text{ Kgf}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot tn \cdot t \cdot fr1)) \cdot S$
 $= (17.5488 + 0.0 + 1. + 0.0 + (7.8203)) \cdot 138$
 $= 37105.32 \text{ Kgf}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw$
 $= (3.1416/2.0) \cdot 117.348 \cdot 10.0 \cdot 0.49 \cdot 138$
 $= 12710. \text{ Kgf}$

Shear, Nozzle Wall [Snw]:
 $= (\pi \cdot (Dlr + Dlo)/4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$
 $= (3.1416 \cdot 49.137) \cdot (20.574 - 1.5) \cdot 0.7 \cdot 138$
 $= 29003. \text{ Kgf}$

Tension, Shell Groove Weld [Tngw]:
 $= (\pi/2) \cdot Dlo \cdot (T - cas) \cdot 0.74 \cdot Sng$
 $= (3.1416/2.0) \cdot 117.348 \cdot (22.0 - 1.5) \cdot 0.74 \cdot 138$
 $= 39348. \text{ Kgf}$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
 $= (SONW + SNW) = (12710 + 29003) = 41712 \text{ Kgf}$

Failure Path [Path 2-2]:
 $= (Sonw + Tpgw + Tngw + Sinw)$
 $= (12710 + 0 + 39348 + 0) = 52058 \text{ Kgf}$

Failure Path [Path 3-3]:
 $= (Sonw + Tngw + Sinw)$
 $= (12710 + 39348 + 0) = 52058 \text{ Kgf}$

Summary of Failure Path Calculations:

Path 1-1 = 41712 Kgf, must exceed W = 19848 Kgf or W1 = 26100 Kgf



Path 2-2 = 52057 Kgf, must exceed W = 19848 Kgf or W2 = 37105 Kgf
Path 3-3 = 52057 Kgf, must exceed W = 19848 Kgf or W3 = 37105 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 25 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 1 bars

The Drop for this Nozzle is : 1.5659 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 223.5909 mm.

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Input, Nozzle Desc: K3D3"

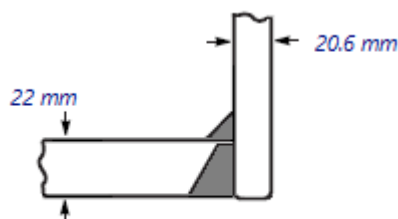
From: 50

Pressure for Reinforcement Calculations	P	23.224	bars
Temperature for Internal Pressure	Temp	80	°C
Design External Pressure	Pext	1.01	bars
Temperature for External Pressure	Tempex	80	°C
Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	138.00	N./mm2
Parent Allowable Stress At Ambient	Sva	138.00	N./mm2
Inside Diameter of Cylindrical Shell	D	2200.00	mm.
Design Length of Section	L	7068.2671	mm.
Shell Finished (Minimum) Thickness	t	22.0000	mm.
Shell Internal Corrosion Allowance	cas	1.5000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		150.80	mm.
User Entered Minimum Design Metal Temperature		-38.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-350 LF2	+IT
Material UNS Number		K03011	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm2
Allowable Stress At Ambient	Sna	138.00	N./mm2
Diameter Basis (for tr calc only)		Inside	
Layout Angle		215.00	deg
Diameter		3.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	20.5740	mm.
Flange Material		SA-350 LF2	+IT
Flange Type		Long Weld Neck	
Corrosion Allowance	can	1.5000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	200.0250	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel Wgnv		22.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: K3D3"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 3.000 in.
 Actual Thickness Used in Calculation 0.810 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 23.22 \cdot 1101.5 / (138 \cdot 1.0 - 0.6 \cdot 23.22)$
 $= 18.7280$ mm.

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 23.22 \cdot 39.6 / (138 \cdot 1.0 - 0.6 \cdot 23.22)$
 $= 0.6733$ mm.

Required Nozzle thickness under External Pressure per UG-28 : 0.4669 mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	158.4000	mm.
Parallel to Vessel Wall, opening length	d	79.2000	mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	47.6850	mm.

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1.0, 1.0)$
 $= 1.000$

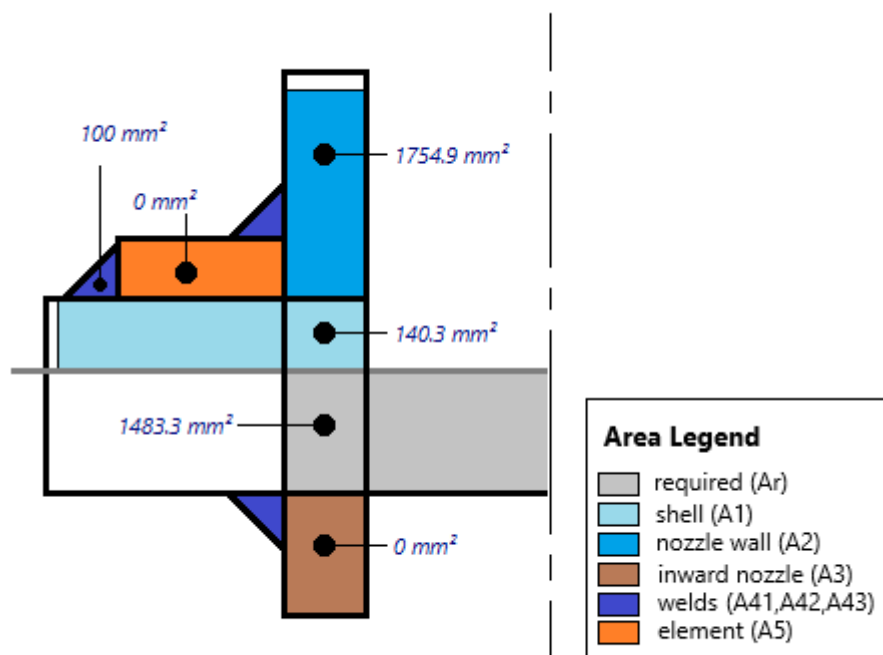


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm2)

AREA AVAILABLE, A1 to A5	Design	External	Mapnc
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Area Required	Ar	14.833	4.564	NA
Area in Shell	A1	1.403	7.109	NA
Area in Nozzle Wall	A2	17.549	17.746	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	1.000	1.000	NA
Area in Element	A5	0.000	0.000	NA
TOTAL AREA AVAILABLE	Atot	19.952	25.854	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$
 $= (79.2 \cdot 18.728 \cdot 1.0 + 2 \cdot 19.074 \cdot 18.728 \cdot 1.0 \cdot (1 - 1.0))$
 $= 14.833 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 79.2(1.0 \cdot 20.5 - 1.0 \cdot 18.728) - 2 \cdot 19.074$
 $(1.0 \cdot 20.5 - 1.0 \cdot 18.728) \cdot (1 - 1.0)$
 $= 1.403 \text{ cm}^2$

Area Available in Nozzle Projecting Outward [A2]:
 $= (2 \cdot tlnp)(tn - trn)fr2$
 $= (2 \cdot 47.68)(19.07 - 0.67)1.0$
 $= 17.549 \text{ cm}^2$

Area Available in Inward Weld + Outward Weld [A41 + A43]:
 $= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$
 $= 10.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0$
 $= 1.000 \text{ cm}^2$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures $ta = 2.1733 \text{ mm.}$
 Wall Thickness per UG16(b), $tr16b = 3.0000 \text{ mm.}$
 Wall Thickness, shell/head, internal pressure $trb1 = 20.2280 \text{ mm.}$
 Wall Thickness $tb1 = \max(trb1, tr16b) = 20.2280 \text{ mm.}$
 Wall Thickness, shell/head, external pressure $trb2 = 2.3091 \text{ mm.}$
 Wall Thickness $tb2 = \max(trb2, tr16b) = 3.0000 \text{ mm.}$
 Wall Thickness per table UG-45 $tb3 = 7.2300 \text{ mm.}$

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[7.23, \max(20.228, 3.0)]$
 $= 7.2300 \text{ mm.}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(2.1733, 7.23)$
 $= 7.2300 \text{ mm.}$

Available Nozzle Neck Thickness = 20.5740 mm. -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note:
This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification -46 °C

Calculated Minimum Design Metal Temperature -104 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Weld Size Calculations, Description: K3D3"

Intermediate Calc. for nozzle/shell Welds Tmin 19.0000 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm.

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
= max(0, (A-A1+2•tn•fr1•(E1•t-tr))•Sv)
= max(0, (14.8326 - 1.4034 + 2 • 19.074 • 1.0 •
(1.0 • 20.5 - 18.728))138)
= 19848.14 Kgf

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
= (A2+A5+A4-(Wi-Can/.707)²•fr2)•Sv
= (17.5488 + 0.0 + 1. - 0.0 • 1.0) • 138
= 26100.91 Kgf

Weld Load [W2]:
= (A2 + A3 + A4 + (2 • tn • t • fr1)) • Sv
= (17.5488 + 0.0 + 1. + (7.8203)) • 138
= 37105.32 Kgf

Weld Load [W3]:
= (A2+A3+A4+A5+(2•tn•t•fr1))•S
= (17.5488 + 0.0 + 1. + 0.0 + (7.8203)) • 138
= 37105.32 Kgf

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
= (π/2) • Dlo • Wo • 0.49 • Snw
= (3.1416/2.0) • 117.348 • 10.0 • 0.49 • 138
= 12710. Kgf

Shear, Nozzle Wall [Snw]:
= (π • (Dlr + Dlo)/4) • (Thk - Can) • 0.7 • Sn
= (3.1416 • 49.137) • (20.574 - 1.5) • 0.7 • 138
= 29003. Kgf

Tension, Shell Groove Weld [Tngw]:
= (π/2) • Dlo • (T - cas) • 0.74 • Sng
= (3.1416/2.0) • 117.348 • (22.0 - 1.5) • 0.74 • 138
= 39348. Kgf

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
= (SONW + SNW) = (12710 + 29003) = 41712 Kgf

Failure Path [Path 2-2]:
= (Sonw + Tpgw + Tngw + Sinw)
= (12710 + 0 + 39348 + 0) = 52058 Kgf

Failure Path [Path 3-3]:
= (Sonw + Tngw + Sinw)
= (12710 + 39348 + 0) = 52058 Kgf

Summary of Failure Path Calculations:

Path 1-1 = 41712 Kgf, must exceed W = 19848 Kgf or W1 = 26100 Kgf



Path 2-2 = 52057 Kgf, must exceed W = 19848 Kgf or W2 = 37105 Kgf
Path 3-3 = 52057 Kgf, must exceed W = 19848 Kgf or W3 = 37105 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 25 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 1 bars

The Drop for this Nozzle is : 1.5659 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 223.5909 mm.

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Input, Nozzle Desc: N82"

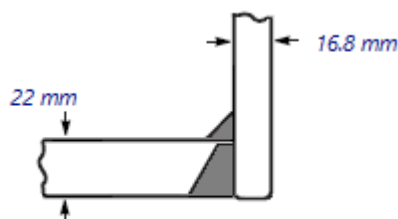
From: 50

Pressure for Reinforcement Calculations	P	23.218	bars
Temperature for Internal Pressure	Temp	80	°C
Design External Pressure	Pext	1.01	bars
Temperature for External Pressure	Tempex	80	°C
Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	138.00	N./mm2
Parent Allowable Stress At Ambient	Sva	138.00	N./mm2
Inside Diameter of Cylindrical Shell	D	2200.00	mm.
Design Length of Section	L	7068.2671	mm.
Shell Finished (Minimum) Thickness	t	22.0000	mm.
Shell Internal Corrosion Allowance	cas	1.5000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		250.80	mm.
User Entered Minimum Design Metal Temperature		-38.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-350 LF2	+IT
Material UNS Number		K03011	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm2
Allowable Stress At Ambient	Sna	138.00	N./mm2
Diameter Basis (for tr calc only)		Inside	
Layout Angle		250.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.7640	mm.
Flange Material		SA-350 LF2	+IT
Flange Type		Long Weld Neck	
Corrosion Allowance	can	1.5000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	206.3750	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	22.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: N82"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45



Actual Inside Diameter Used in Calculation 2.000 in.
Actual Thickness Used in Calculation 0.660 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
= $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $23.22 \cdot 1101.5 / (138 \cdot 1.0 - 0.6 \cdot 23.22)$
= 18.7227 mm.

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
= $R \cdot (\exp([P / (S_n \cdot E)] - 1))$ per Appendix 1-2 (a)(1)
= $26.9 \cdot (\exp([23.22 / (138.0 \cdot 1.0)] - 1))$
= 0.4564 mm.

Required Nozzle thickness under External Pressure per UG-28 : 0.3890 mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	125.3280 mm.
Parallel to Vessel Wall	Rn+tn+t	62.6640 mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	38.1600 mm.

*Taking a UG-36(c)(3)(a) exemption for nozzle: N8_2".
This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 1.9564 mm.
Wall Thickness per UG16(b),	tr16b	= 3.0000 mm.
Wall Thickness, shell/head, internal pressure	trb1	= 20.2227 mm.
Wall Thickness	tb1 = max(trb1, tr16b)	= 20.2227 mm.
Wall Thickness, shell/head, external pressure	trb2	= 2.3091 mm.
Wall Thickness	tb2 = max(trb2, tr16b)	= 3.0000 mm.
Wall Thickness per table UG-45	tb3	= 6.3000 mm.

Determine Nozzle Thickness candidate [tb]:
= min[tb3, max(tb1,tb2)]
= min[6.3, max(20.2227, 3.0)]
= 6.3000 mm.

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
= max(ta, tb)
= max(1.9564, 6.3)
= 6.3000 mm.

Available Nozzle Neck Thickness = 16.7640 mm. -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

*Note:
This Material was specified as being an Impact Tested (Low Temperature) Material.*

Impact Test Temperature provided per Specification	-46 °C
Calculated Minimum Design Metal Temperature	-104 °C
Gov. MDMT of the nozzle to shell joint welded assembly :	-104 °C

Weld Size Calculations, Description: N8_2"

Intermediate Calc. for nozzle/shell Welds	Tmin	15.2640 mm.
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Results Per UW-16.1:

Required Thickness	Actual Thickness
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PV Elite 28 U1 Licensee: SL User
FileName : 401-V-32004 - Copy - 7
Nozzle Calcs.: N8_2"

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Nozzle Weld $6.0000 = \text{Min per Code } 7.0700 = 0.7 * W_o \text{ mm.}$

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 0.8084 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 229.1834 mm.

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Input, Nozzle Desc: K2A1.5"

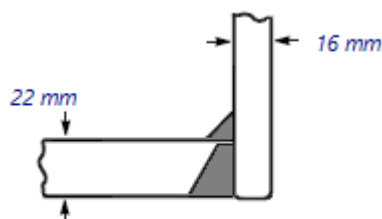
From: 60

Pressure for Reinforcement Calculations	P	23.000	bars
Temperature for Internal Pressure	Temp	80	°C
Design External Pressure	Pext	1.01	bars
Temperature for External Pressure	Tempex	80	°C
Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	138.00	N./mm2
Parent Allowable Stress At Ambient	Sva	138.00	N./mm2
Inside Diameter of Cylindrical Shell	D	2200.00	mm.
Design Length of Section	L	7068.2671	mm.
Shell Finished (Minimum) Thickness	t	22.0000	mm.
Shell Internal Corrosion Allowance	cas	1.5000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		3650.80	mm.
User Entered Minimum Design Metal Temperature		-38.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-350 LF2	+IT
Material UNS Number		K03011	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm2
Allowable Stress At Ambient	Sna	138.00	N./mm2
Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		1.5000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.0020	mm.
Flange Material		SA-350 LF2	+IT
Flange Type		Long Weld Neck	
Corrosion Allowance	can	1.5000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	207.9620	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	22.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: K2A1.5"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45



Actual Inside Diameter Used in Calculation 1.500 in.
Actual Thickness Used in Calculation 0.630 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
= $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $23.0 \cdot 1101.5 / (138 \cdot 1.0 - 0.6 \cdot 23.0)$
= 18.5453 mm.

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
= $R \cdot (\exp([P / (S_n \cdot E)] - 1))$ per Appendix 1-2 (a)(1)
= $20.55 \cdot (\exp([23.0 / (138 \cdot 1.0)] - 1))$
= 0.3454 mm.

Required Nozzle thickness under External Pressure per UG-28 : 0.3519 mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	Dl	111.1040 mm.
Parallel to Vessel Wall	Rn+tn+t	55.5520 mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	36.2550 mm.

*Taking a UG-36(c)(3)(a) exemption for nozzle: K2A1.5".
This calculation is valid for nozzles that meet all the requirements of
paragraph UG-36. Please check the Code carefully, especially for nozzles
that are not isolated or do not meet Code spacing requirements. To force
the computation of areas for small nozzles go to Tools->Configuration
and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 1.8519 mm.
Wall Thickness per UG16(b),	tr16b	= 3.0000 mm.
Wall Thickness, shell/head, internal pressure	trb1	= 20.0453 mm.
Wall Thickness	tb1 = max(trb1, tr16b)	= 20.0453 mm.
Wall Thickness, shell/head, external pressure	trb2	= 2.3091 mm.
Wall Thickness	tb2 = max(trb2, tr16b)	= 3.0000 mm.
Wall Thickness per table UG-45	tb3	= 6.0200 mm.

Determine Nozzle Thickness candidate [tb]:
= min[tb3, max(tb1, tb2)]
= min[6.02, max(20.0453, 3.0)]
= 6.0200 mm.

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
= max(ta, tb)
= max(1.8519, 6.02)
= 6.0200 mm.

Available Nozzle Neck Thickness = 16.0020 mm. -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

*Note:
This Material was specified as being an Impact Tested (Low Temperature) Material.*

Impact Test Temperature provided per Specification	-46 °C
Calculated Minimum Design Metal Temperature	-104 °C
Gov. MDMT of the nozzle to shell joint welded assembly :	-104 °C

Weld Size Calculations, Description: K2A1.5"

Intermediate Calc. for nozzle/shell Welds	Tmin	14.5020 mm.
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Results Per UW-16.1:

Required Thickness	Actual Thickness
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Nozzle Weld $6.0000 = \text{Min per Code } 7.0700 = 0.7 * W_o \text{ mm.}$

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 0.5586 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 230.5206 mm.

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Input, Nozzle Desc: K3A3"

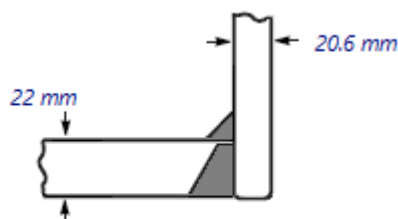
From: 60

Pressure for Reinforcement Calculations	P	23.000	bars
Temperature for Internal Pressure	Temp	80	°C
Design External Pressure	Pext	1.01	bars
Temperature for External Pressure	Tempex	80	°C
Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	138.00	N./mm2
Parent Allowable Stress At Ambient	Sva	138.00	N./mm2
Inside Diameter of Cylindrical Shell	D	2200.00	mm.
Design Length of Section	L	7068.2671	mm.
Shell Finished (Minimum) Thickness	t	22.0000	mm.
Shell Internal Corrosion Allowance	cas	1.5000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		3650.80	mm.
User Entered Minimum Design Metal Temperature		-38.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-350 LF2	+IT
Material UNS Number		K03011	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm2
Allowable Stress At Ambient	Sna	138.00	N./mm2
Diameter Basis (for tr calc only)		Inside	
Layout Angle		90.00	deg
Diameter		3.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	20.5740	mm.
Flange Material		SA-350 LF2	+IT
Flange Type		Long Weld Neck	
Corrosion Allowance	can	1.5000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	200.0250	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	22.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: K3A3"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45



Actual Inside Diameter Used in Calculation 3.000 in.
 Actual Thickness Used in Calculation 0.810 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 23.0 \cdot 1101.5 / (138 \cdot 1.0 - 0.6 \cdot 23.0)$
 $= 18.5453 \text{ mm.}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 23.0 \cdot 39.6 / (138 \cdot 1.0 - 0.6 \cdot 23.0)$
 $= 0.6667 \text{ mm.}$

Required Nozzle thickness under External Pressure per UG-28 : 0.4669 mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	158.4000 mm.
Parallel to Vessel Wall, opening length	d	79.2000 mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	47.6850 mm.

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1.0, 1.0)$
 $= 1.000$

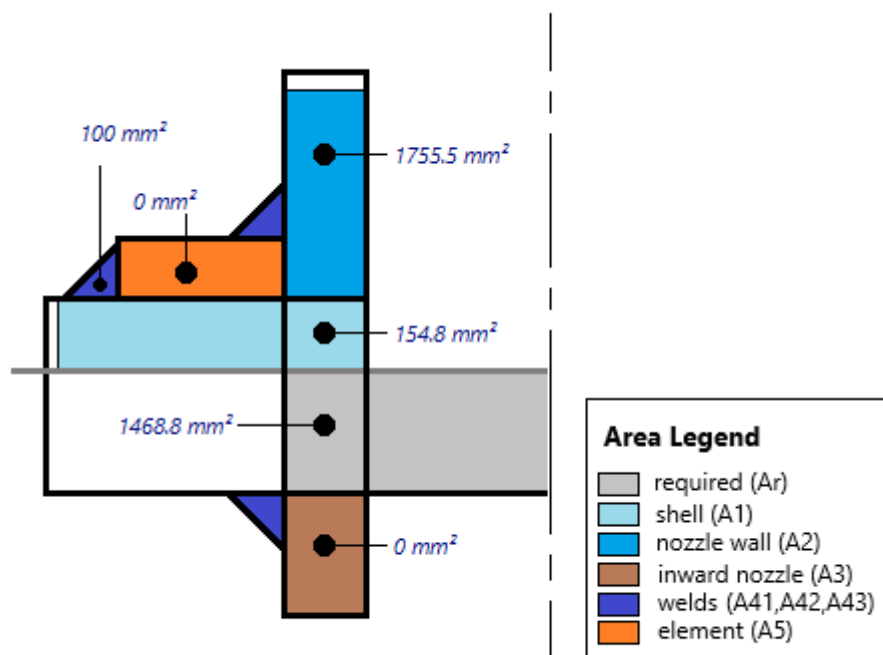


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm2)

AREA AVAILABLE, A1 to A5	Design	External	Mapnc
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Area Required	Ar	14.688	4.564	NA
Area in Shell	A1	1.548	7.109	NA
Area in Nozzle Wall	A2	17.555	17.746	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	1.000	1.000	NA
Area in Element	A5	0.000	0.000	NA
TOTAL AREA AVAILABLE	Atot	20.103	25.854	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$
 $= (79.2 \cdot 18.5453 \cdot 1.0 + 2 \cdot 19.074 \cdot 18.5453 \cdot 1.0 \cdot (1 - 1.0))$
 $= 14.688 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d (E1 \cdot t - F \cdot tr) - 2 \cdot tn (E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 79.2 (1.0 \cdot 20.5 - 1.0 \cdot 18.545) - 2 \cdot 19.074$
 $(1.0 \cdot 20.5 - 1.0 \cdot 18.5453) \cdot (1 - 1.0)$
 $= 1.548 \text{ cm}^2$

Area Available in Nozzle Projecting Outward [A2]:
 $= (2 \cdot tlnp) (tn - trn) fr2$
 $= (2 \cdot 47.68) (19.07 - 0.67) 1.0$
 $= 17.555 \text{ cm}^2$

Area Available in Inward Weld + Outward Weld [A41 + A43]:
 $= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$
 $= 10.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0$
 $= 1.000 \text{ cm}^2$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures $ta = 2.1667 \text{ mm.}$
 Wall Thickness per UG16(b), $tr16b = 3.0000 \text{ mm.}$
 Wall Thickness, shell/head, internal pressure $trb1 = 20.0453 \text{ mm.}$
 Wall Thickness $tb1 = \max(trb1, tr16b) = 20.0453 \text{ mm.}$
 Wall Thickness, shell/head, external pressure $trb2 = 2.3091 \text{ mm.}$
 Wall Thickness $tb2 = \max(trb2, tr16b) = 3.0000 \text{ mm.}$
 Wall Thickness per table UG-45 $tb3 = 7.2300 \text{ mm.}$

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[7.23, \max(20.0453, 3.0)]$
 $= 7.2300 \text{ mm.}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(2.1667, 7.23)$
 $= 7.2300 \text{ mm.}$

Available Nozzle Neck Thickness = 20.5740 mm. -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note:
This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification -46 °C

Calculated Minimum Design Metal Temperature -104 °C



Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Weld Size Calculations, Description: K3A3"

Intermediate Calc. for nozzle/shell Welds Tmin 19.0000 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm.

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - t_r)) \cdot S_v)$
 $= \max(0, (14.6879 - 1.5481 + 2 \cdot 19.074 \cdot 1.0 \cdot (1.0 \cdot 20.5 - 18.5453)) \cdot 138)$
 $= 19538.86 \text{ Kgf}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A5+A4-(W_i-\text{Can}/.707)^2 \cdot f_{r2}) \cdot S_v$
 $= (17.555 + 0.0 + 1. - 0.0 \cdot 1.0) \cdot 138$
 $= 26109.73 \text{ Kgf}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v$
 $= (17.555 + 0.0 + 1. + (7.8203)) \cdot 138$
 $= 37114.14 \text{ Kgf}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S$
 $= (17.555 + 0.0 + 1. + 0.0 + (7.8203)) \cdot 138$
 $= 37114.14 \text{ Kgf}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$
 $= (3.1416/2.0) \cdot 117.348 \cdot 10.0 \cdot 0.49 \cdot 138$
 $= 12710. \text{ Kgf}$

Shear, Nozzle Wall [Snw]:
 $= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n$
 $= (3.1416 \cdot 49.137) \cdot (20.574 - 1.5) \cdot 0.7 \cdot 138$
 $= 29003. \text{ Kgf}$

Tension, Shell Groove Weld [Tngw]:
 $= (\pi/2) \cdot D_{lo} \cdot (T - \text{cas}) \cdot 0.74 \cdot S_{ng}$
 $= (3.1416/2.0) \cdot 117.348 \cdot (22.0 - 1.5) \cdot 0.74 \cdot 138$
 $= 39348. \text{ Kgf}$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
 $= (SONW + SNW) = (12710 + 29003) = 41712 \text{ Kgf}$

Failure Path [Path 2-2]:
 $= (Sonw + Tpgw + Tngw + Sinw)$
 $= (12710 + 0 + 39348 + 0) = 52058 \text{ Kgf}$

Failure Path [Path 3-3]:
 $= (Sonw + Tngw + Sinw)$
 $= (12710 + 39348 + 0) = 52058 \text{ Kgf}$

Summary of Failure Path Calculations:

Path 1-1 = 41712 Kgf, must exceed W = 19538 Kgf or W1 = 26109 Kgf



Path 2-2 = 52057 Kgf, must exceed W = 19538 Kgf or W2 = 37114 Kgf
Path 3-3 = 52057 Kgf, must exceed W = 19538 Kgf or W3 = 37114 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 25 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 1 bars

The Drop for this Nozzle is : 1.5659 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 223.5909 mm.

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Input, Nozzle Desc: K3C3"

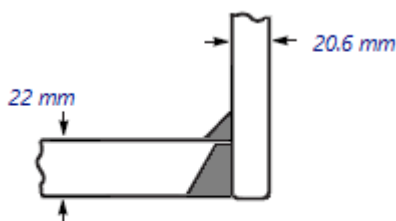
From: 60

Pressure for Reinforcement Calculations	P	23.000	bars
Temperature for Internal Pressure	Temp	80	°C
Design External Pressure	Pext	1.01	bars
Temperature for External Pressure	Tempex	80	°C
Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	138.00	N./mm2
Parent Allowable Stress At Ambient	Sva	138.00	N./mm2
Inside Diameter of Cylindrical Shell	D	2200.00	mm.
Design Length of Section	L	7068.2671	mm.
Shell Finished (Minimum) Thickness	t	22.0000	mm.
Shell Internal Corrosion Allowance	cas	1.5000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		3650.80	mm.
User Entered Minimum Design Metal Temperature		-38.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-350 LF2	+IT
Material UNS Number		K03011	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm2
Allowable Stress At Ambient	Sna	138.00	N./mm2
Diameter Basis (for tr calc only)		Inside	
Layout Angle		180.00	deg
Diameter		3.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	20.5740	mm.
Flange Material		SA-350 LF2	+IT
Flange Type		Long Weld Neck	
Corrosion Allowance	can	1.5000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	200.0250	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	22.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: K3C3"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45



Actual Inside Diameter Used in Calculation 3.000 in.
 Actual Thickness Used in Calculation 0.810 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 23.0 \cdot 1101.5 / (138 \cdot 1.0 - 0.6 \cdot 23.0)$
 $= 18.5453 \text{ mm.}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 23.0 \cdot 39.6 / (138 \cdot 1.0 - 0.6 \cdot 23.0)$
 $= 0.6667 \text{ mm.}$

Required Nozzle thickness under External Pressure per UG-28 : 0.4669 mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	158.4000 mm.
Parallel to Vessel Wall, opening length	d	79.2000 mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	47.6850 mm.

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1.0, 1.0)$
 $= 1.000$

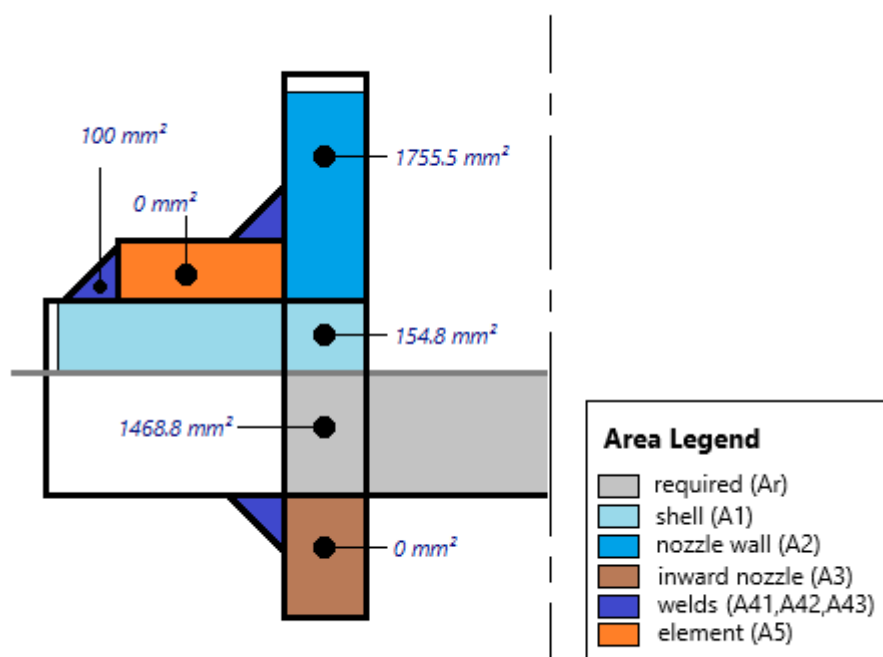


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm2)

AREA AVAILABLE, A1 to A5	Design	External	Mapnc
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Area Required	Ar	14.688	4.564	NA
Area in Shell	A1	1.548	7.109	NA
Area in Nozzle Wall	A2	17.555	17.746	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	1.000	1.000	NA
Area in Element	A5	0.000	0.000	NA
TOTAL AREA AVAILABLE	Atot	20.103	25.854	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$
 $= (79.2 \cdot 18.5453 \cdot 1.0 + 2 \cdot 19.074 \cdot 18.5453 \cdot 1.0 \cdot (1 - 1.0))$
 $= 14.688 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 79.2(1.0 \cdot 20.5 - 1.0 \cdot 18.545) - 2 \cdot 19.074$
 $(1.0 \cdot 20.5 - 1.0 \cdot 18.5453) \cdot (1 - 1.0)$
 $= 1.548 \text{ cm}^2$

Area Available in Nozzle Projecting Outward [A2]:
 $= (2 \cdot tlnp)(tn - trn)fr2$
 $= (2 \cdot 47.68)(19.07 - 0.67)1.0$
 $= 17.555 \text{ cm}^2$

Area Available in Inward Weld + Outward Weld [A41 + A43]:
 $= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$
 $= 10.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0$
 $= 1.000 \text{ cm}^2$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures $ta = 2.1667 \text{ mm.}$
 Wall Thickness per UG16(b), $tr16b = 3.0000 \text{ mm.}$
 Wall Thickness, shell/head, internal pressure $trb1 = 20.0453 \text{ mm.}$
 Wall Thickness $tb1 = \max(trb1, tr16b) = 20.0453 \text{ mm.}$
 Wall Thickness, shell/head, external pressure $trb2 = 2.3091 \text{ mm.}$
 Wall Thickness $tb2 = \max(trb2, tr16b) = 3.0000 \text{ mm.}$
 Wall Thickness per table UG-45 $tb3 = 7.2300 \text{ mm.}$

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[7.23, \max(20.0453, 3.0)]$
 $= 7.2300 \text{ mm.}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(2.1667, 7.23)$
 $= 7.2300 \text{ mm.}$

Available Nozzle Neck Thickness = 20.5740 mm. -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note:
This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification -46 °C

Calculated Minimum Design Metal Temperature -104 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Weld Size Calculations, Description: K3C3"

Intermediate Calc. for nozzle/shell Welds Tmin 19.0000 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm.

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
= max(0, (A-A1+2•tn•fr1•(E1•t-tr))•Sv)
= max(0, (14.6879 - 1.5481 + 2 • 19.074 • 1.0 •
(1.0 • 20.5 - 18.5453)) 138)
= 19538.86 Kgf

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
= (A2+A5+A4-(Wi-Can/.707)²•fr2)•Sv
= (17.555 + 0.0 + 1. - 0.0 • 1.0) • 138
= 26109.73 Kgf

Weld Load [W2]:
= (A2 + A3 + A4 + (2 • tn • t • fr1)) • Sv
= (17.555 + 0.0 + 1. + (7.8203)) • 138
= 37114.14 Kgf

Weld Load [W3]:
= (A2+A3+A4+A5+(2•tn•t•fr1))•S
= (17.555 + 0.0 + 1. + 0.0 + (7.8203)) • 138
= 37114.14 Kgf

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
= (π/2) • Dlo • Wo • 0.49 • Snw
= (3.1416/2.0) • 117.348 • 10.0 • 0.49 • 138
= 12710. Kgf

Shear, Nozzle Wall [Snw]:
= (π • (Dlr + Dlo)/4) • (Thk - Can) • 0.7 • Sn
= (3.1416 • 49.137) • (20.574 - 1.5) • 0.7 • 138
= 29003. Kgf

Tension, Shell Groove Weld [Tngw]:
= (π/2) • Dlo • (T - cas) • 0.74 • Sng
= (3.1416/2.0) • 117.348 • (22.0 - 1.5) • 0.74 • 138
= 39348. Kgf

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
= (SONW + SNW) = (12710 + 29003) = 41712 Kgf

Failure Path [Path 2-2]:
= (Sonw + Tpgw + Tngw + Sinw)
= (12710 + 0 + 39348 + 0) = 52058 Kgf

Failure Path [Path 3-3]:
= (Sonw + Tngw + Sinw)
= (12710 + 39348 + 0) = 52058 Kgf

Summary of Failure Path Calculations:

Path 1-1 = 41712 Kgf, must exceed W = 19538 Kgf or W1 = 26109 Kgf



Path 2-2 = 52057 Kgf, must exceed W = 19538 Kgf or W2 = 37114 Kgf
Path 3-3 = 52057 Kgf, must exceed W = 19538 Kgf or W3 = 37114 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 25 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 1 bars

The Drop for this Nozzle is : 1.5659 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 223.5909 mm.

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Input, Nozzle Desc: K1A3"

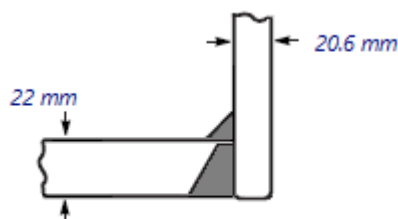
From: 60

Pressure for Reinforcement Calculations	P	23.000	bars
Temperature for Internal Pressure	Temp	80	°C
Design External Pressure	Pext	1.01	bars
Temperature for External Pressure	Tempex	80	°C
Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	138.00	N./mm2
Parent Allowable Stress At Ambient	Sva	138.00	N./mm2
Inside Diameter of Cylindrical Shell	D	2200.00	mm.
Design Length of Section	L	7068.2671	mm.
Shell Finished (Minimum) Thickness	t	22.0000	mm.
Shell Internal Corrosion Allowance	cas	1.5000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		3650.80	mm.
User Entered Minimum Design Metal Temperature		-38.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-350 LF2	+IT
Material UNS Number		K03011	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm2
Allowable Stress At Ambient	Sna	138.00	N./mm2
Diameter Basis (for tr calc only)		Inside	
Layout Angle		120.00	deg
Diameter		3.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	20.5740	mm.
Flange Material		SA-350 LF2	+IT
Flange Type		Long Weld Neck	
Corrosion Allowance	can	1.5000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	200.0250	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	22.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: K1A3"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45



Actual Inside Diameter Used in Calculation 3.000 in.
 Actual Thickness Used in Calculation 0.810 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 23.0 \cdot 1101.5 / (138 \cdot 1.0 - 0.6 \cdot 23.0)$
 $= 18.5453 \text{ mm.}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 23.0 \cdot 39.6 / (138 \cdot 1.0 - 0.6 \cdot 23.0)$
 $= 0.6667 \text{ mm.}$

Required Nozzle thickness under External Pressure per UG-28 : 0.4669 mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	158.4000	mm.
Parallel to Vessel Wall, opening length	d	79.2000	mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	47.6850	mm.

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1.0, 1.0)$
 $= 1.000$

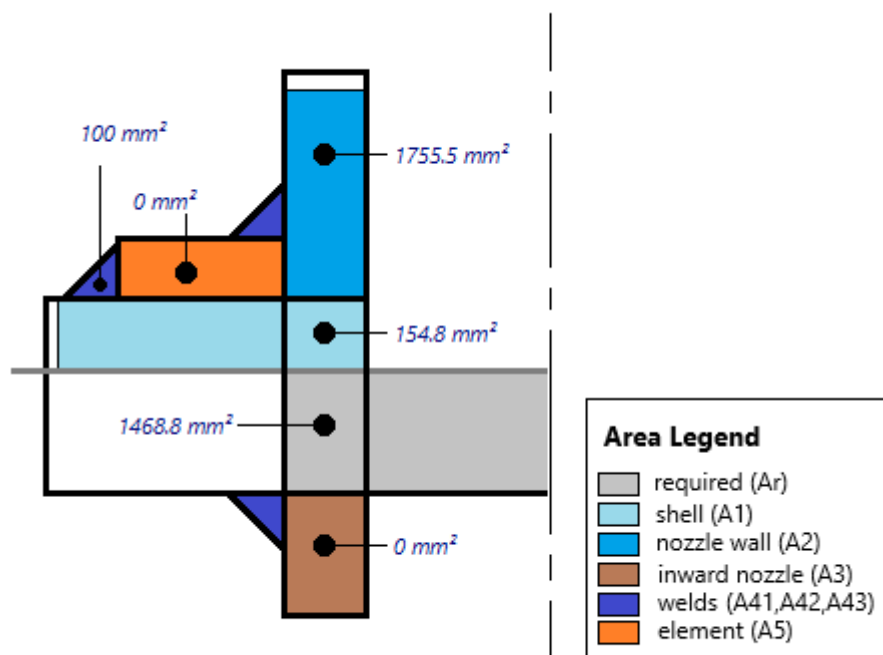


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm2)

AREA AVAILABLE, A1 to A5	Design	External	Mapnc
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Area Required	Ar	14.688	4.564	NA
Area in Shell	A1	1.548	7.109	NA
Area in Nozzle Wall	A2	17.555	17.746	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	1.000	1.000	NA
Area in Element	A5	0.000	0.000	NA
TOTAL AREA AVAILABLE	Atot	20.103	25.854	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$
 $= (79.2 \cdot 18.5453 \cdot 1.0 + 2 \cdot 19.074 \cdot 18.5453 \cdot 1.0 \cdot (1 - 1.0))$
 $= 14.688 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 79.2(1.0 \cdot 20.5 - 1.0 \cdot 18.545) - 2 \cdot 19.074$
 $(1.0 \cdot 20.5 - 1.0 \cdot 18.5453) \cdot (1 - 1.0)$
 $= 1.548 \text{ cm}^2$

Area Available in Nozzle Projecting Outward [A2]:
 $= (2 \cdot tlnp)(tn - trn)fr2$
 $= (2 \cdot 47.68)(19.07 - 0.67)1.0$
 $= 17.555 \text{ cm}^2$

Area Available in Inward Weld + Outward Weld [A41 + A43]:
 $= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$
 $= 10.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0$
 $= 1.000 \text{ cm}^2$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures $ta = 2.1667 \text{ mm.}$
 Wall Thickness per UG16(b), $tr16b = 3.0000 \text{ mm.}$
 Wall Thickness, shell/head, internal pressure $trb1 = 20.0453 \text{ mm.}$
 Wall Thickness $tb1 = \max(trb1, tr16b) = 20.0453 \text{ mm.}$
 Wall Thickness, shell/head, external pressure $trb2 = 2.3091 \text{ mm.}$
 Wall Thickness $tb2 = \max(trb2, tr16b) = 3.0000 \text{ mm.}$
 Wall Thickness per table UG-45 $tb3 = 7.2300 \text{ mm.}$

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[7.23, \max(20.0453, 3.0)]$
 $= 7.2300 \text{ mm.}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(2.1667, 7.23)$
 $= 7.2300 \text{ mm.}$

Available Nozzle Neck Thickness = 20.5740 mm. -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note:
This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification -46 °C

Calculated Minimum Design Metal Temperature -104 °C



Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Weld Size Calculations, Description: K1A3"

Intermediate Calc. for nozzle/shell Welds Tmin 19.0000 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm.

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
= max(0, (A-A1+2•tn•fr1•(E1•t-tr))•Sv)
= max(0, (14.6879 - 1.5481 + 2 • 19.074 • 1.0 •
(1.0 • 20.5 - 18.5453)) 138)
= 19538.86 Kgf

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
= (A2+A5+A4-(Wi-Can/.707)²•fr2)•Sv
= (17.555 + 0.0 + 1. - 0.0 • 1.0) • 138
= 26109.73 Kgf

Weld Load [W2]:
= (A2 + A3 + A4 + (2 • tn • t • fr1)) • Sv
= (17.555 + 0.0 + 1. + (7.8203)) • 138
= 37114.14 Kgf

Weld Load [W3]:
= (A2+A3+A4+A5+(2•tn•t•fr1))•S
= (17.555 + 0.0 + 1. + 0.0 + (7.8203)) • 138
= 37114.14 Kgf

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
= (π/2) • Dlo • Wo • 0.49 • Snw
= (3.1416/2.0) • 117.348 • 10.0 • 0.49 • 138
= 12710. Kgf

Shear, Nozzle Wall [Snw]:
= (π • (Dlr + Dlo)/4) • (Thk - Can) • 0.7 • Sn
= (3.1416 • 49.137) • (20.574 - 1.5) • 0.7 • 138
= 29003. Kgf

Tension, Shell Groove Weld [Tngw]:
= (π/2) • Dlo • (T - cas) • 0.74 • Sng
= (3.1416/2.0) • 117.348 • (22.0 - 1.5) • 0.74 • 138
= 39348. Kgf

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
= (SONW + SNW) = (12710 + 29003) = 41712 Kgf

Failure Path [Path 2-2]:
= (Sonw + Tpgw + Tngw + Sinw)
= (12710 + 0 + 39348 + 0) = 52058 Kgf

Failure Path [Path 3-3]:
= (Sonw + Tngw + Sinw)
= (12710 + 39348 + 0) = 52058 Kgf

Summary of Failure Path Calculations:

Path 1-1 = 41712 Kgf, must exceed W = 19538 Kgf or W1 = 26109 Kgf



Path 2-2 = 52057 Kgf, must exceed W = 19538 Kgf or W2 = 37114 Kgf
Path 3-3 = 52057 Kgf, must exceed W = 19538 Kgf or W3 = 37114 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 25 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 1 bars

The Drop for this Nozzle is : 1.5659 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 223.5909 mm.

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Input, Nozzle Desc: K4A3"

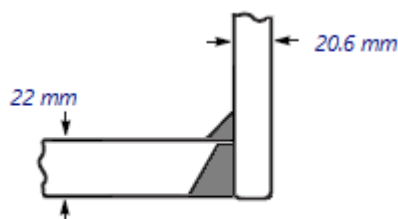
From: 60

Pressure for Reinforcement Calculations	P	23.000	bars
Temperature for Internal Pressure	Temp	80	°C
Design External Pressure	Pext	1.01	bars
Temperature for External Pressure	Tempex	80	°C
Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	138.00	N./mm2
Parent Allowable Stress At Ambient	Sva	138.00	N./mm2
Inside Diameter of Cylindrical Shell	D	2200.00	mm.
Design Length of Section	L	7068.2671	mm.
Shell Finished (Minimum) Thickness	t	22.0000	mm.
Shell Internal Corrosion Allowance	cas	1.5000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		3650.80	mm.
User Entered Minimum Design Metal Temperature		-38.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-350 LF2	+IT
Material UNS Number		K03011	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm2
Allowable Stress At Ambient	Sna	138.00	N./mm2
Diameter Basis (for tr calc only)		Inside	
Layout Angle		160.00	deg
Diameter		3.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	20.5740	mm.
Flange Material		SA-350 LF2	+IT
Flange Type		Long Weld Neck	
Corrosion Allowance	can	1.5000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	200.0250	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	22.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: K4A3"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45



Actual Inside Diameter Used in Calculation 3.000 in.
 Actual Thickness Used in Calculation 0.810 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 23.0 \cdot 1101.5 / (138 \cdot 1.0 - 0.6 \cdot 23.0)$
 $= 18.5453 \text{ mm.}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 23.0 \cdot 39.6 / (138 \cdot 1.0 - 0.6 \cdot 23.0)$
 $= 0.6667 \text{ mm.}$

Required Nozzle thickness under External Pressure per UG-28 : 0.4669 mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	158.4000 mm.
Parallel to Vessel Wall, opening length	d	79.2000 mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	47.6850 mm.

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1.0, 1.0)$
 $= 1.000$

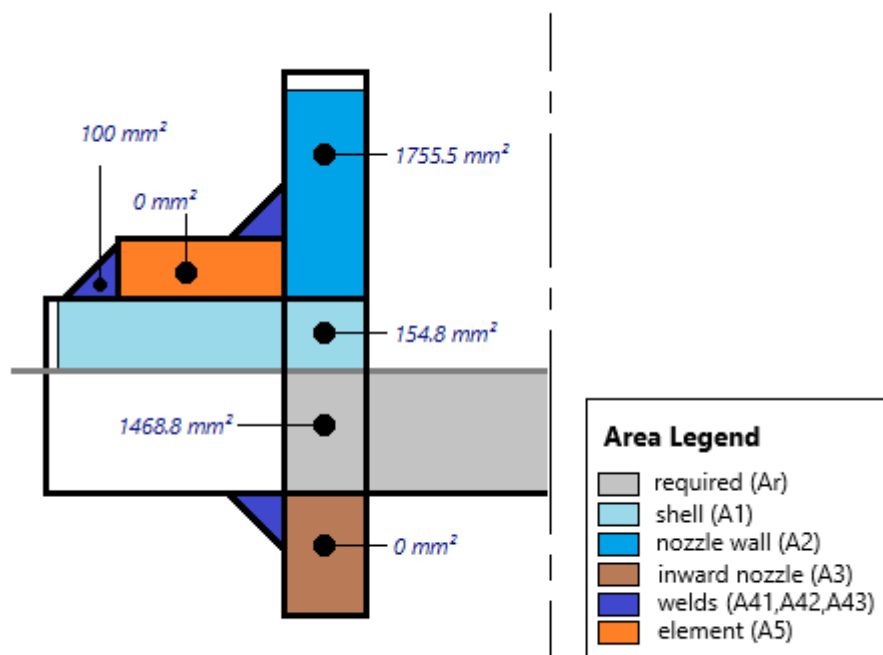


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm2)

AREA AVAILABLE, A1 to A5	Design	External	Mapnc
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Area Required	Ar	14.688	4.564	NA
Area in Shell	A1	1.548	7.109	NA
Area in Nozzle Wall	A2	17.555	17.746	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	1.000	1.000	NA
Area in Element	A5	0.000	0.000	NA
TOTAL AREA AVAILABLE	Atot	20.103	25.854	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$
 $= (79.2 \cdot 18.5453 \cdot 1.0 + 2 \cdot 19.074 \cdot 18.5453 \cdot 1.0 \cdot (1 - 1.0))$
 $= 14.688 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d (E1 \cdot t - F \cdot tr) - 2 \cdot tn (E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 79.2 (1.0 \cdot 20.5 - 1.0 \cdot 18.545) - 2 \cdot 19.074$
 $(1.0 \cdot 20.5 - 1.0 \cdot 18.5453) \cdot (1 - 1.0)$
 $= 1.548 \text{ cm}^2$

Area Available in Nozzle Projecting Outward [A2]:
 $= (2 \cdot tlnp) (tn - trn) fr2$
 $= (2 \cdot 47.68) (19.07 - 0.67) 1.0$
 $= 17.555 \text{ cm}^2$

Area Available in Inward Weld + Outward Weld [A41 + A43]:
 $= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$
 $= 10.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0$
 $= 1.000 \text{ cm}^2$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures $ta = 2.1667 \text{ mm.}$
 Wall Thickness per UG16(b), $tr16b = 3.0000 \text{ mm.}$
 Wall Thickness, shell/head, internal pressure $trb1 = 20.0453 \text{ mm.}$
 Wall Thickness $tb1 = \max(trb1, tr16b) = 20.0453 \text{ mm.}$
 Wall Thickness, shell/head, external pressure $trb2 = 2.3091 \text{ mm.}$
 Wall Thickness $tb2 = \max(trb2, tr16b) = 3.0000 \text{ mm.}$
 Wall Thickness per table UG-45 $tb3 = 7.2300 \text{ mm.}$

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[7.23, \max(20.0453, 3.0)]$
 $= 7.2300 \text{ mm.}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(2.1667, 7.23)$
 $= 7.2300 \text{ mm.}$

Available Nozzle Neck Thickness = 20.5740 mm. -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note:
This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification -46 °C

Calculated Minimum Design Metal Temperature -104 °C



Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Weld Size Calculations, Description: K4A3"

Intermediate Calc. for nozzle/shell Welds Tmin 19.0000 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm.

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
= max(0, (A-A1+2•tn•fr1•(E1•t-tr))•Sv)
= max(0, (14.6879 - 1.5481 + 2 • 19.074 • 1.0 •
(1.0 • 20.5 - 18.5453)) 138)
= 19538.86 Kgf

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
= (A2+A5+A4-(Wi-Can/.707)²•fr2)•Sv
= (17.555 + 0.0 + 1. - 0.0 • 1.0) • 138
= 26109.73 Kgf

Weld Load [W2]:
= (A2 + A3 + A4 + (2 • tn • t • fr1)) • Sv
= (17.555 + 0.0 + 1. + (7.8203)) • 138
= 37114.14 Kgf

Weld Load [W3]:
= (A2+A3+A4+A5+(2•tn•t•fr1))•S
= (17.555 + 0.0 + 1. + 0.0 + (7.8203)) • 138
= 37114.14 Kgf

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
= (π/2) • Dlo • Wo • 0.49 • Snw
= (3.1416/2.0) • 117.348 • 10.0 • 0.49 • 138
= 12710. Kgf

Shear, Nozzle Wall [Snw]:
= (π • (Dlr + Dlo)/4) • (Thk - Can) • 0.7 • Sn
= (3.1416 • 49.137) • (20.574 - 1.5) • 0.7 • 138
= 29003. Kgf

Tension, Shell Groove Weld [Tngw]:
= (π/2) • Dlo • (T - cas) • 0.74 • Sng
= (3.1416/2.0) • 117.348 • (22.0 - 1.5) • 0.74 • 138
= 39348. Kgf

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
= (SONW + SNW) = (12710 + 29003) = 41712 Kgf

Failure Path [Path 2-2]:
= (Sonw + Tpgw + Tngw + Sinw)
= (12710 + 0 + 39348 + 0) = 52058 Kgf

Failure Path [Path 3-3]:
= (Sonw + Tngw + Sinw)
= (12710 + 39348 + 0) = 52058 Kgf

Summary of Failure Path Calculations:

Path 1-1 = 41712 Kgf, must exceed W = 19538 Kgf or W1 = 26109 Kgf



Path 2-2 = 52057 Kgf, must exceed W = 19538 Kgf or W2 = 37114 Kgf
Path 3-3 = 52057 Kgf, must exceed W = 19538 Kgf or W3 = 37114 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 25 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 1 bars

The Drop for this Nozzle is : 1.5659 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 223.5909 mm.

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Input, Nozzle Desc: K3E3"

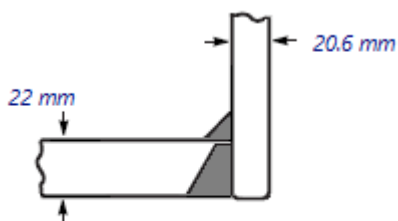
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Pressure for Reinforcement Calculations	P	23.000	bars
Temperature for Internal Pressure	Temp	80	°C
Design External Pressure	Pext	1.01	bars
Temperature for External Pressure	Tempex	80	°C
Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	138.00	N./mm2
Parent Allowable Stress At Ambient	Sva	138.00	N./mm2
Inside Diameter of Cylindrical Shell	D	2200.00	mm.
Design Length of Section	L	7068.2671	mm.
Shell Finished (Minimum) Thickness	t	22.0000	mm.
Shell Internal Corrosion Allowance	cas	1.5000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		3650.80	mm.
User Entered Minimum Design Metal Temperature		-38.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-350 LF2	+IT
Material UNS Number		K03011	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm2
Allowable Stress At Ambient	Sna	138.00	N./mm2
Diameter Basis (for tr calc only)		Inside	
Layout Angle		270.00	deg
Diameter		3.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	20.5740	mm.
Flange Material		SA-350 LF2	+IT
Flange Type		Long Weld Neck	
Corrosion Allowance	can	1.5000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	200.0250	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	22.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: K3E3"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45



Actual Inside Diameter Used in Calculation 3.000 in.
 Actual Thickness Used in Calculation 0.810 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 23.0 \cdot 1101.5 / (138 \cdot 1.0 - 0.6 \cdot 23.0)$
 $= 18.5453 \text{ mm.}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 23.0 \cdot 39.6 / (138 \cdot 1.0 - 0.6 \cdot 23.0)$
 $= 0.6667 \text{ mm.}$

Required Nozzle thickness under External Pressure per UG-28 : 0.4669 mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	158.4000 mm.
Parallel to Vessel Wall, opening length	d	79.2000 mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	47.6850 mm.

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1.0, 1.0)$
 $= 1.000$

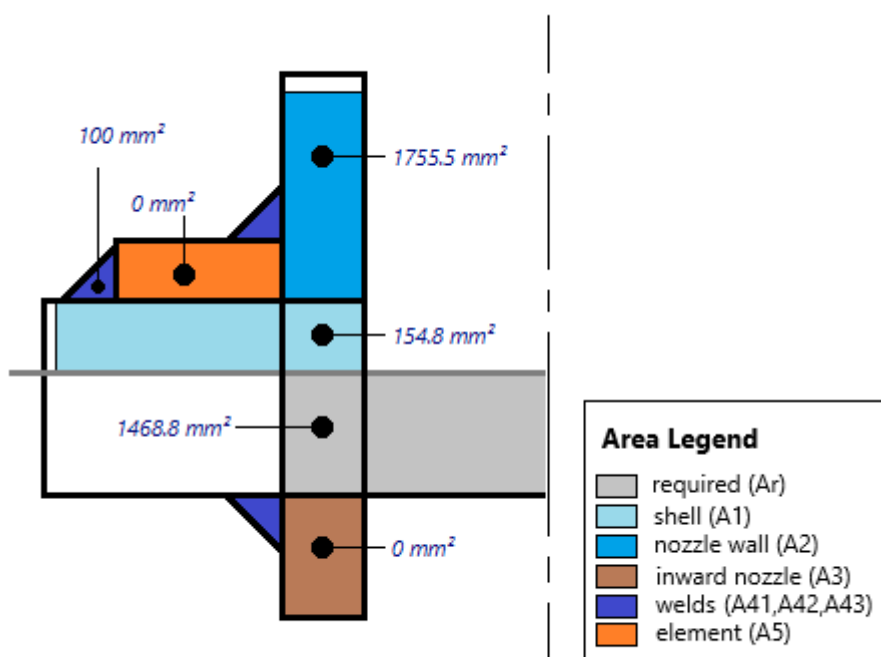


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm2)

AREA AVAILABLE, A1 to A5	Design	External	Mapnc
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Area Required	Ar	14.688	4.564	NA
Area in Shell	A1	1.548	7.109	NA
Area in Nozzle Wall	A2	17.555	17.746	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	1.000	1.000	NA
Area in Element	A5	0.000	0.000	NA
TOTAL AREA AVAILABLE	Atot	20.103	25.854	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$
 $= (79.2 \cdot 18.5453 \cdot 1.0 + 2 \cdot 19.074 \cdot 18.5453 \cdot 1.0 \cdot (1 - 1.0))$
 $= 14.688 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d (E1 \cdot t - F \cdot tr) - 2 \cdot tn (E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 79.2 (1.0 \cdot 20.5 - 1.0 \cdot 18.545) - 2 \cdot 19.074$
 $(1.0 \cdot 20.5 - 1.0 \cdot 18.5453) \cdot (1 - 1.0)$
 $= 1.548 \text{ cm}^2$

Area Available in Nozzle Projecting Outward [A2]:
 $= (2 \cdot tlnp) (tn - trn) fr2$
 $= (2 \cdot 47.68) (19.07 - 0.67) 1.0$
 $= 17.555 \text{ cm}^2$

Area Available in Inward Weld + Outward Weld [A41 + A43]:
 $= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$
 $= 10.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0$
 $= 1.000 \text{ cm}^2$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures $ta = 2.1667 \text{ mm.}$
 Wall Thickness per UG16(b), $tr16b = 3.0000 \text{ mm.}$
 Wall Thickness, shell/head, internal pressure $trb1 = 20.0453 \text{ mm.}$
 Wall Thickness $tb1 = \max(trb1, tr16b) = 20.0453 \text{ mm.}$
 Wall Thickness, shell/head, external pressure $trb2 = 2.3091 \text{ mm.}$
 Wall Thickness $tb2 = \max(trb2, tr16b) = 3.0000 \text{ mm.}$
 Wall Thickness per table UG-45 $tb3 = 7.2300 \text{ mm.}$

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[7.23, \max(20.0453, 3.0)]$
 $= 7.2300 \text{ mm.}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(2.1667, 7.23)$
 $= 7.2300 \text{ mm.}$

Available Nozzle Neck Thickness = 20.5740 mm. -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note:
This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification -46 °C

Calculated Minimum Design Metal Temperature -104 °C



Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Weld Size Calculations, Description: K3E3"

Intermediate Calc. for nozzle/shell Welds Tmin 19.0000 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm.

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot tn \cdot fr1 \cdot (E1 \cdot t-tr)) \cdot Sv)$
 $= \max(0, (14.6879 - 1.5481 + 2 \cdot 19.074 \cdot 1.0 \cdot (1.0 \cdot 20.5 - 18.5453)) \cdot 138)$
 $= 19538.86 \text{ Kgf}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A5+A4-(Wi-Can/.707)^2 \cdot fr2) \cdot Sv$
 $= (17.555 + 0.0 + 1. - 0.0 \cdot 1.0) \cdot 138$
 $= 26109.73 \text{ Kgf}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot tn \cdot t \cdot fr1)) \cdot Sv$
 $= (17.555 + 0.0 + 1. + (7.8203)) \cdot 138$
 $= 37114.14 \text{ Kgf}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot tn \cdot t \cdot fr1)) \cdot S$
 $= (17.555 + 0.0 + 1. + 0.0 + (7.8203)) \cdot 138$
 $= 37114.14 \text{ Kgf}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw$
 $= (3.1416/2.0) \cdot 117.348 \cdot 10.0 \cdot 0.49 \cdot 138$
 $= 12710. \text{ Kgf}$

Shear, Nozzle Wall [Snw]:
 $= (\pi \cdot (Dlr + Dlo)/4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$
 $= (3.1416 \cdot 49.137) \cdot (20.574 - 1.5) \cdot 0.7 \cdot 138$
 $= 29003. \text{ Kgf}$

Tension, Shell Groove Weld [Tngw]:
 $= (\pi/2) \cdot Dlo \cdot (T - cas) \cdot 0.74 \cdot Sng$
 $= (3.1416/2.0) \cdot 117.348 \cdot (22.0 - 1.5) \cdot 0.74 \cdot 138$
 $= 39348. \text{ Kgf}$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
 $= (SONW + SNW) = (12710 + 29003) = 41712 \text{ Kgf}$

Failure Path [Path 2-2]:
 $= (Sonw + Tpgw + Tngw + Sinw)$
 $= (12710 + 0 + 39348 + 0) = 52058 \text{ Kgf}$

Failure Path [Path 3-3]:
 $= (Sonw + Tngw + Sinw)$
 $= (12710 + 39348 + 0) = 52058 \text{ Kgf}$

Summary of Failure Path Calculations:

Path 1-1 = 41712 Kgf, must exceed W = 19538 Kgf or W1 = 26109 Kgf



Path 2-2 = 52057 Kgf, must exceed W = 19538 Kgf or W2 = 37114 Kgf
Path 3-3 = 52057 Kgf, must exceed W = 19538 Kgf or W3 = 37114 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 25 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 1 bars

The Drop for this Nozzle is : 1.5659 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 223.5909 mm.

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Input, Nozzle Desc: MH124"

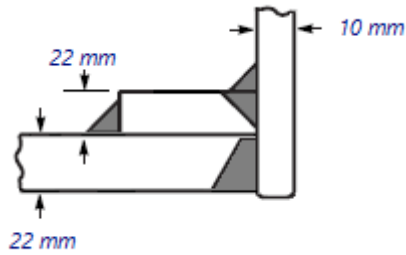
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Pressure for Reinforcement Calculations	P	23.000	bars
Temperature for Internal Pressure	Temp	80	°C
Design External Pressure	Pext	1.01	bars
Temperature for External Pressure	Tempex	80	°C
Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	138.00	N./mm2
Parent Allowable Stress At Ambient	Sva	138.00	N./mm2
Inside Diameter of Cylindrical Shell	D	2200.00	mm.
Design Length of Section	L	7068.2671	mm.
Shell Finished (Minimum) Thickness	t	22.0000	mm.
Shell Internal Corrosion Allowance	cas	1.5000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		5800.80	mm.
User Entered Minimum Design Metal Temperature		-38.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-516 70	+IT
Material UNS Number		K02700	
Material Specification/Type		Plate	
Allowable Stress at Temperature	Sn	138.00	N./mm2
Allowable Stress At Ambient	Sna	138.00	N./mm2
Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		24.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	10.0000	mm.
Flange Material		SA-350 LF2	+IT
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	1.5000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	161.3760	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	22.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Pad Material		SA-516 70	+IT
Pad Allowable Stress at Temperature	Sp	138.00	N./mm2
Pad Allowable Stress At Ambient	Spa	138.00	N./mm2
Diameter of Pad along vessel surface	Dp	1100.0000	mm.
Thickness of Pad	te	22.0000	mm.
Weld leg size between Pad and Shell	Wp	18.0000	mm.
Groove weld depth between Pad and Nozzle	Wgpn	22.0000	mm.
Reinforcing Pad Width		235.2000	mm.
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: MH124"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	24.000 in.
Actual Thickness Used in Calculation	0.394 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 23.0 \cdot 1101.5 / (138 \cdot 1.0 - 0.6 \cdot 23.0)$
 $= 18.5453 \text{ mm.}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 23.0 \cdot 306.3 / (138 \cdot 1.0 - 0.6 \cdot 23.0)$
 $= 5.1570 \text{ mm.}$

Required Nozzle thickness under External Pressure per UG-28 : 1.1328 mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	1225.2000 mm.
Parallel to Vessel Wall, opening length	d	612.6000 mm.
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		43.2500 mm.

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, S_p / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1.0, 1.0)$
 $= 1.000$

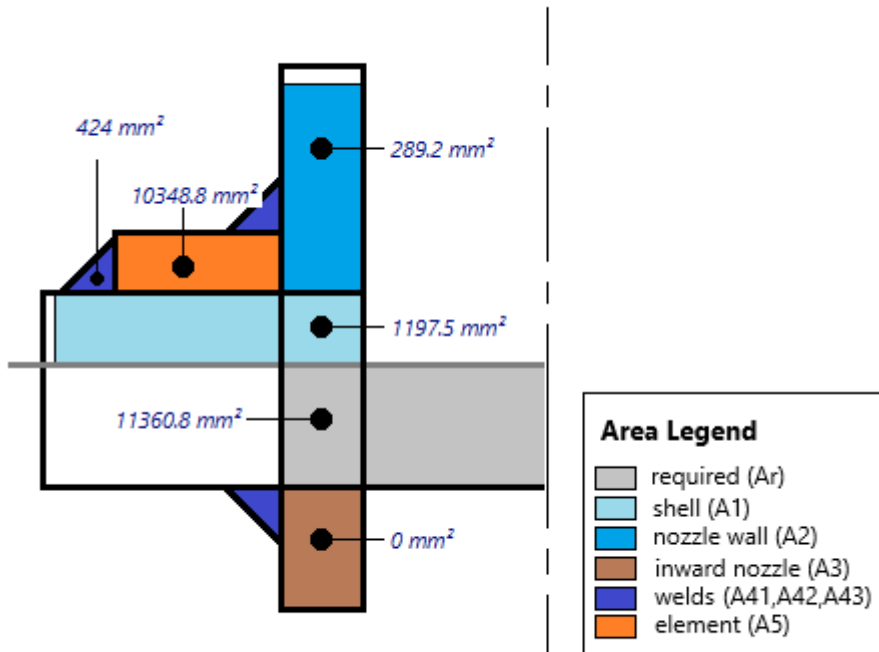


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm2)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	113.608	35.299	NA
Area in Shell	A1	11.975	54.986	NA
Area in Nozzle Wall	A2	2.892	6.373	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	4.240	4.240	NA
Area in Element	A5	103.488	103.488	NA
TOTAL AREA AVAILABLE	Atot	122.594	169.086	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Insufficient.
 The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS: Diameter Thickness
 Based on given Pad Thickness: 1059.1542 22.0000 mm.
 Based on given Pad Diameter: 1100.0000 20.0897 mm.
 Based on the Estimated Diameter Limit: 1223.6125 15.9091 mm.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$
 $= (612.6 \cdot 18.5453 \cdot 1.0 + 2 \cdot 8.5 \cdot 18.5453 \cdot 1.0 \cdot (1 - 1.0))$
 $= 113.608 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 612.6(1.0 \cdot 20.5 - 1.0 \cdot 18.545) - 2 \cdot 8.5$
 $(1.0 \cdot 20.5 - 1.0 \cdot 18.5453) \cdot (1 - 1.0)$
 $= 11.975 \text{ cm}^2$

Area Available in Nozzle Wall Projecting Outward [A2]:
 $= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$
 $= (2 \cdot 43.25) \cdot (8.5 - 5.16) \cdot 1.0$



= 2.892 cm²

Area Available in Welds [A41 + A42 + A43]:
= $W_o^2 \cdot fr3 + (W_i - can/0.707)^2 \cdot fr2 + W_p^2 \cdot fr4$
= $10.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0 + 18.0^2 \cdot 1.0$
= 4.240 cm²

Area Available in Element [A5]:
= $(\min(D_p, DL) - (\text{Nozzle OD})) \cdot (\min(t_p, Tlwp, te)) \cdot fr4$
= $(1100.0 - 629.6) \cdot 22.0 \cdot 1.0$
= 103.488 cm²

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 6.6570 mm.
Wall Thickness per UG16(b),	tr16b = 3.0000 mm.
Wall Thickness, shell/head, internal pressure	trb1 = 20.0453 mm.
Wall Thickness	tb1 = max(trb1, tr16b) = 20.0453 mm.
Wall Thickness, shell/head, external pressure	trb2 = 2.3091 mm.
Wall Thickness	tb2 = max(trb2, tr16b) = 3.0000 mm.
Wall Thickness per table UG-45	tb3 = 9.8312 mm.

Determine Nozzle Thickness candidate [tb]:
= min[tb3, max(tb1, tb2)]
= min[9.831, max(20.0453, 3.0)]
= 9.8312 mm.

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
= max(ta, tb)
= max(6.657, 9.8312)
= 9.8312 mm.

Available Nozzle Neck Thickness = 10.0000 mm. -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Nozzle Neck to Pad Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Nozzle Neck to Pad Weld for Reinf. pad (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Shell to Pad Weld at Pad OD for pad (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Gov. MDMT of the Nozzle	:	-60 °C
Gov. MDMT of the Reinforcement Pad	:	-43 °C
Gov. MDMT of the nozzle to shell joint welded assembly	:	-43 °C

Flange MDMT including Temperature reduction per UCS-66.1:

MDMT of ASME B16.5/47 flange per Matl. Specification	-46 °C
Flange MDMT with Temp reduction per UCS-66(i)(2)	-85 °C

Where the Stress Reduction Ratio per UCS-66(i)(2) is :



Design Pressure/Ambient Rating = 23.00/51.00 = 0.451

Weld Size Calculations, Description: MH124"

Intermediate Calc. for nozzle/shell Welds Tmin 8.5000 mm.
Intermediate Calc. for pad/shell Welds TminPad 19.0000 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	5.9500 = 0.7 * tmin.	7.0700 = 0.7 * Wo mm.
Pad Weld	9.5000 = 0.5 * TminPad	12.7260 = 0.7 * Wp mm.

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
= max(0, (A-A1+2*tn*fr1*(E1*t-tr))Sv)
= max(0, (113.6083 - 11.9747 + 2 * 8.5 * 1.0 *
(1.0 * 20.5 - 18.5453))138)
= 143481.53 Kgf

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
= (A2+A5+A4-(Wi-Can/.707)*fr2)*Sv
= (2.8917 + 103.488 + 4.24 - 0.0 * 1.0) * 138
= 155658.72 Kgf

Weld Load [W2]:
= (A2 + A3 + A4 + (2 * tn * t * fr1)) * Sv
= (2.8917 + 0.0 + 1. + (3.485)) * 138
= 10380.15 Kgf

Weld Load [W3]:
= (A2+A3+A4+A5+(2*tn*t*fr1))*S
= (2.8917 + 0.0 + 4.24 + 103.488 + (3.485)) * 138
= 160562.64 Kgf

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
= (π/2) * Dlo * Wo * 0.49 * Snw
= (3.1416/2.0) * 629.6 * 10.0 * 0.49 * 138
= 68190. Kgf

Shear, Pad Element Weld [Spew]:
= (π/2) * DP * WP * 0.49 * SEW
= (3.1416/2.0) * 1100.0 * 18.0 * 0.49 * 138
= 214448. Kgf

Shear, Nozzle Wall [Snw]:
= (π * (Dlr + Dlo)/4) * (Thk - Can) * 0.7 * Sn
= (3.1416 * 310.55) * (10.0 - 1.5) * 0.7 * 138
= 81684. Kgf

Tension, Pad Groove Weld [Tpgw]:
= (π/2) * Dlo * Wgpn * 0.74 * Seg
= (3.1416/2) * 629.6 * 22.0 * 0.74 * 138
= 226558. Kgf

Tension, Shell Groove Weld [Tngw]:
= (π/2) * Dlo * (T - cas) * 0.74 * Sng
= (3.1416/2.0) * 629.6 * (22.0 - 1.5) * 0.74 * 138
= 211111. Kgf

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
= (SPEW + SNW) = (214448 + 81684) = 296132 Kgf



Failure Path [Path 2-2]:

$$= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw})$$
$$= (68190 + 226558 + 211111 + 0) = 505860 \text{ Kgf}$$

Failure Path [Path 3-3]:

$$= (\text{Spew} + \text{Tngw} + \text{Sinw})$$
$$= (214448 + 211111 + 0) = 425559 \text{ Kgf}$$

Summary of Failure Path Calculations:

Path 1-1 = 296132 Kgf, must exceed W = 143481 Kgf or W1 = 155658 Kgf
Path 2-2 = 505859 Kgf, must exceed W = 143481 Kgf or W2 = 10380 Kgf
Path 3-3 = 425559 Kgf, must exceed W = 143481 Kgf or W3 = 160562 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 24 bars

Nozzle is O.K. for the External Pressure: 1 bars

The Drop for this Nozzle is : 46.0072 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 229.3832 mm.

Percent Elongation Calculations:

% Elongation per Table UG-79-1 $(50 * t_{nom} / R_f * (1 - R_f / R_o))$ 1.614 %

This nozzle or its diameter limit intersects weld seam # [1] on this element. The distance between the nozzle and seam is, 576.73 mm., while the distance from the center of the nozzle to the edge of the diameter limit is: 612.60 mm.

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Input, Nozzle Desc: N31"

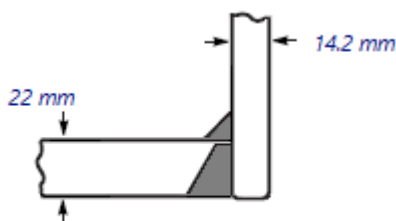
From: 70

Pressure for Reinforcement Calculations	P	23.000	bars
Temperature for Internal Pressure	Temp	80	°C
Design External Pressure	Pext	1.01	bars
Temperature for External Pressure	Tempex	80	°C
Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	138.00	N./mm2
Parent Allowable Stress At Ambient	Sva	138.00	N./mm2
Inside Diameter of Cylindrical Shell	D	2200.00	mm.
Design Length of Section	L	7068.2671	mm.
Shell Finished (Minimum) Thickness	t	22.0000	mm.
Shell Internal Corrosion Allowance	cas	1.5000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		5150.80	mm.
User Entered Minimum Design Metal Temperature		-38.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-350 LF2	+IT
Material UNS Number		K03011	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm2
Allowable Stress At Ambient	Sna	138.00	N./mm2
Diameter Basis (for tr calc only)		Inside	
Layout Angle		90.00	deg
Diameter		1.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	14.2240	mm.
Flange Material		SA-350 LF2	+IT
Flange Type		Long Weld Neck	
Corrosion Allowance	can	1.5000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	200.0250	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	22.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: N31"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45



Actual Inside Diameter Used in Calculation 1.000 in.
Actual Thickness Used in Calculation 0.560 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
= $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $23.0 \cdot 1101.5 / (138 \cdot 1.0 - 0.6 \cdot 23.0)$
= 18.5453 mm.

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
= $R \cdot (\exp([P / (S_n \cdot E)] - 1))$ per Appendix 1-2 (a)(1)
= $14.2 \cdot (\exp([23.0 / (138 \cdot 1.0)] - 1))$
= 0.2387 mm.

Required Nozzle thickness under External Pressure per UG-28 : 0.2953 mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	94.8480 mm.
Parallel to Vessel Wall	Rn+tn+t	47.4240 mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	31.8100 mm.

*Taking a UG-36(c)(3)(a) exemption for nozzle: N3_1".
This calculation is valid for nozzles that meet all the requirements of
paragraph UG-36. Please check the Code carefully, especially for nozzles
that are not isolated or do not meet Code spacing requirements. To force
the computation of areas for small nozzles go to Tools->Configuration
and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 1.7953 mm.
Wall Thickness per UG16(b),	tr16b	= 3.0000 mm.
Wall Thickness, shell/head, internal pressure	trb1	= 20.0453 mm.
Wall Thickness	tb1 = max(trb1, tr16b)	= 20.0453 mm.
Wall Thickness, shell/head, external pressure	trb2	= 2.3091 mm.
Wall Thickness	tb2 = max(trb2, tr16b)	= 3.0000 mm.
Wall Thickness per table UG-45	tb3	= 4.9200 mm.

Determine Nozzle Thickness candidate [tb]:
= min[tb3, max(tb1,tb2)]
= min[4.92, max(20.0453, 3.0)]
= 4.9200 mm.

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
= max(ta, tb)
= max(1.7953, 4.92)
= 4.9200 mm.

Available Nozzle Neck Thickness = 14.2240 mm. -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	:	39.6,	Allowable	:	138.0 N./mm2	Passed
Expansion	:	0.0,	Allowable	:	305.4 N./mm2	Passed
Occasional	:	0.9,	Allowable	:	183.5 N./mm2	Passed
Shear	:	38.5,	Allowable	:	96.6 N./mm2	Passed

*The number of cycles on this nozzle was assumed to be 7000 or less for the determination of
the expansion stress allowable.*

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note:

This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification -46 °C



Calculated Minimum Design Metal Temperature -104 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Weld Size Calculations, Description: N31"

Intermediate Calc. for nozzle/shell Welds Tmin 12.7240 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm.

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 0.3295 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 222.3545 mm.

Input Echo, WRC297 Item 1, Description: N31" :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	1.5000	mm.
Shell External Corrosion Allowance	0.0000	mm.
Shell Diameter	2200.000	mm.
Shell Thickness	22.0000	mm.
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc	138.00 N./mm2
Vessel Hot S.I. Allowable	Smh	138.00 N./mm2

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	1.5000	mm.
Nozzle Diameter	25.400	mm.
Nozzle Thickness	14.2240	mm.
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-350 LF2	
Nozzle Cold S.I. Allowable	SNmc	138.00 N./mm2
Nozzle Hot S.I. Allowable	SNmh	138.00 N./mm2

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Note:

*External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUSstained loads.*

Design Internal Pressure	Dp	23.00	bars
Radial Load	P	-135.00	Kgf
Circumferential Shear	Vc	205.00	Kgf
Longitudinal Shear	Vl	205.00	Kgf
Circumferential Moment	Mc	35.00	Kg-m.
Longitudinal Moment	Ml	45.00	Kg-m.
Torsional Moment	Mt	55.00	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

*Warning - The ratio, Dn/Tv, (2.448) is < 5
 check the Limitations of WRC bulletin 297.*
*Warning - The ratio, Dn/Tn (3.786) is < 10
 check the Limitations of WRC bulletin 297.*

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))}$$

$$= 53.848 + 2 \cdot 1.65 \cdot \sqrt{(1111.75(22.0 - 1.5))}$$

$$= 552.037 \text{ mm.}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter (D) = 2223.500 mm.
 Nozzle Outside Diameter (d) = 53.848 mm.
 Vessel Thickness used (T) = 20.500 mm.
 Nozzle Thickness used (t) = 12.724 mm.
 T / t = 1.611
 d / t = 10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 0.252$$

Nr/P = 0.250
 Mr/P = 0.309
 M0/P = 0.157
 N0/P = 0.187
 MrD/Mc = 0.256
 NrDL/Mc = 0.177
 M0d/Mc = 0.158
 N0DL/Mc = -0.127
 MrD/M1 = 0.251
 NrDL/M1 = 0.180
 M0D/M1 = 0.154
 N0DL/M1 = -0.096

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	0	0	0	0
Outplane Bending (P)	2	-2	2	-2
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	1	1	-1	-1
Outplane Bending (ML)	-18	18	18	-18
Normal Pressure Stress	122	124	122	124
Outplane Stress Summary	109	142	142	102

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	0	0	0	0
Inplane Bending (P)	5	-5	5	-5
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-3	-3	3	3
Inplane Bending (ML)	-29	29	29	-29
Inplane Pressure Stress	61	61	61	61
Inplane Stress Summary	34	82	100	30

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	5	5	5	5
Shear Stress Summary	6	6	4	4

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	110	143	142	102

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	0	0	0	0
Outplane Bending (P)	2	-2	2	-2
Outplane Membrane (Mc)	1	1	-1	-1
Outplane Bending (Mc)	-14	14	14	-14
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	61	61	61	61
Outplane Stress Summary	52	75	77	42

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Inplane Membrane (P)	0	0	0	0
Inplane Bending (P)	5	-5	5	-5
Inplane Membrane (Mc)	-2	-2	2	2
Inplane Bending (Mc)	-23	23	23	-23
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	122	124	122	124
Inplane Stress Summary	103	140	155	99

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-1	-1	1	1
Torsional Shear (Mt)	5	5	5	5
Shear Stress Summary	4	4	6	6

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Two * Max Shear Stress	103	140	155	99

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Hoop Membrane (P)	0	0	0	0
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	1	1	-1	-1
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	2	4	2	4
Hoop Stress Summary	4	7	0	3

Nozzle Stresses (N./mm²):



LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	9	-9	9	-9
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-31	-31	31	31
Axial Bending (ML)	-48	48	48	-48
Axial Pressure Stress	1	1	1	1
Axial Stress Summary	-69	10	90	-24

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)	1	1	-1	-1
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	9	9	9	9
Shear Stress Summary	11	11	7	7

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	77	22	91	32

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	0	0	0	0
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	1	1	-1	-1
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	2	4	2	4
Hoop Stress Summary	4	7	0	3

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	9	-9	9	-9
Axial Membrane (Mc)	-24	-24	24	24
Axial Bending (Mc)	-39	39	39	-39
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	1	1	1	1
Axial Stress Summary	-53	8	74	-22

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-1	-1	1	1
Shear due to Torsion	9	9	9	9
Shear Stress Summary	7	7	11	11



Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	59	15	77	34

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	122	124	122	124	122	124	122	124
Circ. Pl (SUS)	1	1	-1	-1	-2	-2	2	2
Circ. Q (SUS)	-16	16	20	-20	-18	18	28	-28
Long. Pm (SUS)	61	61	61	61	61	61	61	61
Long. Pl (SUS)	-3	-3	3	3	1	1	-1	-1
Long. Q (SUS)	-24	24	34	-34	-12	12	16	-16
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	5	5	5	5	5	5	5	5
Pm (SUS)	122.0	124.0	122.0	124.0	122.0	124.0	122.0	124.0
Pm+Pl (SUS)	123.0	125.0	121.0	123.0	120.0	122.0	124.0	126.0
Pm+Pl+Q (Total)	107.5	141.6	141.4	103.2	102.3	140.2	152.5	98.7

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	124.00	138.00	Passed
Pm+Pl (SUS)	126.02	207.00	Passed
Pm+Pl+Q (TOTAL)	152.47	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	2	4	2	4	2	4	2	4
Circ. Pl (SUS)	1	1	-1	-1	1	1	-1	-1
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	1	1	1	1	1	1	1	1
Long. Pl (SUS)	-31	-31	31	31	-24	-24	24	24
Long. Q (SUS)	-39	39	57	-57	-30	30	48	-48
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	9	9	9	9	9	9	9	9
Pm (SUS)	2.0	4.0	2.0	4.0	2.0	4.0	2.0	4.0
Pm+Pl (SUS)	33.1	35.1	32.0	32.0	26.1	28.1	25.0	25.0
Pm+Pl+Q (Total)	74.7	20.4	89.7	32.2	58.2	16.1	74.7	32.8



Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	4.00	138.00	Passed
Pm+Pl (SUS)	35.06	207.00	Passed
Pm+Pl+Q (TOTAL)	89.72	413.99	Passed

*Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.*

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Input, Nozzle Desc: N116"

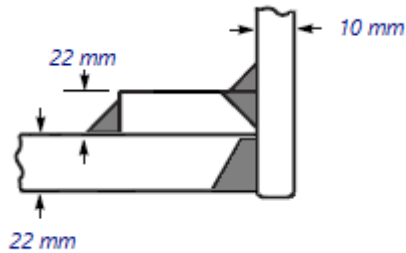
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Pressure for Reinforcement Calculations	P	23.000	bars
Temperature for Internal Pressure	Temp	80	°C
Design External Pressure	Pext	1.01	bars
Temperature for External Pressure	Tempex	80	°C
Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	138.00	N./mm2
Parent Allowable Stress At Ambient	Sva	138.00	N./mm2
Inside Diameter of Cylindrical Shell	D	2200.00	mm.
Design Length of Section	L	7068.2671	mm.
Shell Finished (Minimum) Thickness	t	22.0000	mm.
Shell Internal Corrosion Allowance	cas	1.5000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		4750.80	mm.
User Entered Minimum Design Metal Temperature		-38.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-516 70	+IT
Material UNS Number		K02700	
Material Specification/Type		Plate	
Allowable Stress at Temperature	Sn	138.00	N./mm2
Allowable Stress At Ambient	Sna	138.00	N./mm2
Diameter Basis (for tr calc only)		Outside	
Layout Angle		180.00	deg
Diameter		16.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	10.0000	mm.
Flange Material		SA-350 LF2	+IT
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	1.5000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	83.4741	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	22.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Pad Material		SA-516 70	+IT
Pad Allowable Stress at Temperature	Sp	138.00	N./mm2
Pad Allowable Stress At Ambient	Spa	138.00	N./mm2
Diameter of Pad along vessel surface	Dp	812.8000	mm.
Thickness of Pad	te	22.0000	mm.
Weld leg size between Pad and Shell	Wp	18.0000	mm.
Groove weld depth between Pad and Nozzle	Wgpn	22.0000	mm.
Reinforcing Pad Width		203.2000	mm.
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: N116"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	16.000 in.
Actual Thickness Used in Calculation	0.394 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 23.0 \cdot 1101.5 / (138 \cdot 1.0 - 0.6 \cdot 23.0)$
 $= 18.5453 \text{ mm.}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 23.0 \cdot 203.2 / (138 \cdot 1.0 + 0.4 \cdot 23.0)$
 $= 3.3645 \text{ mm.}$

Required Nozzle thickness under External Pressure per UG-28 : 0.6672 mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	778.8000 mm.
Parallel to Vessel Wall, opening length	d	389.4000 mm.
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		43.2500 mm.

The Pad diameter is greater than the Diameter Limit. The excess will not be considered.

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, S_p / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1.0, 1.0)$
 $= 1.000$

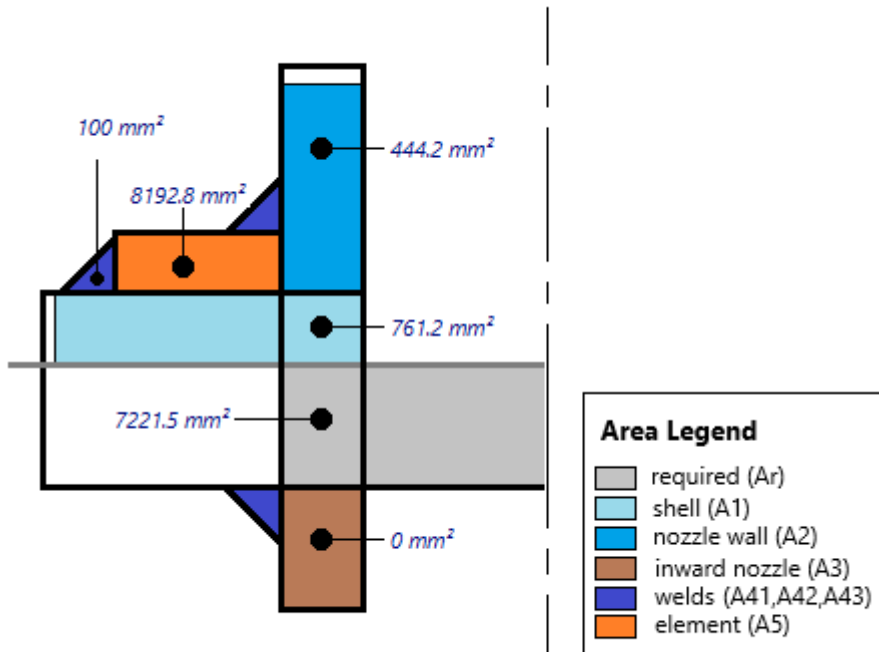


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm2)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	72.215	22.438	NA
Area in Shell	A1	7.612	34.952	NA
Area in Nozzle Wall	A2	4.442	6.775	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	1.000	1.000	NA
Area in Element	A5	81.928	81.928	NA
TOTAL AREA AVAILABLE	Atot	94.982	124.655	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Insufficient.
 The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS: Diameter Thickness
 Based on given Pad Thickness: 675.3152 22.0000 mm.
 Based on given Pad Diameter: 812.8000 15.8865 mm.
 Based on the Estimated Diameter Limit: 777.2125 15.9545 mm.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$
 $= (389.4 \cdot 18.5453 \cdot 1.0 + 2 \cdot 8.5 \cdot 18.5453 \cdot 1.0 \cdot (1 - 1.0))$
 $= 72.215 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 389.4(1.0 \cdot 20.5 - 1.0 \cdot 18.545) - 2 \cdot 8.5$
 $(1.0 \cdot 20.5 - 1.0 \cdot 18.5453) \cdot (1 - 1.0)$
 $= 7.612 \text{ cm}^2$

Area Available in Nozzle Wall Projecting Outward [A2]:
 $= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$
 $= (2 \cdot 43.25) \cdot (8.5 - 3.36) \cdot 1.0$

= 4.442 cm²

Area Available in Welds [A41 + A42 + A43]:
= $W_o^2 \cdot fr_3 + (W_i - can/0.707)^2 \cdot fr_2 + W_p^2 \cdot fr_4$
= $10.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0 + 0.0^2 \cdot 1.0$
= 1.000 cm²

Area Available in Element [A5]:
= $(\min(D_p, DL) - (\text{Nozzle OD})) \cdot (\min(t_p, Tlwp, te)) \cdot fr_4$
= $(778.8 - 406.4) \cdot 22.0 \cdot 1.0$
= 81.928 cm²

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 4.8645 mm.
Wall Thickness per UG16(b),	tr16b = 3.0000 mm.
Wall Thickness, shell/head, internal pressure	trb1 = 20.0453 mm.
Wall Thickness	tb1 = max(trb1, tr16b) = 20.0453 mm.
Wall Thickness, shell/head, external pressure	trb2 = 2.3091 mm.
Wall Thickness	tb2 = max(trb2, tr16b) = 3.0000 mm.
Wall Thickness per table UG-45	tb3 = 9.8312 mm.

Determine Nozzle Thickness candidate [tb]:
= min[tb3, max(tb1, tb2)]
= min[9.831, max(20.0453, 3.0)]
= 9.8312 mm.

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
= max(ta, tb)
= max(4.8645, 9.8312)
= 9.8312 mm.

Available Nozzle Neck Thickness = 10.0000 mm. -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 55.7,	Allowable	: 138.0 N./mm ²	Passed
Expansion	: 0.0,	Allowable	: 289.3 N./mm ²	Passed
Occasional	: 25.8,	Allowable	: 183.5 N./mm ²	Passed
Shear	: 20.2,	Allowable	: 96.6 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Nozzle Neck to Pad Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Nozzle Neck to Pad Weld for Reinf. pad (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Shell to Pad Weld at Pad OD for pad (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Gov. MDMT of the Nozzle : -90 °C
 Gov. MDMT of the Reinforcement Pad : -43 °C
 Gov. MDMT of the nozzle to shell joint welded assembly : -43 °C

Flange MDMT including Temperature reduction per UCS-66.1:

MDMT of ASME B16.5/47 flange per Matl. Specification -46 °C
 Flange MDMT with Temp reduction per UCS-66(i)(2) -85 °C

Where the Stress Reduction Ratio per UCS-66(i)(2) is :
 Design Pressure/Ambient Rating = 23.00/51.00 = 0.451

Weld Size Calculations, Description: N1_16"

Intermediate Calc. for nozzle/shell Welds Tmin 8.5000 mm.
 Intermediate Calc. for pad/shell Welds TminPad 19.0000 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	5.9500 = 0.7 * tmin.	7.0700 = 0.7 * Wo mm.
Pad Weld	9.5000 = 0.5*TminPad	12.7260 = 0.7 * Wp mm.

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - t_r)) \cdot S_v)$
 $= \max(0, (72.2153 - 7.6117 + 2 \cdot 8.5 \cdot 1.0 \cdot (1.0 \cdot 20.5 - 18.5453)) \cdot 138)$
 $= 91374.59 \text{ Kgf}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A5+A4-(W_i-\text{Can}/.707)^2 \cdot f_{r2}) \cdot S_v$
 $= (4.4422 + 81.928 + 1. - 0.0 \cdot 1.0) \cdot 138$
 $= 122943.15 \text{ Kgf}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v$
 $= (4.4422 + 0.0 + 1. + (3.485)) \cdot 138$
 $= 12561.93 \text{ Kgf}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S$
 $= (4.4422 + 0.0 + 1. + 81.928 + (3.485)) \cdot 138$
 $= 127847.07 \text{ Kgf}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$
 $= (3.1416/2.0) \cdot 406.4 \cdot 10.0 \cdot 0.49 \cdot 138$
 $= 44016. \text{ Kgf}$

Shear, Pad Element Weld [Spew]:
 $= (\pi/2) \cdot D_P \cdot W_P \cdot 0.49 \cdot S_{EW}$
 $= (3.1416/2.0) \cdot 812.8 \cdot 18.0 \cdot 0.49 \cdot 138$
 $= 158458. \text{ Kgf}$

Shear, Nozzle Wall [Snw]:
 $= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n$
 $= (3.1416 \cdot 198.95) \cdot (10.0 - 1.5) \cdot 0.7 \cdot 138$
 $= 52330. \text{ Kgf}$

Tension, Pad Groove Weld [Tpgw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_{gpn} \cdot 0.74 \cdot S_{eg}$
 $= (3.1416/2) \cdot 406.4 \cdot 22.0 \cdot 0.74 \cdot 138$
 $= 146241. \text{ Kgf}$

Tension, Shell Groove Weld [Tngw]:
 $= (\pi/2) \cdot Dlo \cdot (T - cas) \cdot 0.74 \cdot Sng$
 $= (3.1416/2.0) \cdot 406.4 \cdot (22.0 - 1.5) \cdot 0.74 \cdot 138$
 $= 136270. \text{ Kg}$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
 $= (SPEW + SNW) = (158458 + 52330) = 210788 \text{ Kg}$

Failure Path [Path 2-2]:
 $= (Sonw + Tpgw + Tngw + Sinw)$
 $= (44016 + 146241 + 136270 + 0) = 326527 \text{ Kg}$

Failure Path [Path 3-3]:
 $= (Spew + Tngw + Sinw)$
 $= (158458 + 136270 + 0) = 294728 \text{ Kg}$

Summary of Failure Path Calculations:

Path 1-1 = 210787 Kg, must exceed W = 91374 Kg or W1 = 122943 Kg
 Path 2-2 = 326526 Kg, must exceed W = 91374 Kg or W2 = 12561 Kg
 Path 3-3 = 294727 Kg, must exceed W = 91374 Kg or W3 = 127847 Kg

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 25 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 1 bars

The Drop for this Nozzle is : 18.9312 mm.
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 124.4053 mm.

Percent Elongation Calculations:

% Elongation per Table UG-79-1 ($50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o)$) 2.523 %

Input Echo, WRC297 Item 1, Description: N116" :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	1.5000	mm.
Shell External Corrosion Allowance	0.0000	mm.
Shell Diameter	2200.000	mm.
Shell Thickness	22.0000	mm.
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc	138.00 N./mm ²
Vessel Hot S.I. Allowable	Smh	138.00 N./mm ²

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Diameter Basis for Nozzle	OD	
Nozzle Corrosion Allowance	1.5000	mm.
Nozzle Diameter	406.400	mm.
Nozzle Thickness	10.0000	mm.
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-516 70	
Nozzle Cold S.I. Allowable	SNmc	138.00 N./mm ²
Nozzle Hot S.I. Allowable	SNmh	138.00 N./mm ²

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Diameter of Reinforcing Pad	812.800	mm.
Thickness of Reinforcing Pad	22.0000	mm.

Note:
 External Forces and Moments in WRC 107/537 Convention:

These loads are assumed to be SUSstained loads.

Design Internal Pressure	Dp	23.00	bars
Radial Load	P	-1105.00	Kgf
Circumferential Shear	Vc	1635.00	Kgf
Longitudinal Shear	Vl	1635.00	Kgf
Circumferential Moment	Mc	1960.00	Kg-m.
Longitudinal Moment	Ml	2614.99	Kg-m.
Torsional Moment	Mt	3265.00	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))}$
 $= 406.4 + 2 \cdot 1.65 \cdot \sqrt{(1122.75(22.0 - 1.5))}$
 $= 907.048 \text{ mm.}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	2245.500 mm.
Nozzle Outside Diameter	(d) =	406.400 mm.
Vessel Thickness used	(T) =	42.500 mm.
Nozzle Thickness used	(t) =	8.500 mm.
T / t	=	5.000
d / t	=	47.812

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 1.316$$

Nr/P	=	0.042
Mr/P	=	0.036
M0/P	=	0.090
N0/P	=	0.295
MrD/Mc	=	0.080
NrDL/Mc	=	0.081
M0d/Mc	=	0.242
N0DL/Mc	=	0.328
MrD/Ml	=	0.064
NrDL/Ml	=	0.065
M0D/Ml	=	0.159
N0DL/Ml	=	0.561

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	1	1	1	1
Outplane Bending (P)	3	-3	3	-3
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (Ml)	-19	-19	19	19
Outplane Bending (Ml)	-33	33	33	-33
Normal Pressure Stress	58	60	58	60
Outplane Stress Summary	10	73	116	45

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	0	0	0	0
Inplane Bending (P)	1	-1	1	-1
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0



Inplane Membrane (ML)	-2	-2	2	2
Inplane Bending (ML)	-13	13	13	-13
Inplane Pressure Stress	29	29	29	29
Inplane Stress Summary	15	39	46	17

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress normal to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	2	2	2	2
Shear Stress Summary	3	3	2	2

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Two * Max Shear Stress	17	73	116	45

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside
Outplane Membrane (P)	1	1	1	1
Outplane Bending (P)	3	-3	3	-3
Outplane Membrane (Mc)	-8	-8	8	8
Outplane Bending (Mc)	-38	38	38	-38
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	29	29	29	29
Outplane Stress Summary	-12	57	80	-1

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses parallel to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside
Inplane Membrane (P)	0	0	0	0
Inplane Bending (P)	1	-1	1	-1
Inplane Membrane (Mc)	-2	-2	2	2
Inplane Bending (Mc)	-12	12	12	-12
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	58	60	58	60
Inplane Stress Summary	45	70	74	49

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Torsional Shear (Mt)	2	2	2	2
Shear Stress Summary	2	2	3	3

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities)	Left Outside	Left Inside	Right Outside	Right Inside



Two * Max Shear Stress 58 70 82 51

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Hoop Membrane (P)	1	1	1	1
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-19	-19	19	19
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	61	64	61	64
Hoop Stress Summary	44	46	83	85

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Axial Membrane (P)	1	1	1	1
Axial Bending (P)	13	-13	13	-13
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-24	-24	24	24
Axial Bending (ML)	-163	163	163	-163
Axial Pressure Stress	30	30	30	30
Axial Stress Summary	-142	156	234	-120

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)	2	2	-2	-2
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	14	14	14	14
Shear Stress Summary	17	17	11	11

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	189	159	234	207

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	1	1	1	1
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-8	-8	8	8
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	61	64	61	64
Hoop Stress Summary	55	57	72	74

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Axial Membrane (P)	1	1	1	1
Axial Bending (P)	13	-13	13	-13



Axial Membrane (Mc)	-18	-18	18	18
Axial Bending (Mc)	-154	154	154	-154
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	30	30	30	30

Axial Stress Summary	-127	154	219	-118
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE				
(Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-2	-2	2	2
Shear due to Torsion	14	14	14	14
Shear Stress Summary	11	11	17	17

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE				
(Stress Intensities	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	183	155	221	196

Stress Computations at the Edge of the Reinforcing Pad:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))}$
 $= 406.4 + 2 \cdot 1.65 \cdot \sqrt{(1111.75(22.0 - 1.5))}$
 $= 904.589 \text{ mm.}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	2223.500 mm.
Nozzle Outside Diameter	(d) =	812.800 mm.
Vessel Thickness used	(T) =	20.500 mm.
Nozzle Thickness used	(t) =	211.700 mm.
T / t	=	0.097
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 3.807$$

Note: Re-pad thickness is added to nozzle thickness to simulate the Re-pad.

Nr/P	=	0.054
Mr/P	=	0.088
M0/P	=	0.025
N0/P	=	0.035
MrD/Mc	=	0.255
NrDL/Mc	=	0.106
M0d/Mc	=	0.076
N0DL/Mc	=	0.060
MrD/Ml	=	0.064
NrDL/Ml	=	0.064
M0D/Ml	=	0.019
N0DL/Ml	=	0.040

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE				
(Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	0	0	0	0
Outplane Bending (P)	3	-3	3	-3
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-3	-3	3	3
Outplane Bending (ML)	-8	8	8	-8
Normal Pressure Stress	122	124	122	124



Outplane Stress Summary 115 127 138 116

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE
 (Stresses parallel to
 longitudinal plane)

	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
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Inplane Membrane (P)	1	1	1	1
Inplane Bending (P)	13	-13	13	-13
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-4	-4	4	4
Inplane Bending (ML)	-28	28	28	-28
Inplane Pressure Stress	61	61	61	61

Inplane Stress Summary 42 73 109 25

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE
 (Shear stress normal to
 longitudinal plane)

	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
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Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	1	1	1	1

Shear Stress Summary 2 2 0 0

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE
 (Stress Intensities)

	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
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Two * Max Shear Stress 115 127 138 116

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE
 (Stresses Normal to
 circumferential plane)

	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
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Outplane Membrane (P)	0	0	0	0
Outplane Bending (P)	3	-3	3	-3
Outplane Membrane (Mc)	-3	-3	3	3
Outplane Bending (Mc)	-25	25	25	-25
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	61	61	61	61

Outplane Stress Summary 36 80 95 35

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE
 (Stresses parallel to
 circumferential plane)

	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
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Inplane Membrane (P)	1	1	1	1
Inplane Bending (P)	13	-13	13	-13
Inplane Membrane (Mc)	-5	-5	5	5
Inplane Bending (Mc)	-86	86	86	-86
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	122	124	122	124

Inplane Stress Summary 45 192 229 32

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE
 (Shear stress normal to
 circumferential plane)

	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
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Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Torsional Shear (Mt)	1	1	1	1

Shear Stress Summary	0	0	2	2
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Two * Max Shear Stress	45	192	229	36

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	58	60	58	60	58	60	58	60
Circ. Pl (SUS)	-18	-18	20	20	-2	-2	2	2
Circ. Q (SUS)	-30	30	36	-36	-11	11	13	-13
Long. Pm (SUS)	29	29	29	29	29	29	29	29
Long. Pl (SUS)	-2	-2	2	2	-7	-7	9	9
Long. Q (SUS)	-12	12	14	-14	-35	35	41	-41
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	2	2	2	2	2	2	2	2
Pm (SUS)	58.0	60.0	58.0	60.0	58.0	60.0	58.0	60.0
Pm+Pl (SUS)	40.0	42.0	78.0	80.0	56.0	58.0	60.0	62.0
Pm+Pl+Q (Total)	15.7	72.1	114.1	44.1	58.1	69.3	79.6	52.2

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	60.00	138.00	Passed
Pm+Pl (SUS)	80.00	207.00	Passed
Pm+Pl+Q (TOTAL)	114.06	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Reinforcing Pad Edge (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	122	124	122	124	122	124	122	124
Circ. Pl (SUS)	-3	-3	3	3	-4	-4	6	6
Circ. Q (SUS)	-5	5	11	-11	-73	73	99	-99
Long. Pm (SUS)	61	61	61	61	61	61	61	61
Long. Pl (SUS)	-3	-3	5	5	-3	-3	3	3
Long. Q (SUS)	-15	15	41	-41	-22	22	28	-28
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0



Shear Q (SUS)	1	1	1	1	1	1	1	1
Pm (SUS)	122.0	124.0	122.0	124.0	122.0	124.0	122.0	124.0
Pm+Pl (SUS)	119.0	121.0	125.0	127.0	118.0	120.0	128.0	130.0
Pm+Pl+Q (Total)	114.0	126.0	136.0	116.0	45.1	193.0	227.0	36.2

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	124.00	138.00	Passed
Pm+Pl (SUS)	130.00	207.00	Passed
Pm+Pl+Q (TOTAL)	227.01	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	61	64	61	64	61	64	61	64
Circ. Pl (SUS)	-18	-18	20	20	-7	-7	9	9
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	30	30	30	30	30	30	30	30
Long. Pl (SUS)	-23	-23	25	25	-17	-17	19	19
Long. Q (SUS)	-150	150	176	-176	-141	141	167	-167
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	2	2	-2	-2	-2	-2	2	2
Shear Q (SUS)	14	14	14	14	14	14	14	14
Pm (SUS)	61.0	64.0	61.0	64.0	61.0	64.0	61.0	64.0
Pm+Pl (SUS)	43.1	46.1	81.2	84.1	54.1	57.1	70.2	73.2
Pm+Pl+Q (Total)	188.7	159.3	232.0	206.4	183.6	155.5	217.7	193.7

Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	64.00	138.00	Passed
Pm+Pl (SUS)	84.14	207.00	Passed
Pm+Pl+Q (TOTAL)	231.95	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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Input, Nozzle Desc: N216"

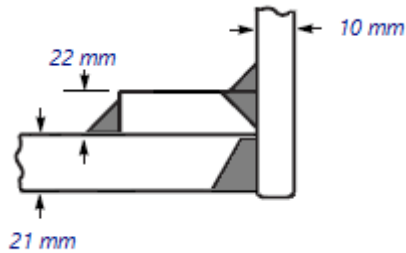
From: 80

Pressure for Reinforcement Calculations	P	23.000	bars
Temperature for Internal Pressure	Temp	80	°C
Design External Pressure	Pext	1.01	bars
Temperature for External Pressure	Tempex	80	°C
Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	138.00	N./mm2
Parent Allowable Stress At Ambient	Sva	138.00	N./mm2
Inside Diameter of Elliptical Head	D	2200.00	mm.
Aspect Ratio of Elliptical Head	Ar	2.00	
Head Finished (Minimum) Thickness	t	21.0000	mm.
Head Internal Corrosion Allowance	cas	1.5000	mm.
Head External Corrosion Allowance	co	0.0000	mm.
Distance from Head Centerline	L1	0.0000	mm.
User Entered Minimum Design Metal Temperature		-38.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-516 70	+IT
Material UNS Number		K02700	
Material Specification/Type		Plate	
Allowable Stress at Temperature	Sn	138.00	N./mm2
Allowable Stress At Ambient	Sna	138.00	N./mm2
Diameter Basis (for tr calc only)		Outside	
Layout Angle		0.00	deg
Diameter		16.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	10.0000	mm.
Flange Material		SA-350 LF2	+IT
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	1.5000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	383.6740	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	21.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Pad Material		SA-516 70	+IT
Pad Allowable Stress at Temperature	Sp	138.00	N./mm2
Pad Allowable Stress At Ambient	Spa	138.00	N./mm2
Diameter of Pad along vessel surface	Dp	700.0000	mm.
Thickness of Pad	te	22.0000	mm.
Weld leg size between Pad and Shell	Wp	18.0000	mm.
Groove weld depth between Pad and Nozzle	Wgpn	22.0000	mm.
Reinforcing Pad Width		146.8000	mm.
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: N216"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	16.000 in.
Actual Thickness Used in Calculation	0.394 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]
 $= P \cdot D \cdot K1 / (2 \cdot Sv \cdot E - 0.2 \cdot P)$ per Appendix 1-4(c)
 $= 23.0 \cdot 2203.0 \cdot 0.8988 / (2 \cdot 138.0 \cdot 1.0 - 0.2 \cdot 23.0)$
 $= 16.5299$ mm.

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot Ro / (Sn \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 23.0 \cdot 203.2 / (138 \cdot 1.0 + 0.4 \cdot 23.0)$
 $= 3.3645$ mm.

Required Nozzle thickness under External Pressure per UG-28 : 1.2600 mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	778.8000 mm.
Parallel to Vessel Wall, opening length	d	389.4000 mm.
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		43.2500 mm.

Weld Strength Reduction Factor [fr1]:
 $= \min(1, Sn/Sv)$
 $= \min(1, 138.0/138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, Sn/Sv)$
 $= \min(1, 138.0/138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, Sp/Sv)$
 $= \min(1, 138.0/138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1.0, 1.0)$
 $= 1.000$

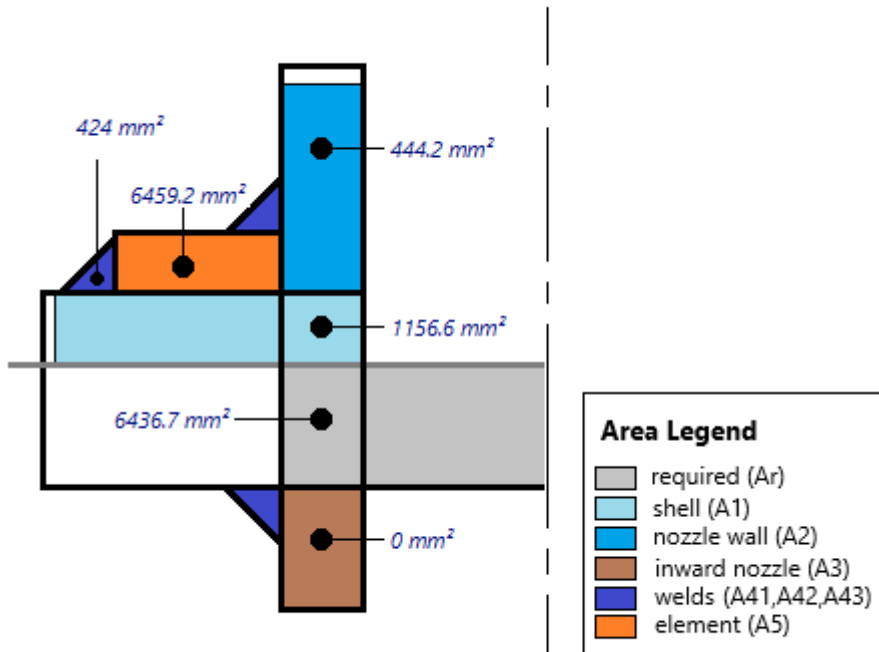


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm2)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	64.367	11.187	NA
Area in Shell	A1	11.566	53.559	NA
Area in Nozzle Wall	A2	4.442	6.263	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	4.240	4.240	NA
Area in Element	A5	64.592	64.592	NA
TOTAL AREA AVAILABLE	Atot	84.840	128.654	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Insufficient.
 The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS: Diameter Thickness
 Based on given Pad Thickness: 606.9440 22.0000 mm.
 Based on given Pad Diameter: 700.0000 15.0271 mm.
 Based on the Estimated Diameter Limit: 777.2125 11.8981 mm.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$
 $= (389.4 \cdot 16.5299 \cdot 1.0 + 2 \cdot 8.5 \cdot 16.5299 \cdot 1.0 \cdot (1 - 1.0))$
 $= 64.367 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 389.4(1.0 \cdot 19.5 - 1.0 \cdot 16.53) - 2 \cdot 8.5$
 $(1.0 \cdot 19.5 - 1.0 \cdot 16.5299) \cdot (1 - 1.0)$
 $= 11.566 \text{ cm}^2$

Area Available in Nozzle Wall Projecting Outward [A2]:
 $= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$
 $= (2 \cdot 43.25) \cdot (8.5 - 3.36) \cdot 1.0$

= 4.442 cm²

Area Available in Welds [A41 + A42 + A43]:
= $W_o^2 \cdot fr_3 + (W_i - can/0.707)^2 \cdot fr_2 + W_p^2 \cdot fr_4$
= $10.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0 + 18.0^2 \cdot 1.0$
= 4.240 cm²

Area Available in Element [A5]:
= $(\min(D_p, DL) - (\text{Nozzle OD})) \cdot (\min(t_p, Tlwp, te)) \cdot fr_4$
= $(700.0 - 406.4) \cdot 22.0 \cdot 1.0$
= 64.592 cm²

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 4.8645 mm.
Wall Thickness per UG16(b),	tr16b = 3.0000 mm.
Wall Thickness, shell/head, internal pressure	trb1 = 19.8571 mm.
Wall Thickness	tb1 = max(trb1, tr16b) = 19.8571 mm.
Wall Thickness, shell/head, external pressure	trb2 = 2.3074 mm.
Wall Thickness	tb2 = max(trb2, tr16b) = 3.0000 mm.
Wall Thickness per table UG-45	tb3 = 9.8312 mm.

Determine Nozzle Thickness candidate [tb]:
= min[tb3, max(tb1, tb2)]
= min[9.831, max(19.8571, 3.0)]
= 9.8312 mm.

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
= max(ta, tb)
= max(4.8645, 9.8312)
= 9.8312 mm.

Available Nozzle Neck Thickness = 10.0000 mm. -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 55.7,	Allowable	: 138.0 N./mm ²	Passed
Expansion	: 0.0,	Allowable	: 289.3 N./mm ²	Passed
Occasional	: 25.8,	Allowable	: 183.5 N./mm ²	Passed
Shear	: 20.2,	Allowable	: 96.6 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Nozzle Neck to Pad Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Nozzle Neck to Pad Weld for Reinf. pad (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Shell to Pad Weld at Pad OD for pad (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Gov. MDMT of the Nozzle : -90 °C
 Gov. MDMT of the Reinforcement Pad : -46 °C
 Gov. MDMT of the nozzle to shell joint welded assembly : -46 °C

Flange MDMT including Temperature reduction per UCS-66.1:

MDMT of ASME B16.5/47 flange per Matl. Specification -46 °C
 Flange MDMT with Temp reduction per UCS-66(i)(2) -85 °C

Where the Stress Reduction Ratio per UCS-66(i)(2) is :
 Design Pressure/Ambient Rating = 23.00/51.00 = 0.451

Weld Size Calculations, Description: N216"

Intermediate Calc. for nozzle/shell Welds Tmin 8.5000 mm.
 Intermediate Calc. for pad/shell Welds TminPad 19.0000 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	5.9500 = 0.7 * tmin.	7.0700 = 0.7 * Wo mm.
Pad Weld	9.5000 = 0.5 * TminPad	12.7260 = 0.7 * Wp mm.

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot t_n \cdot f_r1 \cdot (E1 \cdot t - t_r)) \cdot S_v)$
 $= \max(0, (64.3674 - 11.5656 + 2 \cdot 8.5 \cdot 1.0 \cdot (1.0 \cdot 19.5 - 16.5299)) \cdot 138)$
 $= 75010.77 \text{ Kgf}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A5+A4-(W_i-\text{Can}/.707)^2 \cdot f_r2) \cdot S_v$
 $= (4.4422 + 64.592 + 4.24 - 0.0 \cdot 1.0) \cdot 138$
 $= 103107.91 \text{ Kgf}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_r1)) \cdot S_v$
 $= (4.4422 + 0.0 + 1. + (3.315)) \cdot 138$
 $= 12322.71 \text{ Kgf}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_r1)) \cdot S$
 $= (4.4422 + 0.0 + 4.24 + 64.592 + (3.315)) \cdot 138$
 $= 107772.62 \text{ Kgf}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$
 $= (3.1416/2.0) \cdot 406.4 \cdot 10.0 \cdot 0.49 \cdot 138$
 $= 44016. \text{ Kgf}$

Shear, Pad Element Weld [Spew]:
 $= (\pi/2) \cdot D_P \cdot W_P \cdot 0.49 \cdot S_{EW}$
 $= (3.1416/2.0) \cdot 700.0 \cdot 18.0 \cdot 0.49 \cdot 138$
 $= 136467. \text{ Kgf}$

Shear, Nozzle Wall [Snw]:
 $= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n$
 $= (3.1416 \cdot 198.95) \cdot (10.0 - 1.5) \cdot 0.7 \cdot 138$
 $= 52330. \text{ Kgf}$

Tension, Pad Groove Weld [Tpgw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_{gpn} \cdot 0.74 \cdot S_{eg}$
 $= (3.1416/2) \cdot 406.4 \cdot 22.0 \cdot 0.74 \cdot 138$
 $= 146241. \text{ Kgf}$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot Dlo \cdot (T - cas) \cdot 0.74 \cdot Sng$$

$$= (3.1416/2.0) \cdot 406.4 \cdot (21.0 - 1.5) \cdot 0.74 \cdot 138$$

$$= 129623. \text{ Kgf}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SPEW + SNW) = (136467 + 52330) = 188797 \text{ Kgf}$$

Failure Path [Path 2-2]:

$$= (Sonw + Tpgw + Tngw + Sinw)$$

$$= (44016 + 146241 + 129623 + 0) = 319880 \text{ Kgf}$$

Failure Path [Path 3-3]:

$$= (Spew + Tngw + Sinw)$$

$$= (136467 + 129623 + 0) = 266090 \text{ Kgf}$$

Summary of Failure Path Calculations:

Path 1-1 = 188797 Kgf, must exceed W = 75010 Kgf or W1 = 103107 Kgf
 Path 2-2 = 319879 Kgf, must exceed W = 75010 Kgf or W2 = 12322 Kgf
 Path 3-3 = 266089 Kgf, must exceed W = 75010 Kgf or W3 = 107772 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 24 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 1 bars

The Drop for this Nozzle is : 10.3660 mm.
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 415.0401 mm.

Percent Elongation Calculations:

% Elongation per Table UG-79-1 ($50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o)$) 2.523 %

Input Echo, WRC107/537 Item 1, Description: N216" :

Diameter Basis for Vessel	Vbasis	ID	
Cylindrical or Spherical Vessel	Cylsph	Spherical	
Internal Corrosion Allowance	Cas	1.5000	mm.
Vessel Diameter	Dv	3960.000	mm.
Vessel Thickness	Tv	21.000	mm.
Design Temperature	T1	80.0	C
Vessel Material		SA-516 70	
Vessel UNS Number		K02700	
Vessel Cold S.I. Allowable	Smc	138.00	N./mm ²
Vessel Hot S.I. Allowable	Smh	138.00	N./mm ²

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Attachment Type	Type	Round	
WRC107 Attachment Classification	Holsol	Hollow	
Diameter Basis for Nozzle	Nbasis	OD	
Corrosion Allowance for Nozzle	Can	1.5000	mm.
Nozzle Diameter	Dn	406.400	mm.
Nozzle Thickness	Tn	10.000	mm.
Nozzle Material		SA-516 70	
Nozzle UNS Number		K02700	
Nozzle Cold S.I. Allowable	SNmc	138.00	N./mm ²
Nozzle Hot S.I. Allowable	SNmh	138.00	N./mm ²
Thickness of Reinforcing Pad	Tpad	22.000	mm.
Diameter of Reinforcing Pad	Dpad	700.000	mm.

Design Internal Pressure Dp 23.000 bars
 Include Pressure Thrust No

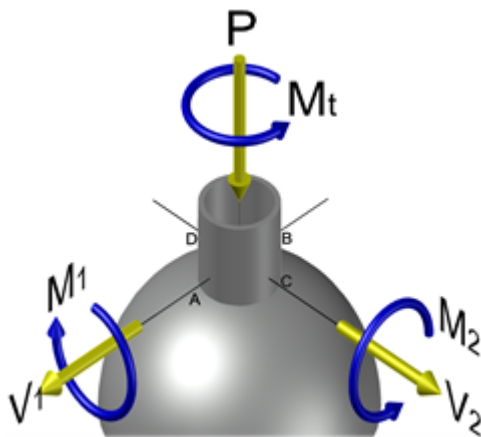
External Forces and Moments in WRC 107/537 Convention:

Radial Load	(SUS)	P	-1105.0	Kgf
Longitudinal Shear	(SUS)	(V1) V1	1635.0	Kgf
Circumferential Shear	(SUS)	(Vc) V2	1635.0	Kgf
Circumferential Moment	(SUS)	(Mc) M1	2615.0	Kg-m.
Longitudinal Moment	(SUS)	(Ml) M2	1960.0	Kg-m.
Torsional Moment	(SUS)	Mt	3265.0	Kg-m.

Use Interactive Control No
 WRC107 Version Version March 1979

Include Pressure Stress Indices per Div. 2 No
 Compute Pressure Stress per WRC-368 No
 Local Loads applied at end of Nozzle/Attachment No

WRC Bulletin 537 provides equations for the dimensionless curves found in bulletin 107. As noted in the foreword to bulletin 537, "537 is equivalent to WRC 107". Where 107 is printed in the results below, "537" can be interchanged with "107".



WRC 107/537 has dimensional limitations that can be found in the bulletin's Foreword and which can also be inferred by review of the non-dimensional curve limits.

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(\text{t} - \text{ca}))}$$

$$= 406.4 + 2 \cdot 1.65 \cdot \sqrt{(1991.25 (21.0 - 1.5))}$$

$$= 1056.671 \text{ mm.}$$

WRC 107 Stress Calculation for SUStained loads:

Radial Load		P	-1105.0	Kgf
Circumferential Shear	(VC)	V2	1635.0	Kgf
Longitudinal Shear	(VL)	V1	1635.0	Kgf
Circumferential Moment	(MC)	M1	2615.0	Kg-m.
Longitudinal Moment	(ML)	M2	1960.0	Kg-m.
Torsional Moment		MT	3265.0	Kg-m.

Dimensionless Parameters: U = 0.70 TAU = 23.41 RHO = 4.88

Dimensionless Loads for Spherical Shells at Attachment Junction:

Curves read for 1979	Figure	Value	Location
$N(x) \cdot T / P$	SP 7	0.02756	(A,B,C,D)
$M(x) / P$	SP 7	0.01255	(A,B,C,D)
$N(x) \cdot T \cdot \text{SQRT}(R_m \cdot T) / MC$	SM 7	0.06249	(A,B,C,D)
$M(x) \cdot \text{SQRT}(R_m \cdot T) / MC$	SM 7	0.02868	(A,B,C,D)
$N(x) \cdot T \cdot \text{SQRT}(R_m \cdot T) / ML$	SM 7	0.06249	(A,B,C,D)
$M(x) \cdot \text{SQRT}(R_m \cdot T) / ML$	SM 7	0.02868	(A,B,C,D)



N(y) * T / P	SP 7	0.26845	(A,B,C,D)
M(y) / P	SP 7	0.04769	(A,B,C,D)
N(y) * T * SQRT(Rm * T) / MC	SM 7	0.35482	(A,B,C,D)
M(y) * SQRT(Rm * T) / MC	SM 7	0.12963	(A,B,C,D)
N(y) * T * SQRT(Rm * T) / ML	SM 7	0.35482	(A,B,C,D)
M(y) * SQRT(Rm * T) / ML	SM 7	0.12963	(A,B,C,D)

Stress Concentration Factors: $K_n = 1.00$, $K_b = 1.00$

Stresses in the Vessel at the Attachment Junction (N./mm²)

Type of Stress Load		Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad.	Memb. P	0.2	0.2	0.2	0.2	0.2	0.2	0.2	0.2
Rad.	Bend. P	0.5	-0.5	0.5	-0.5	0.5	-0.5	0.5	-0.5
Rad.	Memb. MC	0.0	0.0	0.0	0.0	-3.2	-3.2	3.2	3.2
Rad.	Bend. MC	0.0	0.0	0.0	0.0	-8.9	8.9	8.9	-8.9
Rad.	Memb. ML	-2.4	-2.4	2.4	2.4	0.0	0.0	0.0	0.0
Rad.	Bend. ML	-6.7	6.7	6.7	-6.7	0.0	0.0	0.0	0.0
Tot. Rad. Str.		-8.4	3.9	9.7	-4.5	-11.5	5.4	12.8	-6.0
Tang. Memb. P		1.7	1.7	1.7	1.7	1.7	1.7	1.7	1.7
Tang. Bend. P		1.8	-1.8	1.8	-1.8	1.8	-1.8	1.8	-1.8
Tang. Memb. MC		0.0	0.0	0.0	0.0	-18.3	-18.3	18.3	18.3
Tang. Bend. MC		0.0	0.0	0.0	0.0	-40.2	40.2	40.2	-40.2
Tang. Memb. ML		-13.7	-13.7	13.7	13.7	0.0	0.0	0.0	0.0
Tang. Bend. ML		-30.1	30.1	30.1	-30.1	0.0	0.0	0.0	0.0
Tot. Tang. Str.		-40.4	16.3	47.3	-16.5	-55.0	21.7	62.0	-22.0
Shear VC		0.6	0.6	-0.6	-0.6	0.0	0.0	0.0	0.0
Shear VL		0.0	0.0	0.0	0.0	-0.6	-0.6	0.6	0.6
Shear MT		3.0	3.0	3.0	3.0	3.0	3.0	3.0	3.0
Tot. Shear		3.6	3.6	2.4	2.4	2.4	2.4	3.6	3.6
Str. Int.		40.8	17.2	47.5	16.9	55.1	22.1	62.3	22.7

Unitless Prm: U = 1.78 TAU = 0.00 (40.68) RHO = 0.00 (2.29)

Dimensionless Loads for Spherical Shells at Pad edge:

Curves read for 1979	Figure	Value	Location
$N(x) * T / P$	SR 2	0.02955	(A,B,C,D)
$M(x) / P$	SR 2	0.01416	(A,B,C,D)
$N(x) * T * \text{SQRT}(R_m * T) / MC$	SR 3	0.02695	(A,B,C,D)
$M(x) * \text{SQRT}(R_m * T) / MC$	SR 3	0.01832	(A,B,C,D)
$N(x) * T * \text{SQRT}(R_m * T) / ML$	SR 3	0.02695	(A,B,C,D)
$M(x) * \text{SQRT}(R_m * T) / ML$	SR 3	0.01832	(A,B,C,D)
$N(y) * T / P$	SR 2	0.00885	(A,B,C,D)
$M(y) / P$	SR 2	0.00425	(A,B,C,D)
$N(y) * T * \text{SQRT}(R_m * T) / MC$	SR 3	0.00809	(A,B,C,D)
$M(y) * \text{SQRT}(R_m * T) / MC$	SR 3	0.00538	(A,B,C,D)
$N(y) * T * \text{SQRT}(R_m * T) / ML$	SR 3	0.00809	(A,B,C,D)
$M(y) * \text{SQRT}(R_m * T) / ML$	SR 3	0.00538	(A,B,C,D)

Stress Concentration Factors: $K_n = 1.00$, $K_b = 1.00$

Stresses in the Vessel at the Edge of Reinforcing Pad (N./mm²)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	DL
Rad.	Memb. P	0.8	0.8	0.8	0.8	0.8	0.8	0.8	0.8



Rad. Bend. P	2.4	-2.4	2.4	-2.4	2.4	-2.4	2.4	-2.4
Rad. Memb. MC	0.0	0.0	0.0	0.0	-9.2	-9.2	9.2	9.2
Rad. Bend. MC	0.0	0.0	0.0	0.0	-37.6	37.6	37.6	-37.6
Rad. Memb. ML	-6.9	-6.9	6.9	6.9	0.0	0.0	0.0	0.0
Rad. Bend. ML	-28.2	28.2	28.2	-28.2	0.0	0.0	0.0	0.0
Tot. Rad. Str.	-31.8	19.7	38.4	-22.9	-43.6	26.8	50.1	-30.0
Tang. Memb. P	0.3	0.3	0.3	0.3	0.3	0.3	0.3	0.3
Tang. Bend. P	0.7	-0.7	0.7	-0.7	0.7	-0.7	0.7	-0.7
Tang. Memb. MC	0.0	0.0	0.0	0.0	-2.8	-2.8	2.8	2.8
Tang. Bend. MC	0.0	0.0	0.0	0.0	-11.1	11.1	11.1	-11.1
Tang. Memb. ML	-2.1	-2.1	2.1	2.1	0.0	0.0	0.0	0.0
Tang. Bend. ML	-8.3	8.3	8.3	-8.3	0.0	0.0	0.0	0.0
Tot. Tang. Str.	-9.4	5.7	11.3	-6.7	-12.8	7.8	14.8	-8.8
Shear VC	0.7	0.7	-0.7	-0.7	0.0	0.0	0.0	0.0
Shear VL	0.0	0.0	0.0	0.0	-0.7	-0.7	0.7	0.7
Shear MT	2.1	2.1	2.1	2.1	2.1	2.1	2.1	2.1
Tot. Shear	2.9	2.9	1.4	1.4	1.4	1.4	2.9	2.9
Str. Int.	32.2	20.3	38.4	23.0	43.6	26.9	50.3	30.4

WRC 107/537 Stress Summations:

Vessel Stress Summation at Attachment Junction (N./mm²)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad. Pm (SUS)		54.9	54.9	54.9	54.9	54.9	54.9	54.9	54.9
Rad. Pl (SUS)		-2.2	-2.2	2.6	2.6	-3.1	-3.1	3.4	3.4
Rad. Q (SUS)		-6.2	6.2	7.1	-7.1	-8.4	8.4	9.4	-9.4
Long. Pm (SUS)		54.9	54.9	54.9	54.9	54.9	54.9	54.9	54.9
Long. Pl (SUS)		-12.0	-12.0	15.4	15.4	-16.6	-16.6	20.0	20.0
Long. Q (SUS)		-28.3	28.3	31.9	-31.9	-38.4	38.4	42.0	-42.0
Shear Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear Pl (SUS)		0.6	0.6	-0.6	-0.6	-0.6	-0.6	0.6	0.6
Shear Q (SUS)		3.0	3.0	3.0	3.0	3.0	3.0	3.0	3.0
Pm (SUS)		54.9	54.9	54.9	54.9	54.9	54.9	54.9	54.9
Pm+Pl (SUS)		52.7	52.7	70.4	70.4	51.9	51.9	75.0	75.0
Pm+Pl+Q (Total)		46.9	72.2	102.4	50.8	43.8	77.0	117.2	49.7

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	54.93	138.00	Passed
Pm+Pl (SUS)	74.97	207.00	Passed
Pm+Pl+Q (TOTAL)	117.18	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 107/537 Stress Summations:

Vessel Stress Summation at Reinforcing Pad Edge (N./mm²)



Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad. Pm (SUS)		116.9	116.9	116.9	116.9	116.9	116.9	116.9	116.9
Rad. Pl (SUS)		-6.1	-6.1	7.8	7.8	-8.4	-8.4	10.1	10.1
Rad. Q (SUS)		-25.8	25.8	30.6	-30.6	-35.2	35.2	40.0	-40.0
Long. Pm (SUS)		116.9	116.9	116.9	116.9	116.9	116.9	116.9	116.9
Long. Pl (SUS)		-1.8	-1.8	2.3	2.3	-2.5	-2.5	3.0	3.0
Long. Q (SUS)		-7.6	7.6	9.0	-9.0	-10.3	10.3	11.8	-11.8
Shear Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear Pl (SUS)		0.7	0.7	-0.7	-0.7	-0.7	-0.7	0.7	0.7
Shear Q (SUS)		2.1	2.1	2.1	2.1	2.1	2.1	2.1	2.1
Pm (SUS)		116.9	116.9	116.9	116.9	116.9	116.9	116.9	116.9
Pm+Pl (SUS)		115.2	115.2	124.7	124.7	114.4	114.4	127.0	127.0
Pm+Pl+Q (Total)		107.9	137.2	155.3	110.3	104.1	143.8	167.2	108.5

Vessel Stress Summation Comparison (N./mm2):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	116.87	138.00	Passed
Pm+Pl (SUS)	127.01	207.00	Passed
Pm+Pl+Q (TOTAL)	167.21	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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